

OpenAI's mathematical release

OPDP difficulty scores and archive comparison

722

manuscripts

372

result families

17

subjects

Release examined

OpenAI announced this collection on 6 October 2026. This report freezes openai/math at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Every manuscript in that catalogue has an individual entry.

Archive examined

The OPDP comparison uses the 102,819-record v1.8 snapshot at commit `e0e18186a142fb4f61cfc06716aac4fb31bc3b81`. The OpenMath competition archive is assessed separately where result scopes are relevant.

Reference

The format follows the eight-page OpenMath judging reference dated 6 October: mathematical scope, difficulty D and progress-sensitive points. Manuscripts in OpenAI's release have different stages of verification.

Reading the scores

D is the 0-1000 OPDP full-solution difficulty of the stated target. Applicable published values are retained; other values are provisional estimates. Points are conditional calculations subject to statement, novelty, proof and family review.

Authorship

Alejandro Zarzuelo Urdiales, creator and author of OPDP, with ChatGPT-assisted preparation. This research assessment does not represent a signed decision by the OpenMath judges.

[OpenAI announcement](#) | [Pinned collection](#)

Scope comes before score

Archive relation

Same target requires compatible objects, hypotheses, quantifiers and conclusion. A subcase, improvement, related theorem or shared source is recorded separately. An unresolved match establishes neither identity nor absence.

Difficulty basis

Published original CFSD-1000 values cover 5,426 targets. Later archive dimensions can support a reconstructed score with their original confidence limits. New or differently scoped claims receive editorial estimates; a related parent's difficulty is not automatically transferred.

Conditional OpenMath points

For a non-focus original target, $F = m \times p \times D^2 / 1000$. The reference uses $m=1$ for original targets, $m=0.5$ for admitted variations, and p for logical progress. Each entry states the scenario; no accepted event points are assigned.

Progress factors

$p=1$ means completion of the precise target; 0.15 means a substantial structural result or infinite subcase; 0.05 means a quantitative or intermediate advance; 0.01 means a limited construction or obstruction. $p=0$ gives no new open-problem credit when the manuscript expressly reconstructs an acknowledged existing theorem. These are indicative review judgments.

Family accounting

Papers may supply principal results, companions, consequences or alternative proofs. Their points cannot be summed as separate accomplishments. The upstream 372-family grouping may itself need overlap review.

Registration and verification

The handbook requires three independent difficulty assessments for a new admitted target, followed by qualified mathematical and formal review. These provisional estimates support that process. A Lean link records source material; it does not establish a fresh successful kernel replay here.

One reproducible difficulty scale

The fixed CFSD-1000 protocol concerns ChatGPT 5.6 Sol Ultra in a tool-enabled research campaign with a nominal 100-agent-hour ceiling. It combines nine OPDP barriers. New estimates rate the mathematical target on each normalized 0-1 component using the same directions and weights. Unknown human attention receives 0.5. These retrospective editorial estimates have C0 confidence; no solve episodes were run.

AI-relative core - 400 / 1000

Burden on the fixed tool-enabled AI research protocol.

Intrinsic difficulty - 170 / 1000

Conceptual and technical gap in the mathematical target.

Human resistance - 90 / 1000

Human-effort and exposure prior; neutral when unobserved.

Inverse tractability - 150 / 1000

Obstacle to independently checked partial progress.

Verification burden - 45 / 1000

Difficulty of independently checking a candidate result.

Formalization burden - 35 / 1000

Burden of encoding the statement and prerequisites.

Prerequisite load - 40 / 1000

Required preparation and cross-field breadth.

Context and ambiguity - 40 / 1000

Literature/context load and long reasoning dependencies.

Tool constraint - 30 / 1000

Low computational or symbolic tool fit increases difficulty.

R = weighted sum of the nine normalized components.

$D = 1000 \times [1 - (1 - R/1000)^{1.4}]$.

New estimates are rounded to the nearest 25; exact applicable published integers are preserved. The machine-readable companion includes each component and calculation trace. D is an ordinal research-difficulty estimate, not a probability of success.

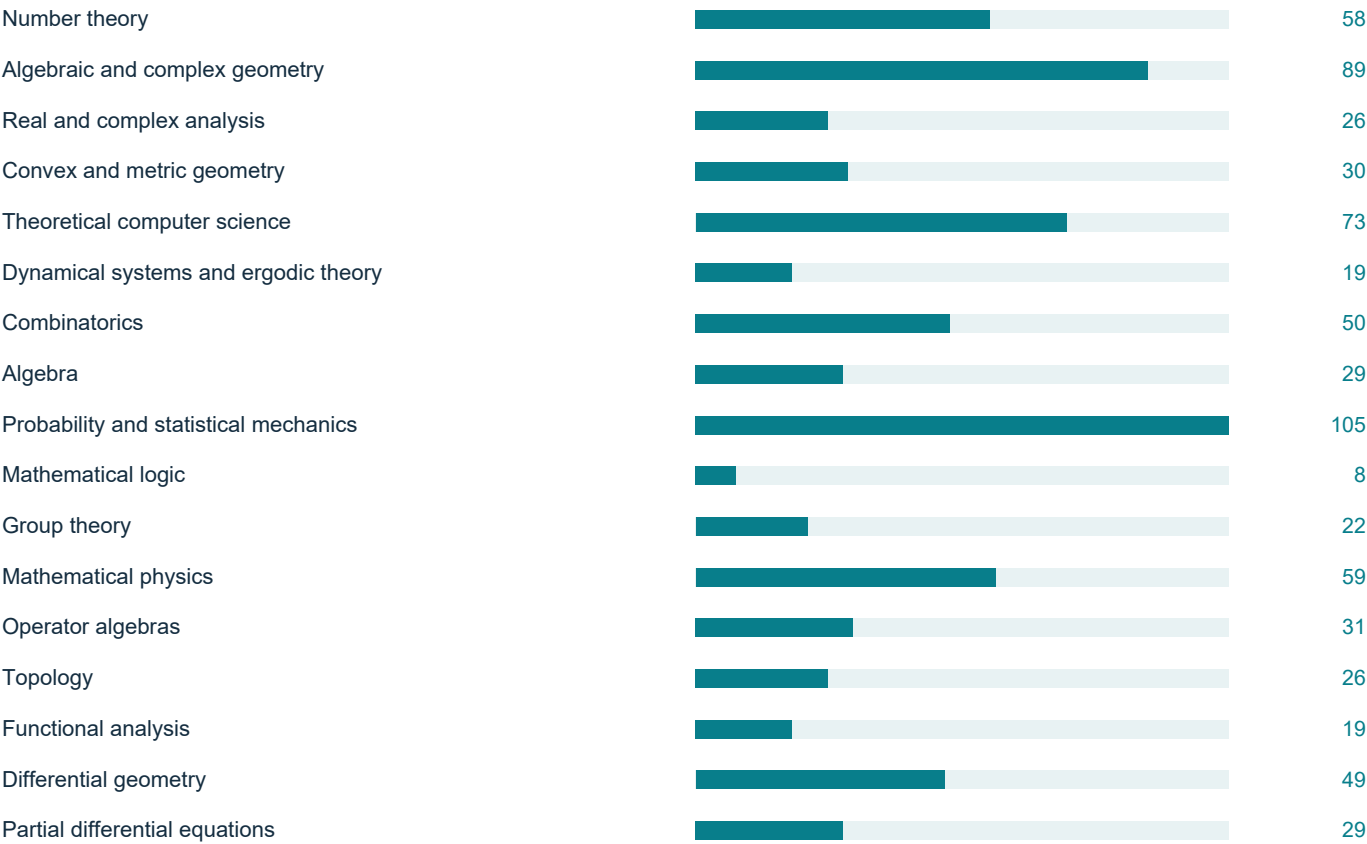
Reading a new solution can introduce hindsight bias. Neither manuscript length nor a source's claim of formalization directly supplies a difficulty rating; the assessment concerns the target before its reported resolution.

Archive comparison

related target	287
lexical candidate	147
unresolved no match	138
partial or special case	91
same target	59

Counts describe manuscript-level review outcomes. Related targets and lexical candidates are not confirmed duplicates; one archived target may connect to several papers.

Manuscripts by subject



No portfolio total is awarded. Per-paper points are scenarios; overlapping contributions require one family decision.

How this relates to OpenMath

The 6 October OpenMath judging reference records nineteen awarded result families. A separate comparison of the OpenAI catalogue and manuscript statements with those award scopes and selected known or held archive results found thirteen OpenAI families with cautious related links. No identical awarded theorem scope was located in that bounded review; this does not prove absence from every archived theorem or proof.

Family 107 - matrix multiplication

The OpenAI family claims asymptotic matrix-multiplication exponent bounds, including $\omega \leq 9/4$. The competition's support-138 construction and one-factor rigidity concern fixed 3×3 schemes with 23 products. These are distinct mathematical objectives; the registered difficulty 724 for the finite schemes is not transferred to the exponent claim.

Family 159 - arithmetic progressions

The OpenAI claim concerns quasipolynomial density bounds and the all-length divergent-reciprocal-sum conclusion. The awarded finite four-term-progression-free set has reciprocal sum above 4.44. Both connect to the broader problem, but an explicit finite lower-bound construction and a general upper-bound theorem are separate contributions.

Family 191 - Heilbronn triangles

The OpenAI family claims an asymptotic construction with a positive power improvement for all sufficiently large point counts. The earlier held project concerned a nine-point construction and an incomplete global-cover argument. The point count, asymptotic quantifiers and delivered proof boundary differ.

Family 283 - quantum synthesis

The OpenAI manuscript uses a target-dependent Boolean oracle and constant approximation error. The awarded generic pivot theorem establishes an exact intermediate existence result; its larger universal CNOT-count claim remained unproved. Oracle access and approximation change both the computational model and the target, so the earlier difficulty 824 is not inherited.

[Pinned OpenMath archive](#)

The machine-readable companion retains all thirteen scoped relationships. Comparisons use the dated judging reference; earlier repository records may represent an earlier review stage.

OAI-M0001 | Family 001

Milne's rationality conjecture for abelian varieties

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims rational compatibility between Hodge classes on a good-reduction abelian variety and divisor intersections on its special fibre. It includes residue characteristic two and asks for one rational pairing value across crystalline and all prime-to-p realizations. An additional algebraic-cycle conclusion uses a companion CM Hodge theorem. The distinction between rational compatibility and algebraicity matters because reduction can introduce divisors absent on the original variety.

Archive: related target. The archived rationality entry is restricted to CM abelian varieties; this manuscript treats arbitrary good-reduction abelian varieties and adds a companion-dependent algebraicity consequence.

Score basis: Simultaneous cohomological realizations and divisor lifting create substantial conceptual and formal context burdens.

[Read manuscript](#) | OPDP ID 40316661

OAI-M0002 | Family 002

Exact Birch-Swinnerton-Dyer Formula from Low Selmer Corank

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the complete BSD leading coefficient formula for rational elliptic curves whenever one full prime-power Selmer group has corank zero or one. It includes finiteness of the entire Tate-Shafarevich group, agreement of analytic and algebraic ranks, and every prime factor in the formula. This restricted-corank result removes auxiliary reduction and residual-representation conditions, while higher-corank curves remain outside its specified scope.

Archive: partial or special case. The full BSD family is archived, but the claimed result requires low Selmer corank and cannot inherit a full-BSD difficulty.

Score basis: Exact leading terms and removal of all local exclusions require more than the restricted rank-equality target.

[Read manuscript](#) | OPDP ID 40315850

OAI-M0003 | Family 002

The Selmer converse for elliptic curves at every prime

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a converse from low prime-power Selmer corank to analytic and algebraic rank for every rational elliptic curve and every prime. Coranks zero and one are covered, together with finiteness of the full Tate-Shafarevich group. A cubic-curve application includes primes congruent to four, seven or eight modulo nine. The result concerns this low-corank regime rather than unrestricted BSD leading coefficients.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniformity at every prime and additive-reduction cases makes the converse technically demanding.

[Read manuscript](#)

OAI-M0004 | Family 002

The two-primary Birch-Swinnerton-Dyer formula in Selmer corank at most one

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the two-primary part of the BSD leading-term formula for rational elliptic curves with two-power Selmer corank at most one. Within that regime, analytic rank, algebraic rank and Selmer corank agree, and the Tate-Shafarevich group is finite. Combining the theorem with twist-distribution inputs supplies a density-one twist consequence. The precise score concerns the two-primary formula, rather than every prime component of BSD.

Archive: partial or special case. The archived target covers every rational elliptic curve's rank equality; this paper adds a two-primary formula within Selmer corank at most one.

Score basis: The restricted primary component narrows the target, but arithmetic comparison and twist dependencies remain substantial.

[Read manuscript](#) | OPDP ID 1140

OAI-M0005 | Family 003

The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane $\text{Re } s > 7/8$

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform zero-free region to the right of real part seven eighths for Dirichlet L-functions and finite-order Hecke L-functions over the Eisenstein quadratic field. Principal poles at one are allowed. This would establish a quasi-Riemann conclusion and exclude exceptional zeros near one, while leaving the critical-line assertion unresolved. Its asserted half-plane is fixed, unlike classical conductor-dependent regions shrinking toward one.

Archive: partial or special case. The seven-eighths zero-free half-plane advances the archived critical-line hypothesis, but it leaves the strip between one half and seven eighths unresolved.

Score basis: A fixed zero-free half-plane confronts a major analytic barrier and demands unusually strong uniform control.

[Read manuscript](#) | OPDP ID 40402699 | [Family formalization notes](#)

OAI-M0006 | Family 003

The Quasi-Riemann Hypothesis (alternate 11/12 proof)

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an alternative argument excluding zeros with real part above eleven twelfths for every Dirichlet L-function and the specified finite-order Hecke family over the Eisenstein quadratic field. The claimed consequence includes absence of Landau-Siegel zeros. Its half-plane is weaker than the companion seven-eighths claim, yet still uniform across conductors. This is an alternative contribution within the same release family, not an independent second solution of the quasi-Riemann problem.

Archive: partial or special case. The eleven-twelfths zero-free region is weaker than the archived Riemann critical-line statement; it is an alternate scoped theorem, not a full RH solution.

Score basis: The weaker fixed half-plane still crosses the central exceptional-zero barrier with extensive analytic dependencies.

[Read manuscript](#) | OPDP ID 2 | [Family formalization notes](#)

OAI-M0007 | Family 003

Uniform exclusion of Landau-Siegel zeros

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an absolute positive constant bounding the conductor-normalized distance of every exceptional real Dirichlet zero from one. It treats primitive nonprincipal real characters and all conductors at least three. Such uniformity would rule out Landau-Siegel zeros rather than merely compute finitely many characters. The target is a quantitative exceptional-zero exclusion, distinct from proving that all complex nontrivial zeros lie on the critical line.

Archive: same target. The source-linked ProofAtlas target asks for an absolute conductor-normalized exclusion interval for primitive quadratic characters. Primitive nonprincipal real Dirichlet characters supply the same exceptional-zero target; no original CFSD value is available here.

Score basis: An effective uniform exceptional-zero exclusion is a deep barrier, but narrower than a full fixed-half-plane theorem.

[Read manuscript](#) | OPDP ID 50000043 | [Family formalization notes](#)

OAI-M0008 | Family 004

Hilbert's tenth problem over the rational numbers

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims undecidability of rational solvability for polynomial equations with integer coefficients. The number of variables is unrestricted and included in the input, so the target is Hilbert's tenth problem over the rationals rather than its integer analogue. A valid reduction would obstruct every possible decision procedure, not merely existing algorithms. The statement does not assert undecidability for a separately fixed small number of variables.

Archive: same target. The archived rational-solvability decision question is addressed negatively with the same unrestricted-variable input. The archived entry is excerpt-based and its estimate is not inherited.

Score basis: Encoding arbitrary computation in rational Diophantine solvability has a large conceptual gap and long verification chain.

[Read manuscript](#) | OPDP ID 40393813

OAI-M0009 | Family 004

A pointwise 2-converse for elliptic curves with rational two-torsion

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a pointwise two-Selmer converse for rational elliptic curves possessing nonzero rational two-torsion. Selmer corank zero or one then determines both relevant ranks and entails finiteness of the Tate-Shafarevich group. Arbitrary behaviour at the prime two is permitted. This gives a precisely restricted arithmetic theorem: curves lacking rational two-torsion and higher-corank cases are not automatically covered by the claimed conclusion.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Rational two-torsion provides structure while unrestricted local reduction still requires a substantial arithmetic argument.

[Read manuscript](#)

OAI-M0010 | Family 005

Catalan's constant is irrational

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that Catalan's constant is irrational, addressing whether the alternating reciprocal-square sum can equal a rational number. The target is qualitative irrationality, with no stated optimal irrationality exponent or approximation bound. Its importance comes from settling a specific classical constant whose defining series is elementary but whose arithmetic nature is difficult. Any stronger approximation-measure statement would require separate evidence beyond this stated claim.

Archive: same target. Both targets ask whether Catalan's constant is irrational. The archived MathDB entry lacks an original CFSD score and remains C0.

Score basis: A simple statement hides a persistent Diophantine barrier with little direct computational feedback toward a proof.

[Read manuscript](#) | OPDP ID 40403659 | [Family formalization notes](#)

OAI-M0011 | Family 006

Goldfeld's analytic density conjecture and the 2-converse for elliptic curves

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims equal limiting densities of analytic ranks zero and one among signed squarefree quadratic twists of every fixed rational elliptic curve. It also asserts a low-corank two-Selmer converse and finiteness of the corresponding Tate-Shafarevich groups. The twist parameters are ordered by absolute value. This convention and fixed-curve quantifier distinguish the theorem from distributions over all elliptic curves ordered by height.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform twist densities plus a two-primary converse combine deep distributional and arithmetic obstacles.

[Read manuscript](#)

OAI-M0012 | Family 006

The mean analytic rank of quadratic twists of elliptic curves

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that the mean analytic rank across signed squarefree quadratic twists of any fixed rational elliptic curve converges to one half. Parameters are ordered by absolute value. This mean statement controls the contribution of exceptional high ranks, beyond a mere density-one assertion that ranks are zero or one. The claim is specific to the stated counting convention and does not describe an unrestricted height-ordered population of curves.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Controlling the average requires tail information beyond the associated rank-zero and rank-one density statements.

[Read manuscript](#)

OAI-M0013 | Family 007

Ordinary two-point correlations of multiplicative functions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims ordinary, rather than logarithmically weighted, two-point correlation decay for Liouville on fixed nonproportional affine forms. It supplies a logarithmic power saving at each cutoff and also treats a corrected Elliott formulation for bounded complex multiplicative functions. That extension assumes uniform nonpretentiousness of one original factor and allows fixed residue classes. The nonpretentiousness condition is essential to the exact correlation target being assessed.

Archive: related target. The archived Elliott correlation topic is present, but the corrected binary formulation, ordinary averaging and exact nonpretentiousness assumptions need separate scope identification.

Score basis: Ordinary-cutoff cancellation and pretentiousness distinctions impose a major analytic barrier beyond logarithmic averages.

[Read manuscript](#) | OPDP ID 40347881 | [Family formalization notes](#)

OAI-M0014 | Family 008

The Deligne-Drinfeld conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that the rational Grothendieck-Teichmüller Lie algebra is a free Lie algebra with exactly one generator in each odd weight starting at three. It also identifies the corresponding completed object. This would settle the Deligne-Drinfeld freeness conjecture, a structural assertion rather than a computation in finitely many weights. Both the absence of hidden relations and the stated all-weight generation are necessary parts of the target.

Archive: same target. The curator-authored archived target is all-odd-weight freeness of the rational Grothendieck-Teichmüller Lie algebra, matching the reported target. The completed formulation supplies additional precision; no original CFSD is stored.

Score basis: All-weight generation and independence demand global structural control beyond finite-degree symbolic checks.

[Read manuscript](#) | OPDP ID 50000470 | [Family formalization notes](#)

OAI-M0015 | Family 009

Reconstruction of Function Fields from Mod- ℓ Milnor K-Theory

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims recovery of a function field, up to perfect closure, from its first two mod- ℓ Milnor K-groups and their multiplication. Transcendence degree is at least two and the algebraically closed constants have characteristic different from ℓ . Compatible isomorphisms must arise from field isomorphisms up to scalar, with Frobenius ambiguity in positive characteristic. Recovering the constants is part of the theorem, not additional input.

Archive: related target. The archived mod- ℓ reconstruction conjecture is directly relevant; this manuscript specifies Milnor K-groups and their product, whereas exact correspondence with the archive's Galois formulation requires its comparison theorem.

Score basis: Reconstructing field geometry from sparse algebraic data requires delicate incidence and positive-characteristic ambiguity analysis.

[Read manuscript](#) | OPDP ID 30002716 | [Family formalization notes](#)

OAI-M0016 | Family 009

Reconstruction from Milnor K-theory modulo the characteristic

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an equal-characteristic reconstruction theorem using mod- p Milnor K-theory for finitely generated fields over algebraically closed characteristic- p constants. An isomorphism preserving degree-two Steinberg relations is asserted to come from a unique field isomorphism, up to nonzero scalar. Transcendence degree must be at least two. The equal-characteristic setting changes the reconstruction mechanism: derivations and projective lines over p th powers carry the stated geometric information.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: The derivation-based characteristic- p reconstruction is narrow but has substantial conceptual and statement-correspondence burdens.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0017 | Family 009

The Bogomolov-Pop reconstruction theorem

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the Bogomolov-Pop reconstruction theorem from pro- ℓ abelian-by-central Galois data. Its fields have transcendence degree at least two over algebraically closed constants whose characteristic differs from ℓ . The claimed recovery includes the perfect closure and constant field, with explicit Frobenius and ℓ -adic unit ambiguities. Those ambiguities are part of the exact classification, so the theorem is not an unqualified uniqueness claim for raw field isomorphisms.

Archive: related target. The mod- ℓ Bogomolov-Pop problem is archived, while the manuscript states a pro- ℓ abelian-by-central reconstruction with explicit ambiguities. Those data should not be identified solely from the title.

Score basis: Galois-data reconstruction requires deep algebraic geometry and explicit control of several natural nonuniqueness mechanisms.

[Read manuscript](#) | OPDP ID 40399379 | [Family formalization notes](#)

OAI-M0018 | Family 010

Unrestricted pro-modularity at the prime two

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims occurrence of every odd absolutely irreducible two-dimensional two-adic rational Galois representation in a completed Hecke algebra at some odd tame level. Only finite ramification is imposed, and scalar or reducible residual representations are included. Auxiliary tame primes may enter the level. The target is completed-Hecke occurrence; without a de Rham assumption, its conclusion does not itself give classical modularity of the representation.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Removing residual restrictions at two creates a strong deformation-theoretic target with extensive automorphic dependencies.

[Read manuscript](#)

OAI-M0019 | Family 010

The Dimension of the Two-Adic Hecke Algebra at Odd Level

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every irreducible component of the full two-adic Hecke algebra at odd level has Krull dimension four. It addresses the prime-two case of Emerton's dimension conjecture across all residual components. The statement fixes Gamma-one level and concerns the entire algebra, rather than a selected well-behaved residual locus. This dimension assertion supplies structural information useful for modularity arguments without itself identifying every classical modular form.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform component dimensions at the exceptional prime require nontrivial deformation and local-global compatibility control.

[Read manuscript](#)

OAI-M0020 | Family 010

Fontaine-Mazur modularity at the prime 2

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims classical modularity, up to Tate twist, for regular odd irreducible two-dimensional two-adic representations of the rational Galois group. Finite ramification and distinct de Rham Hodge-Tate weights are assumed. It includes every residual representation. These geometric conditions are load-bearing: the theorem is stronger than completed-Hecke occurrence within this regime, but does not claim classical modularity for arbitrary non-de-Rham two-adic representations.

Archive: related target. The archived modularity conjecture is over totally real fields at primes at least three; this paper treats the exceptional prime two over the rationals.

Score basis: The regular Fontaine-Mazur case at two requires extensive arithmetic machinery and removal of difficult residual exclusions.

[Read manuscript](#) | OPDP ID 40369401

OAI-M0021 | Family 011

Weighted dilation graphs, smooth shifted primes and totient fibers

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims very large fibres of Euler's totient map: for any positive epsilon, infinitely many values have more than the value to the power one minus epsilon preimages. A second assertion produces almost-linearly many primes with unusually smooth predecessors in a specified interval. The two conclusions link shifted-prime factorization to multiplicity of totients. Their infinitude and asymptotic quantifiers require more than large finite examples or numerical searches.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Simultaneous shifted-prime smoothness and near-maximal totient multiplicities require substantial sieve and combinatorial control.

[Read manuscript](#)

OAI-M0022 | Family 011

The Poisson-Dirichlet law for prime predecessors

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that the normalized logarithms of the ordered prime factors of prime predecessors converge to a Poisson-Dirichlet law with parameter one. The convergence covers every fixed finite joint distribution. The target concerns integers of the special form p minus one, so an analogous distribution for unrestricted integers is insufficient. Its significance is a detailed factorization law for shifted primes rather than merely bounding their largest factor.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Joint factorization statistics of shifted primes require difficult distributional control beyond ordinary integer models.

[Read manuscript](#)

OAI-M0023 | Family 011

Prime Predecessors with an Even Number of Prime Factors

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims infinitely many primes whose predecessor is squarefree and contains an even number of prime factors. Equivalently, the Mobius value at p minus one equals positive one infinitely often. Both squarefreeness and parity are required, so the statement is not just a smoothness theorem or an almost-prime construction. This narrow infinitude result would overcome a particular parity obstruction in the arithmetic of shifted primes.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Prescribing Mobius sign on shifted primes directly challenges a parity barrier despite the narrow conclusion.

[Read manuscript](#)

OAI-M0024 | Family 012

The joint Dickman law for consecutive integers

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims asymptotic independence of the largest prime factors of consecutive integers after logarithmic normalization. Each marginal is Dickman's distribution and the averaging uses ordinary natural density. It also concludes that either neighbouring integer has the larger prime factor with limiting frequency one half. The ordinary-density and joint-distribution assertions go beyond one-variable smooth-number laws or statements derived only along selected subsequences of integer cutoffs.

Archive: related target. The archive contains a largest-prime-factor symmetry conjecture for neighbours; the manuscript claims the stronger full joint Dickman law. The excerpt does not certify theorem identity.

Score basis: Joint ordinary-density behaviour couples two arithmetic sequences and requires strong correlation control.

[Read manuscript](#) | OPDP ID 40376088 | [Family formalization notes](#)

OAI-M0025 | Family 013

The additive indecomposability of the primes

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims additive indecomposability of any set that differs from the primes in only finitely many elements. Such a set cannot be the sum of two nonnegative-integer sets each containing at least two elements. This is Ostmann's inverse Goldbach question, distinct from representing even integers as sums of primes. The finite-modification allowance makes the theorem robust to initial exceptions rather than an assertion about small prime samples.

Archive: related target. The named additive-irreducibility conjecture is present, but the archived excerpt is cut before the full primes conclusion. Identity remains provisional rather than established from the name alone.

Score basis: Excluding every nontrivial additive factorization of an infinite arithmetic set needs a strong global structural argument.

[Read manuscript](#) | OPDP ID 40326237 | [Family formalization notes](#)

OAI-M0026 | Family 014

Global Arthur Enhancements of Cuspidal Excursion Parameters

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims construction of global Arthur enhancements for cuspidal excursion parameters at a prescribed finite level. It assumes the stated finite-level Ramanujan-Arthur decomposition. A commuting Weil-centralizer map and one algebraic SL-two recover the original parameter, including inertia, under diagonal specialization. The conclusion does not include packet classification, ellipticity or a multiplicity formula. Assessing this conditional enhancement target therefore requires preserving its main decomposition input explicitly.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: The conditional construction is narrower than full Arthur classification but still carries deep representation-theoretic dependencies.

[Read manuscript](#)

OAI-M0027 | Family 014

Rationality of the Canonical Unramified Arthur Filtration

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims rationality of the canonical Arthur filtration on finitely supported unramified automorphic functions for split connected semisimple groups. Its characteristic restrictions come from the restricted geometric Langlands framework. Noncuspidal functions and closed invariant nilpotent supports are included. The target asks whether the canonical filtration descends to rational coefficients, not just whether an associated decomposition exists over an ℓ -adic field or after extending scalars to another coefficient field.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Canonical rational descent across noncuspidal supports requires strong categorical and arithmetic compatibility.

[Read manuscript](#)

OAI-M0028 | Family 014

Ramanujan-Arthur Decompositions of Cuspidal Functions at Full Finite Level

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims full-finite-level Ramanujan-Arthur decompositions of cuspidal functions for split semisimple groups over global function fields. Nilpotent dual-group orbits index the summands consistently across unramified places and complex embeddings. A further statement gives unramified temperedness for adjoint absolutely simple groups with a generic local component. Arbitrary level multiplicities are allowed, but ramified-place temperedness is handled in a separate companion target rather than supplied here.

Archive: partial or special case. The archive contains a general generic Ramanujan conjecture. This result is restricted to specified split adjoint groups over function fields and unramified places.

Score basis: Arbitrary finite levels and orbit-uniform decompositions demand broad automorphic and categorical coordination.

[Read manuscript](#) | OPDP ID 40357865

OAI-M0029 | Family 014

Temperedness at ramified places for globally generic exceptional groups

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims temperedness at every place for globally generic cuspidal representations of split adjoint exceptional groups over global function fields. It permits arbitrary characteristic and ramification depth. The work extends a companion unramified assertion to ramified local components, so its target is a specific all-place generic Ramanujan theorem. The exceptional-group, split and globally generic qualifiers distinguish it from unrestricted Ramanujan statements for every group and automorphic representation.

Archive: partial or special case. All-place temperedness is claimed for globally generic split adjoint exceptional groups over function fields. This is a scoped case of a broader generic Ramanujan target.

Score basis: Ramified exceptional-group local analysis is highly technical even with the companion unramified theorem available.

[Read manuscript](#) | OPDP ID 40357865

OAI-M0030 | Family 014

The Restricted Geometric Langlands Equivalence in Positive Characteristic

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the restricted geometric Langlands equivalence for reductive groups on smooth projective curves in positive characteristic. Four explicit characteristic hypotheses are required over algebraic closures of finite fields; arbitrary algebraically closed bases add very-good-characteristic and Weyl-order restrictions. Coefficients are ℓ -adic with ℓ different from the characteristic. This precisely qualified categorical equivalence is broader than isolated eigensheaf constructions, but its stated characteristic regimes must remain part of the target.

Archive: partial or special case. The restricted geometric Langlands target is archived; the manuscript's characteristic regimes and additional group restrictions delimit the exact reported conclusion.

Score basis: A general categorical equivalence has exceptional context, prerequisite and proof-correspondence burdens.

[Read manuscript](#) | OPDP ID 40403332

OAI-M0031 | Family 014

Constructible tame Hecke eigensheaves in positive characteristic

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims existence of nonzero constructible perverse Hecke eigensheaves with Borel level at one marked point. The curve has genus at least two over an algebraic closure of a finite field, the group is simple and simply connected, and the dense local-system parameter has tame regular-unipotent monodromy. Nilpotent singular support and full tensor/fusion compatibility are included. The explicit characteristic hypotheses and monodromy conditions narrow the construction substantially.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: The scoped eigensheaf construction has demanding categorical compatibility but fewer regimes than a full Langlands equivalence.

[Read manuscript](#)

OAI-M0032 | Family 014

Tame Hecke Eigensheaves with Several Marked Points

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims tame Hecke eigensheaves with Borel structure at at least two marked points for special linear groups in characteristic greater than the rank. Dense projective-linear local systems may have any unipotent Jordan type, including trivial logarithm. The sheaves retain parabolic nilpotent support and tensor/fusion structure. These simultaneous multi-point and monodromy assertions extend the companion one-point construction, without asserting the corresponding full categorical equivalence for arbitrary groups or all characteristic regimes.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Several ramification points and arbitrary unipotent types add technical complexity to a controlled group setting.

[Read manuscript](#)

OAI-M0033 | Family 014

Frobenius Structures on Tame Hecke Eigensheaves

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a Frobenius-compatible arithmetic enhancement of tame Hecke eigensheaves for special linear groups. It permits a finite extension of the constant field and assumes dense arithmetic projective-linear parameters with regular-unipotent monodromy at one puncture. Genus is at least two and characteristic exceeds rank. The exact eigenvalue and multi-leg compatibility are retained, so the claim is an arithmetic descent/structure theorem beyond merely constructing an underlying geometric perverse sheaf.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Arithmetic Frobenius descent is substantial, while the regular-unipotent one-point setting limits the range of cases.

[Read manuscript](#)

OAI-M0034 | Family 015

Equidistribution of Prime-Degree Torus Packets with Arbitrary Local Type

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims higher-dimensional Duke-type equidistribution for volume-weighted torus packets attached to totally real fields of fixed prime degree at least five. Full lattices can have arbitrary prescribed local homothety types. As the multiplier-order discriminant grows, packet measures converge to Haar probability without mass escaping. The target concerns complete weighted packets rather than equidistribution of every individual periodic orbit, a distinction essential to its mathematical scope and difficulty estimate.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: High-degree packet equidistribution requires strong arithmetic estimates and prevention of escape of mass.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0035 | Family 015

Equidistribution of primitive quartic torus packets for arbitrary orders

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Haar equidistribution of volume-weighted periodic diagonal packets arising from invertible ideal classes in arbitrary orders of primitive totally real quartic fields. Primitive means there is no intermediate field. Discriminants tend to infinity and escape of mass is excluded. Uniform control at primes dividing the order index is part of the claimed route. The arbitrary-order qualifier makes this stronger than a maximal-order-only quartic packet result.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniformity across nonmaximal orders and local bad primes adds difficulty to arithmetic measure rigidity.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0036 | Family 015

Equidistribution of Primitive Sextic Torus Packets

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims equidistribution for complete ideal-class packets of maximal orders in primitive totally real sextic fields. The discriminant grows, each packet includes coordinate-sign translates, and weights are diagonal-orbit volumes. Haar probability is the limiting measure with no escaped mass. The maximal-order and no-intermediate-field conditions are essential, so this does not automatically supply an arbitrary-order sextic theorem or conclusions for all degree-six fields and selected individual orbits.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Composite-degree arithmetic and no-escape estimates require more structure than lower-dimensional packet problems.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0037 | Family 016

The abelian Zilber-Pink conjecture

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims finiteness of maximal atypical subvarieties for every irreducible subvariety of an abelian variety over the algebraic numbers. Atypicality is measured relative to the smallest containing torsion coset. This is the abelian Zilber-Pink target, with its ambient-category and base-field restrictions. It does not by itself settle every mixed-Shimura or semiabelian version over arbitrary complex parameters, although it addresses a major unlikely-intersection setting within this arithmetic category.

Archive: partial or special case. The full archived Zilber-Pink target permits mixed Shimura or semiabelian ambient varieties over the complex numbers. This paper treats the abelian setting over algebraic numbers.

Score basis: Uniform unlikely-intersection finiteness for arbitrary abelian subvarieties is a major geometric and arithmetic target.

[Read manuscript](#) | OPDP ID 9400091

OAI-M0038 | Family 016

The E x CM component of Zilber-Pink for curves in A2

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the product-with-CM part of Zilber-Pink for Hodge-generic curves in the Siegel threefold over algebraic numbers. Only finitely many points can represent surfaces isogenous to a product of elliptic curves with at least one CM factor. This specifies a particular exceptional locus rather than all atypical intersections. Its significance lies in closing one component of a broader curve theorem when combined with other separately stated finiteness results.

Archive: partial or special case. The paper settles the specified elliptic-times-CM locus on arithmetic Siegel curves, a particular component of the broader archived conjecture.

Score basis: A focused exceptional-locus finiteness theorem is narrower than general Zilber-Pink but retains difficult arithmetic geometry.

[Read manuscript](#) | OPDP ID 9400091

OAI-M0039 | Family 016

Quaternionic division points on curves in the Siegel threefold

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that a Hodge-generic algebraic curve in the Siegel threefold contains only finitely many quaternionic-division points. The full rational geometric endomorphism algebra must be an indefinite quaternion division algebra over the rationals. The curve is defined over algebraic numbers, and no boundary or reduction restriction is assumed. This targets one endomorphism-locus component of Zilber-Pink rather than every exceptional abelian-surface type or an unrestricted higher-dimensional unlikely-intersection statement.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: The narrow endomorphism locus limits scope while the lack of boundary assumptions leaves a serious finiteness problem.

[Read manuscript](#)

OAI-M0040 | Family 016

Elliptic squares and Zilber-Pink for curves in A2

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims finiteness of non-CM elliptic-square points on Hodge-generic curves in the Siegel threefold over algebraic numbers. A qualifying abelian surface is isogenous to the square of one elliptic curve. Combining this component with the CM-product and quaternionic companion results gives the stated curve-level Zilber-Pink conclusion without a boundary assumption. The combined conclusion depends on those companion inputs; the individual theorem's own target is the elliptic-square locus.

Archive: partial or special case. The claimed elliptic-square component and companion-dependent Siegel-curve theorem are narrower than the full mixed-Shimura/semiabelian Zilber-Pink target.

Score basis: Closing the remaining locus and assembling companion finiteness results raises the burden beyond an isolated special component.

[Read manuscript](#) | OPDP ID 9400091

OAI-M0041 | Family 017

The irrationality exponent of pi is 2

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that pi has irrationality exponent exactly two, the smallest value possible for an irrational real number. It also deduces convergence of the Flint-Hills series involving reciprocal squared sines, with arguments in radians. The main target specifies the strength of rational approximation to pi, not merely its known irrationality. Qualitative convergence of the series is a consequence, so it should not be counted as a separate independent solution family.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Optimal irrationality exponent requires a much sharper Diophantine barrier than proving a constant irrational.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0042 | Family 018

The Margulis-Platonov conjecture over global function fields

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the Margulis-Platonov normal-subgroup classification over every global function field, including characteristic two. For absolutely almost simple simply connected groups, noncentral abstract normal subgroups of rational points arise from open normal subgroups in the finite product of anisotropic local factors. This describes all eligible abstract normal subgroups through arithmetic local data. The simple-connectivity and algebraic-group hypotheses delimit the classification rather than an unrestricted theorem about arbitrary arithmetic groups.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Global normal-subgroup classification with characteristic-two uniformity has deep group-theoretic and arithmetic burdens.

[Read manuscript](#)

OAI-M0043 | Family 018

The Margulis-Platonov conjecture over number fields

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the number-field Margulis-Platonov conjecture for absolutely almost simple simply connected algebraic groups. Every noncentral abstract normal subgroup of rational points is described using an open normal subgroup of the product of anisotropic nonarchimedean local groups. The local factors and noncentral qualifier matter to the precise result. This complements the function-field manuscript but concerns a separate arithmetic base category, not merely a different exposition of the same field-specific proof.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform arithmetic normal-subgroup structure requires strong global-to-local arguments across all number fields.

[Read manuscript](#)

OAI-M0044 | Family 019

Étale covers with a prescribed exterior sheet

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims construction of a connected finite étale cover of a smooth proper hyperbolic curve over an algebraic closure of a p -adic field. Given finitely many disjoint small disks, one sheet reproduces their whole exterior, while each connected component over a disk has degree divisible by p . The global connectedness and exterior-sheet requirements coexist. This is a geometric covering tool, not itself the full rational-point section conjecture used elsewhere in the family.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A controlled connected cover must reconcile local divisibility with global exterior geometry.

[Read manuscript](#)

OAI-M0045 | Family 019

The p -adic section conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the p -adic section conjecture for every smooth proper geometrically connected curve of genus at least two over a finite extension of the p -adic rationals. Rational points correspond to the asserted arithmetic-fundamental-group sections. Global consequences additionally use finite-descent inputs and equality of a descent locus with rational points. These extra conditions cover specified modular curves but do not turn the local theorem into an unrestricted number-field section result.

Archive: related target. The named local section conjecture is archived. Its excerpt is not enough to certify all section conventions and hypotheses, so full statement identity is withheld.

Score basis: Recovering rational points from fundamental-group sections is a major arithmetic reconstruction target.

[Read manuscript](#) | OPDP ID 40322496

OAI-M0046 | Family 020

Squarefree values of quartics and power-free values of polynomials

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims positive Euler-product densities of squarefree values for irreducible integral quartics without a fixed prime-square divisor. It also handles degree-minus-two power-free values in degrees four through eight under the appropriate local condition. A cited higher-degree input extends that latter conclusion further. The fixed-divisor exclusion is necessary: the score concerns polynomials satisfying the local admissibility condition, rather than impossible density assertions for every integral polynomial.

Archive: partial or special case. The archived squarefree polynomial-values target is broader; quartics and the specified power-free degree ranges are precise scoped advances.

Score basis: Quartic squarefree densities require difficult large-square-divisor control and precise local-to-global counting.

[Read manuscript](#) | OPDP ID 40352709 | [Family formalization notes](#)

OAI-M0047 | Family 021

A quadratic bound for Jacobsthal's function

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an upper bound for Jacobsthal's function of order k squared divided by the square of $\log \log$ of three k . Here k limits the number of distinct prime divisors of the modulus, and the function measures intervals guaranteed to contain a coprime integer. The bound is uniform over qualifying moduli. It addresses quadratic growth in k , rather than estimating one primorial or checking individual intervals by exhaustive search.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform sieve-gap control improves the exponent across arbitrary prime sets rather than selected moduli.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0048 | Family 022

The Weak Inhomogeneous Duffin-Schaeffer Conjecture

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the weak inhomogeneous Duffin-Schaeffer conclusion for every real shift and finite-valued approximation function. Divergence of the totient-weighted sum forces infinitely many shifted approximation inequalities for almost every real point. Numerators need not be coprime to denominators, and no Diophantine condition is placed on the shift. These features distinguish the weak theorem from a stronger reduced-numerator formulation and preserve its exact metric-approximation scope under assessment.

Archive: related target. The named weak inhomogeneous Duffin-Schaeffer conjecture is present, but the archive excerpt does not certify the exact unreduced-numerator and shift conventions.

Score basis: Arbitrary shifts and approximation functions require strong metric and correlation estimates.

[Read manuscript](#) | OPDP ID 40363703

OAI-M0049 | Family 023

An unconditional first moment for cubic Gauss sums

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Patterson's first-moment formula for normalized cubic Gauss sums over primary Eisenstein primes, without conditional analytic assumptions. Its main term has exponent five sixths and coefficient involving Gamma of two thirds, with a sharp cutoff. Nonzero angular Fourier modes are asserted to cancel at the same scale. The target is a specific cubic-prime average and angular refinement, rather than a universal distribution theorem for all orders of Gauss sums.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: An unconditional prime-weighted first moment and angular cancellation impose advanced analytic-number-theory requirements.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0050 | Family 024

An asymptotic formula for the number of totients

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit asymptotic equivalent for the count of distinct totient values below a growing bound. The coefficient is built as a uniform limit of finite arithmetic approximations rather than asserted to be a single constant. It also gives linear scaling under every fixed positive dilation. This resolves the stated Erdős-Hall scaling question while preserving the phase-dependent structure needed for the sharper counting formula and its interpretation.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: An asymptotic equivalent needs fine arithmetic multiplicity control beyond order-of-magnitude totient estimates.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0051 | Family 025

Short Egyptian fractions

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every proper rational with sufficiently large denominator can be expressed using order $\log \log$ of the denominator distinct positive unit fractions. It asserts that this order is optimal when numerators vary. Additional claims concern doubly exponential growth of fixed-length expansions of one and missing denominators. Distinctness of denominators and the uniform numerator quantifier are essential, making the target stronger than existence of some Egyptian-fraction representation.

Archive: related target. The retrieved Erdős305 entry optimizes the largest denominator, whereas this manuscript primarily optimizes the number of distinct terms; these are different Egyptian-fraction objectives.

Score basis: Uniform near-optimal expansion length and matching combinatorial growth create a substantial constructive arithmetic target.

[Read manuscript](#) | [OPDP ID 40390863](#) | [Family formalization notes](#)

OAI-M0052 | Family 026

Positive lower density of large prime gaps

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims positive lower density of prime gaps exceeding any prescribed constant times the logarithm of the smaller prime. The density constant may depend on that prescribed threshold, but applies to every sufficiently long initial segment. A consequence gives positive lower density of indices where the ratio of the n th prime to n increases. This strengthens isolated large-gap infinitude and does not claim one threshold-independent proportion for arbitrarily large constants.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Positive lower density requires distributional control beyond constructing rare extreme gaps.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0053 | Family 027

Integral points on character varieties of curves

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims potential Zariski density of integral points on every component of special-linear character varieties of curves, in arbitrary rank. A single finite number-field extension supplies its full ring of integers. Boundary monodromy can be prescribed with quasi-unipotent, even nonsemisimple, conjugacy conditions. Integrality is taken in the ambient character variety. This combination distinguishes the target from density of ordinary rational points or a theorem only on generic semisimple components.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Integral density with exact boundary conditions combines arithmetic construction and singular character-variety geometry.

[Read manuscript](#)

OAI-M0054 | Family 028

Bounded-Step Walks on Gaussian Primes

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every bounded-step Gaussian-prime graph has uniformly bounded connected components, independent of the starting prime. This implies that no infinite distinct-prime walk can use a fixed finite step bound, settling the stated Gaussian moat question negatively. Coordinate-axis primes are included and the bound need not be explicit. The uniform-component conclusion is stronger than merely blocking a walk from one chosen starting point or producing finite graphical evidence.

Archive: related target. The archived bounded-step walk problem is answered negatively, but this manuscript's target strengthens it to a uniform size bound for every component and starting prime. The original difficulty⁵³⁷ is a parent anchor, not this stronger target's inherited score.

Score basis: Uniform bounds for every starting prime are stronger than the archived single-walk question and need new global sieve obstructions.

[Read manuscript](#) | OPDP ID 1869 | [Family formalization notes](#)

OAI-M0055 | Family 029

Primitive roots for every admissible integer base

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims infinitely many primes for which any admissible fixed integer is a primitive root, with a lower count of order x divided by $\log^2 x$ in each sufficiently large dyadic interval. The base excludes minus one and squares. This settles the infinitude part of Artin's question but does not provide its expected precise density. The weaker quantitative count must therefore be kept distinct from the full asymptotic conjecture.

Archive: partial or special case. The archive asks how many primes admit the fixed primitive root. The manuscript proves infinitude with a lower bound and does not establish Artin's expected asymptotic density; original difficulty 596 is not inherited.

Score basis: Infinitude for every fixed base confronts a deep prime-order barrier without supplying the full density formula.

[Read manuscript](#) | OPDP ID 1183

OAI-M0056 | Family 029

Simultaneous primitive roots: a conditional lower bound for prime bases

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims simultaneous primitive-root lower bounds for each fixed finite collection of positive prime bases. In every sufficiently large dyadic interval, at least a constant times x divided by $\log^2 x$ primes make every base primitive. Four analytic and sieve inputs from a companion are assumed. The target is that conditional implication, so the reported multi-base conclusion cannot be treated as an unconditional theorem merely because the manuscript gives a complete argument under its inputs.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Several simultaneous bases complicate the sieve, while explicit companion assumptions substantially narrow the proof obligation.

[Read manuscript](#)

OAI-M0057 | Family 030

Modularity of elliptic curves over imaginary quadratic fields

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims modularity of every elliptic curve over every imaginary quadratic field. It includes agreement of local parameters at all places rather than merely selected unramified data. This is a field-specific modularity target extending beyond the rational case. The elliptic-curve and imaginary-quadratic qualifications remain essential: the claim does not say every abelian variety over every number field is automorphic or settle all higher-dimensional modularity questions by the same assertion.

Archive: related target. The imaginary-quadratic elliptic modularity target is named in the archive, but exact local-parameter conventions and field quantifiers are not fully recoverable from its excerpt.

Score basis: Uniform modularity across imaginary quadratic fields has deep local-global and automorphic barriers.

[Read manuscript](#) | OPDP ID 40337605

OAI-M0058 | Family 031

Open Homomorphisms of Global Solvably Closed Galois Groups

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that continuous open homomorphisms between the specified global solvably closed Galois groups arise from unique equivariant field embeddings in the opposite direction. No extra kernel restriction or compatibility condition is imposed. The reversal of arrow direction is intrinsic to the anabelian formulation. Its significance is recovery of arithmetic field maps from group maps, rather than reconstructing only the abstract isomorphism class of an isolated Galois group.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Open-homomorphism rigidity requires exact recovery of embeddings with delicate Galois compatibility.

[Read manuscript](#)

OAI-M0059 | Family 032

The rational Hodge conjecture for products of K3 surfaces

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims algebraicity of every rational Hodge class on every finite product of projective complex K3 surfaces. Factors may be repeated or distinct and have arbitrary Picard ranks, periods and endomorphism fields. The all-product and all-class quantifiers greatly exceed a theorem about divisors on one surface. Its target remains the K3-product category; it does not state the rational Hodge conjecture for every smooth projective complex variety.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Arbitrary products and all cohomological degrees demand control of tensor Hodge classes and algebraic correspondences.

[Read manuscript](#)

OAI-M0060 | Family 032

Algebraicity of Kuga-Satake Correspondences for K3 Surfaces

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims algebraicity of the prescribed Kuga-Satake correspondence for every projective complex K3 surface. The rational cycle must induce the exact transcendental-cohomology embedding with fixed normalization and full even-Clifford target. Isogenous models are included with transported embeddings. The target is stronger than existence of some nonzero cohomological correspondence: preserving the designated tensor and normalization is part of what the asserted proof must establish across all qualifying K3 surfaces.

Archive: related target. Kuga-Satake algebraicity is archived; this manuscript's exact prescribed tensor and normalization exceed what the truncated source record can confirm.

Score basis: Exact prescribed correspondences impose a stronger fidelity burden than an unspecified algebraicity witness.

[Read manuscript](#) | OPDP ID 40366509

OAI-M0061 | Family 032

The rational Hodge conjecture for CM abelian varieties

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the rational Hodge conjecture for complex abelian varieties with complex multiplication in all codimensions. It further derives generalized Hodge, finite-field abelian Tate, and abelian Hodge-standard conclusions. The CM assumption restricts the principal characteristic-zero target, while the broader listed consequences depend on the asserted deductions. These implications belong to one mathematical result family rather than automatically creating several independent competition awards from the same underlying theorem or companion arguments.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: All CM Hodge classes and consequential transfers across characteristics create an exceptional structural target.

[Read manuscript](#)

OAI-M0062 | Family 032

Algebraic Kuga-Satake correspondences and Hodge conjectures on a K3 quadratic locus

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Hodge and generalized Hodge results for all powers of K3 surfaces on a specified rational quadratic-embedding locus. An explicit P-polarized family and its Picard jumps are included. Algebraic cycles realize prescribed even-Clifford Kuga-Satake tensors there. A more general implication assumes one exact algebraic tensor and concludes both Hodge assertions for every self-power. The quadratic-locus theorem and the broader conditional reduction should remain separately visible in assessing this manuscript's scope.

Archive: partial or special case. The special K3 quadratic locus and conditional exact-tensor implication are scoped parts of the broader archived Kuga-Satake Hodge target.

Score basis: A restricted quadratic locus narrows the target, while all powers and exact tensor realization remain demanding.

[Read manuscript](#) | OPDP ID 40360408

OAI-M0063 | Family 032

Weil classes and Hodge classes on abelian powers

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Hodge algebraicity on all powers of certain imaginary-quadratic abelian varieties. One class consists of sixfolds with a hyperbolic signature-three-three Hermitian form; another covers dimension at most five with the stated action. Nonsimple varieties and additional endomorphisms are allowed. The argument uses a companion CM Hodge theorem. These dimensional and polarization restrictions distinguish the result from unrestricted abelian Hodge algebraicity in every dimension and arbitrary endomorphism type.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Special endomorphism structure and a strong companion input reduce scope while leaving nontrivial tensor-cycle construction.

[Read manuscript](#)

OAI-M0064 | Family 032

Abelian covers, Gale correspondences, and the Hodge conjecture for powers

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims rational Hodge algebraicity on all powers of Jacobians at tensor-Hodge-generic points of connected abelian-cover variations. Base genus and compatible branch patterns are unrestricted, and products with CM abelian varieties are included. A second statement treats very general diagonal complete intersections cut by at most two equal-degree equations. Generic-locus qualifiers are crucial: the theorem does not assert the same algebraicity for every special fibre in those families.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Broad families and all self-powers require substantial tensor and correspondence control even on generic loci.

[Read manuscript](#)

OAI-M0065 | Family 032

Algebraicity of Weil classes on split abelian eightfolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims algebraicity of the entire rational Weil-class space in codimension four on split abelian eightfolds of Weil type. Every imaginary quadratic field and compatible polarization type is allowed, including special members with extra endomorphisms. The relevant rational space is two-dimensional. This is a specific cycle-algebraicity target on a structured eight-dimensional family, not a statement that every Hodge class on every abelian eightfold is thereby algebraic without further argument.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Split Weil structure limits the target, but uniform explicit algebraic realization is still substantial.

[Read manuscript](#)

OAI-M0066 | Family 032

A Conditional Reduction for Algebraic Kuga-Satake Correspondences

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a conditional route to exact Kuga-Satake algebraicity for a polarized K3 surface with full orthogonal Hodge group. One algebraic nonzero correspondence to an abelian variety is assumed; equivalent one-form criteria and propagation within a polarized component are supplied. The existence of that first correspondence remains unresolved in the target. Consequently, the reduction is useful infrastructure, but it cannot be scored as an unconditional proof of general Kuga-Satake algebraicity.

Archive: partial or special case. The manuscript assumes an initial algebraic correspondence and proves a recovery implication. That conditional reduction leaves the archived unconditional existence obstacle explicit.

Score basis: The central algebraic existence input is assumed, leaving a narrower but nontrivial tensor-recovery implication.

[Read manuscript](#) | OPDP ID 40366509

OAI-M0067 | Family 033

Orbifold and logarithmic litaka subadditivity

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims orbifold litaka subadditivity for rational simple-normal-crossing boundaries on compact manifolds in Fujiki class C. Boundary coefficient one is permitted. Ordinary and logarithmic subadditivity are then included as consequences. The target combines compact nonprojective geometry with orbifold boundary data, while retaining the specified smooth-crossing setup. It is not a claim of the same inequality for arbitrary singular spaces, arbitrary boundary coefficients, or unrelated morphisms outside the stated geometric category.

Archive: related target. The orbifold litaka target is named in the archive, while the manuscript explicitly broadens to Fujiki-class compact manifolds with coefficient-one rational crossing boundaries.

Score basis: Orbifold boundaries and nonprojective compact geometry create broad conceptual and analytic difficulties.

[Read manuscript](#) | OPDP ID 40377651 | [Family formalization notes](#)

OAI-M0068 | Family 033

Logarithmic Kodaira dimension and whole-fiber variation

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a logarithmic Iitaka-Viehweg inequality for projective connected-fibre maps between smooth complex quasi-projective varieties. The base must have nonnegative logarithmic Kodaira dimension. Variation is defined through the birational field of definition of the whole geometric generic fibre. This precise variation notion matters: the theorem addresses a superadditivity target with boundary geometry and fibre variation, rather than only ordinary subadditivity for a compact smooth projective fibration or its naive fibre moduli.

Archive: related target. The logarithmic variation problem is archived, but the exact whole-fibre variation definition and nonnegative-base condition require a full canonical statement comparison.

Score basis: Whole-fibre variation and logarithmic boundary control deepen the standard subadditivity problem.

[Read manuscript](#) | OPDP ID 40356335 | [Family formalization notes](#)

OAI-M0069 | Family 033

The reverse logarithmic Kodaira inequality and additivity

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the reverse logarithmic Kodaira inequality for connected-fibre maps of smooth projective reduced-crossing pairs. Pullback boundary support is compatible, and the total space together with every boundary stratum is smooth away from the base boundary. Combined with subadditivity, this yields additivity, including negative-infinity cases. The smooth-stratum condition is essential to the target and prevents treating the result as unrestricted logarithmic additivity for arbitrary singular or degenerating fibrations.

Archive: partial or special case. The strong boundary-stratum smoothness hypothesis narrows the logarithmic additivity target.

Score basis: The reverse inequality is difficult, but strong boundary-stratum smoothness restricts the admissible geometric cases.

[Read manuscript](#) | OPDP ID 40352702 | [Family formalization notes](#)

OAI-M0070 | Family 033

Projective Hodge lines and ordinary Iitaka subadditivity

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims ordinary Iitaka subadditivity for projective surjective connected-fibre morphisms between smooth connected projective varieties in characteristic zero. The total Kodaira dimension is at least the sum of those of the geometric generic fibre and the base. This concerns every qualifying fibration rather than a selected dimension or fibre class. Logarithmic boundaries, orbifold data and positive characteristic are separate targets, so their broader companion assertions do not automatically enter this manuscript's exact conclusion.

Archive: same target. The full curator-authored source target is characteristic-zero projective connected-fibre Iitaka subadditivity, matching the direct claimed theorem. No original CFSD value is available in this append record.

Score basis: Uniform characteristic-zero fibration subadditivity is a major geometric positivity target.

[Read manuscript](#) | OPDP ID 50000113 | [Family formalization notes](#)

OAI-M0071 | Family 033

B-semiampleness for compact log-smooth Kähler fibrations

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims b-semiampleness for compact log-smooth Kahler fibrations satisfying a rational linear-equivalence condition on the adjoint divisor. A smooth compact modification makes the threshold-moduli line stable under further modifications, and a multiple is globally generated. Boundary coefficient one is allowed; projectivity and an orbifold litaka assumption are not required. The exact compact Kahler hypotheses and divisor identity form part of the assertion, rather than optional explanatory context for an unrestricted moduli-divisor theorem.

Archive: related target. The archive's klt-trivial-fibration b-semiampleness target differs from the compact Kahler rational-boundary setting and permits less scope verification from its excerpt.

Score basis: Birational stability of moduli lines in the nonprojective Kahler setting requires delicate positivity and descent.

[Read manuscript](#) | OPDP ID 40326649 | [Family formalization notes](#)

OAI-M0072 | Family 034

Log abundance for compact Kähler spaces under logarithmic litaka subadditivity

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims log abundance for normal compact Kahler spaces in every dimension under an assumed logarithmic litaka-subadditivity statement for smooth projective pairs. Log canonical rational-boundary adjoints must be rational Cartier, and analytic nefness then implies semiample. The subadditivity input is explicitly conditional. Thus the scored target is a cross-category implication with singularity and positivity hypotheses, not an independently unconditional resolution of all compact Kahler abundance cases from this manuscript alone.

Archive: partial or special case. Compact Kahler abundance is archived, but the manuscript assumes a major subadditivity theorem; only the conditional implication is the scored target.

Score basis: A major assumed input reduces the target to a still substantial projective-to-Kahler reduction.

[Read manuscript](#) | OPDP ID 40374040

OAI-M0073 | Family 034

Uniform indices for semi-log-canonical log Calabi-Yau pairs

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform Cartier-and-triviality index for connected projective semi-log-canonical log Calabi-Yau pairs in each fixed dimension at least four. Boundary coefficients come from a fixed finite rational set, and the bound ignores the number of components. Lower dimensions are supplied by an established input. Uniformity across reducible pairs is essential; the result does not state one numerical index valid simultaneously across every dimension and every possible boundary coefficient set.

Archive: related target. The same index-conjecture family is present; exact coefficient and reducibility conventions remain unresolved in the archive excerpt.

Score basis: Uniform index control across singular reducible pairs creates a demanding boundedness problem.

[Read manuscript](#) | OPDP ID 40341557

OAI-M0074 | Family 034

Conditional good minimal models for compact Kähler fourfolds

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims good minimal models for specified pseudo-effective compact Kahler klt fourfold pairs, assuming orbifold litaka subadditivity, a fourfold minimal-model-program input, and abundance in a stated nef range. Its additional asserted step is nonvanishing, including algebraic-dimension-zero cases. Strong rational factoriality and actual rational-Cartier adjoints are required. The central assumed program and abundance statements remain visible, so this conditional theorem cannot stand in for their independent unconditional proofs.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Several major program inputs are assumed, narrowing the new burden mainly to difficult nonvanishing reductions.

[Read manuscript](#)

OAI-M0075 | Family 034

Log abundance in characteristic zero

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims rational-boundary log abundance in every dimension in characteristic zero. On projective log canonical pairs, nef rational-Cartier adjoints are asserted to be semiample. Over the complex numbers it also claims canonical nonvanishing for smooth projective varieties. This removes a fixed-dimension restriction but retains projectivity and rational boundaries. Its scope is substantially broader than individual fourfold or numerical-dimension-one companion cases, and those special cases should not be counted as separate completions of this parent theorem.

Archive: related target. Log-canonical abundance is archived, but its truncated statement cannot certify all rational-boundary and base-field conventions of the all-dimensional claim.

Score basis: All-dimensional log abundance confronts a central birational-geometry barrier with exceptional prerequisite and verification demands.

[Read manuscript](#) | OPDP ID 40320332

OAI-M0076 | Family 034

Minimal metrics and interior injectivity for nef adjoints

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims zero-Lelong minimal semipositive metrics on pulled-back nef adjoints of projective complex klt pairs, on any log resolution. A second statement gives first-cohomology injectivity for suitable rational interior boundaries when both endpoint line bundles carry such metrics. The positivity and metric hypotheses are essential. These analytic tools can support abundance arguments, but the manuscript's direct targets are metric existence and a scoped injectivity theorem rather than semiample for every nef adjoint in isolation.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Singular metric regularity and cohomological injectivity require sophisticated analytic geometry under controlled hypotheses.

[Read manuscript](#)

OAI-M0077 | Family 034

Fourfold nonvanishing by minimal metrics and moving jets

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims canonical nonvanishing for smooth projective complex fourfolds: pseudo-effectivity of the canonical class entails a nonzero pluricanonical section at some positive multiple. The assertion is existence of a section rather than full finite generation or semiample. Its four-dimensional and smooth-projective qualifications narrow the broader nonvanishing problem. A valid result supplies an important input to minimal-model arguments, but does not itself resolve every singular pair or higher-dimensional canonical case.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Fourfold section existence remains a major positivity obstruction despite the restricted dimension.

[Read manuscript](#)

OAI-M0078 | Family 034

Lifting sections from the reduced support of an adjoint

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims positive Iitaka dimension for a projective rationally factorial dlt pair when an effective adjoint multiple is supported on the coefficient-one boundary and becomes semiample on that whole reduced support. The nonzero support and restriction conditions are load-bearing. A low-dimensional abundance consequence follows after nonvanishing is supplied. This is a precise lifting theorem from boundary geometry, rather than an unconditional assertion that every pseudo-effective log adjoint has sections or is semiample.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Strong boundary and restriction hypotheses narrow the target to a sophisticated section-lifting mechanism.

[Read manuscript](#)

OAI-M0079 | Family 034

Schnell fiber spaces and good canonical models

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a Campana-Peternell inequality comparing canonical Kodaira dimension with an effective divisor under a pseudo-effective difference condition. For fibre spaces, an analogous comparison involving a pulled-back ample base divisor yields additivity and sections at sufficiently divisible multiples. The specified pseudo-effectivity assumptions are essential. This targets positivity-sensitive inequalities and canonical models, not unconditional Kodaira additivity for arbitrary fibre spaces without those supporting divisor relations or adequate canonical positivity on the total variety.

Archive: related target. The named Schnell fibre-space problem is present; the manuscript's divisor inequalities and eventual-section refinements are not fully comparable to the excerpt.

Score basis: Comparing pseudo-effective divisors and deriving uniform sections requires difficult birational positivity control.

[Read manuscript](#) | OPDP ID 40366421

OAI-M0080 | Family 034

Uniform log litaka fibrations and bounded moduli denominators

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform rounded pluricanonical system generating the litaka field for log canonical pairs of fixed dimension at least five and fixed finite rational boundary coefficients. Nonnegative Kodaira dimension is required. The degree depends only on the dimension and coefficient set; a further complex canonical-bundle assertion bounds moduli denominators. Neither uniformity over all dimensions nor independence from every coefficient set is claimed, and those distinctions materially determine this boundedness target's mathematical scope.

Archive: related target. Uniform effectiveness of litaka fibrations is archived; the fixed coefficient-set and moduli-denominator assertions require more exact statement evidence.

Score basis: Uniform pluricanonical effectiveness and moduli-denominator bounds combine two demanding high-dimensional boundedness problems.

[Read manuscript](#) | OPDP ID 40328952

OAI-M0081 | Family 034

Uniform Pluricanonical litaka Fibrations

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform pluricanonical degree defining litaka fibrations for all smooth integral projective varieties in each fixed characteristic-zero dimension. Nonnegative Kodaira dimension is required, and section ratios must generate the full litaka function field. The degree may depend on dimension. This is an effective uniformity assertion across varieties, not merely existence of some variety-dependent large multiple. Singular log pairs and positive-characteristic versions remain outside the stated smooth characteristic-zero target.

Archive: related target. The effective litaka family is archived, but the source excerpt does not certify the smooth all-dimensional and full-field-generation conventions.

Score basis: Uniform effective generation across all varieties requires difficult boundedness beyond ordinary fibration existence.

[Read manuscript](#) | OPDP ID 40328952

OAI-M0082 | Family 034

Relative denominators and effective systems for log Calabi-Yau fibrations

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims bounded denominators and effective linear systems for log Calabi-Yau fibrations from projective complex log canonical pairs of dimension at most four. Boundaries use a fixed finite rational coefficient set and the base has positive dimension. The moduli b-divisor receives a uniform Cartier multiple. If the base divisor is big, section ratios of a uniformly rounded adjoint generate its function field. The bigness condition is specific to that final effectiveness assertion.

Archive: related target. The relative log-Calabi-Yau boundedness topic is present; low-dimensional denominator and effective-system conclusions need exact source comparison.

Score basis: Low dimension narrows the problem while relative denominator descent and effective systems remain technically demanding.

[Read manuscript](#) | OPDP ID 40354237

OAI-M0083 | Family 034

Arithmetic Stein-degree bounds for log Calabi-Yau pairs

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a bound on the arithmetic Stein degree of large-coefficient boundary components in ordinary projective log Calabi-Yau pairs. Dimension and a positive coefficient threshold are fixed; the characteristic-zero base field need not be algebraically closed. Each component's constant-field extension degree is bounded only by those parameters. This is the contraction-to-a-point version of the stated Stein-degree conjecture, rather than a uniform bound for every coefficient size or a general positive-dimensional-base fibration.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Arithmetic constant-field degrees require subtle boundedness, but a point base and coefficient threshold restrict scope.

[Read manuscript](#)

OAI-M0084 | Family 034

Uniform effective log litaka fibrations for fourfolds

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims effective log litaka systems for projective complex log canonical fourfolds with finite rational boundary coefficients. A uniform rounded reflexive system is nonempty and generates the full litaka field under pseudo-effectivity of the rational-Cartier adjoint. A separate assertion bounds canonical indices for rationally trivial klt fourfolds. The fixed-dimension and coefficient-set constraints are essential to the uniform degree, distinguishing this from unrestricted all-dimensional effectiveness with arbitrary real boundaries or every field characteristic.

Archive: partial or special case. The dimension-four finite-coefficient assertion narrows the broader effective log-litaka target.

Score basis: Fourfold uniform effectiveness couples nonvanishing and boundedness with singular-pair technical burdens.

[Read manuscript](#) | OPDP ID 40341978

OAI-M0085 | Family 034

Abundance after nonvanishing for compact Kähler fourfolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims abundance after nonvanishing for compact Kahler klt fourfolds. An analytically nef rational-Cartier adjoint becomes semiample once a positive Cartier multiple has a nonzero section. Effective rational boundaries are allowed, without assuming projectivity or rational factoriality. At litaka dimension zero, a multiple is actually trivial as a holomorphic line bundle. The existence of the initial section is assumed, so this theorem does not alone close the fourfold nonvanishing obstacle.

Archive: partial or special case. The compact Kahler abundance target is archived, but nonvanishing is assumed and dimension is four in the manuscript.

Score basis: Assumed nonvanishing removes a central obstacle, while nonprojective semiamplessness remains technically difficult.

[Read manuscript](#) | OPDP ID 40363014

OAI-M0086 | Family 035

Log abundance in numerical dimension one for threefolds in positive characteristic

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims semiample for nef numerical-dimension-one log adjoints on projective threefolds in characteristic greater than three. Boundaries are effective and rational, singularities log canonical, and the adjoint rational Cartier. Terminality and rational factoriality are not required. The target extends the controlled numerical-dimension-one range under weaker singularity assumptions, but does not settle abundance for every numerical dimension or every positive characteristic, where different singularity and positivity phenomena may arise.

Archive: related target. The archive names positive-characteristic threefold abundance with characteristic above five and klt hypotheses. This manuscript instead specifies numerical dimension one, characteristic above three and log-canonical singularities.

Score basis: The low numerical dimension helps, while log-canonical singularities and small positive characteristic increase technical difficulty.

[Read manuscript](#) | OPDP ID 40330088

OAI-M0087 | Family 035

Abundance in numerical dimension one for terminal threefolds in positive characteristic

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims abundance in numerical dimension one for projective rationally factorial terminal threefolds over algebraically closed fields of characteristic greater than three. A nef canonical divisor is asserted to be semiample with Kodaira dimension one. The terminal and factorial assumptions distinguish this paper from its more general log-canonical companion. Its contribution is therefore a specific canonical threefold case rather than an all-pair or all-numerical-dimension statement across arbitrary field characteristic.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Strong terminal-factorial structure narrows the numerical-dimension-one abundance target.

[Read manuscript](#)

OAI-M0088 | Family 036

Numerical semiample of nef adjoint classes on compact Kähler manifolds

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims numerical semiample of nef adjoint classes on smooth compact Kähler manifolds with rational crossing boundaries and an added nef line bundle. The canonical boundary adjoint must be pseudo-effective and the sum nef. A semiample rational bundle represents the resulting real Bott-Chern class. A flat change of bundle is permitted, so the theorem does not assert semiample of the exact original holomorphic adjoint; that distinction controls both significance and score.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Numerical positivity in nonprojective Kähler geometry is substantial, but allowing flat changes weakens exact semiample.

[Read manuscript](#)

OAI-M0089 | Family 036

Numerical Semiampleness of Nef Adjoint Divisors

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims generalized numerical abundance for projective characteristic-zero klt rational pairs. If the boundary adjoint is pseudo-effective and adding a nef rational divisor produces a nef sum, that sum is numerically equivalent to a semiample divisor. Numerical equivalence is the conclusion, not linear equivalence or semiampleness of the same divisor. This exact weakening is essential when comparing with ordinary abundance: the result controls a numerical class under the stated positivity assumptions.

Archive: related target. The generalized abundance topic is archived; the exact numerical-equivalence conclusion cannot be certified from its excerpt.

Score basis: Uniform generalized numerical abundance has broad scope but a weaker conclusion than actual adjoint semiampleness.

[Read manuscript](#) | OPDP ID 40355267

OAI-M0090 | Family 036

Minimal models and Mori fibre spaces for generalized log canonical $\mathbb{Q}$ -pairs

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims existence of minimal models for pseudo-effective projective generalized log canonical rational pairs in characteristic zero, and Mori fibre spaces otherwise. The nef rational-Cartier b -divisor is held fixed during the process. Setting it to zero gives ordinary log canonical pairs. The target is an existence dichotomy, distinct from termination of every arbitrary flip sequence. Generalized boundary data and singularities make correspondence with the exact claimed model a central assessment requirement.

Archive: related target. The archive contains model-existence targets, while this manuscript fixes generalized log-canonical rational pairs and their nef b -divisor. The full target conventions require source recovery.

Score basis: Generalized log-canonical model existence across all dimensions is a central high-context birational target.

[Read manuscript](#) | OPDP ID 40394766

OAI-M0091 | Family 036

Minimal models in numerical dimension one

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims minimal models for smooth projective complex varieties of dimension at least three whose canonical divisor has section-growth numerical dimension one. Pseudo-effectivity is assumed. The resulting model is projective, terminal and rationally factorial. The definition of numerical dimension uses an ample twist, so it should not be silently replaced by a different convention. The theorem settles this restricted numerical range rather than arbitrary pseudo-effective varieties or every generalized pair.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: The numerical-one restriction reduces scope while arbitrary ambient dimension keeps substantial birational complexity.

[Read manuscript](#)

OAI-M0092 | Family 037

The ordinary-double-point gap in every dimension

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the normalized-volume ordinary-double-point gap for complex algebraic klt germs with zero boundary in every dimension. A singular n -dimensional germ has volume at most twice the n th power of n minus one, and equality characterizes an analytic ordinary double point. Both inequality and sharp rigidity are required. The zero-boundary condition matters: the claim does not state the same numerical bound for arbitrary boundary-weighted singularity pairs or smooth germs.

Archive: related target. The ODP volume-gap conjecture is present; all-dimensional zero-boundary inequality and exact analytic equality need a complete archive statement for identity certification.

Score basis: An all-dimensional sharp local invariant bound with equality classification has major geometric rigidity demands.

[Read manuscript](#) | OPDP ID 40359228

OAI-M0093 | Family 037

The normalized-volume gap in dimension four

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the four-dimensional normalized-volume bound of 162 for singular complex algebraic klt germs without boundary. Equality is asserted exactly for ordinary double points in the analytic category. The target combines a sharp maximum with rigidity of extremizers, not merely a nonoptimal upper estimate. It is a dimension-specific member of the broader family, so its conclusion should not be used as proof of the gap in every dimension without the separate companion theorem.

Archive: partial or special case. The fourfold bound 162 is a fixed-dimensional case of the broader volume-gap target.

Score basis: Fixed dimension reduces the gap problem, while sharp equality still requires substantial singularity analysis.

[Read manuscript](#) | OPDP ID 40359228

OAI-M0094 | Family 038

Fujita's freeness conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Fujita freeness for every smooth complex projective variety: adding at least dimension plus one copies of an ample line bundle to the canonical bundle makes the result globally generated. The threshold is sharp in the conjectured general form. This is freeness, not the separate stronger very-ample assertion. All dimensions and all ample polarizations are included, distinguishing the target from proven low-dimensional or positivity-strengthened cases within adjoint linear-system theory.

Archive: related target. Fujita freeness is explicitly named in the archive; its excerpt-based record does not supply a preserved original CFSD or complete conventions for score inheritance.

Score basis: Sharp all-dimensional adjoint freeness is a major positivity barrier with little finite computational feedback.

[Read manuscript](#) | OPDP ID 40405146

OAI-M0095 | Family 039

Nagata's conjecture for plane curves

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the strict nonhomogeneous form of Nagata's conjecture for plane curves through at least ten very general complex points. Total multiplicity is strictly below degree times the square root of the number of points. The assertion is simultaneous for all effective curves, including reducible and nonreduced ones. The very-general qualification is essential; the theorem does not state the same bound for arbitrary specially arranged points or merely equal multiplicities.

Archive: related target. The older Nagata summary is broad and does not specify the exact strict nonhomogeneous inequality. Its difficulty646 is not inherited for the strengthened precise target.

Score basis: Simultaneous strict multiplicity bounds across all curves require difficult global geometric control.

[Read manuscript](#) | OPDP ID 7200013 | [Family formalization notes](#)

OAI-M0096 | Family 039

Maximal Seshadri constants on arbitrary polarized surfaces

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims eventual maximal multipoint Seshadri constants on every smooth complex projective polarized surface. For sufficiently many very general points, the constant equals the square-root volume bound. The threshold may depend on the surface and line bundle and need not be explicit. This qualitative eventual assertion is distinct from prescribing the exact threshold or establishing equality at every small number of points for every polarization, where separate obstructions can occur.

Archive: partial or special case. The eventual maximal multipoint Seshadri question is archived; the manuscript treats surfaces and permits a polarization-dependent threshold.

Score basis: Arbitrary surfaces and eventual volume saturation are difficult, but unspecified thresholds weaken the effective target.

[Read manuscript](#) | OPDP ID 40377849 | [Family formalization notes](#)

OAI-M0097 | Family 039

Maximal Multipoint Seshadri Constants in Higher Dimensions

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims eventual equality with the volume bound for multipoint Seshadri constants in every complex projective dimension at least three. The ample line bundle and smooth integral variety are fixed before choosing a sufficiently large number of very general points. The threshold depends on that polarization. Its scope is qualitative asymptotic maximality across dimensions, not an explicit universal number of points or equality for special point configurations and arbitrary nonample bundles.

Archive: related target. The named multipoint Seshadri family is present, but full eventual-equality quantifiers are not confirmed from the truncated target.

Score basis: High-dimensional polarization-uniform arguments create broad difficulty despite allowing variety-dependent thresholds.

[Read manuscript](#) | OPDP ID 40364781 | [Family formalization notes](#)

OAI-M0098 | Family 039

Maximal multipoint Seshadri constants in positive characteristic

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims positive-characteristic eventual maximality of multipoint Seshadri constants at geometric generic tuples. Smooth integral projective varieties and ample bundles are considered, including surfaces and dimensions at least three. The number of points must exceed a polarization-dependent threshold. Geometric generic tuples replace the complex very-general formulation, an essential field-sensitive distinction. The score therefore concerns the stated generic-point theorem, not equality for every rationally chosen tuple over a finite or countable field.

Archive: related target. The manuscript's positive-characteristic geometric-generic-tuple formulation needs distinct scope comparison with the archived multipoint conjecture.

Score basis: Positive characteristic adds inseparability and generic-point subtleties to a broad asymptotic positivity claim.

[Read manuscript](#) | OPDP ID 40364781 | [Family formalization notes](#)

OAI-M0099 | Family 040

Bloch's conjecture for surfaces with $p_g=0$

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Bloch's conjecture for smooth connected projective complex surfaces with geometric genus zero. The integral degree-zero cycle group maps isomorphically to the Albanese variety. This includes injectivity and surjectivity at the integral cycle level, rather than only a rational-coefficient statement or a selected class of surfaces. The geometric-genus and smooth-projective hypotheses restrict the category, but within it the asserted result describes all eligible zero-cycle classes and their Albanese equivalence.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Integral zero-cycle control across all genus-zero surfaces is a major algebraic-cycle target.

[Read manuscript](#)

OAI-M0100 | Family 041

Projective-space bases of Lagrangian fibrations

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that the normal projective base of a projective Lagrangian fibration from a compact irreducible holomorphic symplectic Kahler manifold is projective space. Every dimension and deformation type is included. The base need not initially be assumed smooth, while projectivity of the fibration remains part of the setup. This is a classification of bases, not an assertion that every isotropic class automatically produces a Lagrangian fibration without an additional existence or semiampleness theorem.

Archive: related target. The projective-space-base conjecture is present; exact projectivity and singular-base conventions remain unavailable in its excerpt.

Score basis: Classifying possibly singular bases uniformly across hyperkahler types requires strong geometric structure.

[Read manuscript](#) | OPDP ID 40359597

OAI-M0101 | Family 041

The strong hyperkähler SYZ conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the strong hyperkahler SYZ statement: a line bundle with nonzero nef isotropic first Chern class on a compact irreducible holomorphic symplectic Kahler manifold is semiample. The class conditions are essential and the conclusion concerns the actual holomorphic line bundle. This would supply fibrations in the intended regime, not merely numerical equivalence to some semiample bundle. The target spans arbitrary deformation types rather than one fixed K3-derived family or dimension.

Archive: related target. The nef parabolic hyperkahler/SYZ target is archived, but the exact actual-bundle semiamplessness claim is not certified by the excerpt.

Score basis: Actual semiamplessness across all hyperkahler types is a major geometric existence problem.

[Read manuscript](#) | OPDP ID 40334494

OAI-M0102 | Family 042

Every complex K3 surface is Oka

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every complex K3 surface is Oka. In a concrete approximation formulation, holomorphic maps from neighborhoods of compact convex subsets of finite-dimensional complex affine space can be uniformly approximated by entire maps into the surface. The domain dimension is arbitrary. This analytic flexibility statement is distinct from projective K3 Hodge algebraicity: it applies to all complex K3 surfaces and concerns holomorphic mapping behaviour rather than cohomological cycles or algebraic correspondences.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform entire-map approximation on all K3 surfaces requires substantial analytic flexibility arguments.

[Read manuscript](#)

OAI-M0103 | Family 043

 $P=W$ in composite rank for fixed determinant

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims P equals W in composite rank for fixed-determinant trace-free Higgs moduli on complex curves of genus at least two. Rank and degree must be coprime and the equality covers full rational cohomology. Established prime-rank results supply the remaining coprime ranks. The coprimality, determinant and trace conditions delimit the exact moduli problem, so this does not automatically assert the same filtration equality for every singular noncoprime moduli space or unrestricted determinant variant.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Composite-rank filtration comparison requires deep moduli geometry and exact global cohomological compatibility.

[Read manuscript](#)

OAI-M0104 | Family 044

The equivariant cohomological Hikita conjecture for arbitrary quivers

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims equivariant cohomological Hikita for arbitrary finite quivers, allowing loops, multiple arrows and flavour tori. Stability must equal semistability and the gauge action must be free. The canonical graded-algebra isomorphism identifies Nakajima cohomology with a scheme-theoretic fixed locus on the deformed Coulomb branch, preserving nilpotents and even the empty case. Those scheme and stability conditions make the target more precise than an equality of reduced point sets or graded dimensions.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Canonical scheme-level compatibility for arbitrary quivers imposes high categorical and singular-geometry burdens.

[Read manuscript](#)

OAI-M0105 | Family 046

A projective fourfold with large fundamental group and non-Stein universal cover

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a smooth projective complex fourfold with large fundamental group whose universal cover is neither Stein nor holomorphically convex. The cover contains no positive-dimensional compact analytic subvariety. Such a construction would refute the relevant Shafarevich prediction even after imposing the large-fundamental-group condition. The claimed counterexample must establish all these properties together; merely exhibiting a non-Stein nonprojective cover would not meet the exact mathematical target as stated.

Archive: related target. The large-fundamental-group Shafarevich target is archived; the manuscript gives a structured fourfold counterexample whose all properties require separate proof review.

Score basis: Coordinating projectivity, group largeness and nonconvex analytic covering geometry makes the counterexample difficult.

[Read manuscript](#) | OPDP ID 40372341

OAI-M0106 | Family 046

A surface counterexample to Shafarevich holomorphic convexity

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a smooth connected projective complex surface whose universal cover fails holomorphic convexity. This would refute the unrestricted Shafarevich universal-cover conjecture already in complex dimension two. The target requires a genuine projective surface and its ordinary universal cover, not an arbitrary complex surface or a nonuniversal covering construction. Its significance is a low-dimensional counterexample; a separate fourfold manuscript treats stronger large-fundamental-group restrictions in a different geometric scope.

Archive: related target. The broad Shafarevich holomorphic-convexity target is archived; this is a proposed surface counterexample, rather than an accepted status update.

Score basis: A low-dimensional projective counterexample must reconcile strong algebraic constraints with analytic nonconvexity.

[Read manuscript](#) | OPDP ID 40392792

OAI-M0107 | Family 047

An explicit failure of complex affine-space cancellation

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit complex affine fourfold that becomes affine five-space after multiplication by one affine line, yet is not affine four-space. It also derives stable-coordinate and nontrivial affine-fibration counterexamples. Dimension four is essential to the cancellation statement. These consequences may be grouped through the same construction, but they do not settle affine three-space cancellation merely by sharing the word cancellation or by considering a different fibre dimension in companion applications.

Archive: related target. The archived ProofAtlas target is affine three-space cancellation; this manuscript's affine fourfold counterexample is a different dimension and cannot inherit that target's score.

Score basis: An explicit characteristic-zero cancellation counterexample confronts a major algebraic structure barrier.

[Read manuscript](#) | OPDP ID 51000233 | [Family formalization notes](#)

OAI-M0108 | Family 048

A singular normal affine surface with free tangent sheaf

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a singular normal affine surface over the complex numbers with a free rank-two tangent sheaf. Such an example would refute the characteristic-zero Lipman-Zariski conjecture that tangent-sheaf freeness forces smoothness. Normality, singularity and the globally specified free sheaf are all required simultaneously. This is a two-dimensional construction rather than an assertion about arbitrary singular varieties or merely locally generated derivations at selected points, which would not establish the same counterexample.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Normality and free tangent structure are tightly constrained, making a genuine singular example conceptually difficult.

[Read manuscript](#)

OAI-M0109 | Family 049

A stable coordinate that is not a coordinate in four variables

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit degree-five polynomial in four complex variables that becomes a coordinate after one variable is added, while remaining noncoordinate originally. Every fibre is affine three-space but its embedding cannot be rectified. The construction would refute both the stated stable-coordinate and embedding predictions at that ambient dimension. The exact original-variable noncoordinate proof is essential; a stable change of variables alone would not distinguish the counterexample from an ordinary coordinate presentation.

Archive: related target. The stable-coordinate conjecture is archived; the manuscript specifies a degree-five four-variable counterexample and extra fibre claims not confirmed by the excerpt.

Score basis: Proving nonrectifiability despite affine fibres and stable triviality requires delicate algebraic invariants.

[Read manuscript](#) | OPDP ID 40320163 | [Family formalization notes](#)

OAI-M0110 | Family 049

An explicit noncoordinate polynomial with affine three-space zero fibre

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a noncoordinate polynomial in four complex variables whose zero fibre is nevertheless affine three-space. This targets the Abhyankar-Sathaye embedding conjecture by separating intrinsic fibre geometry from the ambient coordinate property. Adding variables gives higher-dimensional examples. The score concerns that explicit embedding obstruction rather than the separate affine-space cancellation problem in dimension three, and the asserted fibre identification and failure of coordinate status must both be verified for the exact polynomial.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: An intrinsic affine-space fibre can conceal a difficult ambient embedding obstruction.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0111 | Family 050

Ample rank-two bundles on the quadric surface without Griffiths-positive metrics

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims ample rank-two vector bundles on the quadric surface without strictly Griffiths-positive smooth Hermitian metrics. The family is built from coordinatewise power pullbacks and a fixed twist; ampleness holds at every positive power, while metric nonexistence is asserted for sufficiently large powers. This would refute the ample-bundle positivity conjecture. The distinction between algebraic ampleness and differential-geometric curvature positivity is the central mathematical content, not a failure of the bundle's ordinary ampleness.

Archive: same target. The preserved ProofAtlas target is whether all ample bundles admit Griffiths-positive metrics. A qualifying ample rank-two counterexample refutes that universal assertion; its special construction remains the reported theorem scope.

Score basis: Separating algebraic positivity from every smooth curvature metric requires substantial differential and algebraic analysis.

[Read manuscript](#) | OPDP ID 50000291 | [Family formalization notes](#)

OAI-M0112 | Family 051

Canonical ampleness of compact hyperbolic Kähler manifolds

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims canonical ampleness and projectivity for every positive-dimensional compact connected Kahler manifold with no nonconstant entire curve. This is the stated smooth compact Kobayashi canonical-ampleness target. The absence of entire curves is the crucial hyperbolicity hypothesis, while the conclusion strengthens numerical positivity to an actual ample canonical bundle. The theorem does not assert the same result for arbitrary compact complex manifolds without Kahler structure or for singular spaces lacking the specified smooth setup.

Archive: same target. The source-linked archived target is canonical ampleness of hyperbolic compact Kahler manifolds, matching the direct claimed conclusion in the smooth compact category. No original CFSD is stored.

Score basis: Deriving ample canonical positivity from analytic hyperbolicity is a major cross-category geometric target.

[Read manuscript](#) | OPDP ID 50000376

OAI-M0113 | Family 052

Universal-cover splitting for compact Kähler manifolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims compatible splitting of the ordinary universal cover when a compact connected Kähler manifold's tangent bundle decomposes into two integrable positive-rank holomorphic summands. The cover becomes a product whose tangent factors match those prescribed summands. Integrability is assumed, not deduced for every tangent decomposition. The two-factor restriction also matters, making the target a particular Beauville splitting assertion rather than an automatic decomposition for any finite collection of nonintegrable holomorphic subbundles.

Archive: partial or special case. The manuscript addresses two prescribed integrable tangent summands; broader tangent-splitting variants need separate hypotheses and scope review.

Score basis: Global product decomposition must respect prescribed integrable distributions on a possibly nonprojective compact manifold.

[Read manuscript](#) | OPDP ID 40349903 | [Family formalization notes](#)

OAI-M0114 | Family 052

Integrability of split tangent bundles on rationally connected manifolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims integrability of both positive-rank summands in a specified tangent-bundle splitting on any smooth rationally connected projective complex manifold. A cited product theorem then gives a decomposition compatible with that splitting. Rational connectedness provides strong geometric structure and is a necessary hypothesis in this target. The new integrability assertion should be distinguished from the companion universal-cover theorem, which assumes integrability for a more general compact Kähler setting rather than deriving it there.

Archive: related target. The named rationally-connected integrability target is present, but exact summand and product-theorem conventions are not certified by the excerpt.

Score basis: Rational connectedness supplies structure, while automatic foliation integrability remains a significant global geometric obstacle.

[Read manuscript](#) | OPDP ID 40373181 | [Family formalization notes](#)

OAI-M0115 | Family 053

A high-arity counterexample to Pixton completeness in Chow

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a counterexample to completeness of Pixton's original tautological relation system in both Chow and rational cohomology. A formal class lies outside the original relation span but vanishes after realization. Its genus is ten to the sixtieth, with an explicitly enormous marking count. The arity is part of the construction, not evidence of a finite feasible enumeration. The target concerns the original relation system, so a different expanded relation family requires separate comparison.

Archive: related target. Pixton completeness is archived; the original-system qualification and enormous explicit arity matter to the proposed counterexample.

Score basis: A high-arity independence witness requires exact symbolic control far beyond practical brute-force checking.

[Read manuscript](#) | OPDP ID 40398246

OAI-M0116 | Family 054

Irrational cubic fourfolds with geometric K3 categories

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims irrationality of very general cubic fourfolds at every sufficiently large admissible Hassett discriminant, even when their Kuznetsov categories are geometric K3 categories. Untwisted Hodge-theoretic K3 associations also occur. This would refute the sufficiency directions of the associated categorical and Hodge rationality predictions. The threshold is ineffective and the fibres very general, so the claim does not identify every individual rationality status at each small discriminant or every special cubic fourfold.

Archive: related target. The cubic-fourfold categorical rationality problem is archived, but the manuscript specifies geometric untwisted K3 categories at very-general high discriminants; the excerpt does not establish exact identity.

Score basis: Irrationality with strong K3 associations crosses a major birational and categorical obstruction.

[Read manuscript](#) | OPDP ID 40357651

OAI-M0117 | Family 055

A Gepner stability condition on every smooth quintic threefold

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a numerical Bridgeland stability condition on every smooth complex quintic threefold with the normalized Gepner symmetry. Tensoring by the hyperplane bundle and then twisting at the structure sheaf shifts phase by two fifths. The precise functor, normalization and phase are part of Toda's target. This is an existence theorem for the specified stability structure, not merely construction of one central charge or a formal numerical pattern without the full stability axioms.

Archive: related target. The quintic Gepner target is present; its precise normalized phase/functor conventions need full statement recovery.

Score basis: The support and stability axioms with exact Gepner symmetry create a deep derived-category construction target.

[Read manuscript](#) | OPDP ID 40381162

OAI-M0118 | Family 055

Prescribed large-volume charges on threefolds with trivial canonical bundle

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims numerical Bridgeland conditions on smooth projective complex threefolds with trivial canonical bundle at sufficiently large volumes. It prescribes both ordinary and square-root-Todd charges, full support and stable point sheaves, uniformly on an open set of twists and ample directions. A separate tilt inequality treats arbitrary smooth projective threefolds under a volume threshold. The special canonical-bundle condition belongs to the stability claim and must not be dropped when generalizing the auxiliary inequality.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Exact large-volume charges and uniform support control are technically demanding derived and numerical geometry.

[Read manuscript](#)

OAI-M0119 | Family 056

Termination of generalized log canonical flips on compact Kähler fourfolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims termination of every existing generalized log-canonical flip sequence on specified compact Kahler fourfolds. The models are globally Weil rationally factorial, boundaries rational, and the fixed nef b-divisor comes from an analytically nef rational-Cartier divisor on a projective modification. The small diagrams are projective with opposite ample signs. No scaling or pseudo-effectivity assumption is imposed, but existence of all required flips and contractions is explicitly outside the stated theorem.

Archive: related target. Kahler flip termination is archived, while the log-canonical singularities and existing-projective-diagram boundary here need source-specific comparison.

Score basis: Arbitrary-sequence termination in the compact Kahler category is difficult even with controlled projective diagrams.

[Read manuscript](#) | OPDP ID 40376282

OAI-M0120 | Family 056

Termination of generalized-canonical flips on compact Kähler fourfolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims termination for existing generalized-canonical flip sequences on compact Kahler fourfolds with rational boundaries and fixed rational nef b-divisors. Models are normal and globally Weil rationally factorial, boundary coefficients are below one, and projective small diagrams have the required ample signs. Exceptional discrepancies equal to one are allowed. The claim does not construct every flip; its canonical singularity range is a scoped version of the broader log-canonical termination statement in the companion manuscript.

Archive: partial or special case. The canonical fourfold setting is a specified case of the broader flip-termination question.

Score basis: Canonical singularities narrow the range, but termination without scaling retains a long global birational burden.

[Read manuscript](#) | OPDP ID 40377483

OAI-M0121 | Family 056

Termination for projective log canonical fourfolds with rational boundary

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims termination of every permitted minimal-model program on projective characteristic-zero log canonical fourfolds with rational boundary. Negative extremal rays may be selected arbitrarily and birational steps may be mixed on the given models. Initial rational factoriality and pseudo-effectivity are not assumed. Its target is broad termination within dimension four, while the word permitted retains the program's specified geometric validity conditions and does not authorize treating arbitrary non-MMP transformations as included steps.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Arbitrary negative-ray choices and weak singularity assumptions create a substantial fourfold termination problem.

[Read manuscript](#)

OAI-M0122 | Family 056

Finite ordinary minimal model programs on compact Kähler fourfolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every maximal ordinary negative-ray program from a specified compact Kahler klt fourfold terminates. Effective rational boundaries and global Weil-divisor rational factoriality are required. The endpoint is nef in the pseudo-effective case and a projective Mori fibre space otherwise. This all-program assertion is stronger than existence of one finite chosen route. The exact category and endpoint distinction must therefore survive comparison with companion finite-program existence statements.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Termination for every allowed route is harder than selecting one successful finite program.

[Read manuscript](#)

OAI-M0123 | Family 056

Finite ordinary minimal model programs on compact Kähler fourfolds

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims existence of a finite ordinary minimal-model program beginning on each specified compact Kahler klt fourfold pair. Global Weil rational factoriality, effective rational boundary and a canonical rational line bundle are assumed. The endpoint is nef or a Mori fibre space according to pseudo-effectivity. This is an existential finite-route statement; it should not be identified with the companion assertion that every maximal negative-ray program terminates merely because the two manuscripts share a title.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Constructing one finite valid program narrows the arbitrary-sequence termination burden.

[Read manuscript](#)

OAI-M0124 | Family 057

The abelianity conjecture for special compact Kähler manifolds

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims virtual abelianity of the ordinary fundamental group of every special compact Kahler manifold, in arbitrary dimension. Kodaira-dimension-zero manifolds are included as a consequence. Specialness is Campana's geometric condition, not an informal synonym for a convenient example. The target is the full ordinary fundamental group rather than only its linear representation images or abelianization, so a weaker monodromy theorem does not establish the same all-dimensional abelianity assertion under review.

Archive: related target. Campana's compact Kahler abelianity target is explicitly named, but the exact specialness and group conventions are not certified by the archive excerpt.

Score basis: Entire-group virtual abelianity is stronger than linear-image control and requires deep special-variety structure.

[Read manuscript](#) | OPDP ID 40321834

OAI-M0125 | Family 057

Two-step monodromy of special quasi-projective varieties

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a separate proof that linear representations of fundamental groups of special smooth quasi-projective complex varieties have virtually two-step nilpotent image. The conclusion had been previously announced, so its exact prior status must remain visible. A second construction gives special open surfaces whose general quasi-Albanese fibres are not special. The representation-image theorem does not itself classify the whole fundamental group, and new formalization eligibility would require separate library and contribution review.

Archive: related target. The manuscript treats linear representation images rather than the entire fundamental group. It also reports an earlier announcement, so independent original novelty is not inferred.

Score basis: A known monodromy theorem still has technical proof burden, while its scope is weaker than full group classification.

[Read manuscript](#) | OPDP ID 40373620

OAI-M0126 | Family 057

A conditional abelianity theorem for special fourfold pairs with a half-weight divisor

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims virtual abelianity of an order-two root-orbifold group associated with a special projective fourfold pair carrying half a smooth connected divisor. It assumes the all-dimensional compact Kahler abelianity theorem. A constructed special eightfold maps onto the required group. This is a conditional transfer to a precisely weighted orbifold, not an independent proof of general orbifold abelianity or the assumed compact theorem; both weight and divisor conditions shape the scored target.

Archive: partial or special case. The compact abelianity theorem is a load-bearing assumption for a special half-divisor root-orbifold transfer, not independently proved here.

Score basis: A strong companion theorem supplies the main group barrier, leaving a scoped geometric transfer construction.

[Read manuscript](#) | OPDP ID 40321834

OAI-M0127 | Family 058

Semialgebraic universal covers of normal projective varieties

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a characterization of normal projective complex varieties with semialgebraic universal covers. Such a cover occurs exactly when it decomposes into a bounded symmetric domain, complex affine space and a simply connected normal projective factor. Both directions and the specified product category are part of the target. The theorem concerns semialgebraic realizability, rather than every universal cover or an arbitrary complex-analytic domain lacking the projective and definability hypotheses.

Archive: related target. The semialgebraic-cover classification is present by name, but exact normal-variety and product-factor conventions are not recoverable from the excerpt.

Score basis: A global analytic-algebraic cover classification requires extensive structure and rigidity arguments.

[Read manuscript](#) | OPDP ID 40321437 | [Family formalization notes](#)

OAI-M0128 | Family 058

Symmetry of semialgebraic bounded domains with compact quotient

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that a connected semialgebraic bounded domain in a complex affine variety with a properly discontinuous cocompact biholomorphism group is smooth and biholomorphic to a bounded symmetric domain. Semialgebraicity and compact quotient are crucial. The statement addresses a specific rigidity question rather than every bounded domain with some automorphism. Its smoothness conclusion also matters when the affine ambient setup permits singularities, so classification cannot be inferred from symmetry alone without establishing that regularity.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Definability plus cocompactness must force strong analytic regularity and domain symmetry.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0129 | Family 059

Ambiently homeomorphic isolated hypersurfaces of multiplicities two and three

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims reduced isolated hypersurface germs that are ambiently homeomorphic yet have multiplicities two and three. They lie in a common complex affine dimension divisible by eight, and defining functions are also topologically right equivalent. This would refute embedded Zariski multiplicity and the stated corank expectation. Exact ambient equivalence is stronger than intrinsic homeomorphism of the zero sets, so the asserted counterexample requires both the prescribed equivalence and genuinely unequal multiplicities.

Archive: related target. The archived statement asks multiplicity invariance under topological equivalence. The proposed ambiently homeomorphic unequal-multiplicity examples directly address that family, but the archive's full equivalence convention is truncated.

Score basis: Coordinating ambient topological equivalence with distinct analytic orders challenges a major singularity invariant conjecture.

[Read manuscript](#) | OPDP ID 30002104

OAI-M0130 | Family 059

Ambiently homeomorphic isolated hypersurface germs in $\mathbb{C}^4$ with multiplicities four and five

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims two convergent holomorphic function germs in four complex variables with isolated critical points and multiplicities four and five, whose zero-set germs are ambiently homeomorphic. It targets Zariski's multiplicity question at a fixed small ambient dimension. The reducedness and isolated-critical-point assertions are essential. This more specific construction is not merely a restatement of the higher-dimensional companion example, and its ambient topology must be checked independently for the exact germs claimed here.

Archive: related target. The four-variable multiplicity counterexample concerns the archived invariance family; ambient-homeomorphism versus other topological-equivalence conventions require exact source verification.

Score basis: A low-ambient-dimension multiplicity counterexample has restrictive singularity and topology requirements.

[Read manuscript](#) | OPDP ID 30002104

OAI-M0131 | Family 060

Global Spherical Shells on Minimal Surfaces of Class VII

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the Global Spherical Shell conjecture for minimal compact complex class-VII surfaces with positive second Betti number. The shell is a holomorphically embedded neighborhood of the standard three-sphere with connected complement. This concrete global structure concerns all qualifying surfaces, not a finite list of known examples. Minimality, class VII and the positive-Betti condition delimit the target; it does not assert analogous shells on arbitrary compact complex surfaces or every nonminimal modification.

Archive: related target. The class-VII spherical-shell target is present; source recovery is needed to confirm every minimality and Betti convention.

Score basis: Uniform shell existence on class-VII surfaces requires difficult global non-Kähler surface geometry.

[Read manuscript](#) | OPDP ID 40346558

OAI-M0132 | Family 062

Contact Fano manifolds and the LeBrun-Salamon conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims contact-Fano homogeneity with the specified contact distribution, identifying qualifying projective manifolds with simple-Lie-algebra adjoint varieties. It also claims the positive quaternionic-Kähler LeBrun-Salamon conclusion in real dimension at least eight, up to homothety. These are linked complex and Riemannian classification targets. The distribution-preserving identification and metric scaling are part of the exact assertions; a weaker underlying-manifold classification would not establish the complete claimed results in this manuscript's scope.

Archive: related target. The LeBrun-Salamon classification is archived. Exact contact-Fano and metric normalization statements are not fully supplied by the excerpt.

Score basis: A linked complex-contact and Riemannian classification confronts a major structural geometry barrier.

[Read manuscript](#) | OPDP ID 40398728

OAI-M0133 | Family 063

The generalized Mukai conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the generalized Mukai inequality for every positive-dimensional smooth complex Fano manifold: Picard number times pseudoindex minus one is at most dimension. Equality is classified as a product of the corresponding projective spaces. Pseudoindex is the relevant rational-curve invariant, not automatically interchangeable with the divisibility index. Both the sharp inequality and equality description are assessed, rather than a weaker numerical estimate on selected homogeneous or low-dimensional Fano varieties.

Archive: related target. The generalized Mukai target is named, while exact pseudoindex equality conventions need the full archive statement.

Score basis: A sharp all-Fano numerical inequality and equality rigidity require broad rational-curve geometry.

[Read manuscript](#) | OPDP ID 40362400

OAI-M0134 | Family 064

Topological triviality of mu-constant families of surface singularities

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims topological right-triviality of any one-parameter holomorphic family of isolated hypersurface singularities in complex three-space with constant Milnor number. The trivialization fixes both parameter and origin section and preserves the defining functions. This is the surface-dimensional mu-constant target, not just homeomorphism of individual zero fibres. The isolated-singularity and single-parameter conditions matter when distinguishing it from broader equimultiplicity predictions or arbitrary deformations in a different ambient dimension.

Archive: related target. The mu-constant/equimultiplicity family is archived; right-triviality with parameter and defining-function preservation is a distinct precise scope.

Score basis: Function-preserving triviality in the exceptional surface case has deep local topology and deformation burdens.

[Read manuscript](#) | OPDP ID 40330823

OAI-M0135 | Family 065

Virasoro Constraints under Projectivization

D 725/1000 | p 1 | m 1 | Conditional points 525.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that full ordinary descendant Virasoro constraints transfer from a smooth projective complex base to any projective bundle of rank at least two. The vector bundle need not split or satisfy positivity. Every genus, individual curve class and primitive or odd insertion is included, with iteration along towers. The base constraints are assumed, making the target a transfer theorem rather than an unconditional proof for every projective variety without verified base input.

Archive: partial or special case. The parent Virasoro prediction is archived. This manuscript proves transfer assuming the base constraints and does not establish them for every smooth projective variety.

Score basis: Assumed base constraints narrow the task, while nonsplit bundles and all descendant data increase technical context.

[Read manuscript](#) | OPDP ID 7200020

OAI-M0136 | Family 065

Virasoro Constraints for Projective Complete Intersections

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims ordinary descendant Virasoro constraints for every smooth projective complete intersection, at every genus and integral curve class. Cohomology insertions can be primitive or odd, and semisimplicity is not assumed. These qualifiers make the assertion stronger than an ambient-cohomology or semisimple special case. The result is still specific to complete intersections in projective space; a separate projective-bundle transfer may extend it only after its hypotheses and base theorem are applied correctly.

Archive: partial or special case. Complete intersections form a specified ambient category within the broader archived Virasoro conjecture; primitive/odd and all-genus qualifiers belong to this precise claim.

Score basis: All genera, primitive insertions and lack of semisimplicity create a major enumerative-geometry target.

[Read manuscript](#) | OPDP ID 7200020

OAI-M0137 | Family 066

Bounded klt complements for Fano contractions

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims bounded-index klt complements near every closed base point of characteristic-zero epsilon-log-canonical Fano contractions. The bound depends only on dimension and epsilon. A complex version gives monotone complements for Fano-type pairs with a nef anticanonical adjoint and finite rational boundary coefficients. Locality, fixed coefficient sets and epsilon constraints are essential: the theorem does not promise one index for arbitrary dimension or unrestricted singularity depth and boundary values.

Archive: related target. Relative bounded klt complements are named in the archive, while epsilon and finite-coefficient restrictions need exact source comparison.

Score basis: Uniform complement indices across contractions require substantial singularity and boundedness theory.

[Read manuscript](#) | OPDP ID 40361924

OAI-M0138 | Family 066

Uniform Cartier sections for Fano type contractions

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims uniform Cartier divisors through prescribed points of positive-dimensional bases of epsilon-log-canonical Fano-type contractions. Their pullbacks have log canonical thresholds bounded below only by dimension and epsilon under a nef negative-adjoint condition. Rational boundaries are treated in characteristic zero and real boundaries over the complex numbers. The theorem is a controlled relative-section statement, not merely choosing any Cartier divisor or bounding thresholds for arbitrary non-Fano fibrations without positivity assumptions.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform relative sections and threshold control require delicate boundedness across singular fibrations.

[Read manuscript](#)

OAI-M0139 | Family 067

The Campana-Peternell conjecture in dimension six

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims rational homogeneity of every smooth complex projective Fano sixfold with nef tangent bundle. A compact Kahler consequence describes universal covers when dimension minus maximal irregularity is at most six. The six-dimensional restriction applies to the principal Campana-Peternell target. This gives a dimensional case, rather than solving the conjecture for all Fano dimensions, and its cover consequence additionally depends on the specified irregularity and tangent-nef structure hypotheses of the compact manifold.

Archive: partial or special case. The archived Campana-Peternell target is all-dimensional; this manuscript's principal Fano classification is dimension six.

Score basis: Dimension six restricts a broad classification conjecture but retains substantial tangent-positivity geometry.

[Read manuscript](#) | OPDP ID 50000457

OAI-M0140 | Family 068

Anticanonical nonvanishing from smooth semipositivity

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims anticanonical nonvanishing for every smooth connected projective complex variety whose anticanonical bundle has a smooth semipositive metric. Some positive power then has a nonzero section. Smooth semipositivity is stronger data than numerical nefness and must be preserved. The statement asserts existence of sections, not semiample or ampleness, so it should not be upgraded into a general abundance theorem for every nef anticanonical divisor or singular log pair.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Smooth curvature semipositivity supplies structure, yet converting it into sections across all dimensions is difficult.

[Read manuscript](#)

OAI-M0141 | Family 068

Invariant anticanonical indices and conversion of twisted differentials

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims polynomial behaviour of invariant Euler characteristics for anticanonical powers under compact-torus actions, on a divisible sequence with the actual exponent-zero value retained. A separate removal-of-error argument concerns semipositive anticanonical bundles on smooth projective varieties. Together they yield nonvanishing through descent and twisted differential forms. The invariant-index calculation and effectivity transfer are concrete targets, while the final nonvanishing conclusion depends on combining them with the stated geometric structure rather than extrapolating from an index polynomial alone.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Index asymptotics and error removal are technical transfer mechanisms rather than unrestricted positivity breakthroughs by themselves.

[Read manuscript](#)

OAI-M0142 | Family 068

Bounded anticanonical metrics on klt pairs and torus quotients

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims anticanonical sections for projective rationally factorial klt pairs under nefness, bounded semipositive metric weights on a resolution, and nonzero structure-sheaf Euler characteristic. The boundary is effective and rational. A rationally connected smooth consequence gives invariant plurisecions for torus actions. The metric and Euler assumptions are necessary to this target; it is not an unqualified section-existence theorem for every klt pair with nef negative log canonical divisor.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Strong metric and Euler inputs narrow the effectivity target to a difficult but controlled singular setting.

[Read manuscript](#)

OAI-M0143 | Family 068

Cohomological transfer and equivariant anticanonical sections

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a cohomological transfer removing a fixed negative pseudo-effective error from invariant anticanonical sections in unbounded degrees. It treats smooth projective complex varieties with smoothly semipositive anticanonical bundle and arbitrary torus linearization. With invariant-index and compact-monodromy inputs, it derives a positive untwisted anticanonical section. The target keeps the existence of the error-twisted sequence explicit, rather than assuming an Euler calculation alone produces the required invariant sections automatically.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: The error-removal implication has precise inputs and a substantial equivariant analytic proof burden.

[Read manuscript](#)

OAI-M0144 | Family 068

Metric descent and rank-preserving contractions

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims descent of corrected anticanonical metrics through specified toroidal models and rank-one generic adjoint data. A further transfer yields boundary-pole sections when finite-volume and bounded-weight metric conditions hold. For semipositive projective varieties without positive-degree holomorphic forms, it gives invariant anticanonical sections and rank-preserving contractions under further hypotheses. Those geometric and metric qualifiers are load-bearing; the direct target is a bundle of scoped descent implications, not unrestricted anticanonical effectivity on arbitrary bases.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Several controlled metric and rank hypotheses reduce generality while requiring complicated analytic and birational correspondence.

[Read manuscript](#)

OAI-M0145 | Family 068

Integrable metrics and effectivity with controlled boundary

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims rational effectivity of pseudo-effective divisors after a boundary-supported correction when a log-anticanonical metric has a strict curvature lower bound and an integrability property. A second implication removes a fixed pseudo-effective error from sections at unbounded anticanonical exponents under smooth semipositivity. The boundary and metric conditions are essential. This is an effectivity-transfer target, not an assertion that every pseudo-effective divisor on every smooth projective variety already has an effective representative.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Strong curvature and existing-section hypotheses make this a scoped analytic effectivity reduction.

[Read manuscript](#)

OAI-M0146 | Family 068

Exact orders and invariant anticanonical linear systems

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that a zero asymptotic valuation on a curvature-null face is actually attained by an effective rational anticanonical divisor. The variety is smooth, connected, complex projective and has a smoothly semipositive anticanonical bundle; the auxiliary line bundle is arbitrary. Attainment is stronger than merely approaching an infimum. The result concerns the specified curvature-face condition, so it should not be read as unconditional finite generation of all anticanonical linear systems or every possible divisor valuation.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Exact attainment of an asymptotic geometric order requires delicate metric-to-divisor transfer.

[Read manuscript](#)

OAI-M0147 | Family 069

Global quantum geometric Langlands at irrational level

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims global unramified quantum geometric Langlands at every irrational complex level, including nonreal levels, for connected simple complex groups. Full twisted D-module categories on bundle moduli are identified with those of the dual group at the specified reciprocal shifted level involving the lacing number. Global forms and connected components are retained. This all-category equivalence is stronger than a correspondence of selected objects or numerical invariants, while rational levels and ramified variants remain separate targets.

Archive: related target. The quantum geometric Langlands family is archived. Full-category irrational complex levels and global forms are not established as identical from the excerpt.

Score basis: A full quantum categorical equivalence carries exceptional prerequisite, context and formalization burdens.

[Read manuscript](#) | OPDP ID 40316387

OAI-M0148 | Family 071

Removable Boundaries and Rigidity of Circle Domains

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims conformal rigidity of any circle domain whose boundary is conformally removable, without limiting its number of complementary components. It establishes one direction of the He-Schramm prediction. Removability is assumed and is the exact rigidity-producing condition. The statement does not itself characterize which arbitrary boundaries are removable or prove the converse implication, so its score concerns this directional theorem rather than the entire bidirectional conjectural framework surrounding circle-domain uniformization.

Archive: partial or special case. Only removability implies rigidity is claimed, under the assumed removable boundary; the whole bidirectional framework is not scored as completed.

Score basis: Arbitrary connectivity increases analytic complexity, while removability supplies a strong rigidity hypothesis.

[Read manuscript](#) | OPDP ID 40373633 | [Family formalization notes](#)

OAI-M0149 | Family 071

Koebe's Circle-Domain Conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Koebe circle-domain uniformization for every domain in the Riemann sphere. A conformal equivalence must send the domain to one with circular or point complementary components, covering unrestricted connectivity. This is an existence theorem rather than a uniqueness classification of every such equivalence. Boundary regularity or countability limitations are not imposed in the stated target, so finite-connectivity examples and known restricted uniformization cases would not suffice to establish its claimed full scope.

Archive: related target. Koebe circle uniformization is named in the archive; unrestricted connectivity and exact complementary-component conventions need full source confirmation.

Score basis: Unrestricted-connectivity conformal uniformization is a deep analytic construction and compactness target.

[Read manuscript](#) | OPDP ID 40373632 | [Family formalization notes](#)

OAI-M0150 | Family 072

Brennan's conjecture and sharp inverse-square integral means

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Brennan integrability for derivatives of conformal maps from simply connected plane domains to the disk, throughout the open exponent range between four thirds and four. It also gives the sharp inverse-square universal integral-means value for normalized schlicht maps. Endpoints are not included in the integrability assertion. The range, mapping direction and area measure matter to the statement, preventing accidental conversion into a different derivative-integrability conjecture or an endpoint estimate.

Archive: related target. Brennan's conjecture is archived, but the mapping direction, area measure and strict range should not be inferred solely from the named excerpt.

Score basis: Sharp conformal distortion exponents require difficult uniform analytic control near irregular boundaries.

[Read manuscript](#) | OPDP ID 40393538 | [Family formalization notes](#)

OAI-M0151 | Family 072

A strict inverse-first-power bound for univalent functions

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an inverse-first-power integral-means upper exponent strictly smaller than one quarter for normalized univalent disk maps. It implies the bounded universal spectrum lies below the Kraetzer prediction at negative one. The target is a strict improvement at one specific spectral parameter, not a complete formula for the spectrum at every parameter. Keeping the normalized class and bounded-spectrum consequence separate matters when comparing to other inverse-power or meromorphic conformal estimates.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A single strict spectrum improvement is narrower than a complete spectrum conjecture but needs nontrivial distortion analysis.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0152 | Family 073

The Falconer distance conjecture in all dimensions

D 729/1000 | p 1 | m 1 | Conditional points 531.441

Published original CFSD-1000 target difficulty

The manuscript claims the Falconer distance conjecture in every Euclidean dimension at least two. A compact set with Hausdorff dimension strictly greater than half the ambient dimension determines distances of positive Lebesgue measure. The strict threshold and compactness are essential. This is a measure conclusion for fractal sets rather than counting distances in finite point configurations, so results about discrete distinct distances or weaker dimension thresholds do not establish the exact claimed theorem.

Archive: same target. The complete original archived statement has the same compact-set, strict half-dimension and positive-distance-measure conclusion. Original CFSD729 is preserved rather than replacing it with a new rounded estimate.

Score basis: The sharp all-dimensional fractal distance threshold is a major harmonic-analysis barrier.

[Read manuscript](#) | [OPDP ID 1164](#) | [Family formalization notes](#)

OAI-M0153 | Family 074

Theakeya maximal conjecture in three dimensions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the three-dimensionalakeya maximal estimate at its conjectured exponent. Unit-tube averages of radius δ map L^3 space to L^3 direction space with at most an arbitrary ϵ power loss in δ . The constant may depend on ϵ . This quantitative maximal theorem is stronger information than a bare dimension assertion, but it remains specific to three dimensions and does not automatically state identical maximal estimates in every Euclidean dimension.

Archive: related target. The named three-dimensionalakeya maximal target is archived; exact tube, norm and ϵ -loss conventions remain excerpt-limited.

Score basis: Sharp tube-overlap control with only ϵ loss is a major geometric harmonic-analysis target.

[Read manuscript](#) | [OPDP ID 40365971](#)

OAI-M0154 | Family 074

Every four-dimensional Kakeya set has full Hausdorff dimension

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every four-dimensional Kakeya set has full Hausdorff dimension four. It requires a unit segment in each direction but imposes no compactness or regularity condition on the set or the witnessing segment family. The dimension target differs from the companion three-dimensional maximal bound. Hausdorff dimension is the stated invariant, so neither Minkowski-only estimates nor a claim about smooth or compact selected line families would establish this full assertion.

Archive: partial or special case. The archived Kakeya target spans every dimension. The manuscript claims dimension-four Hausdorff dimension only; original difficulty685 is not inherited.

Score basis: Full Hausdorff dimension without set regularity confronts a central four-dimensional incidence barrier.

[Read manuscript](#) | OPDP ID 32

OAI-M0155 | Family 075

Almost-everywhere Fourier convergence in $L \log L$

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims almost-everywhere convergence of ordinary symmetric Fourier sums for every complex circle function in $L \log L$. Convergence holds along the whole sequence, not only a subsequence or selected lacunary cutoffs. This is a sufficiency theorem at the stated endpoint-type integrability class. It does not claim ordinary Fourier convergence for every L -one function, so the precise logarithmic improvement in the function space is an essential part of the target.

Archive: partial or special case. The broader archived question characterizes L -one Fourier convergence; this manuscript proves sufficiency of the particular L -log- L class along full ordinary symmetric sums.

Score basis: Full-sequence almost-everywhere control at the L -log- L threshold requires difficult endpoint analysis.

[Read manuscript](#) | OPDP ID 30002499

OAI-M0156 | Family 076

Ultraflat real Littlewood polynomials

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims ultraflat real Littlewood polynomials at every sufficiently large integer length. Consecutive coefficients are plus or minus one, and modulus stays within any prescribed relative epsilon of square-root length on the entire unit circle. Signs may change with the length. The real-sign restriction and all-circle uniformity are central, distinguishing the target from complex unimodular coefficients, average norm flatness, or constructions available only on a sparse sequence of degrees.

Archive: related target. The archived L -one-flatness question is weaker than whole-circle real ultraflatness at every sufficiently large length; exact target scores cannot be inherited.

Score basis: Uniform flatness with real signs has a substantial combinatorial and harmonic obstruction.

[Read manuscript](#) | OPDP ID 40331235 | [Family formalization notes](#)

OAI-M0157 | Family 076

Nearly minimal maxima and positive minima of Littlewood polynomials

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims real Littlewood polynomials whose maximum modulus is asymptotically at most square-root length while their minimum stays at least one sixteenth of that scale. Every sufficiently large integer length is covered. The positive lower bound is fixed, rather than tending to the same sharp scale as the maximum, so this target is weaker than ultraflatness. It still controls the whole circle and goes beyond average-norm or sparse-length constructions.

Archive: related target. A positive minimum and almost-optimal maximum differ from the archive's L-one-flatness question.

Score basis: The fixed positive minimum weakens ultraflatness, while near-optimal maxima at every length retain substantial construction difficulty.

[Read manuscript](#) | OPDP ID 40331235 | [Family formalization notes](#)

OAI-M0158 | Family 076

Asymptotically minimal maxima of real Littlewood polynomials

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that the smallest possible maximum modulus of real-sign Littlewood polynomials is asymptotic to square-root length along all integer lengths. It derives unbounded binary merit factors and refutes the stated relative-gap and bounded-merit expectations. This target controls the maximum, not a uniformly near-optimal minimum. The companion ultraflat theorem is stronger; the merit-factor consequence should remain grouped with the same underlying maximum construction rather than counted independently by changing the presentation.

Archive: related target. All-length maximum-modulus optimality and unbounded merit factors are stronger/different objectives than L-one-flatness.

Score basis: All-length optimal maxima and merit-factor consequences require strong global sign constructions without full minimum control.

[Read manuscript](#) | OPDP ID 40331235 | [Family formalization notes](#)

OAI-M0159 | Family 077

Elliptic capacity propagation and Fourier restriction to the sphere

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Fourier extension from bounded functions on the two-sphere into every L^p space with p above three in three-dimensional Euclidean space. The open exponent threshold is asserted sharp. This is the bounded-data sphere restriction target, not the endpoint p equal to three or every possible curved surface and input space. The range and input norm are therefore essential when comparing to diagonal extension results and local-smoothing companions elsewhere in the release family.

Archive: partial or special case. The bounded-data three-dimensional sphere open range is a specified formulation of the broader restriction target.

Score basis: The full sharp open restriction range is a central oscillatory-integral and incidence target.

[Read manuscript](#) | OPDP ID 40403784

OAI-M0160 | Family 077

Diagonal Fourier extension for positively curved surfaces in three dimensions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims diagonal Fourier extension from L^p surface data to L^p ambient space for every p above three on compact smooth positively curved surfaces in three dimensions. Smooth boundary is allowed. A compact-paraboloid consequence gives two-dimensional Schrodinger local smoothing above the stated Sobolev threshold of two minus six over p . These are open-range assertions; neither endpoint equality nor extension for arbitrary nonpositively curved surfaces is automatically included in the exact claimed target.

Archive: related target. The curved-surface restriction topic is archived, but exact diagonal exponent and boundary conventions require full statement comparison.

Score basis: General curved surfaces and diagonal norms deepen the sharp three-dimensional restriction burden.

[Read manuscript](#) | OPDP ID 40345688

OAI-M0161 | Family 078

Bochner-Riesz multipliers in three dimensions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims three-dimensional Bochner-Riesz multiplier boundedness at every positive order on L^3 . Interpolation and duality then recover the conjectured strict exponent range. Strict order is essential: an order-zero endpoint assertion is not included merely because arbitrarily small positive orders are allowed. This is an operator-bound theorem for the specified dimension, and neither higher-dimensional cases nor endpoint conclusions outside the derived strict range are automatically established by the stated interpolation argument.

Archive: partial or special case. The general Bochner-Riesz target is archived; this manuscript specifies dimension three and the strict-order range.

Score basis: Sharp positive-order multiplier control is a central harmonic-analysis target with demanding oscillatory cancellation.

[Read manuscript](#) | OPDP ID 50000165

OAI-M0162 | Family 079

Critical local smoothing for the three-dimensional wave equation

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims critical L^3 local smoothing for the wave equation in three spatial dimensions, allowing every positive Sobolev loss. It targets Sogge's Euclidean three-dimensional conjecture. The arbitrarily small loss is still part of the conclusion, so a genuinely lossless endpoint bound is not asserted. The wave evolution and spatial dimension also distinguish this paper from free Schrodinger smoothing or variable-coefficient manifold theorems involving different propagation geometry and exponent ranges.

Archive: partial or special case. The Euclidean wave-smoothing topic is archived; the manuscript addresses three dimensions with every positive Sobolev loss.

Score basis: Critical wave smoothing with minimal losses requires difficult multilinear and geometric frequency control.

[Read manuscript](#) | OPDP ID 40403427

OAI-M0163 | Family 080

Endpoint convergence for the planar Schrodinger equation

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims almost-everywhere planar free-Schrodinger convergence at Sobolev regularity one third. Gaussian regularization is removed before taking time to zero, and on one full-measure spatial set the regularized limit exists for every time between zero and one. The resulting evolution is continuous at initial time. This specifies the representative and order of limits, rather than an informal pointwise assertion that ignores how rough-data evolution is defined at individual spatial points.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: True endpoint convergence needs delicate maximal estimates and simultaneous control of regularization and time limits.

[Read manuscript](#)

OAI-M0164 | Family 080

Endpoint pointwise convergence for the Schrodinger equation in higher dimensions

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims almost-everywhere free-Schrodinger convergence at Sobolev exponent n divided by twice n plus one in every spatial dimension at least three. Gaussian regularization is removed uniformly over positive times below one on a single full-measure set. The critical exponent is included, unlike a theorem requiring additional regularity. The exact representative matters for rough data, while the claim remains an initial-time convergence statement rather than full nonlinear evolution or unrestricted pointwise differentiability.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: All-dimensional endpoint regularity and uniform representatives increase the analytic and formal fidelity burdens.

[Read manuscript](#)

OAI-M0165 | Family 081

Riesz transforms and uniform rectifiability in higher codimension

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that bounded higher-codimension Riesz transforms force uniform rectifiability for Ahlfors-David regular measures. Dimensions satisfy ambient d at least four and two at most n at most d minus two. Bounds must hold for every positive hard truncation from scalar to vector-valued L -two. The conclusion gives uniform large Lipschitz-image pieces, not mere rectifiability almost everywhere. These regularity and truncation assumptions are essential to the remaining David-Semmes target under assessment.

Archive: partial or special case. The Riesz-transform rectifiability family is present; the precise higher-codimension dimension range and hard-truncation assumptions define this reported target.

Score basis: Higher codimension removes codimension-one tools and requires strong quantitative geometric measure theory.

[Read manuscript](#) | [OPDP ID 40397235](#) | [Family formalization notes](#)

OAI-M0166 | Family 082

Annular variation of the triangular Hilbert transform at the symmetric point

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims annular r -variation control for the triangular Hilbert transform at the symmetric L -three input exponents, for every r above two. Partitions may depend on the output point and run through all positive scales. Consequences include two-endpoint maximal estimates and joint norm and almost-everywhere principal values. This variation target is stronger than one fixed truncation bound; the symmetric exponent and scalar complex-input setup remain explicit rather than an unrestricted full-range multilinear assertion.

Archive: related target. The archive names triangular Hilbert boundedness; all-scale output-dependent variation is a stronger scoped operator claim.

Score basis: Output-dependent all-scale variation requires more control than ordinary boundedness of a singular integral.

[Read manuscript](#) | OPDP ID 40401120 | [Family formalization notes](#)

OAI-M0167 | Family 082

An L^3 bound for the dyadic triangular Hilbert form

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a dyadic triangular Hilbert-form estimate for unrestricted real L -three inputs. Absolute local contributions at arbitrary finite scales sum to at most forty times the product of norms. Independently varying bounded coefficients on interval triples are allowed. This is an absolute dyadic-form theorem, distinct from the continuous maximal or variation statements in companion papers. The exact dyadic discretization and local summation structure must therefore be preserved when judging what the estimate actually establishes.

Archive: related target. The absolute dyadic form is not identical to the continuous triangular-Hilbert boundedness target.

Score basis: Absolute local summation strengthens the dyadic target, while discretization reduces continuous-scale difficulties.

[Read manuscript](#) | OPDP ID 40401120 | [Family formalization notes](#)

OAI-M0168 | Family 082

The maximal triangular Hilbert transform at the symmetric point

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a pointwise maximal triangular Hilbert estimate from two complex L -three inputs to L -three-halves, taking a supremum over both hard truncation endpoints. It gives joint almost-everywhere and norm convergence as lower and upper endpoints approach zero and infinity. The symmetric exponent is fixed. This maximal claim is stronger than a single truncated bound but weaker in stated structure than the companion all-partition variation theorem, so the two targets need separate scope-sensitive estimates.

Archive: related target. Two-endpoint pointwise maximal control is stronger than the archived ordinary boundedness formulation; scope identity is not inferred.

Score basis: Two-endpoint maximal control creates a major singular-integral barrier beyond fixed-scale estimates.

[Read manuscript](#) | OPDP ID 40401120 | [Family formalization notes](#)

OAI-M0169 | Family 083

A uniform Hilbert transform estimate for Lipschitz directions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform planar L-two Hilbert-transform estimate along Lipschitz unit vector fields depending on both coordinates. Integration lengths are limited to an absolute multiple of the reciprocal Lipschitz seminorm, and the bound is uniform in the inner truncation. This supports Stein's weak-two-two conclusion at that scale. The short-scale condition is essential: the claim is not an arbitrary long-integration theorem for every merely measurable direction field or an unrestricted endpoint estimate in higher dimensions.

Archive: partial or special case. The manuscript imposes a short integration scale proportional to the reciprocal Lipschitz seminorm; the broad direction-field target requires those conventions to be matched.

Score basis: Two-coordinate Lipschitz directions present a major variable-direction barrier, moderated by the prescribed short scale.

[Read manuscript](#) | OPDP ID 50000436 | [Family formalization notes](#)

OAI-M0170 | Family 084

The geometric case of the Erdős similarity conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims large-measure compact subsets of the unit interval avoiding all nontrivial affine copies of a fixed geometric progression. Every ratio strictly between zero and one is permitted, and both signs of dilation are excluded. The constructed set may depend on the ratio. This is a geometric-sequence case of Erdős similarity, not one simultaneous avoidance set for every ratio or a resolution for every infinite sequence with unrelated spacing and accumulation behaviour.

Archive: partial or special case. Only fixed geometric sequences are treated, with a ratio-dependent avoidance set; arbitrary infinite sequences remain outside this target.

Score basis: Uniform avoidance across translations and dilations is difficult, while fixing one geometric ratio narrows the parent conjecture.

[Read manuscript](#) | OPDP ID 40365065 | [Family formalization notes](#)

OAI-M0171 | Family 084

The dyadic case of the Erdős similarity conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims compact sets of measure arbitrarily close to one in the unit interval containing no affine copy of the dyadic sequence. Every translation and every nonzero real dilation is covered, including negative dilations. The ratio is fixed at one half, making this narrower than the companion arbitrary-geometric-ratio result. High measure and total affine avoidance must coexist; excluding only a finite set of scales or one selected translation would not establish the exact target.

Archive: partial or special case. The dyadic sequence is one specified case of the broader similarity problem.

Score basis: The dyadic structure gives leverage, but high-measure avoidance of every affine placement remains nontrivial.

[Read manuscript](#) | OPDP ID 40365065 | [Family formalization notes](#)

OAI-M0172 | Family 085

An Endpoint Gradient Bound for the Centered Disk Maximal Operator

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an L^1 -one gradient bound for the planar centered-disk maximal operator on real Sobolev $W^{1,1}$ functions. Its weak gradient is globally integrable and the maximal function locally belongs to the same Sobolev class. Disk centering and dimension two define the exact operator. The theorem does not automatically include other averaging shapes or every spatial dimension, and local function integrability should not be silently upgraded into an unsupported global maximal-function L^1 assertion.

Archive: partial or special case. The manuscript's dimension-two centered disks form a specified endpoint Sobolev case; other shapes and dimensions are not supplied.

Score basis: Endpoint Sobolev control of a nonlinear centered maximal operator has a substantial analytic obstacle.

[Read manuscript](#) | OPDP ID 40399173 | [Family formalization notes](#)

OAI-M0173 | Family 086

An L^3 bound for the trilinear Hilbert transform

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that the trilinear Hilbert transform with fixed slopes one, two and three maps three L^3 functions to L^1 . The slopes and exponents specify this case of the conjecture. This is not the same operator as the triangular Hilbert transform on planar inputs, despite similar naming. Nor does the target establish every possible slope configuration or full conjectured exponent range; those variants require their own source statements and assessment beyond this claimed symmetric instance.

Archive: partial or special case. The trilinear-Hilbert target is restricted here to slopes 1,2,3 and symmetric L^3 exponents; it differs from the triangular operator.

Score basis: Three interacting modulation directions create a deep multilinear singular-integral target.

[Read manuscript](#) | OPDP ID 40359602

OAI-M0174 | Family 087

The symmetric Mahler conjecture and its equality cases

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the symmetric Mahler lower bound for polar volume products in every dimension and classifies equality as invertible linear images of Hanner polytopes. Origin symmetry is required. The exact bound is four to dimension divided by dimension factorial. Equality rigidity adds scope beyond merely proving that cubes attain the minimum. This differs from the nonsymmetric companion involving a Santalo translation and simplex equality, so neither theorem should inherit the other's hypotheses or extremal classification.

Archive: related target. The original Mahler target asks symmetric minimum volume product; this manuscript adds a full Hanner equality classification. Original CFSD730 is a parent anchor, not an inherited score for the stronger exact reported scope.

Score basis: Sharp all-dimensional volume products plus Hanner equality classification create a major convex-geometry target.

[Read manuscript](#) | OPDP ID 1122 | [Family formalization notes](#)

OAI-M0175 | Family 087

The Mahler Conjecture for General Convex Bodies

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Mahler's lower bound for arbitrary convex bodies after translation to the Santalo point. The volume product is at least dimension-plus-one to that power divided by the square of dimension factorial, and equality occurs exactly for simplices. This is the nonsymmetric version, with its specific centering operation and extremizers. The symmetric Hanner-polytopal theorem is a different target, and the present assertion does not imply an arbitrary-origin polar bound without the required translation.

Archive: related target. The general Mahler target is named, while Santalo-centering and simplex equality conventions cannot be certified from the excerpt alone.

Score basis: Nonsymmetric sharp volume products require controlling translation as well as global convex shape.

[Read manuscript](#) | OPDP ID 40401981 | [Family formalization notes](#)

OAI-M0176 | Family 087

Symplectic Balls in Symmetric Polar Products

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Gromov width four for interiors of symmetric convex bodies times their polar bodies in every dimension at least two. Smooth symplectic embeddings of balls at every capacity below four are constructed. Volume preservation then yields the symmetric Mahler lower bound. The exact symplectic target is stronger structural information than the volume inequality alone; it should therefore be distinguished from the companion direct Mahler theorem and its separate equality-classification assertion.

Archive: related target. Polar-product symplectic width and ball embeddings imply the symmetric volume bound but are a distinct stronger structural target than archived Mahler minimization.

Score basis: Explicit near-sharp symplectic embeddings in arbitrary polar products combine difficult convex and symplectic geometry.

[Read manuscript](#) | OPDP ID 1122 | [Family formalization notes](#)

OAI-M0177 | Family 088

Petty's projection-volume conjecture in dimensions at least four

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims unique minimization of projection-body volume by ellipsoids among convex bodies of fixed volume in dimensions at least four. This addresses Petty's projection-volume target in that dimensional range. The object is the projection body, and the assertion includes uniqueness, so it should not be confused with a polar projection inequality or a maximum problem with simplex extremizers. Lower dimensions and different normalizations need independent comparison rather than automatic transfer from the stated theorem.

Archive: related target. The named Petty projection inequality is present; projection-body volume minimization is not automatically identical to polar projection formulations.

Score basis: Sharp projection-body minimization and rigidity in high dimensions require substantial convex analytic control.

[Read manuscript](#) | OPDP ID 40400020 | [Family formalization notes](#)

OAI-M0178 | Family 088

A product counterexample to the simplex maximum for projection-body volume

D 600/1000 | p 1 | m 1 | Conditional points 360

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that a product of two ten-dimensional simplices beats a twenty-dimensional simplex in normalized projection-body volume. This is a concrete counterexample to the proposed simplex maximum, not an identification of the true maximizing convex body in every dimension. The product and normalization must be evaluated exactly. The construction's value is to eliminate a specific extremal prediction and redirect the optimization question, while global maximization and the remaining dimensional range stay separate targets.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A structured finite-dimensional counterexample has computational leverage and narrower scope than full extremal classification.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0179 | Family 089

Planar Graph Metrics Embed into L1 with Constant Distortion

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims bounded-distortion embeddings into real L-one for every finite connected planar graph equipped with arbitrary positive edge lengths. One universal constant controls the distortion. The claim concerns weighted shortest-path metrics, not merely unweighted graphs or a chosen planar subclass. It also differs from decomposing graphs into paths or cycles. Its significance is a general metric representation theorem without demanding an isometric embedding or specifying an optimal numerical value of the universal distortion constant.

Archive: related target. The planar metric L-one target is named, but arbitrary positive edge weights and uniform distortion conventions remain excerpt-limited.

Score basis: Universal weighted-planar metric embeddings require a major structural metric argument.

[Read manuscript](#) | OPDP ID 40334126 | [Family formalization notes](#)

OAI-M0180 | Family 089

L1 Embeddings of Graphs of Bounded Treewidth

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims real L-one embeddings of finite connected weighted graphs with uniformly bounded distortion for each fixed treewidth bound. Edge lengths can be arbitrary positive reals. The constant may depend on the treewidth bound, so the theorem does not imply one universal distortion across all finite graphs. This is a metric consequence of bounded combinatorial width, distinct from induced-subgraph classification conjectures that merely use the same treewidth terminology in their statements.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A fixed-width decomposition offers leverage while controlling arbitrary weights remains technically substantial.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0181 | Family 090

An atomic certificate for triangular-lattice universal optimality

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims universal optimality of the density-one triangular lattice for all nonnegative completely monotone potentials of squared distance. Competitors are locally finite planar configurations with centered-disk density one, and infinite energies are included. A sharp Gaussian Fourier minorant is supplied through an atomic interpolation certificate. The target is energy optimality over the stated configuration class, not only comparison against other lattices or proof of the already known optimal circle-packing density.

Archive: related target. The archived hexagonal universal-optimality target is directly relevant; the manuscript's centered-density and infinite-energy conventions are more precise than a title match.

Score basis: A universal Fourier certificate must control arbitrary configurations and every admissible completely monotone potential.

[Read manuscript](#) | OPDP ID 50000248 | [Family formalization notes](#)

OAI-M0182 | Family 090

Universal optimality of the triangular lattice

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims triangular-lattice universal energy optimality among centered-density-one planar configurations, including infinite energies. It also treats logarithmic and renormalized Riesz energies for exponents strictly between zero and two, with corresponding jellium minima. These extra energy categories require normalization and limiting procedures beyond a lattice-only comparison. The broad result should not be reduced to a finite packing experiment, and the listed consequences remain scientifically linked within one release family rather than independent awards by default.

Archive: related target. The manuscript adds logarithmic, Riesz and jellium energy formulations beyond the archive's basic universal-potential target.

Score basis: Universal and renormalized energy categories add difficult limit and normalization control to the Fourier-certificate target.

[Read manuscript](#) | OPDP ID 50000248 | [Family formalization notes](#)

OAI-M0183 | Family 090

A sharp Fourier certificate for planar circle packing

D 700/1000 | p 1 | m 1 | Conditional points 490

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a radial Schwartz certificate making the planar Cohn-Elkies Fourier bound attain the exact triangular circle-packing density. Global sign conditions are supported by rigorous interval arithmetic and analytic estimates. Periodic equality cases recover classical triangular uniqueness. This targets sharpness of a proof method, not discovery of the already known optimal packing density. The full all-radius sign certification is required; checking a finite numerical grid without controlled tails would not establish the stated certificate.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Analytic tails and exact sign certification are demanding, but the target has strong computational verification feedback.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0184 | Family 090

Triangular minimality for planar Coulomb renormalized energy

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims triangular-lattice minimization of planar Coulomb renormalized energy over admissible curl-free fields with unit background. A cited asymptotic formula transfers it to the linear term in optimal spherical logarithmic energy. The claimed route uses Voronoi comparison and rigorous interval arithmetic. This is a renormalized infinite-configuration statement, not simply an ordinary finite energy sum or lattice-only calculation; the admissible-field and ordered-pair conventions remain essential to the exact geometric conclusion.

Archive: related target. Renormalized-energy minimization is archived; exact field admissibility and normalization require full source comparison.

Score basis: Global renormalized-field minimization requires controlling infinite limits beyond finite Voronoi computations.

[Read manuscript](#) | OPDP ID 40397446 | [Family formalization notes](#)

OAI-M0185 | Family 091

The logarithmic Brunn-Minkowski conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims logarithmic Brunn-Minkowski for arbitrary origin-symmetric convex bodies in every dimension. Symmetric L_p inequalities for positive p below one follow. With a transfer theorem and support reduction, it extends to even log-concave measures and the scalar-dilation B-conjecture. Symmetry and the stated transfer inputs matter. The target does not assert the identical logarithmic inequality for all nonsymmetric bodies or measures without the prescribed evenness, and linked implications should remain grouped appropriately.

Archive: related target. The logarithmic endpoint and even-measure transfer are stronger/different precise statements than a generic L_p Brunn-Minkowski excerpt.

Score basis: All-dimensional sharp convex interpolation plus measure transfer crosses a major geometric inequality barrier.

[Read manuscript](#) | OPDP ID 40401987 | [Family formalization notes](#)

OAI-M0186 | Family 092

A single-lattice covering bound of order $n \log n$

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every convex body in dimension at least two can be covered by translates along a single full-rank lattice with density at most an absolute constant times $n \log n$. No symmetry or boundary smoothness is assumed. The single-lattice condition is stronger than permitting arbitrary translative centres. The theorem gives an order bound rather than an exact best constant or a complete classification of covering-optimal lattices for individual convex bodies.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Requiring one lattice for arbitrary bodies adds a significant structure constraint to optimal-order covering.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0187 | Family 092

Translative covering densities of order $n \log n$

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims centrally symmetric convex bodies in every sufficiently large dimension whose translative covering density is at least an absolute multiple of $n \log n$. This refutes a proposed universal linear upper bound and matches the order of Rogers' upper estimate. It is an existential lower-bound construction, rather than saying every body requires that density. The target also concerns unrestricted translative centres, so a lattice-only obstruction would not by itself establish the same bound.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: An obstruction against every translative covering is substantial, though existential bad bodies narrow the target.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0188 | Family 093

A dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a dimension-independent logarithmic Sobolev inequality for centered log-concave probability densities with a bounded linear subgaussian parameter. The constant scales with the square of that parameter, uniformly over ambient dimension. Test functions are smooth and compactly supported. This is stronger than concentration alone, but assumes subgaussian one-dimensional marginals; it should not be described as the same inequality for every log-concave distribution without that input or every unrelated coarse-curvature problem.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Dimension-free functional inequality under marginal subgaussian control requires a deep concentration-to-entropy argument.

[Read manuscript](#)

OAI-M0189 | Family 094

Subpolynomial dimension reduction in L_p

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims subpolynomial dimension reduction for each fixed L_p exponent strictly between one and infinity and each distortion above one. Any n -point real L_p subset embeds in finite-dimensional ℓ_p with dimension n to a vanishing exponent. Nonlinear embeddings are allowed and the target retains the original p . This is not linear Johnson-Lindenstrauss in every norm, nor an embedding into a different exponent; those differences fundamentally change the claimed metric statement.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Same-exponent low-distortion reduction beyond Hilbert geometry is a major metric target.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0190 | Family 095

Hyperbolicity Cones Without Semidefinite Lifts

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims hyperbolicity cones that admit no finite affine semidefinite lift, even after arbitrary finitely many auxiliary variables and real coefficients are introduced. This would refute the projected Lax assertion and hence the ordinary generalized Lax prediction. The absence of any lift is stronger than failure of one matrix representation in the original coordinates. This distinction is crucial because a separate companion explicitly gives a lift for a different nonspectrahedral example within the same broad family.

Archive: related target. The projected Lax no-lift target is archived; exact closed-cone and affine-lift conventions cannot be fully confirmed from the excerpt.

Score basis: Ruling out every auxiliary-variable lift requires a strong global convex-algebraic obstruction.

[Read manuscript](#) | OPDP ID 40402662 | [Family formalization notes](#)

OAI-M0191 | Family 095

A nonspectrahedral hyperbolicity cone

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit degree-sixteen homogeneous polynomial in twenty-three real variables whose hyperbolicity cone has no finite homogeneous symmetric linear-matrix representation in its original coordinates. This refutes the geometric generalized Lax assertion. The target is nonspectrahedrality, not absence of semidefinite lifts. Indeed, its companion supplies an exact lift using additional variables, so the two manuscripts are compatible rather than contradictory when their representation models are stated precisely and their cones identified correctly.

Archive: related target. The archive's generalized Lax question is directly contradicted by a qualifying nonspectrahedral cone, but its source text is truncated and the exact reported construction includes further fixed data.

Score basis: A concrete nonspectrahedral witness is narrower than excluding all possible semidefinite lifts.

[Read manuscript](#) | OPDP ID 30002559 | [Family formalization notes](#)

OAI-M0192 | Family 095

An Exact Semidefinite Lift of a Nonspectrahedral Hyperbolicity Cone

D 600/1000 | p 1 | m 1 | Conditional points 360

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an exact semidefinite lift for the companion twenty-three-variable nonspectrahedral cone. A symmetric homogeneous pencil of size one hundred and 307 auxiliary variables represents the entire closed cone, including singular boundary cases. Those sizes are not claimed minimal. The assertion is a positive lift construction for one specific cone, not proof of the projected Lax conjecture for every hyperbolicity cone or a contradiction of another example lacking any lift.

Archive: related target. An exact lift of this one nonspectrahedral cone illustrates the broader shadow question but neither proves it for every cone nor refutes the separate no-lift example.

Score basis: An explicit finite lift has strong algebraic checking leverage, while singular boundary fidelity still matters.

[Read manuscript](#) | OPDP ID 40400872 | [Family formalization notes](#)

OAI-M0193 | Family 096

The Gaussian propeller bound in every dimension

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the Gaussian propeller upper bound for every finite measurable partition of Euclidean space, in every dimension. The summed squared Gaussian first moments do not exceed nine over eight pi. In dimension at least two with enough cells, three equal-angle planar sectors times an orthogonal factor attain it. The cell-count and dimension qualifications belong to attainment, while the upper inequality is general; neither part should be inferred from a finite numerical partition search.

Archive: related target. The propeller conjecture is named in the archive; precise all-partition and attainment conventions are unavailable from the truncated statement.

Score basis: Arbitrary measurable partitions require a global Gaussian extremal argument with sharp geometry.

[Read manuscript](#) | OPDP ID 40341560 | [Family formalization notes](#)

OAI-M0194 | Family 097

The Euclidean Steinitz-Bergström theorem

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims dimension-square-root signed-prefix control for every prescribed-order finite sequence of Euclidean unit vectors. A consequence reorders any zero-sum family with the same bound on ordinary partial sums. Constants are absolute and independent of sequence length. This determines the Euclidean Steinitz order up to constants, not its exact sharp constant. Prescribed-order signing is stronger structure than the reordering conclusion alone, and the two should remain linked within one contribution family.

Archive: related target. The Steinitz order target is archived, while prescribed-order signing and reordering are distinct stronger outputs of this manuscript.

Score basis: Simultaneous control of all prefixes requires deep vector-balancing structure with limited direct search scale.

[Read manuscript](#) | OPDP ID 40368422 | [Family formalization notes](#)

OAI-M0195 | Family 098

A doubling Hilbert subset with no finite-dimensional bi-Lipschitz embedding

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a compact doubling subset in every infinite-dimensional real Banach space that admits no bilipschitz embedding into any finite-dimensional real normed space. Its doubling constant is universally bounded, independently of the ambient space. Hilbert space is included, giving a negative Lang-Plaut answer. The obstruction concerns nonlinear bilipschitz embeddings, not merely linear dimension reduction; compactness and doubling must both survive the construction rather than appearing only in separate illustrative examples.

Archive: related target. The finite-dimensional bilipschitz embedding problem is present; the manuscript strengthens the counterexample to compact universally doubling subsets of every infinite-dimensional Banach space.

Score basis: A universal compact doubling obstruction must defeat all finite-dimensional nonlinear embeddings.

[Read manuscript](#) | OPDP ID 40384349 | [Family formalization notes](#)

OAI-M0196 | Family 099

Edit Distance in $\mathbb{I}^1$: Matching Bounds up to Constants in the Exponent

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims matching exponential-scale bounds on L-one distortion of ordinary edit distance for strings of length at most d . Both bounds have exponent proportional to the square root of $\log d$ times $\log \log d$, uniformly over finite alphabets with at least two symbols. Binary equal-length strings already realize the lower obstruction. The constants in the exponent are not identified, so the target is order-sharp logarithmic distortion rather than an exact distortion formula.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Matching uniform upper and lower metric bounds require substantial coding and embedding arguments.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0197 | Family 099

Finite-Circle Obstructions, Binary Codes, and Histogram Embeddings for Edit Distance

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims two finite-circle binary-string constructions forcing the stated exponential L-one distortion lower bound, together with constant-distortion coding transfers. It also completes a finite overlapping-substring histogram embedding at the matching scale for arbitrary finite alphabets and strings up to the length cap, including empty strings. Each lower construction uses one common length. This separates concrete obstruction families from the general upper embedding and does not assert exact optimal constants in either direction.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Explicit binary obstructions and a complete histogram embedding combine constructive combinatorics with difficult metric estimates.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0198 | Family 099

Tree Constructions for the l1 Distortion of Binary Edit Distance

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims two tree-based lower-bound constructions for ordinary binary edit-distance embeddings into L-one. Strings have one common length at most the cap for every sufficiently large cap, and a constant-distortion binary conversion is included. The lower exponent is the square root of $\log d$ times $\log \log d$. Matching-order completeness uses a companion upper theorem, so that upper estimate is an input to the combined conclusion rather than independently supplied by this lower-bound target.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Constructive lower bounds have structural leverage, while uniform length caps and coding fidelity remain substantial.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0199 | Family 100

Finite angular cylinder covers below the half-area bound

D 600/1000 | p 1 | m 1 | Conditional points 360

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a finite cylinder covering of a regular tetrahedron whose total perpendicular-base area is below half its smallest orthogonal projection area. Compact triangular bases are used. Affine invariance gives a normalized version for every nondegenerate tetrahedron. This refutes a specific half-area prediction, not a full solution of optimal cylinder-covering density for arbitrary bodies. Exact area and coverage certification are necessary because a visually plausible collection of cylinders alone would not prove the inequality.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A finite structured counterexample offers strong geometry and computation leverage compared with universal covering classification.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0200 | Family 100

Finite cylinder approximation of ruled sets

D 600/1000 | p 1 | m 1 | Conditional points 360

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims finite-cylinder approximation of compact ruled segment families at integral projection cost plus any positive error. Velocity fields are continuously differentiable and satisfy explicit differential eigenvalue constraints relative to segment length, including zero and square-zero regimes with shear. Labels may form any compact set. These regularity and spectral hypotheses define a scoped approximation theorem; they do not assert every arbitrary ruled set can be covered at that cost without verifying the stated local geometry conditions.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Controlled differential hypotheses permit approximation, but singular regimes and finite coverage need careful analysis.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0201 | Family 100

Slope-field perturbations of the two-cylinder covering

D 600/1000 | p 1 | m 1 | Conditional points 360

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims perturbations of a tetrahedron's two-cylinder covering that yield finite covers with total perpendicular-base area strictly below half its minimum projection area. The construction refutes both the half-area prediction and a directionwise normalized bound. Affine invariance extends the latter conclusion to all nondegenerate tetrahedra. The target is a concrete strict improvement and coverage certificate, not a formula for the smallest possible cylinder-covering area of every convex body or a complete classification of optimal covers.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A perturbative finite counterexample offers constructive leverage but needs rigorous area and coverage control.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0202 | Family 100

Finite triangular approximation of radial sweeps

D 600/1000 | p 1 | m 1 | Conditional points 360

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims finite triangular-base cylinder approximations to radially aligned segment sweeps. Tagged partitions produce covers whose total base area converges to a weighted parameter area as mesh shrinks. A tetrahedron application crosses the conjectured half-area threshold. The approximation theorem and strict application are connected but distinct assertions. They require rigorous coverage and convergence, not merely an illustrative limiting sweep, and they do not characterize optimal cylinder covers for arbitrary ruled families without the radial-alignment setup.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Radial alignment supplies structure while passing from an infinite sweep to finite certified covers remains delicate.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0203 | Family 101

A sharp entropy bound and the simplex inequality for isotropic constants

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims simplices uniquely maximize the isotropic constant in each dimension. It also gives a sharp entropy lower bound for log-concave densities involving covariance determinant, with equality at affine images of products of one-sided exponential laws. These are strong extremal and equality statements, exceeding an unspecified universal slicing bound. The body and density formulations must preserve their normalizations and extremizers; neither equality classification follows merely from observing a known example reaches the proposed numerical value.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Exact all-dimensional isotropic extremizers and entropy equality create a major convex functional target.

[Read manuscript](#)

OAI-M0204 | Family 102

The Unique Games Theorem

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims deterministic polynomial reductions from three-SAT to fixed-alphabet Unique Games for every prescribed near-one completeness and small positive soundness. This is the Unique Games conjecture's stated gap formulation. Alphabet size may depend on the fixed accuracy parameters, rather than being universal across them. The target is hardness via a valid uniform reduction, not an algorithm deciding Unique Games or a proof that every instance has low value without its promised gap conditions.

Archive: related target. The Unique Games conjecture is present in the original cohort. Full gap-parameter and alphabet conventions are not established from the abbreviated record, so its CFSD622 is not automatically inherited.

Score basis: Near-perfect completeness and arbitrarily low soundness require a major approximation-hardness construction.

[Read manuscript](#) | [OPDP ID 43](#) | [Family formalization notes](#)

OAI-M0205 | Family 102

A Direct Proof of Optimal Max-Cut Hardness

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims NP-hardness of approximating Max-Cut beyond the Goemans-Williamson ratio on simple unweighted graphs. Every fixed factor above that threshold is covered. The word direct describes its proof route, while the theorem's target is the sharp hardness barrier in this graph model. It does not assert an exact complexity classification for every approximation ratio or provide a matching algorithm as a new result, and the underlying reduction must establish the specified promise rigorously.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: A sharp threshold reduction is technically demanding, but the restricted approximation target is narrower than Unique Games itself.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0206 | Family 102

The Factor-Two Hardness Threshold for Vertex Cover

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims NP-hardness of approximating minimum Vertex Cover within any fixed factor strictly below two, on simple unweighted graphs. Strictness is essential: the assertion does not rule out the known factor-two method. The target concerns worst-case polynomial-time approximability of the classical covering objective. It is distinct from eternal vertex-cover variants or special graph classes, and simply sharing a hardness keyword with such archived entries does not establish equivalent mathematical hypotheses or conclusion.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Near-two hardness requires a strong gap reduction with precise completeness and soundness.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0207 | Family 102

Constant-factor hardness of Min-UnCut

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims NP-hardness of every fixed constant-factor approximation to Min-UnCut, even for simple undirected unweighted graphs. The optimization counts edges left uncut, so it differs from approximating Max-Cut by a ratio of its large cut value. The factor is fixed above one. This is a family of worst-case hardness gaps, not an assertion about the objective's exact value on every graph or an impossibility result outside the usual polynomial-time reduction assumptions.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Changing to the small uncut objective demands strong gap amplification rather than ordinary Max-Cut hardness alone.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0208 | Family 102

Constant-factor hardness of directed feedback vertex set

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims constant-factor approximation hardness for minimum directed feedback vertex set on unweighted digraphs. Every fixed approximation constant is covered. Directedness is essential because cycle-breaking behaviour differs from undirected versions and from planar girth restrictions. The target is a worst-case reduction theorem, not a claim that no useful approximation algorithm exists at nonconstant ratios. Source and comparison must therefore retain graph direction, vertex deletion and the constant-factor quantifier when linking archive records.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Broad constant-factor exclusion requires substantial directed-cycle gap construction.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0209 | Family 103

Exact derandomization of logarithmic space: $L = RL = BPL$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims deterministic logarithmic-space simulation of polynomial-time randomized logarithmic-space computation, yielding equality of L , RL and BPL . Both one-sided and bounded two-sided randomization are included. The space, time and error conventions define the classes being compared. This is an exact derandomization theorem, not a larger-space simulation, a nonuniform circuit statement or a claim that all polynomial-time randomness can be removed with no resource cost for arbitrary computation models.

Archive: same target. The curator-authored RL -versus- L target is settled by the claimed $L=RL$ equality; the BPL extension is additional scope. No original CFSD score is available for inheritance.

Score basis: Exact log-space derandomization confronts a major computational-complexity barrier with global uniformity requirements.

[Read manuscript](#) | OPDP ID 50000249

OAI-M0210 | Family 104

Turn-Based Stochastic Mean-Payoff Games in Deterministic Quasipolynomial Time

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims deterministic quasipolynomial algorithms for turn-based stochastic mean-payoff games with signed binary rewards and rational binary probabilities. It exactly identifies nonnegative values, including zero, under expectation of pathwise liminf payoff. Complexity is measured in complete binary input length using exact rational arithmetic. These semantic and bit-model choices are essential: the target is not polynomial-time simple stochastic games, nor a pseudopolynomial algorithm hiding reward magnitude in its running-time bound.

Archive: partial or special case. Quasipolynomial stochastic mean-payoff solving addresses related simple stochastic games but leaves the archived polynomial-time bound unresolved.

Score basis: Stochastic zero-threshold semantics and binary probabilities complicate uniform exact quasipolynomial game solving.

[Read manuscript](#) | OPDP ID 50000465 | [Family formalization notes](#)

OAI-M0211 | Family 104

Mean-payoff parity games in quasipolynomial time

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims deterministic quasipolynomial-time solution of mean-payoff parity games with arbitrary signed binary rewards and unrestricted binary priorities. It finds vertices ensuring both nonnegative liminf payoff and the parity condition. The bit bound depends on complete input length. This conjunctive objective differs from plain parity or plain mean payoff; the quasipolynomial result also leaves a polynomial-time parent target unresolved rather than settling it by relabeling the running-time class or ignoring binary encodings.

Archive: related target. The paper combines parity with mean payoff and gives quasipolynomial rather than polynomial complexity; its exact target differs from the plain archived game question.

Score basis: Combining quantitative rewards and parity with full binary complexity requires a substantial strategy algorithm.

[Read manuscript](#) | OPDP ID 40400734 | [Family formalization notes](#)

OAI-M0212 | Family 104

Deterministic quasipolynomial-time mean-payoff games

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims deterministic quasipolynomial-time mean-payoff solving for signed integer binary weights. It computes the complete zero-threshold winning set, then exact rational vertex values and optimal positional strategies for both players. The bound is expressed in full explicit input length. This is stronger output than approximate values but weaker time than polynomial. The target therefore advances the game-solving programme without establishing the broader archived polynomial-time or strongly polynomial conjecture for all reward magnitudes.

Archive: partial or special case. The archived polynomial-time mean-payoff question is advanced by an exact quasipolynomial algorithm but is not resolved at its requested complexity.

Score basis: Exact values and globally optimal strategies add algorithmic and arithmetic burden to a nonpolynomial upper bound.

[Read manuscript](#) | OPDP ID 40400734 | [Family formalization notes](#)

OAI-M0213 | Family 104

Randomized quasipolynomial-time mean-payoff games

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a randomized quasipolynomial-time mean-payoff algorithm for arbitrary signed binary weights. Every random tape respects the time bound, with correctness probability at least seven eighths. A polynomial certificate check accepts correct regions and positional strategies or reports failure, allowing always-correct repetition with the same expected order. The worst-case randomized and expected Las Vegas guarantees are distinct, and neither should be described as a deterministic polynomial-time resolution of the broader game-solving question.

Archive: partial or special case. The randomized/expected quasipolynomial result does not supply the archived deterministic polynomial-time game-solving target.

Score basis: Certificate feedback helps, but uniform binary-weight strategy construction and randomized guarantees remain substantial.

[Read manuscript](#) | OPDP ID 40400734 | [Family formalization notes](#)

OAI-M0214 | Family 105

Perfect completeness for 2-to-1 games

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the two-to-one Games hardness conjecture with perfect completeness. For any fixed positive rational soundness threshold below one, satisfiable fixed-alphabet instances are hard to distinguish from low-value instances. Constraints are explicitly listed as an unweighted multiset. Perfect satisfiability is the differentiating requirement from imperfect-completeness formulations. The claim is a promise-gap reduction rather than an exact algorithm for all games, and its alphabet and encoding choices remain part of the assessed mathematical target.

Archive: related target. The retrieved archive record explicitly concerns imperfect completeness; this manuscript's perfect-completeness target is a distinct stronger gap assertion.

Score basis: Perfect completeness removes a major hardness-construction relaxation and requires delicate consistency control.

[Read manuscript](#) | OPDP ID 40401953 | [Family formalization notes](#)

OAI-M0215 | Family 106

Hardness of finding large independent sets in three-colorable graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims hardness of distinguishing three-colourable graphs from graphs whose every independent set has density below any fixed threshold less than one third. It then derives hardness of finding any fixed finite c -colouring, for c at least three, on three-colourable inputs. The promise and density thresholds matter. This is a worst-case graph-colouring reduction, not a statement about random dense graphs, Euclidean colouring, or numerical searches for one particular difficult finite graph.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Arbitrary constant-colour hardness requires strong independent-set gaps under exact three-colourable completeness.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0216 | Family 107

An Upper Bound of $9/4$ for the Matrix Multiplication Exponent

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims matrix-multiplication exponent at most nine fourths over the complex numbers. The exponent is an asymptotic arithmetic-operation quantity, so the result is distinct from optimizing coefficients in one small rank-twenty-three three-by-three scheme. It also does not settle the parent exponent-two conjecture. The score concerns this specific asymptotic upper-bound target, with its field restriction, rather than counting an exact finite tensor certificate as a direct proof of an exponent value.

Archive: partial or special case. The archived parent asks exponent2; an upper bound $9/4$ leaves a positive gap. The parent's CFSD469 is not inherited for this narrower upper-bound target.

Score basis: A substantial asymptotic exponent improvement requires global tensor-extraction structure beyond fixed-size sparsity optimization.

[Read manuscript](#) | OPDP ID 3100086 | [Family formalization notes](#)

OAI-M0217 | Family 107

Complex Matrix Multiplication Below 2.258 and Rectangular Bounds

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims square matrix exponent below 2.258 over characteristic-zero fields, dual exponent above 0.465, and a specific rectangular exponent below 2.092. Most characteristic exceptions are confined to one finite set of primes. This secondary writeup is a companion contribution within the same family. The square bound is weaker than nine fourths, and its rectangular refinements have their own parameters; neither an unrestricted all-field bound nor exponent two follows from the stated numerical inequalities.

Archive: partial or special case. The square upper bound and rectangular improvements do not prove exponent2; the characteristic exclusions and secondary-writeup status remain explicit.

Score basis: Several rectangular bounds and controlled characteristic exclusions require demanding tensor asymptotics.

[Read manuscript](#) | OPDP ID 3100086 | [Family formalization notes](#)

OAI-M0218 | Family 107

Staggered extraction for exact matrix multiplication over every field

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims matrix-multiplication exponent below 2.371054886006746 over every fixed field, including all positive characteristics. The bound concerns exact arithmetic operations asymptotically, not numerical floating-point time or one finite matrix size. Its field-uniform scope differs from the sharper characteristic-zero companions. Both the weaker exponent and broader field class matter in comparing targets, and the result leaves exponent two unresolved even when it improves an older upper bound for a particular field.

Archive: partial or special case. An all-field exponent below 2.371054886006746 improves an upper bound while leaving the exponent2 parent unresolved.

Score basis: All-characteristic extraction is technically delicate, while the less aggressive exponent reduces the sharpness target.

[Read manuscript](#) | OPDP ID 3100086 | [Family formalization notes](#)

OAI-M0219 | Family 108

A cubic lower bound for border determinantal complexity of the permanent

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims cubic lower bounds for border determinantal complexity of the complex permanent. Affine-linear determinant representations and coefficientwise limits are allowed. Exact determinantal complexity and fixed-budget affine algebraic branching programs receive related lower bounds. Border formulations make the obstruction robust to degeneration. The result is a cubic bound rather than the conjectured superpolynomial separation, so it cannot be treated as a complete resolution of the permanent-versus-determinant parent merely because it establishes a nontrivial asymptotic barrier.

Archive: partial or special case. A cubic determinant-complexity lower bound advances but does not settle the superpolynomial permanent parent target.

Score basis: Border degeneration adds substantial algebraic complexity beyond exact-representation lower bounds.

[Read manuscript](#) | OPDP ID 40358961 | [Family formalization notes](#)

OAI-M0220 | Family 109

Integer multiplication below $n \log n$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims exact deterministic integer multiplication below $n \log n$ worst-case time on a fixed finite-alphabet multi-tape Turing machine. The saving is a log exponent of one minus two to the negative 182, for every input length. The model and tiny fixed saving are essential. It refutes the corresponding optimality prediction but does not give linear bit complexity, so the stronger archived linear-time target remains unresolved despite the asymptotic improvement in this specified machine model.

Archive: partial or special case. Sub- $n \log n$ complexity remains above the original archived linear-bit target; original CFSD557 is not inherited.

Score basis: A genuine sub- $n \log n$ bit-model algorithm needs deep asymptotic and implementation correspondence despite a tiny exponent gain.

[Read manuscript](#) | OPDP ID 8400041

OAI-M0221 | Family 110

Squared-logarithmic randomized k-server on arbitrary metrics

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims squared-logarithmic randomized k-server competitiveness on every metric space against oblivious request sequences. One policy works for all finite sequences from a fixed initial configuration, even on infinite unbounded spaces. Distinct initial positions eliminate an additive term. The target is an existence/competitiveness theorem, not automatically a uniform efficient implementation; the adversary restriction and metric quantifiers must remain visible when comparing it with algorithmic companions or other versions of the k-server conjecture.

Archive: related target. Randomized squared-logarithmic competitiveness against oblivious sequences differs from deterministic k-server and other adversary formulations; the exact reported target is scored separately.

Score basis: Arbitrary metrics and uniform request-sequence guarantees require deep online metric structure.

[Read manuscript](#) | OPDP ID 50000234 | [Family formalization notes](#)

OAI-M0222 | Family 110

Uniform computation of the squared-logarithmic k-server bound

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform computable randomized k-server algorithm on finite rational metrics with squared-logarithmic competitiveness. Preprocessing and per-request bit work are polynomial in the stated input and counter lengths. The instance-dependent additive movement constant may be enormous. A companion competitiveness existence theorem is used. Thus computational uniformity is the new target, while the finite rational setting and additive allowance make it distinct from an unrestricted efficient policy on every infinite metric space.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform bit implementation adds a meaningful algorithmic bridge while relying on a strong existence theorem.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0223 | Family 111

One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a constant-competitive matroid prophet rule using just one independent sample per element. The arrival adversary sees samples, online values and the complete random seed. No distribution description is needed, and the stated absolute ratio is two to the negative 310. This unusually strong adversary and weak-information setup defines the target. It differs from the matroid secretary problem's sampling/order model, so shared matroid terminology is insufficient for theorem identity or inherited difficulty.

Archive: related target. Matroid secretary and one-sample prophet models have different information and adversary assumptions, so the named matroid family is related rather than identical.

Score basis: One sample against a fully informed adversary creates a major information and combinatorial-selection obstacle.

[Read manuscript](#) | OPDP ID 50000438 | [Family formalization notes](#)

OAI-M0224 | Family 112

Beyond the Square-Root Exponent for Depth-Three Boolean Circuits

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a polynomial-time decidable language requiring more than exponential square-root-size depth-three OR-AND-OR circuits at every sufficiently large input length. All gates, including the bottom layer, are counted and fan-in is unbounded. The lower bound is super-square-root in the exponent, not an arbitrary-depth or superpolynomial general circuit separation. The exact depth and computational class must therefore remain explicit when interpreting the claimed advance and comparing it with unrelated parity or higher-depth circuit conjectures.

Archive: related target. The source names a depth-three super-square-root lower-bound target for a particular function; the manuscript constructs a polynomial-time language with its own all-length guarantee.

Score basis: Explicit uniform depth-three lower bounds beyond the square-root barrier require substantial complexity insight.

[Read manuscript](#) | OPDP ID 40399995 | [Family formalization notes](#)

OAI-M0225 | Family 113

A Fully Polynomial Randomized Approximation Scheme for Perfect Matchings in General Graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an FPRAS for counting perfect matchings in arbitrary finite simple undirected graphs. It returns zero certainly when no perfect matching exists, otherwise gives relative accuracy with polynomial bit time in input length, inverse tolerance and logarithmic inverse failure probability. The general nonbipartite setting is essential. This approximate counting target differs from existence theorems for bridgeless cubic graphs or finding one maximum matching, even though those problems share the same combinatorial objects.

Archive: related target. Perfect-matching approximate counting in arbitrary graphs differs from the Lovasz-Plummer lower-count conjecture in cubic bridgeless graphs.

Score basis: General-graph counting requires difficult distribution and mixing control beyond decision or matching construction.

[Read manuscript](#) | [OPDP ID 40396821](#) | [Family formalization notes](#)

OAI-M0226 | Family 113

Entropy and Face Dimension of the Perfect-Matching Polytope

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims sharp entropy inequalities for matching laws with prescribed feasible edge marginals, throughout a loopless multigraph perfect-matching polytope including its boundary. It also bounds support size minus minimal-face dimension by three m minus two for two m vertices. Marginal feasibility and boundary cases are essential. The target is a geometric/entropic theorem, not itself the full approximate-counting algorithm, although it may supply ingredients for algorithmic companions in the same result family.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Boundary entropy and face-dimension rigidity require detailed matching-polytope geometry under exact marginal constraints.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0227 | Family 114

An FPRAS for Common Integer Polymatroid Bases with Binary Capacities

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims approximate counting of common integer bases of two polymatroids with equal rank, using exact rank-value oracles. Ranks and capacities are binary encoded and each integer vector is counted once. Oracle calls and external bit work are polynomial on every execution in the stated accuracy parameters. Direct binary handling avoids expanding capacities into labelled copies. These counting and encoding conventions distinguish the target from ordinary matroid bases or a pseudopolynomial unit-capacity enumeration.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Binary capacities and oracle access create demanding counting and bit-complexity requirements.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0228 | Family 114

Approximate counting of common bases of two matroids

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an FPRAS for common bases of any two same-rank matroids supplied only by independence oracles. No explicit representation is assumed, and every execution has polynomial oracle and bit bounds. The target includes general pairs, not merely two graphic matroids or a special representable class. Approximate counting is distinct from deciding whether a common base exists, and the oracle guarantees must be maintained when translating the claimed algorithm into an implementable competition artifact.

Archive: related target. Counting common bases of two arbitrary oracle matroids is distinct from predicting the number of bases of one matroid.

Score basis: General oracle matroid intersection counting requires substantial combinatorial and sampling structure.

[Read manuscript](#) | OPDP ID 40327359 | [Family formalization notes](#)

OAI-M0229 | Family 115

Exact Uniform Sampling of Contingency Tables with Arbitrary Margins

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims exact uniform sampling of integer contingency tables with arbitrary prescribed nonnegative margins. Both dimensions vary, numerical data are binary encoded, and expected bit complexity is polynomial; termination holds almost surely. Positivity, balance and sparsity are not assumed. Exact uniformity and expected time distinguish this result from an FPRAS or a sampler restricted to dense balanced tables. An expected bound should not be reported as a polynomial worst-case guarantee on every random execution.

Archive: related target. Semigroup normality of tables differs from exact uniform sampling with arbitrary margins; the source is a shared combinatorial topic, not equivalent target.

Score basis: Exact sampling with arbitrary binary margins requires difficult high-dimensional discrete distribution control.

[Read manuscript](#) | OPDP ID 40396555 | [Family formalization notes](#)

OAI-M0230 | Family 115

An FPRAS for Cell-Bounded Contingency Tables

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an FPRAS for counting integer tables with prescribed row and column sums plus individual cell bounds. Both matrix dimensions and all binary numerical data vary, and zero cell bounds are allowed. Only unbiased random bits are used, with polynomial bit work on every execution. This adds hard capacity constraints to the table setting but gives approximate counting rather than exact sampling, so the output and runtime guarantees differ from the companion arbitrary-margin sampler.

Archive: related target. Approximate cell-bounded table counting is an algorithmic target distinct from the archive's normality property.

Score basis: Cell constraints and full binary encodings complicate uniform approximate counting.

[Read manuscript](#) | OPDP ID 40396555 | [Family formalization notes](#)

OAI-M0231 | Family 116

Uniform Matrix Hitting Points in Every Positive Characteristic

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims one deterministic matrix substitution detecting every nonzero bounded-size division-free noncommutative formula in each specified positive characteristic. It is generated in polynomial bit time from a binary prime and unary size parameters, and also works over extension fields. Matrix dimension is order n cubed times size to the sixth, including characteristic two. The target is a universal hitting tuple in that formula model, not unrestricted rational formulas or arbitrary algebraic circuits with division.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Uniform all-positive-characteristic hitting points need strong algebraic design and exact computational fidelity.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0232 | Family 116

One Rational Matrix Hitting Point for Noncommutative Formulas

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a single rational matrix tuple detecting every nonzero bounded-size division-free noncommutative formula over all characteristic-zero fields. One deterministic polynomial-bit procedure outputs matrices of dimension order variables times size squared. The same tuple is universal for that formula size. Division-free tree structure is crucial, so the result should not be extended without evidence to unrestricted circuits, arbitrary inverse nesting or identities whose coefficients and evaluation domain require another hitting-set construction.

Archive: related target. Noncommutative inverse realization differs from a universal hitting point for bounded division-free formulas.

Score basis: A universal single evaluation has strong algebraic structure, within a controlled division-free formula model.

[Read manuscript](#) | OPDP ID 40347996 | [Family formalization notes](#)

OAI-M0233 | Family 116

Polynomial Hitting Lists for Noncommutative Rational Formulas

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims polynomial-size rational matrix hitting lists for bounded-tree-size noncommutative rational formulas. Every nonzero admissible formula has a defined invertible value somewhere on the list. Inverse nesting and rational-constant heights need no separate limits, while matrix dimensions and output bit lengths remain polynomial. Definedness is essential: a tuple where a formula's denominator fails is not a valid hit. The target differs from a single universal point for division-free formulas and preserves that broader rational-evaluation burden.

Archive: related target. A polynomial hitting list for rational formulas is related to noncommutative realizations but has a different complexity and definedness criterion.

Score basis: Inverse definedness and invertible hits complicate derandomization beyond polynomial-only evaluation.

[Read manuscript](#) | OPDP ID 40347996 | [Family formalization notes](#)

OAI-M0234 | Family 117

Constant-factor hardness of uniform sparsest cut

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims NP-hardness of every fixed constant-factor approximation to Uniform Sparsest Cut. Demand is one for every vertex pair, and capacities are nonnegative rationals. Uniform demand is a specific structural restriction, distinct from arbitrary weighted-demand sparsest cut or cut-norm approximation. The assertion concerns valid worst-case gap reductions, not an exact optimal cut for one graph or a proof that all instances have a large semidefinite integrality gap without constructing the stated hardness family.

Archive: related target. Uniform sparsest cut and cut-norm approximation differ in their optimization objectives despite shared constant-factor hardness vocabulary.

Score basis: Uniform-demand gap amplification is a substantial approximation-hardness target.

[Read manuscript](#) | OPDP ID 40392619 | [Family formalization notes](#)

OAI-M0235 | Family 117

Near-square-root logarithmic integrality gaps for uniform sparsest cut

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Goemans-Linial semidefinite gaps for Uniform Sparsest Cut growing at least square-root $\log n$ divided by $\log \log n$ cubed along a size sequence. Demands are all one and capacities nonnegative real numbers. This nearly matches the known upper order, with a remaining polyloglog loss. It is a relaxation-gap construction rather than an NP-hardness theorem; the distinction matters because poor performance of one semidefinite method does not exclude all approximation algorithms.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Near-optimal SDP gaps need delicate metric constructions but offer structured certificate feedback.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0236 | Family 118

Additive hardness and unbounded configuration gaps in bin packing

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims unbounded additive gaps between bin-packing configuration linear programming and integral optimum. It also gives hardness of every fixed additive-error decision threshold. Rational item sizes exceed one sixth, so bins hold at most five items. These restrictions make the counterexamples specific and inspectable while still refuting the modified integer round-up prediction. The target is additive gap and hardness, not an exact asymptotic multiplicative approximation ratio or a claim that every instance exhibits the obstruction.

Archive: related target. The modified integer round-up conjecture is explicitly named; exact additive-gap and rational-item conventions require complete source recovery before identity certification.

Score basis: Unbounded gaps under a five-item restriction require strong combinatorial optimization structure.

[Read manuscript](#) | OPDP ID 40396909 | [Family formalization notes](#)

OAI-M0237 | Family 119

Sharp binary-information contraction on the discrete cube

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims sharp information contraction for binary channels under symmetric noise on a uniform discrete cube. At fixed initial information, a noisy coordinate is extremal, yielding the Boolean Courtade-Kumar conclusion with an output-entropy refinement and a mean-dependent production estimate. The target includes arbitrary binary-channel information, not just checking Boolean dictators numerically. Its noise model, input law and information constraint define the claimed extremal problem and must remain explicit when assessing the stated stronger refinements.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Sharp noisy-information optimization and dimension induction create a major discrete functional-inequality target.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0238 | Family 119

Hellinger contraction with arbitrary Boolean output bias

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the Boolean Hellinger contraction bound on the uniform cube for arbitrary output bias and every noise parameter between minus one and one. Signed coordinates achieve equality. A Courtade-Kumar information consequence follows. The asserted proof combines asymmetric induction, a noise-semigroup estimate and exact finite certificates. The target is a specific functional inequality with bias retained, not a general assertion that all divergences contract optimally for every nonuniform distribution or unrelated channel model.

Archive: related target. The Hellinger inequality is named in the archive, but arbitrary-bias and full noise-parameter conventions remain excerpt-limited.

Score basis: Arbitrary bias and sharp equality require difficult functional calibration with useful finite-certificate feedback.

[Read manuscript](#) | OPDP ID 40367118 | [Family formalization notes](#)

OAI-M0239 | Family 120

Almost-Linear-Time Maximum-Cardinality Matching in General Graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform randomized almost-linear-time maximum-cardinality matching algorithm on general simple graphs. Every computation path respects the logarithmic-word time bound and success probability is at least two thirds. A reduction handles spanning factors with prescribed degrees. The result supplies an explicit optimum matching rather than approximate cardinality, but it is neither a deterministic guarantee nor a counting algorithm. The graph model and pathwise runtime must survive interpretation of the added factor-construction consequences.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: General-graph exact matching in almost-linear time requires sophisticated combinatorial algorithms and careful randomization.

[Read manuscript](#)

OAI-M0240 | Family 121

An Almost-Linear Approximation Scheme for Edit Distance

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a randomized near-linear approximation scheme for unit-cost edit distance between explicitly stored strings. Each fixed rational accuracy yields factor one plus epsilon with success probability at least two thirds and worst-case expected logarithmic-word-RAM time. Integer alphabets are polynomially bounded, and equal strings return zero deterministically. Expected time and fixed-accuracy asymptotics are essential; the target is not exact edit distance or a uniform linear bound independent of accuracy and alphabet encoding.

Archive: lexical candidate. Retrieved candidates do not establish equivalent hypotheses and conclusion for this exact manuscript target; archive absence is not proved.

Score basis: Near-linear high-accuracy edit distance has a major algorithmic barrier with delicate embedding and runtime composition.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0241 | Family 122

Uniform quasipolynomial-time trace reconstruction

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a binary word can be recovered uniformly from deletion samples in quasipolynomial time for fixed rational retention, and polynomial time for sufficiently rare deletions. The result concerns worst-case reconstruction with an explicit probabilistic guarantee.

Archive: partial or special case. The archived question asks the necessary trace count; the manuscript supplies an upper bound rather than a matching characterization.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [OPDP ID 145](#) | [Family formalization notes](#)

OAI-M0242 | Family 122

A latest-anchor induction with spectrally compact masks for worst-case trace reconstruction

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that worst-case deletion reconstruction needs only quasipolynomially many observations, with a sharper logarithmic sample exponent and polynomial sampling for small deletion probabilities. Sample efficiency is distinguished from computational efficiency and optimality.

Archive: partial or special case. The trace model agrees, but an improved upper bound does not settle the archived optimal-sample question.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [OPDP ID 145](#) | [Family formalization notes](#)

OAI-M0243 | Family 122

Quantitative lower bounds for trace reconstruction

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every fixed nontrivial deletion rate forces superpolynomial samples for exact worst-case binary reconstruction, even with unlimited computation. This would exclude polynomial sampling without imposing an algorithmic model.

Archive: partial or special case. The lower bound addresses the same sampling problem but does not determine its exact complexity together with the upper bound.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [OPDP ID 145](#) | [Family formalization notes](#)

OAI-M0244 | Family 124

A Polynomial-Time Algorithm for Three-Machine Unit-Job Scheduling

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that unit-duration jobs with arbitrary precedence constraints admit exact minimum-makespan scheduling in polynomial time on three identical processors. The fixed number of machines is essential to the complexity claim.

Archive: unresolved no match. Retrieved algorithm titles do not identify this scheduling target.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0245 | Family 125

Single-exponential recovery and bounded-price strictness for metric k-median

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that metric k-median admits exact-budget recovery from approximate anchors and a randomized approximation ratio strictly below two on arbitrary finite rational metrics. Facility-budget preservation and the metric model are substantive restrictions.

Archive: unresolved no match. No retrieved record establishes the same approximation or anchor-recovery question.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0246 | Family 125

The approximation threshold for metric k-median

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite rational metric k-median has a deterministic polynomial-time approximation approaching $1+2/e$ for every fixed positive accuracy allowance without opening extra facilities. Under the stated complexity hypothesis this meets the asserted threshold.

Archive: unresolved no match. Nearby approximation records concern different optimization objectives.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0247 | Family 126

Exponential PSD rank of positively shifted matching matrices

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that positively shifted matching slack matrices have exponential real semidefinite rank for every fixed shift strictly between zero and one. The conclusion would obstruct small exact semidefinite representations of matching.

Archive: related target. The archive studies semidefinite rank, but its spectrahedron characterization is not this exponential matching lower bound.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40332169 | [Family formalization notes](#)

OAI-M0248 | Family 127

Average sensitivity of polynomial threshold functions

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that degree-d Boolean polynomial thresholds on n variables have average sensitivity at most $8d$ times the square root of n . The uniform bound resolves an asymptotic formulation, rather than an exact extremal constant.

Archive: lexical candidate. The named conjecture matches, but the truncated archive passage does not expose which exact or asymptotic formulation it contains.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40351529 | [Family formalization notes](#)

OAI-M0249 | Family 128

A Polynomial-Time 2-Approximation for Shortest Common Superstring

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that explicit ordinary strings admit a deterministic polynomial-time common superstring whose length is no more than twice optimal. The guarantee belongs to a new algorithm and does not certify classical Greedy.

Archive: unresolved no match. Superpatterns and longest common subsequences are different targets.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0250 | Family 129

An exponential state lower bound for two-way nondeterministic complementation

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that complementation of two-way nondeterministic automata requires exponentially many states when alphabet size is unrestricted. The separation concerns state complexity rather than language decidability, with finite alphabets allowed to grow with machine size.

Archive: unresolved no match. Retrieved two-way lower bounds involve unrelated mathematical objects.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0251 | Family 129

An exponential two-way deterministic state lower bound for one-way liveness

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that one-way liveness languages have small nondeterministic automata but require exponentially larger two-way deterministic automata over growing finite alphabets. The alphabet dependence limits the claimed simulation obstruction.

Archive: unresolved no match. No retrieved record states this automata simulation problem.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0252 | Family 130

An explicit power saving for the exact discrete Fourier transform

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that exact complex Fourier transforms can be computed uniformly with a fixed power saving below $n \log n$ in an unrestricted arithmetic model. Unit-cost arithmetic and coefficient permissions are part of the theorem.

Archive: related target. The archive asks an unrestricted linear-circuit lower bound; this is an upper-bound claim in an explicitly specified algorithmic model.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000286 | [Family formalization notes](#)

OAI-M0253 | Family 130

Finite tensor savings and exact Fourier circuits

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that exact nonuniform Fourier circuits beat $n \log n$ along infinitely many lengths through finite tensor savings. The result would refute that lower bound, without supplying a uniform all-length bit algorithm.

Archive: related target. Sub- $n \log n$ size does not refute the weaker archived request for any superlinear lower bound.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000286 | [Family formalization notes](#)

OAI-M0254 | Family 131

Polynomial mixing of the switch chain for every graphical degree sequence

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the lazy switch chain mixes in polynomial time for every simple undirected graphical degree sequence and supports exact uniform sampling in expected polynomial time. Mixing and almost-sure termination are separate guarantees.

Archive: unresolved no match. The retrieved mixing topics do not establish the canonical Kannan-Tetali-Vempala target with this graph model.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0255 | Family 132

A superquadratic separation between sensitivity and block sensitivity

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that total Boolean functions can have block sensitivity larger than a fixed power strictly exceeding the square of sensitivity. This targets the quadratic strengthening and preserves the distinction from polynomial boundedness.

Archive: unresolved no match. The reported stronger separation requires its own target rather than an unqualified sensitivity-conjecture score.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0256 | Family 133

Parity lifts and bounded-treewidth witnesses for Weisfeiler-Leman equivalence

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite-domain compatible-choice instances can be encoded by low-round Weisfeiler-Leman equivalence using parity lifts and bounded-treewidth witnesses. Conditional runtime exclusions depend on the stated positive-rate ETH assumption.

Archive: unresolved no match. The retrieved graph-complexity records do not identify this reduction with its update conventions.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0257 | Family 133

The complexity of identifying a graph by Weisfeiler-Leman refinement

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that identification by Weisfeiler-Leman refinement is EXPTIME-complete when dimension is supplied in binary with an explicit finite graph. Identification quantifies over all comparison graphs, unlike pairwise equivalence.

Archive: unresolved no match. No retrieved record exposes the same identification decision problem.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0258 | Family 133

Unconditional time lower bounds for Weisfeiler-Leman equivalence

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed high-dimensional Weisfeiler-Leman equivalence has unconditional worst-case sequential runtime growing as n to a constant times the dimension. The result fixes explicit input and machine models rather than assuming ETH.

Archive: related target. Resolution lower bounds for clique formulas are a different computational resource and target.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40401167 | [Family formalization notes](#)

OAI-M0259 | Family 133

Variable-dimension Weisfeiler-Leman equivalence on general and subcubic graphs

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that variable-dimensional joint-update Weisfeiler-Leman equivalence is EXPTIME-complete even for connected uncolored subcubic graphs of equal positive order. Binary dimension encoding, explicit adjacency matrices, and bounded graph degree all matter.

Archive: unresolved no match. Subcubic coloring candidates do not establish equivalence to this complexity question.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0260 | Family 134

Finite Monoid Computations and a Uniform Generalized Star-Height Bound

D 676/1000 | p 1 | m 1 | Conditional points 456.976

Published original CFSD-1000 target difficulty

The manuscript reports that every regular language admits a generalized regular expression of star height at most thirteen over its original finite alphabet. A uniform finite ceiling would answer the bounded-height existence question.

Archive: same target. The numerical ceiling supplies the uniform boundedness requested by the archived target, provided generalized complementation conventions coincide.

Score basis: Preserves the complete original target score.

Scoring target: Can all regular languages be expressed with generalized regular expressions of bounded star height?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1221 | [Family formalization notes](#)

OAI-M0261 | Family 134

Generalized Star Height at Most Four

D 676/1000 | p 1 | m 1 | Conditional points 456.976

Published original CFSD-1000 target difficulty

The manuscript reports that every regular language has generalized star height at most four without extending its alphabet. The constructive bound uses a finite clock and controlled periodic-word boundaries, and strengthens mere existence of some uniform ceiling.

Archive: same target. A bound of four would fully answer the archived bounded-star-height question; it is not credited as an additional independent family.

Score basis: Preserves the complete original target score.

Scoring target: Can all regular languages be expressed with generalized regular expressions of bounded star height?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1221 | [Family formalization notes](#)

OAI-M0262 | Family 134

Generalized Star Height at Most Three

D 676/1000 | p 1 | m 1 | Conditional points 456.976

Published original CFSD-1000 target difficulty

The manuscript reports that every regular language has generalized star height at most three with complement interpreted in the original free monoid. The complementation universe is necessary when comparing generalized expressions.

Archive: same target. The asserted ceiling resolves the archived existence-of-a-uniform-bound target under the stated generalized-expression convention.

Score basis: Preserves the complete original target score.

Scoring target: Can all regular languages be expressed with generalized regular expressions of bounded star height?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1221 | [Family formalization notes](#)

OAI-M0263 | Family 135

Homogeneous depth-five lower bounds for iterated matrix multiplication

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that homogeneous depth-five circuits for iterated n -by- n matrix multiplication require n to a constant times square-root- n gates in characteristic zero. Matching growth exponents depend on depth and syntactic homogeneity.

Archive: partial or special case. The unrestricted algebraic-complexity conjecture is broader than this depth-five homogeneous circuit class.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40375152 | [Family formalization notes](#)

OAI-M0264 | Family 136

The PCP-for-PPAD conjecture: a quasilinear reduction

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that End-of-Line admits a quasilinear-size reduction to generalized circuits whose approximate gate satisfaction robustly yields an original solution. Robust recovery and polynomial rational encoding are essential to PCP-for-PPAD.

Archive: unresolved no match. Quasilinear cycle sets and stochastic equations do not identify this reduction problem.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0265 | Family 137

Simulating One-Tape Time in Two-Fifths-Power Space

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a fixed one-work-tape machine can be simulated within time-cap to the two-fifths power times polylogarithmic workspace, with unrestricted runtime. The accessor condition and one writable head constrain the simulation model.

Archive: unresolved no match. The archive candidates do not establish the same time-to-space simulation question.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0266 | Family 138

Subset Sum in Time $O(2^{0.49n})$

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that worst-case Subset Sum on polynomial-bit inputs has a bounded-error classical randomized algorithm running in $O(2^{0.49n})$ time. The improvement concerns worst-case exponential time rather than polynomial solvability.

Archive: unresolved no match. No retrieved record identifies this specific exponent and machine model.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0267 | Family 138

A Low-Space Algorithm for Worst-Case Subset Sum

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that worst-case Subset Sum can be decided with one-sided error using roughly $2^{n/2}$ time and $2^{n/5}$ writable words. The claim improves storage while retaining an exponential running-time scale.

Archive: related target. The archived routine conjecture concerns a particular search structure, not this general time-space guarantee.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40334685

OAI-M0268 | Family 139

Subpolynomial query complexity for well-conditioned log-concave sampling

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that well-conditioned smooth log-concave laws with a known minimizer can be sampled using subpolynomially many exact first-order queries. Unrestricted between-query computation separates oracle complexity from overall efficiency.

Archive: related target. The zigzag algorithm record has a specific method and different performance target.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40381588 | [Family formalization notes](#)

OAI-M0269 | Family 140

Memory and precision in noiseless Gaussian regression

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that bounded-memory learners need dimension times log inverse accuracy Gaussian measurements to estimate a uniform spherical signal reliably. Persistent bits, exact measurements, and angular accuracy define the lower-bound model.

Archive: related target. Model misspecification in Gaussian regression is distinct from noiseless memory-sample complexity.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 20000272 | [Family formalization notes](#)

OAI-M0270 | Family 140

Posterior replicas and conditional information in Gaussian regression

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite-message information conditioned on a fresh projection is controlled sufficiently to force sharp memory-sample lower bounds for spherical regression. The auxiliary projection is analyst-side information, not an extra learner oracle.

Archive: related target. The existing regression example concerns a different signal model and question.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [OPDP ID 20000272](#) | [Family formalization notes](#)

OAI-M0271 | Family 140

Localization costs and information growth for exact Gaussian observations

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite messages from blocks of exact Gaussian observations reveal only dimension-times-block-count information under a quadratic-memory cap and specified localization geometry. The result isolates geometric localization costs in a learning lower bound.

Archive: unresolved no match. Gaussian time-frequency localization candidates concern different objects.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0272 | Family 140

Projection moments, positive cap domination, and Riesz estimates on the sphere

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that projection-moment and cap-domination estimates imply sharp Gaussian sample lower bounds for learners with subquadratic persistent memory. The arguments use ball-controlled measures and positive spherical caps; distinct stopping assumptions remain explicit across the three proofs.

Archive: unresolved no match. Riesz projection extremizer questions are not this spherical inference theorem.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0273 | Family 140

Replacing Gaussian observations in memory-constrained inference

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that replacing measurement rows by fresh Gaussian rows increases residual conditional information by only a dimension-proportional amount under bounded message entropy. The replacement theorem supplies a controlled step in a streaming lower-bound argument.

Archive: unresolved no match. Gaussian channel-entropy candidates do not establish this observation-replacement target.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0274 | Family 140

Subsphere methods for memory-sample lower bounds in noiseless Gaussian regression

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that subquadratic-memory finite-state learners need an explicit dimension-times-log-accuracy number of exact Gaussian samples for spherical estimation. Residual subsphere geometry supplies uniform small-error control with explicit coefficient 2^{-16} in sufficiently large dimensions.

Archive: related target. Polynomial-time regression-risk bounds differ from persistent-memory and noiseless sample lower bounds.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40342240 | [Family formalization notes](#)

OAI-M0275 | Family 141

Existential-universal real sentences in the counting hierarchy

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that existential and existential-universal real sentences lie in a fixed level of the counting hierarchy even with circuit-encoded integer polynomials. Quantifier order and succinct polynomial representation are central to the complexity inclusion.

Archive: unresolved no match. Retrieved Whitham hierarchy questions are unrelated to the computational hierarchy.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0276 | Family 142

Deterministic Polynomial Factorization over Prime Fields

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that dense univariate polynomials over binary-encoded prime fields can be completely factored deterministically in polynomial bit complexity. The arithmetic object is a polynomial, not an integer, and a companion theorem is used.

Archive: lexical candidate. The finite-field factorization title aligns, but excerpt evidence alone does not certify the precise unconditional input model.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40393154

OAI-M0277 | Family 143

Uniform bounds for planar polynomial limit cycles

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that each degree admits a finite uniform ceiling on isolated periodic orbits of real planar polynomial vector fields. This addresses uniform boundedness without determining sharp maxima or curve topology.

Archive: partial or special case. Hilbert sixteen in the archived row bundles exact cycle counts, configurations, and algebraic-curve topology; boundedness is a narrower component.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 46 | [Family formalization notes](#)

OAI-M0278 | Family 143

Two limit cycles for quintic Liénard systems

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that classical Lienard systems with polynomial damping of degree at most five have at most two isolated cycles, with equality achievable. The restriction to the classical restoring force distinguishes the sharp degree-five result.

Archive: related target. The archived target asks specific degree-three and degree-four Dulac proofs, not the stated degree-five theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 4700004 | [Family formalization notes](#)

OAI-M0279 | Family 144

A smooth three-torus diffeomorphism with simple Lebesgue spectrum

D 659/1000 | p 1 | m 1 | Conditional points 434.281

Published original CFSD-1000 target difficulty

The manuscript reports that a smooth volume-preserving diffeomorphism of the three-torus has simple Lebesgue spectrum on its whole mean-zero square-integrable space. An explicit smooth example would satisfy the broader existence problem.

Archive: same target. The archived existence question is answered by the proposed probability-preserving example; unrelated length-spectrum records are excluded.

Score basis: Preserves the complete original target score.

Scoring target: Does there exist an ergodic measure-preserving transformation whose Koopman operator has simple Lebesgue spectrum?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 4900003 | [Family formalization notes](#)

OAI-M0280 | Family 145

Rokhlin's multiple-mixing problem for one transformation

D 703/1000 | p 1 | m 1 | Conditional points 494.209

Published original CFSD-1000 target difficulty

The manuscript reports that every invertible strongly mixing probability-preserving transformation is mixing at every finite order. The theorem would settle the order-three obstruction as part of a universal assertion for one transformation rather than an action of several generators.

Archive: same target. The claimed all-orders conclusion includes the archived order-three target, conditional on the shared strong-mixing convention.

Score basis: Preserves the complete original target score.

Scoring target: Is every strongly mixing measure-preserving transformation strongly mixing of order three?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 4900015 | [Family formalization notes](#)

OAI-M0281 | Family 146

Positive Metric Entropy for the Standard Map at Large Parameters

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the toral standard sine map has positive area-metric entropy throughout a sufficiently large positive parameter tail. An entire tail is stronger than a positive-measure parameter-set existence claim.

Archive: lexical candidate. The conjecture title aligns, but a full archival statement is needed to confirm its exact parameter requirement.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40345911 | [Family formalization notes](#)

OAI-M0282 | Family 147

Continuous Phase Foliations Create Analytic Caustic Collars

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a continuous invariant foliation of a full grazing billiard annulus forces analytic boundary and analytic caustic-collar structure for a smooth positively curved table. The regularity upgrade is a distinct intermediate step toward ellipse rigidity.

Archive: related target. The archive's boundary-integrability discussion is related but does not certify this regularity statement.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 20002471

OAI-M0283 | Family 147

Rigidity of Smooth Billiards with a Continuous Caustic Collar

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a smooth positively curved strictly convex planar billiard with a continuous collar of smooth closed caustics must have elliptical boundary. Leafwise smoothness is retained without assuming differentiability across the foliation.

Archive: related target. Two nested compatible caustics do not automatically provide the full collar assumed here.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 5200007

OAI-M0284 | Family 148

The entropy-rate dimension formula for self-similar measures on the line

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite real self-similar measures have dimension given by entropy rate divided by Lyapunov exponent, capped at one, even with exact overlaps. Allowing unequal and negative contractions extends the stated dimension formula's scope.

Archive: lexical candidate. The dimension-formula record is a plausible target, but its excerpt does not establish matching overlap and contraction hypotheses.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40370581 | [Family formalization notes](#)

OAI-M0285 | Family 149

Uniform Permanence in Weakly Reversible Mass-Action Systems

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that weakly reversible mass-action networks with fixed positive rates admit classwise common compact attracting invariant sets for all positive trajectories. Uniform classwise permanence is stronger than trajectory-dependent persistence.

Archive: lexical candidate. The permanence name agrees; the complete record is needed to distinguish classwise uniformity from weaker persistence formulations.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40356593 | [Family formalization notes](#)

OAI-M0286 | Family 149

Boundedness and persistence of weakly reversible mass-action systems

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that positive trajectories of weakly reversible constant-rate mass-action systems remain globally bounded above and separated from zero. The asserted bounds may depend on the initial state rather than the whole compatibility class.

Archive: lexical candidate. The persistence title is relevant, but excerpt evidence does not certify the exact accompanying boundedness claim.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40329460 | [Family formalization notes](#)

OAI-M0287 | Family 150

Weak mixing of triangular billiards with an irrational angle

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every nondegenerate triangular billiard with at least one irrational angle has weakly mixing unit-speed flow in Liouville measure. The universal irrational-angle condition avoids genericity or extra Diophantine assumptions.

Archive: related target. Weak mixing would imply the archived ergodicity claim, but the paper's scored theorem is the stronger weak-mixing statement.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000447 | [Family formalization notes](#)

OAI-M0288 | Family 150

Ergodicity of triangular billiards with an irrational angle

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every triangular billiard with an irrational angle is ergodic after excluding the null set of vertex-hitting trajectories. The universal quantifier is stronger than an almost-everywhere statement over triangle shapes.

Archive: lexical candidate. The formulation appears aligned, but the ProofAtlas normalization remains unverified source-level evidence.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000447 | [Family formalization notes](#)

OAI-M0289 | Family 151

A C^1 Counterexample to the Entropy Conjecture

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a noninvertible continuously differentiable compact-manifold map has zero entropy but homological spectral radius two. The obstruction appears in second real homology; regularity and noninvertibility constrain the entropy-conjecture counterexample.

Archive: unresolved no match. The returned entropy records concern other systems and do not identify this self-map formulation.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0290 | Family 152

A zero-entropy system without a smooth positive-volume model

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that one zero-entropy ergodic probability transformation admits no smooth model preserving positive smooth volume in any finite dimension. The construction simultaneously excludes all finite-dimensional smooth realization spaces.

Archive: related target. The archive's smooth-realization question is related, but scope and measure conventions require independent confirmation.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000477 | [Family formalization notes](#)

OAI-M0291 | Family 153

Arithmetic classification and non-Pisot singularity for Bernoulli convolutions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that unbiased Bernoulli convolutions admit an arithmetic singularity-versus-continuity criterion, including explicit non-Pisot singular parameters. The criterion is an infinite approximation condition rather than a finite decision procedure.

Archive: related target. The reciprocal-Pisot formulation is narrower in parameter terminology and must be compared to the exact arithmetic criterion.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000424

OAI-M0292 | Family 154

Pointwise Multiple Ergodic Averages for Mixing Transformations

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that mixing invertible probability transformations have almost-everywhere convergence of every fixed-length consecutive multiple ergodic average to the product of means. The mixing hypothesis is stronger than the assumptions in general pointwise convergence questions.

Archive: partial or special case. The archived row includes arbitrary systems and an additional polynomial-time-pattern average; those broader parts are not settled here.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 4800011

OAI-M0293 | Family 154

Pointwise convergence of fourfold ergodic averages for mixing transformations

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fourfold averages at consecutive integer multiples converge pointwise to the product of means for mixing invertible probability transformations. No quantitative mixing rate or standard-space assumption is imposed.

Archive: related target. The archived triple and polynomial averages are different targets; neither their generality nor their conclusion is inherited.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 4800011

OAI-M0294 | Family 154

Triple ergodic averages with distinct integer slopes

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that mixing transformations have pointwise product-limit convergence for triple averages with distinct nonzero integer slopes of either sign. Signed slopes and all averaging lengths remain part of the theorem's scope.

Archive: partial or special case. The result treats a mixing subclass and a linear-pattern claim, leaving the archived arbitrary-system and polynomial-pattern questions open.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 4800011

OAI-M0295 | Family 154

Pointwise convergence of triple ergodic averages for mixing transformations

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that triple consecutive linear ergodic averages converge almost everywhere to the product of means under invertible mixing on arbitrary probability spaces. The theorem provides a precise mixing-case product limit rather than merely existence of a limit.

Archive: partial or special case. The archived target has no mixing restriction and also includes a nonlinear pattern, so the paper is not a full resolution of that bundle.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 4800011

OAI-M0296 | Family 155

A translational tile with no fully periodic tiling in dimension three

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a finite translational tile in three-dimensional integer space admits tilings but none fully periodic, with the same obstruction after thickening to real space. The dimension and arbitrary real translations are essential to the claimed counterexample.

Archive: lexical candidate. The periodic-tiling title matches, but the truncated archive text leaves dimensional and translation hypotheses unresolved.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40364582 | [Family formalization notes](#)

OAI-M0297 | Family 156

A nine-dimensional counterexample to Borsuk's covering assertion

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that rank-one projectors from four-dimensional real space form a nine-dimensional compact set requiring more than ten smaller-diameter covering pieces. The result supplies a dimension-specific obstruction rather than a new score for an unrestricted parent question.

Archive: partial or special case. The archive asks the all-dimensions Borsuk assertion; the manuscript's specifically nine-dimensional bound receives its own difficulty.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 1224 | [Family formalization notes](#)

OAI-M0298 | Family 157

A counterexample to Hadwiger's conjecture

D 835/1000 | p 1 | m 1 | Conditional points 697.225

Published original CFSD-1000 target difficulty

The manuscript reports that finite graphs can have chromatic number exceeding clique-minor order, even with independence number at most two. Such examples would directly refute the universal chromatic-to-minor implication, and fractional coloring gives a still stronger separation.

Archive: same target. The reported counterexamples negate the archived universal Hadwiger statement; fractional coloring is an additional consequence.

Score basis: Preserves the complete original target score.

Scoring target: Every graph with chromatic number k has a K_k minor (where K_k is the complete graph on k vertices).

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 14 | [Family formalization notes](#)

OAI-M0299 | Family 157

A counterexample to the Colin de Verdière chromatic conjecture

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that graphs with independence number at most two can have fractional chromatic number larger than their Colin de Verdière invariant plus one. The fractional separation is stronger than the ordinary chromatic counterexample.

Archive: lexical candidate. The invariant-based conjecture is a candidate, but the excerpt does not fix the exact inequality needed for identity.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40368619 | [Family formalization notes](#)

OAI-M0300 | Family 157

A linear list-coloring bound in terms of the Hadwiger number

D 819/1000 | p 1 | m 1 | Conditional points 670.761

Published original CFSD-1000 target difficulty

The manuscript reports that every finite nonempty graph has list chromatic number bounded by a universal constant times its clique-minor order. The bound is uniform over graph size and specializes to minor-exclusion classes.

Archive: same target. The claimed $h(G)$ bound is equivalent up to an absolute constant to the archived linear list-coloring bound for K_t -minor-free graphs.

Score basis: Preserves the complete original target score.

Scoring target: Conjecture Every K_t -minor-free graph is c -list-colourable for some constant $c \geq 1$.

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 3253 | [Family formalization notes](#)

OAI-M0301 | Family 158

The Euclidean plane is not five-colorable

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that five colors cannot properly color the Euclidean plane at unit distance, without assumptions on measurable color classes. The claim improves the lower bound while leaving the six-versus-seven alternative unsettled.

Archive: partial or special case. The archived question asks the exact plane chromatic number; ruling out five colors alone does not determine it.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 3900003 | [Family formalization notes](#)

OAI-M0302 | Family 159

Quasipolynomial Bounds for Arithmetic Progressions

D 674/1000 | p 1 | m 1 | Conditional points 454.276

Published original CFSD-1000 target difficulty

The manuscript reports that divergence of reciprocal sums forces arbitrarily long arithmetic progressions, via quantitative stretched-exponential density bounds for fixed lengths. The quantitative statement supplies the asserted classical existence consequence.

Archive: same target. The positive-integer reciprocal-sum statement agrees with this complete original target; a duplicate archived formulation has a different frozen score.

Score basis: Preserves the complete original target score.

Scoring target: If the sum of reciprocals of a set of positive integers diverges, does the set contain arbitrarily long arithmetic progressions?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1262 | [Family formalization notes](#)

OAI-M0303 | Family 160

Quantitative Superexponential Bounds for van der Waerden Numbers

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed-color van der Waerden numbers grow faster than every fixed exponential in progression length, with an explicit color-dependent lower bound. The result concerns asymptotic growth rather than unknown exact numbers.

Archive: related target. Exact-value questions and mixed-number concavity do not become resolved by a superexponential lower bound.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 3200015 | [Family formalization notes](#)

OAI-M0304 | Family 161

A counterexample to Sidorenko's conjecture

D 546/1000 | p 1 | m 1 | Conditional points 298.116

Published original CFSD-1000 target difficulty

The manuscript reports that a connected bipartite pattern on thirty-five vertices violates the conjectured homomorphism-density lower bound and also defeats forcing at a fixed density. A finite counterexample would negate the universal Sidorenko inequality.

Archive: same target. The reported density violation directly negates the archived universal homomorphism-count inequality.

Score basis: Preserves the complete original target score.

Scoring target: Conjecture For any bipartite graph H and graph G , the number of homomorphisms from H to G is at least $\leq \text{ft}((2|E(G)|)/(|V(G)|^2))^{|E(H)|} |V(G)|^{|V(H)|}$.

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 3140 | [Family formalization notes](#)

OAI-M0305 | Family 162

Balanced counterexamples to Ryser's conjecture at prime orders

D 530/1000 | p 1 | m 1 | Conditional points 280.9

Published original CFSD-1000 target difficulty

The manuscript reports that large prime orders admit balanced intersecting multipartite uniform hypergraphs whose cover number equals the number of parts. Since the matching number is one, these examples would violate Ryser's universal bound.

Archive: same target. Intersecting examples with cover number r negate the archived tau-at-most- $(r-1)$ -nu inequality.

Score basis: Preserves the complete original target score.

Scoring target: For r -partite r -uniform hypergraphs, is the vertex cover number at most $(r-1)$ times the matching number?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1440 | [Family formalization notes](#)

OAI-M0306 | Family 162

A counterexample to Ryser's covering conjecture

D 530/1000 | p 1 | m 1 | Conditional points 280.9

Published original CFSD-1000 target difficulty

The manuscript reports that prime-power multipartite intersecting hypergraphs can have cover number equal to their number of parts under the specified prime and odd-exponent restrictions. The parameter restrictions still furnish counterexamples to a universally quantified covering inequality.

Archive: same target. Existence within this restricted infinite family suffices to refute the archived universal Ryser statement.

Score basis: Preserves the complete original target score.

Scoring target: For r -partite r -uniform hypergraphs, is the vertex cover number at most $(r-1)$ times the matching number?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1440 | [Family formalization notes](#)

OAI-M0307 | Family 164

Monochromatic finite sums and products in the positive integers

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every finite coloring of positive integers admits arbitrarily large finite sets whose nonempty subset sums and products share one color. The same color must work simultaneously for additive and multiplicative structures.

Archive: lexical candidate. The named Hindman record is plausible, but the excerpt does not establish this precise finite simultaneous formulation.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40354766

OAI-M0308 | Family 165

The crossing number of complete graphs

D 644/1000 | p 1 | m 1 | Conditional points 414.736

Published original CFSD-1000 target difficulty

The manuscript reports that complete graphs have ordinary crossing number given exactly by the four-factor Harary-Hill floor formula at every order. The assertion is an exact drawing optimum rather than an asymptotic estimate.

Archive: same target. The full original formula and ordinary planar crossing-number convention agree with the reported theorem.

Score basis: Preserves the complete original target score.

Scoring target: The crossing number $cr(G)$ of G is the minimum number of crossings in all drawings of G in the plane. Conjecture
$$cr(K_n) = \frac{1}{4} \lfloor \frac{n^2}{2} \rfloor \lfloor \frac{n-1}{2} \rfloor \lfloor \frac{n-2}{2} \rfloor \lfloor \frac{n-3}{2} \rfloor$$

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 3318 | [Family formalization notes](#)

OAI-M0309 | Family 165

The crossing number of complete bipartite graphs

D 551/1000 | p 1 | m 1 | Conditional points 303.601

Published original CFSD-1000 target difficulty

The manuscript reports that complete bipartite graphs have ordinary crossing number equal to the four-floor Zarankiewicz expression for all part sizes. Uniformity in both parameters is required to resolve the brickyard formulation.

Archive: same target. The complete original statement gives the same formula and graph family.

Score basis: Preserves the complete original target score.

Scoring target: The crossing number $cr(G)$ of G is the minimum number of crossings in all drawings of G in the plane. Conjecture
$$cr(K_{m,n}) = \lfloor \frac{m^2}{2} \rfloor \lfloor \frac{m-1}{2} \rfloor \lfloor \frac{n^2}{2} \rfloor \lfloor \frac{n-1}{2} \rfloor$$

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 3319 | [Family formalization notes](#)

OAI-M0310 | Family 166

The higher-dimensional Erdős distinct-distances conjecture

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that n points in fixed Euclidean dimension at least three determine at least a dimension-dependent constant times $n^{2/d}$ distinct distances. The dimension is fixed and the constant may depend on it.

Archive: lexical candidate. The distinct-distances title is relevant, but the archive excerpt does not certify the exact higher-dimensional exponent.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40316174

OAI-M0311 | Family 167

The weak pinned planar distance theorem

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that almost all points of any large planar finite set see at least $n^{1-\epsilon}$ distinct distances for each fixed positive ϵ . The exceptional set is negligible rather than absent, preserving the weak-pinned formulation.

Archive: lexical candidate. The weak-pinned title aligns, but complete quantifiers are needed before confirming scope identity.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40368356 | [Family formalization notes](#)

OAI-M0312 | Family 167

A power saving for planar unit distances

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that planar n -point sets determine at most n to an absolute exponent strictly below four-thirds unit-distance pairs. The theorem claims a power saving without asserting the conjectured near-linear optimum.

Archive: unresolved no match. Squared distance multiplicities and rational-angle restrictions do not establish this unrestricted unit-distance bound.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0313 | Family 168

Combinatorial invariance of Kazhdan-Lusztig polynomials

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that isomorphic Bruhat intervals in arbitrary Coxeter groups have the same equal-parameter Kazhdan-Lusztig polynomial. The invariant depends only on the interval's order structure; unrestricted Coxeter systems distinguish full combinatorial invariance from special cases.

Archive: lexical candidate. The name and object agree, but a complete archive statement is required to confirm parameter conventions and unrestricted scope.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40343126 | [Family formalization notes](#)

OAI-M0314 | Family 169

Elementary positivity of chromatic quasisymmetric functions

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that natural unit interval graphs have elementary-positive chromatic quasisymmetric functions with coefficients in nonnegative q -polynomials. This isolates the positivity component of a larger conjectural package, requiring the elementary basis rather than merely monomial positivity.

Archive: lexical candidate. The positivity target is plausible, but the excerpt leaves graph-class and coefficient-ring conventions incomplete.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40356860 | [Family formalization notes](#)

OAI-M0315 | Family 170

The sharp logarithmic exponent of $r(5,t)$

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the fifth off-diagonal Ramsey parameter has fourth-power growth with logarithmic correction exponent three up to vanishing exponent error. The claim fixes a logarithmic exponent rather than an exact Ramsey asymptotic constant.

Archive: unresolved no match. The retrieved sharp-exponent records do not state this Ramsey target.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0316 | Family 170

Sharp logarithmic exponents for fixed off-diagonal Ramsey numbers

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every fixed off-diagonal Ramsey parameter at least six has the claimed sharp logarithmic correction exponent s -minus-two. The first parameter is fixed while the second grows, unlike diagonal limits.

Archive: related target. The archived diagonal root-limit question has different parameter scaling and is not resolved by these fixed-parameter estimates.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 3069 | [Family formalization notes](#)

OAI-M0317 | Family 171

The hypercube Ramsey number has linear order

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that two-color Ramsey numbers of binary hypercubes grow linearly in their vertex count with a universal constant. The claim retains dimension-uniform control without determining an exact constant.

Archive: lexical candidate. The hypercube Ramsey title aligns, but crossing-number and exact $R(5,5)$ records are different questions. Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40390769

OAI-M0318 | Family 172

A classification of finite Euclidean Ramsey configurations

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite Euclidean Ramsey configurations have a necessary-and-sufficient tensor criterion that includes subtransitive sets and all cyclic sets of at most five points. The criterion would invalidate a proposed necessity condition while preserving original-scale congruence.

Archive: lexical candidate. The classification title is relevant, but the archive excerpt does not specify the same coordinate-field tensor criterion.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398079 | [Family formalization notes](#)

OAI-M0319 | Family 173

A proof of Seymour's second-neighborhood conjecture

D 532/1000 | p 1 | m 1 | Conditional points 283.024

Published original CFSD-1000 target difficulty

The manuscript reports that every nonempty finite oriented graph contains a vertex whose second out-neighborhood is at least as large as its first. The directed-distance convention excludes loops and bidirected pairs implicit in oriented graphs.

Archive: same target. The full original second-neighborhood formulation agrees with the reported theorem.

Score basis: Preserves the complete original target score.

Scoring target: Resolve the archived assertion that every nonempty finite oriented graph contains a vertex whose second out-neighborhood is at least as large as its first.

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 3100021 | [Family formalization notes](#)

OAI-M0320 | Family 174

The strong thin tree conjecture

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every loopless k -edge-connected finite multigraph admits a spanning tree using at most a universal constant divided by k of each cut. The inverse-connectivity rate strengthens qualitative existence of thin spanning trees.

Archive: related target. The paper gives a stronger quantitative rate than the archived existence-of-some-connectivity-threshold question, so its own theorem is scored.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 3000039 | [Family formalization notes](#)

OAI-M0321 | Family 174

A polynomial-time construction of strong thin trees

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that strong thin spanning trees can be constructed deterministically in polynomial bit time even with binary-encoded parallel-edge multiplicities. The guarantee combines inverse-connectivity cut control with succinct input handling.

Archive: related target. The archive asks qualitative thin-tree existence; the algorithmic inverse-connectivity guarantee is a stronger separately scored theorem.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 3000039 | [Family formalization notes](#)

OAI-M0322 | Family 175

Graph Decompositions at the Integral Expectation Threshold

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every graph splits into a universally bounded number of fixed pieces whose ordinary containment thresholds track its integral expectation threshold. Piece embeddings need not agree on shared vertices, which limits the conclusion.

Archive: related target. The fractional-expectation record is related but does not establish identity with this fixed-piece integral decomposition.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40369863 | [Family formalization notes](#)

OAI-M0323 | Family 175

Integral and fractional expectation thresholds are equivalent

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that integral and fractional expectation thresholds differ by at most a universal multiplicative constant at the same covering budget. The budget convention distinguishes this equivalence from general threshold estimates.

Archive: lexical candidate. The equivalence title agrees, but full archival quantifiers are needed to certify the same covering convention.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40344317 | [Family formalization notes](#)

OAI-M0324 | Family 175

Talagrand's discrete-convexity conjecture

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a universal number of high-probability family members yields a low-cost cover for sets outside their unions at the original Bernoulli density. The conclusion concerns discrete covering cost rather than a numerical anti-concentration inequality.

Archive: unresolved no match. Anti-concentration and convolution candidates do not identify this discrete-convexity statement.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0325 | Family 176

The second Kahn-Kalai conjecture

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that ordinary graph-copy thresholds are bounded by expectation thresholds times a universal logarithmic edge factor. The expectation parameter tests every subgraph and uses a fixed half-copy budget.

Archive: lexical candidate. The graphic Kahn-Kalai record is plausible, but its excerpt does not fix the same expectation-threshold definition.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40404497 | [Family formalization notes](#)

OAI-M0326 | Family 177

Bounded-degree coboundary expanders in every dimension

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that arbitrarily large bounded-vertex-degree simplicial complexes exist with uniform binary-field coboundary expansion below every fixed dimension. Expansion is required in every degree below the top dimension, not only for graph edges or spectral gaps.

Archive: related target. The archive concerns dimension two; the paper's main construction addresses dimensions at least three.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40399934 | [Family formalization notes](#)

OAI-M0327 | Family 178

Deterministic nonbipartite Ramanujan graphs in every fixed degree

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every fixed degree at least three admits deterministically constructed nonbipartite Ramanujan graphs at all sufficiently large even orders. Degree-dependent polynomial exponents and full adjacency-list output qualify the algorithmic claim.

Archive: lexical candidate. The infinite-family title aligns only partially with an all-large-orders deterministic construction; full scope requires review.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398322

OAI-M0328 | Family 179

The circulant Hadamard conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that real circulant Hadamard matrices have only orders one and four, yielding the listed finite Barker-sequence lengths. Circulant structure is essential and does not settle unrestricted Hadamard existence.

Archive: related target. The archived Hadamard question permits arbitrary matrices of every multiple-of-four order, unlike the circulant nonexistence target.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 1102 | [Family formalization notes](#)

OAI-M0329 | Family 180

Paired states and Hamiltonian cycles in cubic bipartite planar graphs

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that simple cubic bipartite planar graphs that are three-vertex-connected always have Hamiltonian cycles. All four structural assumptions define Barnette's assertion; other bipartite graph classes remain outside this claimed universal Hamiltonicity guarantee.

Archive: related target. The Tutte record involves a different graph-class conjecture; matching Hamiltonicity terminology is insufficient.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40347600 | [Family formalization notes](#)

OAI-M0330 | Family 181

A linear cycle-and-edge decomposition of every graph

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that edges of any finite simple graph can be partitioned into at most a universal constant times vertex count cycles and single edges. The allowance of individual leftover edges distinguishes the decomposition guarantee.

Archive: lexical candidate. The named decomposition target is plausible, but the excerpt needs full edge-piece and uniform-constant conventions.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40376329 | [Family formalization notes](#)

OAI-M0331 | Family 182

A power saving for intersective polynomial differences with an exponent depending only on the degree

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that sets avoiding nonzero differences from any intersective polynomial have a power density saving whose exponent depends only on degree. Polynomial-specific constants remain allowed while the exponent is uniform across coefficients.

Archive: unresolved no match. The retrieved polynomial bounds concern different equations and extremal sets.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0332 | Family 182

A Power Saving for Polynomial Differences at Prime Arguments

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that polynomials having unit roots at every modulus force a fixed density power saving when forbidden differences are evaluated only at primes. The local unit-root condition and companion analytic estimate are indispensable hypotheses.

Archive: related target. Fractional-power prime progressions have a different pattern and do not identify this polynomial-difference theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40365960 | [Family formalization notes](#)

OAI-M0333 | Family 182

A power saving for square-difference-free sets

D 535/1000 | p 1 | m 1 | Conditional points 286.225

Published original CFSD-1000 target difficulty

The manuscript reports that integer sets containing no nonzero square difference have cardinality at most a fixed power below the ambient interval size. An absolute power saving would settle the existence of a fixed density exponent.

Archive: same target. After harmless exponent adjustment to absorb constants, the assertion supplies the fixed-power square-difference property in the complete original record.

Score basis: Preserves the complete original target score.

Scoring target: Is there $c > 0$ such that whenever A subset $[N]$ has size N^{1-c} , the difference set $A - A$ contains a nonzero square?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 126 | [Family formalization notes](#)

OAI-M0334 | Family 183

A power saving for planar halving lines

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that general-position planar sets have fewer than $n^{4/3}$ halving pairs by an unspecified absolute power saving. The theorem concerns large even cardinalities with no three collinear; neither improved exponent nor implied constant is explicitly quantified numerically.

Archive: partial or special case. A better upper bound does not determine the archived extremal maximum or higher-dimensional sharp bounds.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 3800006 | [Family formalization notes](#)

OAI-M0335 | Family 184

Correspondence coloring graphs with a forbidden clique

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed-clique-free graphs of large maximum degree have correspondence chromatic number on the degree-over-log-degree scale. Correspondence coloring strengthens ordinary and list-coloring consequences without removing the fixed forbidden graph condition.

Archive: lexical candidate. The coloring conjecture is a candidate, but correspondence-coloring scope and constants require full-text comparison.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40351267 | [Family formalization notes](#)

OAI-M0336 | Family 184

A logarithmic independence bound for clique-free graphs

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed-clique-free graphs contain independent sets of size proportional to vertex count times log average degree divided by average degree. The constant may depend on the excluded clique order but not on graph size.

Archive: unresolved no match. Clique-cover candidates do not establish this independent-set lower-bound target.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0337 | Family 185

A Counterexample to the Infinite Matroid Packing/Covering Conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that two countable self-dual partitional matroids in ZFC fail both unrestricted packing-covering partitioning and intersection-witness existence. The examples exclude finitary and cofinitary hypotheses from the claimed counterexample class.

Archive: lexical candidate. The packing-covering name matches, but excerpt evidence does not certify the unrestricted infinite-matroid formulation.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40327726 | [Family formalization notes](#)

OAI-M0338 | Family 186

A uniform influence bound for hypergraph properties

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that permutation-invariant uniform-hypergraph Boolean properties obey a uniform influence inequality at every edge density without monotonicity. Only the threshold-width consequence imposes increasingness, preserving the stronger analytic statement.

Archive: unresolved no match. Hereditary-property and quasirandom-cut records concern different conditions and conclusions.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0339 | Family 186

A Sharp Threshold Bound for Monotone Graph Properties

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that increasing permutation-invariant graph properties have transition-window width bounded by a universal inverse-square logarithm of vertex count. The accuracy parameter contributes a separate logarithmic factor to the sharp-threshold estimate.

Archive: lexical candidate. The named graph sharp-threshold conjecture is plausible, but the exact width normalization needs complete archival text.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40338309 | [Family formalization notes](#)

OAI-M0340 | Family 187

Snaky in 21 Maker moves

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Maker can force the Snaky hexomino in twenty-one moves against arbitrary legal Breaker play on a fixed finite board. The bounded board and actual Maker-move count give a finite verifiable strategy target.

Archive: unresolved no match. The retrieved move-game record concerns clique construction rather than the Snaky configuration.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0341 | Family 188

The sharp terminal leave in random triangle removal

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that random greedy triangle removal leaves an $n^{3/2}$ -scaled edge count converging in squared mean to one over twice square-root-two. The assertion identifies a sharp constant and implies expectation and probability convergence.

Archive: lexical candidate. The final-size record is relevant, but its excerpt does not establish the same limit mode or constant.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40391566 | [Family formalization notes](#)

OAI-M0342 | Family 189

Cycle--clique Ramsey numbers

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that cycle-versus-clique Ramsey numbers equal the proposed product formula apart from the specified triangle exception. The proof claim includes a finite exact case audit rather than solely asymptotic reasoning.

Archive: lexical candidate. The named conjecture aligns, but parameter restrictions and exception handling require a complete archived formulation.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40339753 | [Family formalization notes](#)

OAI-M0343 | Family 190

Polynomial removal fails for ordered binary matrices

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a fixed sixty-six-by-sixty-six binary pattern prevents any polynomial bound in ordered matrix removal. Copies preserve row and column orders and match both entry values, with bidirectional edits allowed.

Archive: lexical candidate. The polynomial-removal target is plausible; induced versus one-valued copies and edit conventions require confirmation.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40369912 | [Family formalization notes](#)

OAI-M0344 | Family 191

A power improvement in the Heilbronn triangle lower bound

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that unit-square point sets can have every determined triangle area bounded below by $n^{-(2+\epsilon)}$ for a fixed positive ϵ . The power improvement falsifies a proposed scale but leaves the exact extremal optimum unknown.

Archive: partial or special case. The archive asks the largest minimum area, whereas the paper provides a new lower bound.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | [OPDP ID 1215](#) | [Family formalization notes](#)

OAI-M0345 | Family 192

Unbounded Violations of the Square-Root Degree Bound

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that linear Fourier coefficient sums of Boolean functions can exceed every constant times the square root of real polynomial degree. The construction violates even a constant-factor weakening of the stated square-root bound.

Archive: unresolved no match. The retrieved square-root estimates involve unrelated independence or polynomial-system questions.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0346 | Family 193

Positivity of Serre's Intersection Multiplicity

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that complementary-dimensional finite modules over regular local rings have positive intersection multiplicity when their tensor product has finite length. The claim explicitly includes ramified mixed characteristic and nonzero-module hypotheses.

Archive: related target. The original prime-ideal formulation omits an essential finite-length qualification, so this review does not transfer its frozen score automatically.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 1469

OAI-M0347 | Family 194

Lech's multiplicity conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Hilbert-Samuel multiplicity cannot decrease through a flat local map of nonzero Noetherian local rings in any characteristic. Residue fields and dimension remain unrestricted in the asserted monotonicity statement.

Archive: lexical candidate. Lech's conjecture is a plausible match, but the excerpt lacks the full flat-local hypotheses.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40400689 | [Family formalization notes](#)

OAI-M0348 | Family 195

A Complete Local Domain without a Small Cohen-Macaulay Module

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a three-dimensional complete normal local complex domain has no nonzero finitely generated maximal Cohen-Macaulay module. The example targets small modules rather than the distinct existence of big Cohen-Macaulay objects.

Archive: lexical candidate. The small-module conjecture is a candidate; the domain and completeness version needs exact scope confirmation.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40352069

OAI-M0349 | Family 196

A Torsion-Free Group Algebra with Zero Divisors

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a finitely presented torsion-free group has zero divisors in its group algebra over the two-element field. The finite two-dimensional classifying space adds structural control to the counterexample.

Archive: lexical candidate. The ProofAtlas target looks compatible, but its independently unverified normalization does not certify canonical identity.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000202 | [Family formalization notes](#)

OAI-M0350 | Family 197

A Torsion-Free Group Algebra That Is Not Directly Finite

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a finitely presented torsion-free group's algebra over the two-element field fails direct finiteness. The characteristic and finite classifying complex are part of the separately assessed construction.

Archive: unresolved no match. Torsion-free divisor or Whitehead-group candidates do not identify the direct-finiteness target.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0351 | Family 197

A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a finite characteristic-two field and finitely presented nonsofic group admit group-ring elements with ab equal to one but ba different from one. The same witness produces an injective nonsurjective cellular automaton, linking direct finiteness to surjectivity.

Archive: lexical candidate. The direct-finiteness title agrees, but characteristic and group restrictions require complete scope review.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40399681 | [Family formalization notes](#)

OAI-M0352 | Family 197

A Counterexample to the Group-Ring Determinant Conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that an integer group-ring matrix invertible over the rational group ring has finite logarithmic determinant strictly between zero and one. The counterexample targets the unrestricted Fuglede-Kadison determinant inequality rather than ordinary finite matrix determinants.

Archive: lexical candidate. The determinant-conjecture record is plausible, but its excerpt does not establish the precise group-ring formulation.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40341505 | [Family formalization notes](#)

OAI-M0353 | Family 197

A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Odd Characteristic

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a specified odd prime admits a fourth-degree finite field and torsion-containing group whose algebra fails direct finiteness. The same one-sided inverse gives an injective nonsurjective cellular automaton, but no all-primes assertion is made.

Archive: lexical candidate. The universal conjecture is a candidate; this review scores the specified odd-characteristic construction rather than transferring a parent score.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40399681 | [Family formalization notes](#)

OAI-M0354 | Family 198

An algebra of infinite little finitistic dimension

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a finite-dimensional complex algebra has finite-dimensional modules with arbitrarily large finite projective dimension. The construction distinguishes infinite little finitistic dimension from modules whose projective dimension is already infinite.

Archive: lexical candidate. The finitistic-dimension title aligns, but the excerpt leaves little-versus-big and coefficient-field scope unresolved.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40393228 | [Family formalization notes](#)

OAI-M0355 | Family 199

An explicit counterexample to the Auslander-Reiten conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that an explicit finite-dimensional characteristic-two rational-function algebra has a nonprojective Gorenstein-projective module with all positive self and algebra extensions zero. The vanishing survives every base-field extension and would obstruct two representation-theoretic conjectures.

Archive: related target. The retrieved Auslander-Gorenstein subclass record has extra hypotheses not established for this counterexample.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40369966 | [Family formalization notes](#)

OAI-M0356 | Family 199

A counterexample to Tachikawa's second conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a finite-dimensional symmetric algebra has a nonprojective module with every positive self-extension group zero over the specified characteristic-two field. Associated endomorphism constructions claim additional Nakayama and tilting consequences, which are not treated as independent credit here.

Archive: related target. Tachikawa's first conjecture is distinct from the second conjecture addressed by the main example.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40362587 | [Family formalization notes](#)

OAI-M0357 | Family 200

The Artinian Lex-Plus-Powers Betti Theorem

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that characteristic-zero ideals containing homogeneous regular sequences are bounded in every graded Betti number by matching lex-plus-powers ideals. Preserving Hilbert functions while controlling every homological and internal degree strengthens the pure existence assertion.

Archive: lexical candidate. The graded-Betti lex-plus-powers record is plausible, but sequence and characteristic scope need exact comparison.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40350205

OAI-M0358 | Family 200

Commuting Division-Coefficient Forms and the Artinian Eisenbud--Green--Harris Conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that homogeneous ideals containing regular sequences in characteristic zero have the Hilbert function of monomial ideals containing the corresponding pure powers. The sequence has arbitrary positive length with all degrees at least two, rather than only the quadratic case.

Archive: related target. The archived quadratic Artinian formulation is narrower than the general-degree theorem being scored.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40319748

OAI-M0359 | Family 201

A Counterexample to Kurosh's Division-Ring Problem

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a countable characteristic-zero division ring is algebraic over its center and two-generated over that center while remaining infinite-dimensional. The example separates elementwise algebraicity from finite-dimensionality in the division-ring Kurosh question.

Archive: related target. The weak Kurosh candidate must not be equated with this precise central-algebraicity formulation without full text.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40342641

OAI-M0360 | Family 202

The Blockwise Alperin Weight Conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every finite-group prime block has as many irreducible Brauer characters as conjugacy classes of its weights. The statement is numerical and blockwise, not merely an inductive sufficient condition for selected group families.

Archive: lexical candidate. The blockwise Alperin title aligns; the archive excerpt does not establish the complete numerical conventions.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40347851

OAI-M0361 | Family 203

Donovan's Conjecture over Algebraically Closed Fields

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that bounded defect-group order yields finitely many linear Morita classes of finite-group blocks over every fixed algebraically closed positive-characteristic field. Nonabelian defects and characteristic two are included, with the field fixed when counting classes.

Archive: unresolved no match. The retrieved algebraically closed-field records concern torsors or finite geometry, not Donovan finiteness.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0362 | Family 203

Integral Donovan Finiteness over Witt Vectors

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that blocks with bounded defect order have finitely many Morita classes over Witt vectors and each fixed qualifying ramified valuation ring. The integral coefficient ring is retained, making this stronger than residue-field equivalence alone.

Archive: related target. The Donovan record is related, but its excerpt does not distinguish field-valued and integral formulations.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40400087

OAI-M0363 | Family 204

Tensor saturation for even spin groups

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that root-lattice-compatible dominant weights for even spin groups already have a tensor invariant whenever some positive dilation has one. The result concerns saturation at the original integral weights and keeps the lattice condition essential.

Archive: partial or special case. The simply-laced saturation candidate includes other types; this theorem treats $\text{Spin}(2n)$.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40339026

OAI-M0364 | Family 205

Universal Tensor Squares for Symmetric Groups

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that symmetric groups outside orders two, four, and nine admit an irreducible complex representation whose tensor square contains every irreducible. The existential representation may depend on the order and need not always be the staircase representation.

Archive: lexical candidate. The tensor-square title matches, but exceptional orders and existence quantifiers require full archival text. Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40370369 | [Family formalization notes](#)

OAI-M0365 | Family 205

A Cyclic Polytabloid Proof of Saxl's Conjecture

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every staircase-partition irreducible complex symmetric-group representation has a tensor square containing all irreducibles of the same group. The specified representation makes this sharper than merely finding some universal tensor square.

Archive: lexical candidate. This candidate is visible in the companion-family workbench; cyclic-Specht terminology alone does not confirm target identity.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40382327 | [Family formalization notes](#)

OAI-M0366 | Family 206

Finite congruence lattices: characterization and undecidability

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite lattices representing full congruence lattices of finite algebras admit a colored-graph characterization but no algorithmic recognition procedure. The paper also claims undecidability of full subgroup-interval recognition, rather than a finite forbidden-list characterization.

Archive: related target. The archive asks universal representability; characterization and undecidability form a stronger separately scored target here.

Score basis: C0 estimate weighs a new complexity barrier or sharp threshold, long reductions, and limited experimental feedback; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 1453 | [Family formalization notes](#)

OAI-M0367 | Family 206

A negative solution to the finite lattice representation problem

D 581/1000 | p 1 | m 1 | Conditional points 337.561

Published original CFSD-1000 target difficulty

The manuscript reports that some finite nonempty lattice fails to be the full congruence lattice of every finite algebra of any finite signature. A single such example would negate universal finite-algebra representation, without needing a recognition algorithm.

Archive: same target. The counterexample would directly answer the complete original universal representation question negatively.

Score basis: Preserves the complete original target score.

Scoring target: Is every finite lattice isomorphic to the congruence lattice of some finite algebra?

Progress: Conditional on correctness, novelty and acceptance, the manuscript would complete the explicitly scored target; no broader-family fraction or official award is asserted.

[Read manuscript](#) | OPDP ID 1453 | [Family formalization notes](#)

OAI-M0368 | Family 207

The ℓ^1 -Bass Conjecture for Discrete Groups

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that idempotent matrices over complex ℓ^1 group algebras have Hattori-Stallings traces supported on finitely many finite-order conjugacy classes. The assertion covers every discrete group and concerns the completed algebra rather than only finitely supported group-ring elements.

Archive: lexical candidate. The ℓ^1 -Bass title is plausible, but its excerpt leaves exact support and matrix conventions incomplete.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40319781 | [Family formalization notes](#)

OAI-M0369 | Family 207

The Bass trace conjecture and the characteristic-zero Kaplansky idempotent conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that complex group-ring projective traces are supported on finite-order classes, forcing only trivial idempotents over characteristic-zero domains for torsion-free groups. The characteristic-zero idempotent consequence is distinct from zero-divisor and unit conjectures in the same archive bundle.

Archive: partial or special case. The archived row bundles three conjectures over broader coefficients; this paper does not resolve the whole package.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 21 | [Family formalization notes](#)

OAI-M0370 | Family 208

Fiber functors for finite symmetric tensor categories in positive characteristic

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that finite symmetric tensor categories over algebraically closed positive-characteristic fields admit exact faithful symmetric monoidal functors to finite higher Verlinde categories. The level may depend on the category, and characteristic two is explicitly included.

Archive: partial or special case. The candidate uses an infinite-level formulation; finiteness and target-category conventions need scope comparison.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40345932

OAI-M0371 | Family 209

An integral counterexample to Gersten's conjecture

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a two-dimensional ramified regular local mixed-characteristic ring has a noninjective integral map from its fifth K-group to the fraction field's. The characteristic pair is zero and five; the construction targets unrestricted integral Gersten injectivity rather than rational coefficients.

Archive: related target. Motivic Gersten exactness is related but cannot be identified automatically with this specific Quillen K-group counterexample.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40333901

OAI-M0372 | Family 209

An integral degree-three Gersten counterexample

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a two-dimensional ramified regular local ring in characteristic pair zero and five has a kernel in its third integral K-group's map to the fraction field. The degree-three example isolates a low-degree obstruction without claiming all Gersten formulations fail.

Archive: lexical candidate. The regular-local Gersten candidate is plausible, but exact coefficient and cohomological-degree conventions require review.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40337223

OAI-M0373 | Family 210

Foulkes' conjecture for the sixth symmetric power

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the sixth symmetric power of a higher symmetric power embeds equivariantly into the reversed composition over complex vector spaces. The outer comparison parameter is at least six, preserving this fixed-sixth-power Foulkes case rather than all cases.

Archive: partial or special case. The broader Foulkes candidate includes other outer degrees; this is the sixth-degree case.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40369201 | [Family formalization notes](#)

OAI-M0374 | Family 210

Quadratic stabilization of the canonical Foulkes--Howe map

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that canonical Foulkes-Howe maps are surjective once the larger index exceeds the smaller index times one less than itself. The quadratic threshold is independent of vector-space dimension, but it does not assert surjectivity at every ordered parameter pair.

Archive: related target. The rectangular-partition candidate is related to this map but has a different conclusion.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40329157 | [Family formalization notes](#)

OAI-M0375 | Family 211

Random Walks on Critical FK-Ising Maps and Liouville Brownian Motion

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that stationary walks on critical spherical FK-Ising maps converge with their path laws to Brownian motion on the unit-area $\sqrt{3}$ quantum sphere. Using every incident half-edge fixes the corner stationary measure and exactly linear edge-count clock.

Archive: related target. The broader map-scaling candidate does not by itself identify the timed random-walk convergence target.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40360718 | [Family formalization notes](#)

OAI-M0376 | Family 211

Spectral convergence for critical FK-Ising planar maps

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical FK-Ising map spectra, heat traces, and measured geometry converge jointly to the corresponding quantum-sphere Brownian objects. The clock is exactly edge count under unit attempt rate, with loops and edge multiplicities retained.

Archive: related target. Geometric scaling alone does not establish this strengthened spectral and positive-time heat-trace statement.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40360718 | [Family formalization notes](#)

OAI-M0377 | Family 211

A Linear Clock for Random Walk on Tree-Weighted Planar Maps

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that walks on spanning-tree-weighted spherical maps converge jointly with measured geometry to Brownian motion on the $\sqrt{2}$ quantum sphere. The degree stationary law and continuum Dirichlet-form normalization determine an exact edge-count time scaling.

Archive: unresolved no match. Tree displacement and planar Martin-boundary questions have different scaling objects and conclusions.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0378 | Family 211

Canonical conformal limits of subcritical FK planar maps

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that subcritical FK spherical maps converge jointly in conformal embedding, area, all rescaled distances, and nested interfaces to decorated quantum spheres. The FK parameter is fixed strictly between zero and four, with loop convergence modulo reparameterization.

Archive: lexical candidate. The conformal-embedding candidate is relevant, but the full joint convergence and parameter scope are not certified by an excerpt.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371024 | [Family formalization notes](#)

OAI-M0379 | Family 211

The critical Liouville quantum sphere and geometric limits of FK maps at $q=4$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical FK maps at parameter four converge to a constructed critical quantum sphere with intrinsic metric and independent nested CLE-four interfaces. The endpoint requires a critical limiting geometry rather than substituting a subcritical parameter in an existing theorem.

Archive: related target. The broad quantum-gravity map-scaling title does not establish this critical endpoint and metric construction.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371023 | [Family formalization notes](#)

OAI-M0380 | Family 211

Metric-measure limits of subcritical FK and spanning-tree planar maps

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed subcritical FK and spanning-tree-decorated spherical maps have rescaled metric-measure limits given by their unit-area quantum spheres. Distances include every primal edge and the limiting law uses degree-weighted probability measure over all positive edge counts.

Archive: unresolved no match. Spanning-two-tree existence and entropy records are distinct from this metric-measure scaling theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0381 | Family 211

Brownian continuum random tree limits of finite Fortuin-Kasteleyn maps above four

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical FK maps with any fixed parameter above four converge after square-root-size metric scaling to the Brownian continuum random tree. The measure is normalized degree measure and convergence uses all integer edge counts, distinguishing this regime from quantum-sphere limits.

Archive: lexical candidate. The scaling-limit record is relevant, but the above-four parameter regime and finite-volume metric statement need complete comparison.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40360718 | [Family formalization notes](#)

OAI-M0382 | Family 212

No bigeodesics in planar first-passage percolation

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that planar first-passage percolation with continuous independent nonnegative weights has no doubly infinite geodesics under the stated second-moment condition. The conclusion excludes all bigeodesics simultaneously without assumptions on limit-shape regularity.

Archive: partial or special case. The archived record quantifies over dimensions and natural models; this paper fixes the plane and a specific moment class.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 10000073 | [Family formalization notes](#)

OAI-M0383 | Family 212

Strict convexity and differentiability of the planar exponential first-passage limit shape

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that planar exponential first-passage shapes are strictly convex with continuously differentiable boundary, while Gamma weights also yield differentiable time constants. Geometric regularity is distinguished from computing the shape explicitly, which remains a different target.

Archive: partial or special case. The archived target requests an explicit shape, not only convexity and differentiability.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 10100007 | [Family formalization notes](#)

OAI-M0384 | Family 213

Critical bond and site percolation on the cubic lattice

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that nearest-neighbor bond and site percolation on the cubic lattice have no infinite cluster exactly at their respective critical probabilities. The lattice dimension is three, with both bond and site models included in the finite-scale exploration argument.

Archive: partial or special case. The original bond-percolation target asks every dimension; dimension three and a site extension do not resolve that whole scope.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 10100001 | [Family formalization notes](#)

OAI-M0385 | Family 213

No percolation at criticality on quasi-transitive graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical bond percolation has no infinite cluster on locally finite quasi-transitive connected infinite graphs whose critical probability is strictly below one. The threshold qualification is essential and cannot be dropped when relating the result to a universal graph statement.

Archive: partial or special case. The archive omits the critical-probability-below-one condition; this review preserves that restriction instead of transferring its original score.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 10000033 | [Family formalization notes](#)

OAI-M0386 | Family 214

Nonuniqueness of percolation on nonamenable quasi-transitive graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that nonamenable locally finite quasi-transitive graphs have a nonempty bond-percolation nonuniqueness interval and a strictly separated L2 operator threshold. The inequality places critical probability below the operator threshold and at most the uniqueness threshold.

Archive: lexical candidate. The operator-threshold conjecture is plausible, but the excerpt needs the same graph assumptions and threshold definitions.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40342417 | [Family formalization notes](#)

OAI-M0387 | Family 215

The canonical massive continuum limit of the two-dimensional $O(3)$ model

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the two-dimensional nearest-neighbor $O(3)$ model has a canonical interacting massive continuum limit with a unique vacuum and gap on its whole complement. Susceptibility and correlation-length normalization fix the limit without subsequences, while separated four-point correlations certify the claimed interaction.

Archive: unresolved no match. The retrieved massive-fluid and constrained-spin candidates do not identify this constructive continuum model.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0388 | Family 215

An Isolated Particle Pole for the Two-Dimensional $O(3)$ Spin Field

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the continuum $O(3)$ spin field from the prescribed lattice limit has an isolated positive spectral atom at its lowest vector mass. The pole is separated from every other vector spectral contribution, rather than merely asserting a positive Hamiltonian gap.

Archive: unresolved no match. No retrieved record establishes the same spectral-pole target for the specified $O(3)$ construction.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0389 | Family 215

Exact mass asymptotics for the two-dimensional $O(4)$ lattice model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the full square-lattice $O(4)$ transfer mass is asymptotic to $32 \exp(\pi/4 - 1/2)$ times square-root-beta times $\exp(-\pi \beta)$. The mass includes rotation-invariant observable sectors and is measured in the original lattice-time units, fixing the normalization of the asymptotic.

Archive: unresolved no match. Sandpile covariance and categorical lattice-model records address different quantities.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0390 | Family 215

Sharp mass bounds for the two-dimensional $O(4)$ model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the two-dimensional $O(4)$ periodic state has a positive full transfer gap at every finite inverse coupling and sharp matching low-temperature mass bounds. Continuum spectral consequences require additional convergence assumptions; lattice bounds alone do not supply them.

Archive: unresolved no match. The retrieved sharp-bound records concern other stochastic or differential-equation models.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0391 | Family 215

Exponential decay in two-dimensional classical $O(n)$ models

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that square-lattice classical $O(n)$ spin correlations decay exponentially at every positive finite inverse temperature for all n at least three. The estimates are uniform over finite free-boundary graphs and bounded nonnegative bond strengths, excluding the planar XY value n equals two.

Archive: partial or special case. The archive also requests perturbative correctness and critical indices; exponential decay settles only a specified component of that bundle.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 7800001 | [Family formalization notes](#)

OAI-M0392 | Family 216

The critical logarithmic correction for the planar XY model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical planar cosine-XY correlations decay with distance exponent negative one-quarter and a log-distance factor of exponent one-eighth. The positive model-specific amplitude is defined by free-box thermodynamic limits followed by axial separation, fixing the critical observable.

Archive: unresolved no match. One-arm probability corrections and analytic endpoint examples do not identify this XY two-point asymptotic.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0393 | Family 216

Critical Center Magnetization in the Planar XY Model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that under the stated critical-height, local-renormalization, and spin-field companion hypotheses, critical planar XY center magnetization with aligned boundary spins scales as $n^{-(1/8)}$ times $\log n$ to the one-sixteenth power. The amplitude is finite, positive, and model-specific; the assessed assertion is conditional on those inputs rather than an unconditional theorem.

Archive: unresolved no match. Planar center-focus and Ising Kertesz-line records are different targets.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0394 | Family 216

The Critical Spin Field of the Planar XY Model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the critical square-domain XY spin field converges to an imaginary exponential of a zero-boundary Gaussian free field at stiffness two over π . Exact magnetization normalization accompanies distributional convergence and all smeared mixed moments under the stated companion inputs.

Archive: unresolved no match. Discrete-XY metastability and noncritical Potts states do not identify this critical spin-field limit.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0395 | Family 216

Essential Singularity of the Correlation Length in the Planar XY Model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the planar XY correlation length has a positive finite BKT essential-singularity constant when approached from the massive side. The theorem fixes the free-box limit before large axial separation and measures the inverse mass through that prescribed order of limits.

Archive: unresolved no match. The retrieved XY metastability formulation does not establish the same correlation-length singularity target.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0396 | Family 216

The critical correlation exponent of the planar XY model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that ordinary square-lattice XY critical correlations decay as distance to the negative one-fourth power up to vanishing exponent error. The mass-defined critical parameter and free-box-before-separation limit convention are retained; this statement does not determine a prefactor or logarithmic correction.

Archive: unresolved no match. Critical exponents for other loop or Ising models are different mathematical claims.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0397 | Family 216

BKT universality for height and planar spin fields

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that rough discrete Gaussian height models including their threshold have Gaussian limits, and sufficiently cold Villain and XY spin fields have imaginary-Gaussian limits. Finite square-symmetric interaction sets qualify the height model, while the spin-field assertions remain low-temperature statements rather than all-temperature results.

Archive: unresolved no match. Order-theoretic and cyclotomic height-universality records concern different objects.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0398 | Family 217

The low-temperature Sherrington-Kirkpatrick free-energy limiting law

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that low-temperature zero-field Gaussian SK free energy has a full-sequence nondegenerate standardized limit and a positive $n^{1/3}$ -scaled variance limit. The inverse temperature is fixed above one, distinguishing the asserted limiting law from critical-temperature bounds or overlap-distribution questions.

Archive: related target. The candidate concerns critical temperature, whereas this paper treats the entire fixed low-temperature regime.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371244

OAI-M0399 | Family 217

The low-temperature Sherrington-Kirkpatrick fluctuation scale

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that low-temperature zero-field Gaussian SK log-partition fluctuations have standard deviation and typical centered magnitude on the $n^{1/6+o(1)}$ scale. The result determines an exponent at each fixed inverse temperature above one, but not an explicit limiting distribution.

Archive: related target. Overlap atomicity is a different observable from free-energy fluctuation scale.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40386308

OAI-M0400 | Family 218

Quenched SLE₃ limits for general weak random-bond Ising models

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that weak bounded independent random bonds in planar Ising have quenched chordal SLE-three interface limits at the spontaneous-magnetization threshold. The disorder strength is sufficiently small, and convergence is in environmental probability for oriented curves in deterministic domain approximations.

Archive: unresolved no match. Closure under pointwise limits does not establish this quenched interface-scaling theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0401 | Family 218

Quenched SLE₃ Universality for the Weak Random-Bond Ising Model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that planar Ising models with sufficiently weak independent symmetric two-valued ferromagnetic bonds have quenched chordal SLE-three interface limits. Criticality is defined by spontaneous magnetization, and the conditional oriented-curve laws converge in environmental probability within deterministic domain approximations.

Archive: related target. Three-dimensional Ising universality is a different dimension and observable from this planar quenched interface theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 51000062 | [Family formalization notes](#)

OAI-M0402 | Family 218

Logarithmic Relative Fluctuations in the Weakly Disordered Planar Ising Model

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that under the stated deterministic critical-reference estimates, critical spin correlations in weak fair-bond planar Ising have relative disorder second moment growing as log-distance to the one-fourth power. The thermodynamic limit precedes disorder moments and large separation, distinguishing this conditional relative-fluctuation result from mean correlation decay.

Archive: unresolved no match. Four-dimensional susceptibility and generic Tracy-Widom records do not identify this quenched relative-moment observable.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0403 | Family 218

Conformal universality of bulk Ising correlations under weak interactions

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that weak square-symmetric finite-range even Ising interactions preserve conformal limits of mixed bulk spin and centered energy correlations. The same critical branch and field normalizations work across the stated boundary conditions and smooth Jordan domains, including perturbations of either sign.

Archive: lexical candidate. The spin-correlation invariance candidate is relevant, but mixed energy fields and weak-interaction scope need exact comparison.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398021 | [Family formalization notes](#)

OAI-M0404 | Family 218

SLE3 universality for weak finite-range Ising interactions

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that sufficiently small finite-range square-symmetric planar Ising contour perturbations have a common critical parameter with chordal SLE-three interface limits. Signed interactions and arbitrary admissible exterior contours are allowed; convergence concerns full oriented curves under uniform Jordan approximation.

Archive: unresolved no match. Long-range state classification and three-dimensional universality records do not identify this contour-interface theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0405 | Family 218

Buffered comparison and stopping-band resolution in critical Ising

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical nearest-neighbor Ising conditional laws admit connection-probability comparison and finite stopping-band approximation even with arbitrary common pinned spins. Deleted pinned vertices enter the FK comparison graph, while finitely many signed barriers control total-variation errors at fine meshes.

Archive: unresolved no match. Critical mixing-time questions do not establish this conditional-law comparison target.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0406 | Family 219

Fixed-energy universality for weak Anderson disorder on random regular graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed-degree random regular graphs with sufficiently weak independent uniform diagonal disorder have fixed-energy GOE bulk point-process limits. The energy stays away from clean spectral edges, and unfolding uses the corresponding infinite-tree density rather than the undisordered graph density.

Archive: related target. An edge-universality record does not identify weak-disorder fixed-energy bulk statistics.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40353920

OAI-M0407 | Family 219

GOE bulk universality for fixed-degree random regular graphs

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that fixed-degree simple random regular graphs have GOE microscopic eigenvalue statistics at every fixed energy inside the Kesten-McKay bulk. The result includes all admissible sizes without additional conditioning and concerns an unfolded point process rather than only individual eigenvalues.

Archive: lexical candidate. The local-statistics title aligns; full archival energy and graph-model hypotheses are still needed.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40402537

OAI-M0408 | Family 220

A directional zero-one law for finite-range-dependent random environments

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that uniformly elliptic random walks in stationary finite-range-dependent environments have directional escape probabilities equal to zero or one in dimensions at least three. Independence is imposed on full transition rows, including collections over infinite deterministic sets; a fixed direction is assessed at a time.

Archive: partial or special case. The candidate directional-escape conjecture must be checked for its environmental independence and ellipticity hypotheses.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40372441 | [Family formalization notes](#)

OAI-M0409 | Family 220

A directional zero-one law under strict ellipticity

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that independent strictly elliptic environments in dimensions at least three force zero-or-one directional escape probability for nearest-neighbor random walks. Transition probabilities need only be positive, with no uniform lower bound or moment hypothesis, distinguishing this extension from uniform ellipticity.

Archive: lexical candidate. The dimensional zero-one title is plausible, but the weaker ellipticity convention requires full comparison.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40404039 | [Family formalization notes](#)

OAI-M0410 | Family 220

Directional transience implies ballisticity

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that directional transience in uniformly elliptic independent environments implies deterministic positive directional speed for nearest-neighbor lattice random walks in dimensions at least two. The theorem retains uniform ellipticity and fixed direction, excluding arbitrary stationary environments from its ballistic conclusion.

Archive: lexical candidate. The transience-ballistic target appears aligned, but the excerpt does not certify all assumptions.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40395951 | [Family formalization notes](#)

OAI-M0411 | Family 221

The Mézard-Parisi formula for diluted spin glasses

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that diluted even-arity Ising models in the specified Panchenko-Talagrand class have pressure equal to the hierarchical cavity variational infimum. Only first interaction and field moments are assumed; the infimum ranges over finite hierarchy depths and trial laws rather than a fixed replica ansatz.

Archive: related target. The finite-dimensional Edwards-Anderson phase-transition bundle is distinct from this diluted-model pressure formula.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 7800011 | [Family formalization notes](#)

OAI-M0412 | Family 222

The free energy of the Ising random perceptron

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Gaussian-pattern Ising perceptrons have an explicit limiting free-energy variational formula for every bounded Borel log-potential at fixed positive temperature and density. The claimed convergence holds in both expectation and probability, with no smoothness condition on the log-potential.

Archive: unresolved no match. Antiferromagnetic graph and tensor-network energy candidates do not identify this perceptron model.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0413 | Family 222

Microscopic jamming in the negative spherical perceptron

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that negative-margin spherical perceptrons have sharp feasibility and microscopic gap-force laws with numerically certified jamming exponent intervals. The system-size, low-temperature, and near-threshold limits occur in a specified order, and disorder universality is restricted to the stated Gaussian mixture.

Archive: related target. The archived fullRSB identity is related but is not the same feasibility threshold and exponent-certificate theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40315606 | [Family formalization notes](#)

OAI-M0414 | Family 222

The spherical perceptron with bi-orthogonally invariant disorder

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that spherical perceptrons with bounded continuous activation and bi-orthogonally invariant disorder have an explicit limiting free-energy variational formula. The singular-value distribution is compactly supported without outliers; those spectral assumptions are part of the claim rather than arbitrary disorder.

Archive: unresolved no match. Bi-invariant geometric metrics and scissors-congruence questions concern different mathematical structures.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0415 | Family 222

The free energy of the spherical random perceptron

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that spherical Gaussian perceptrons with arbitrary bounded continuous single-pattern potential have an exact limiting-pressure variational formula. Convergence in probability and expectation is asserted at every fixed positive density and inverse temperature, retaining boundedness and continuity of the potential.

Archive: unresolved no match. Spherical random-matrix laws and matching-energy records do not identify this perceptron pressure target.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0416 | Family 223

Square-lattice FK interfaces and nested loops for $1 \leq q < 4$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical square-lattice random-cluster interfaces and nested loops have the predicted SLE and CLE limits for fixed cluster weights from one up to four excluding four. Uniform marked Jordan-domain approximation and retention of loop multiplicities specify the full curve limits; the percolation case yields Cardy crossings.

Archive: related target. The archive's up-to-four formulation includes an endpoint omitted by this manuscript.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000187

OAI-M0417 | Family 223

Self-dual random-cluster interfaces below one

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that self-dual square-lattice random-cluster interfaces below cluster weight one converge to chordal SLE with parameter strictly between six and eight. The domains have uniformly convergent marked boundary parametrizations, and oriented-curve convergence holds along the full mesh sequence.

Archive: lexical candidate. The generic interface-scaling title is relevant, but below-one and self-dual hypotheses need full-text confirmation.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40338281

OAI-M0418 | Family 223

Quenched SLE Universality for Weakly Disordered FK-Ising Interfaces

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that weak symmetric two-valued disorder in critical FK-Ising bonds preserves quenched chordal SLE-sixteen-thirds interface limits. Disorder stays fixed as the mesh shrinks, with conditional curve-law convergence in probability over environments rather than only annealed convergence.

Archive: related target. Double-Ising interfaces are different objects from these quenched FK interfaces.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40399490

OAI-M0419 | Family 223

Thermal FK-Ising interfaces and massive SLE

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that thermal FK-Ising interfaces converge to lattice-independent massive SLE laws at each fixed nonzero mass on angle-bounded isoradial lattices. Positive mass uses a prescribed boundary-value drift; negative mass follows by duality and reversal, preserving the distinction between thermal and critical limits.

Archive: unresolved no match. Double-Ising and abstract thermal-clustering records do not establish this massive-interface formulation.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0420 | Family 223

Natural Occupation Measures for Critical Square-Lattice FK Interfaces

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical FK interface counting measures converge jointly with curves to Minkowski-content measures for fixed cluster weights below four and at least one. A deterministic normalization recovers total step-count scaling as well as local occupation, making the theorem stronger than unparametrized curve convergence.

Archive: unresolved no match. Strip-crossing and bias-ratio candidates do not identify natural interface occupation measures.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0421 | Family 223

Conformal Limits of Critical Square-Lattice Random-Cluster Interfaces

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical square-lattice random-cluster interfaces and complete nested loops converge to their predicted SLE and CLE laws including cluster weight four. The theorem retains loop multiplicities and traversals under marked Jordan approximation, and includes free-edge bond-percolation Cardy crossings at weight one.

Archive: lexical candidate. The curated up-to-four target is plausible, but source-normalization verification remains outstanding.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000187

OAI-M0422 | Family 224

From critical crossings to quenched near-critical universality in Voronoi percolation

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that assuming critical Voronoi Cardy crossings, near-critical quadrilateral threshold laws converge quenched to the triangular-lattice law after model-specific pivotal normalization. The claim is conditional and joint across rational polygonal quadrilaterals, not an independent proof of the input critical crossing formula.

Archive: related target. Quenched critical crossing probabilities differ from normalized near-critical threshold laws.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40399866

OAI-M0423 | Family 224

A Pivotal Amplitude for Voronoi Percolation from Cardy's Formula

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Voronoi Cardy crossing asymptotics imply a positive pivotal-cell amplitude on the mesh to the negative three-fourths scale. The expected unit-square pivotal count is determined without requiring a convergence rate in the assumed critical crossing formula.

Archive: related target. Cardy's formula is an explicit input here, so the manuscript does not itself resolve that archived conjecture.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40318187

OAI-M0424 | Family 224

Cardy's formula for critical Poisson-Voronoi percolation

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical Poisson-Voronoi percolation satisfies Cardy's annealed crossing formula in every bounded Jordan quadrilateral. The result concerns averaged crossing probabilities, distinguishing it from quenched law convergence and full interface convergence that require additional arguments.

Archive: lexical candidate. The generic planar Cardy candidate needs model-specific complete text before Voronoi identity can be confirmed.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40318187

OAI-M0425 | Family 225

The Gaussian free field limit of the balanced six-vertex model with variance multiplier $1/\arcsin(c/2)$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that balanced-torus plane states of the six-vertex model with equal first two weights have Gaussian-free-field height limits throughout the stated third-weight range. Unit height jumps and the logarithmic Green-kernel convention fix the squared amplitude as one over arcsine of half the third weight.

Archive: unresolved no match. Arctic curves and edge fluctuations are different observables from the bulk height field.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0426 | Family 226

The curve scaling limit of half-plane double dimers

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that half-plane Temperleyan square-lattice double-dimer loops converge to nested CLE-four with every macroscopic loop matched as an unparametrized curve. The full mesh limit is retained, while the half-plane boundary convention distinguishes this statement from free-boundary monomer ensembles.

Archive: related target. Boundary-touching ALE curves and complete double-dimer CLE loops have different ensembles and boundaries.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40349913

OAI-M0427 | Family 227

Universality of critical quench autocorrelations in the Sherrington-Kirkpatrick model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical SK heat-bath dynamics have common stationary and quench autocorrelation limits for Gaussian and Rademacher couplings after the stated time and amplitude scaling. The fair-spin quench retains a random functional limit and relaxes toward its stationary counterpart as scaled waiting time grows.

Archive: related target. Reluctant-dynamics runtime does not identify critical heat-bath autocorrelation universality.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371621 | [Family formalization notes](#)

OAI-M0428 | Family 227

Functional universality of critical SK autocorrelations

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical SK stationary autocorrelations converge to a common random functional law for Gaussian and Rademacher disorder. Rate-one clocks produce time scale $n^{2/3}$ and amplitude scale $n^{1/3}$; the limiting functions retain environmental randomness and decay at large times.

Archive: related target. Critical free-energy variance is a different observable from the functional autocorrelation process.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371244 | [Family formalization notes](#)

OAI-M0429 | Family 227

A spectral gap throughout the high-temperature Sherrington-Kirkpatrick phase

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that high-temperature Gaussian SK single-site heat-bath dynamics have an unscaled spectral gap bounded away from zero with high disorder probability. The inverse temperature is fixed below one, and the equivalent dimension-free Poincare inequality applies to every function rather than selected observables.

Archive: unresolved no match. Hardcore-tree and piston-model gaps do not identify the SK dynamics theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0430 | Family 227

Cutoff throughout the high-temperature Sherrington-Kirkpatrick phase

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that high-temperature Gaussian SK heat-bath dynamics exhibit worst-case total-variation cutoff at a deterministic logarithmic location with high disorder probability. Per-spin rate-one clocks and uniformly chosen single-site attempts differ by a factor of system size, which is retained in the reported normalization.

Archive: related target. Reluctant-dynamics runtime is not this heat-bath cutoff statement.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371621 | [Family formalization notes](#)

OAI-M0431 | Family 227

Critical slowing down in the Sherrington-Kirkpatrick model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical Gaussian SK heat-bath dynamics remain far from equilibrium before $n^{2/3}$ per-site time for almost every equilibrium-sampled fixed starting configuration. Uniform-site attempt counts scale as $n^{5/3}$, and disorder probability controls the Gibbs mass of slow initial states rather than a single worst example.

Archive: related target. Reluctant dynamics use a different update process; a shared SK label does not establish the same slow-mixing target.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371621 | [Family formalization notes](#)

OAI-M0432 | Family 227

Stretched-exponential barriers for typical SK initial states

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that low-temperature SK dynamics face a stretched-exponential mixing obstruction from a Gibbs-typical fixed initial configuration. The stated time is $\exp(n^{1/10000})$, and the result concerns large Gibbs mass of bad states with high disorder probability rather than arbitrary initial states.

Archive: unresolved no match. Permutation-count and adiabatic-invariant candidates do not identify this spin-glass mixing barrier.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0433 | Family 227

A typical-start upper bound for low-temperature SK Glauber dynamics

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that low-temperature Gaussian SK heat-bath dynamics mix from a typical fixed equilibrium sample within $\exp(n^{1-1/40000000})$ time. The bound is subexponential but close to exponential, and success is joint over the disorder and starting-state draw rather than worst-start uniformity.

Archive: related target. Ordinary low-temperature Ising with plus boundary conditions is a different model and initial-state question.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40396376 | [Family formalization notes](#)

OAI-M0434 | Family 227

Critical mixing in the Sherrington-Kirkpatrick model

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical Gaussian SK worst-start mixing has exponent two-thirds in per-spin rate-one time and five-thirds in single-site update attempts. The assertion determines an exponent in disorder probability, not an exact deterministic mixing-time constant or a cutoff window.

Archive: related target. The archived reluctant-dynamics runtime conjecture has a different dynamical rule.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40371621 | [Family formalization notes](#)

OAI-M0435 | Family 228

A continuum temperature singularity for a radial pair potential

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a stable three-dimensional radial pair potential produces one common finite-temperature derivative jump across an interval of positive densities. The construction includes repulsive-core divergence, attraction, and an integrable tail while keeping canonical free energy finite in free cubes.

Archive: unresolved no match. Potential-kernel and Schwarz singularity records do not identify this continuum thermodynamic construction.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0436 | Family 228

A radial continuum phase transition with algebraic decay

D 750/1000 | p 1 | m 1 | Conditional points 562.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a bounded continuous stable radial potential with specified algebraic tail decay has a canonical derivative jump at a common temperature over a density interval. The bounded-core variant distinguishes this example from constructions needing a divergent repulsion, while preserving finite free energy at all positive parameters.

Archive: unresolved no match. Other phase-transition models do not establish the same radial-potential target.

Score basis: C0 estimate weighs a finite combinatorial construction or inequality, with useful exact checking but nontrivial discovery; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0437 | Family 229

The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that four-state ferromagnetic tree broadcasting reconstructs exactly above the Kesten-Stigum inequality, including nonreconstruction at equality. Both regular trees and observed Poisson trees are treated; the Poisson advantage averages without survival conditioning, and exact polynomial checks support the proof.

Archive: related target. The below-threshold Potts candidate may use a different state count; four-state and tree-conditioning conventions require full text.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40378496 | [Family formalization notes](#)

OAI-M0438 | Family 229

A Capacity Criterion for Four-State Potts Reconstruction on Trees

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that four-state ferromagnetic reconstruction on bounded-degree deterministic trees is equivalent to positivity of a specific cubic capacity. The resistance depends on channel contraction and edge depth, covering irregular trees and the exponential critical boundary rather than merely fixed branching rates.

Archive: unresolved no match. Purity of random-tree measures and combinatorial edge reconstruction are distinct targets.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0439 | Family 229

The exact reconstruction threshold for the three-state symmetric channel

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that three-state symmetric tree broadcasting has an exact signed-channel reconstruction threshold, with the stated sparse three-community recovery consequence. The regular and observed Poisson models include equality-case nonreconstruction without survival conditioning; graph recovery uses additional known transfer and algorithmic results.

Archive: unresolved no match. Character-reconstruction and interference-channel candidates do not identify this broadcast or block-model statement.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0440 | Family 230

An exact Hausdorff gauge for SLE

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that every chordal SLE parameter strictly between zero and eight admits a deterministic Hausdorff gauge giving positive finite measure to each compact positive-time segment. The result establishes existence of a gauge with finite expected measure in bounded boxes, but this version does not specify its formula.

Archive: unresolved no match. Hausdorff distance and DNLS gauge questions are different meanings of the shared terminology.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0441 | Family 230

An explicit exact Hausdorff gauge for SLE

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that chordal SLE has an explicit exact Hausdorff gauge with dimension-power scaling and iterated-log correction exponent half of two minus the dimension. Each fixed parameter lies below eight; the proposed exponent-one logarithmic gauge is instead nonsigmafinite on positive-time trace segments.

Archive: unresolved no match. Rational-map Julia-set measures concern a different random curve and dimension problem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0442 | Family 231

The free uniform spanning forest is a factor of IID

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that free uniform spanning forests are generated from independent labels by one root-free Borel rule on all qualifying infinite graphs. The additional strongly-Rayleigh and determinantal factor assertions extend to countable group actions without amenability, keeping invariance requirements explicit.

Archive: lexical candidate. The factor-of-IID forest question appears aligned, but full graph and equivariance scope need confirmation.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398618 | [Family formalization notes](#)

OAI-M0443 | Family 232

Gaussian free-field limits of weighted integer Lipschitz heights

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that weighted integer Lipschitz height fields on triangular lattices have zero-boundary Gaussian limits for weights from reciprocal square-root-two to one. A weight-dependent variance normalization includes both the uniform model and the critical endpoint under inside smooth-domain approximation.

Archive: related target. Lozenge tilings form a different height ensemble from these weighted integer Lipschitz fields.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40338058

OAI-M0444 | Family 232

The Gaussian free field limit of integer Lipschitz heights with two-arc boundary data

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that centered odd integer triangular-lattice heights with two-arc boundary values converge to a universal multiple of the Dirichlet Gaussian free field. Neighboring heights differ by zero or two, and the normalization has an absolutely convergent finite-volume expression rather than an unspecified arbitrary scale.

Archive: related target. Random-metric geodesics built from a Gaussian field are different from proving convergence of this discrete height field.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 10000078

OAI-M0445 | Family 232

Uniform real Lipschitz surfaces on the triangular lattice

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that uniform real triangular-lattice Lipschitz surfaces have Gaussian field limits and tuned two-arc zero interfaces converging to SLE-four. The boundary amplitude is linked implicitly to field variance through stationary tangent flux, so interface tuning is part of the assessed theorem.

Archive: unresolved no match. Lipschitz-extension constants and unrelated lattice convergence records do not identify the surface-scaling problem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#)

OAI-M0446 | Family 233

The joint scaling limit of critical Ashkin-Teller currents

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical Ashkin-Teller heights and both current-cluster collections converge jointly to a Gaussian field and recursive local sets at every nesting depth. The theorem includes the four-state Potts endpoint and a distinguished wired boundary cluster, retaining all admissible domain approximations.

Archive: related target. Magnetization-field decomposition is related but is not automatically the same joint current-cluster local-set theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40372991

OAI-M0447 | Family 234

All-temperature pressure for orthogonally invariant Ising spin glasses

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that orthogonally invariant Ising spin glasses have all-temperature pressure limits with optional deterministic external fields and a ground-state formula. Compact spectral support, no outliers, and first-moment field convergence constrain the ensemble; these are not arbitrary nearest-neighbor spin glasses.

Archive: related target. The original Edwards-Anderson phase-transition bundle concerns a different coupling geometry and questions.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 7800011 | [Family formalization notes](#)

OAI-M0448 | Family 235

A Limiting Satisfiability Threshold for Every Fixed Clause Size

D 800/1000 | p 0 | m 1 | Conditional points 0

Historical independent-proof burden estimate (C0)

The manuscript reports that every fixed random clause size at least three has a positive finite limiting satisfiability threshold under independent signed-clause sampling with replacement. The manuscript supplies an alternative existence proof and explicitly credits an earlier resolution, so novelty cannot be inferred from this result's appearance.

Archive: related target. The selected archive target concerns a large-clause-size asymptotic threshold formula; this manuscript establishes existence for each fixed clause size. It credits Carenini with the same existence conclusion, so this alternative proof receives no new-resolution credit.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

Progress: Zero new open-problem credit: the source explicitly credits prior resolution of this same threshold-existence target and presents an alternative proof.

[Read manuscript](#) | OPDP ID 40398785 | [Family formalization notes](#)

OAI-M0449 | Family 235

Linear Variance of the Random 3-SAT Hitting Time

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the first unsatisfiable prefix index in uniform random three-SAT has variance proportional to variable count. The theorem removes a logarithmic upper-bound loss while retaining distinct-variable clauses and sampling with replacement; it does not compute the threshold constant.

Archive: related target. An exact satisfiability-threshold record differs from the hitting-time variance target.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000221 | [Family formalization notes](#)

OAI-M0450 | Family 235

Variance of the Random k-SAT Hitting Time

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that random k-SAT first-unsatisfiable-prefix indices have linear variance for each fixed clause size at least four and stated near-linear three-SAT bounds. The theorem also controls clipped indices, while a companion supplies the sharper three-clause result; none of these claims specifies an exact threshold value.

Archive: related target. Threshold existence or numerical location is not the same claim as hitting-time variance.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398785 | [Family formalization notes](#)

OAI-M0451 | Family 235

Computing the Random 3-SAT Threshold

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that one finite deterministic procedure approximates the limiting random three-SAT satisfiability threshold to arbitrary prescribed accuracy. Computability follows from certificates searched without a known finite-size convergence rate; existence is attributed to an earlier result rather than claimed as new.

Archive: related target. Computability does not supply the exact numerical or closed-form threshold requested by a broader threshold-location target.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 50000221 | [Family formalization notes](#)

OAI-M0452 | Family 236

The sharp factor-of-IID threshold for the free Ising model on regular trees

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that free zero-field Ising measures on regular trees are factors of independent labels exactly at and below the stated square-root branching threshold. The claimed new direction includes equality and full automorphism equivariance on every input label configuration, while the strict converse is already known.

Archive: related target. Uniform spanning forests and free Ising measures are different factor-of-IID targets.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398618 | [Family formalization notes](#)

OAI-M0453 | Family 237

Radial transfer estimates and polygon length laws for honeycomb walks

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb polygon ensembles have sharp diameter-tail and truncated length-moment exponents implying three-fourths length-to-diameter scaling. Unrooted and length-weighted polygon measures have different tail exponents, which are retained rather than merged into one unqualified walk statement.

Archive: unresolved no match. Trie transfer and discretized stability records do not identify these self-avoiding polygon observables.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0454 | Family 237

Critical honeycomb chords with prescribed boundary endpoints

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb self-avoiding paths between separated boundary ports have length exponent four-thirds in several specified geometries. Mean-length conclusions use additional half-plane or strip qualifications, including density-one heights and an aligned weighted mixture, rather than every endpoint ensemble.

Archive: related target. Surface-fugacity criticality is a different target from prescribed-endpoint path-length laws.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40324203 | [Family formalization notes](#)

OAI-M0455 | Family 237

Cylinder loop weights and planar nesting

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb separating-loop partition functions have determined cylinder exponents at every positive fugacity, with explicit nesting exponents at fugacity two. Cylinder and planar nesting normalizations differ by a factor of two in the reported exponent, so the underlying geometry is retained.

Archive: unresolved no match. Quantum loop algebras and loop-erased-walk probabilities do not identify this polygon nesting law.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0456 | Family 237

Mass and covering exponents for fixed-length honeycomb walks

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that uniform fixed-length honeycomb self-avoiding walks have diameter exponent three-fourths and simultaneous local-mass and covering exponents four-thirds. The estimates hold at every sufficiently large integer length with arbitrary positive exponent slack and polynomially small failure probability, rather than only selected lengths.

Archive: related target. Quadrant walks have different constraints and critical observables from these unrestricted fixed-length honeycomb walks.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40337067 | [Family formalization notes](#)

OAI-M0457 | Family 237

Signed cylinder propagation and marked polygons on the honeycomb lattice

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb separating polygons have the stated cylinder and planar nesting exponents, together with sharp marked-chord and length-moment laws. Boundary endpoints are summed for the chord partition, and the diameter cutoff and translation quotient distinguish the several observables being controlled.

Archive: unresolved no match. Algebraic gonality of lattice polygons is unrelated to this random-walk nesting target.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0458 | Family 237

Cylinder amplitudes and logarithmic bridge-length windows on the honeycomb lattice

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb bridges place substantial mass in an explicit logarithmic window around the four-thirds height-to-length scale. The bridge height is summed through a mildly enlarged cutoff, and the lower mass retains a separate logarithmic loss rather than a normalized probability bound.

Archive: unresolved no match. Bridge trisections and bridge-girth stabilization refer to different mathematical objects.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0459 | Family 237

Marked polygon correlations and one-arc bounds

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb polygons have a sharp squared-length mass bound together with uniform two-mark cylinder and one-arc estimates. Polygons are counted modulo translation, and the arc estimate allows highly unequal marked intervals; both counting conventions constrain the assertion.

Archive: unresolved no match. Straight-skeleton complexity and arc representations do not identify these stochastic polygon correlations.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0460 | Family 237

Disk transfer representations and confined bridge mass

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb strip-crossing bridges have mean length of four-thirds width scale and mass of inverse-fourth-root width scale. The initial port is fixed and the terminal port summed; confinement preserves the mean exponent while a separate arch kernel has exponent negative five-fourths.

Archive: unresolved no match. Representation transfer and two-bridge link rigidity are different targets.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0461 | Family 237

Polynomial vacuum representations and bridge mass for honeycomb walks

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that all-length critical honeycomb bridge measures have four-thirds strip-height mean-length growth and hexagon chords have the stated central-visit exponent. A fixed starting port, free terminal port, and critical weighting distinguish this ensemble from uniform fixed-length walks, even where the scaling exponents agree.

Archive: unresolved no match. Knot representation and vacuum geometry records do not identify this bridge-mass theorem.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0462 | Family 237

Renewal and changes of law for critical honeycomb walks

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb irreducible bridges have three-fourths spatial scaling with thermal moment laws and local endpoint conditioning bounds. Unrestricted uniform-walk endpoint lower bounds are stated only on a common density-one set of lengths, preserving a significant qualification among the claimed laws.

Archive: related target. A general strong-renewal criterion is not the same spatial-exponent theorem for self-avoiding honeycomb bridges.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40330987 | [Family formalization notes](#)

OAI-M0463 | Family 237

Critical strip-crossing mass on the honeycomb lattice

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that critical honeycomb strip-crossing path mass scales as height to the negative one-fourth power and return displacement moment as height to the three-fourths. These are critically weighted path sums rather than normalized length-conditioned probabilities, and the strip geometry is retained in both estimates.

Archive: related target. Square-lattice strip parity conjectures involve a different lattice and observable.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40376590 | [Family formalization notes](#)

OAI-M0464 | Family 237

Uniform marked-polygon estimates and sharp finite bridge moments

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that marked honeycomb polygons obey periodic-staircase separation bounds, while finite strip bridges satisfy sharp mass and length-moment estimates. The marked bound preserves an explicit period-to-separation ratio; the strip lower mass controls bridges reaching the four-thirds length scale rather than every bridge.

Archive: unresolved no match. Viscous Burgers estimates and hypergraph embeddings concern different quantities.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0465 | Family 237

Cap-selected amplitudes and triangle chords for honeycomb walks

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that large-diameter critical honeycomb chords in an equilateral triangle have three-fourths partition growth and four-thirds mean length. Both endpoints are summed, a macroscopic diameter cutoff is imposed, and a separate two-cap cylinder amplitude has exponent one-sixth, preventing ensemble conflation.

Archive: unresolved no match. Disk-packing and arrangement-triangle candidates do not identify these honeycomb path observables.

Score basis: C0 estimate weighs deep analytic or geometric prerequisites, global control, and substantial formalization burden; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0466 | Family 238

Optimal-order mixing of the Thorp shuffle

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the Thorp shuffle on powers-of-two deck sizes mixes in a number of physical shuffles proportional to the logarithm of deck size. The explicit upper constant is sixteen hundred and a support count provides a linear lower bound in the binary logarithm, uniformly over starting decks.

Archive: related target. Adjacent transposition shuffling is a different transition mechanism and mixing problem.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40398712 | [Family formalization notes](#)

OAI-M0467 | Family 238

Random coordinate frames and partial permutation laws

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that partial permutation laws in random coordinate frames yield explicit linear-in-log-deck upper bounds for worst-start Thorp mixing. A standalone bound of 32800 binary logarithms improves to 512 when combined with a companion Fourier transfer, so the dependency is retained.

Archive: unresolved no match. Random Gabor frames and partial Sylvester sums do not identify this card-shuffle theorem.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0468 | Family 238

From partial permutation information to Fourier bounds

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that near-uniform images of seven-eighths of permutation labels plus vanishing mean sign imply that two independent draws multiply to an asymptotically uniform permutation. The input assumption concerns averaged injection-law total variation, not just a few marginal coordinates; deck size runs through multiples of eight.

Archive: unresolved no match. Partial Schubert permutations and Fourier restricted isometry are different uses of the terminology.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0469 | Family 238

Conditional information under deterministic coordinate sweeps

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that revealing selected Thorp card paths leaves further endpoint laws near uniform after a bounded number of coordinate sweeps under a label-reserve condition. The information estimate averages over observed trajectories; a separate Fourier transfer yields a stated full-deck physical-shuffle bound rather than an immediate marginal-to-joint inference.

Archive: unresolved no match. Conditional Renyi inequalities and empirical-risk information bounds do not identify this conditional permutation law.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0470 | Family 238

Conditional permutations in a revealed switching environment

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that after revealing half of the Thorp card trajectories, remaining conditional permutations approach uniform in expected total variation after bounded sweeps. The estimate is uniform over initial layouts and supports full-deck mixing in logarithmically many physical steps, distinguishing a conditional law from unconditional marginal symmetry.

Archive: unresolved no match. Quasirandom permutations and graph-switching reconstruction do not establish this revealed-environment theorem.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0471 | Family 238

Routing densities and representation contraction for Thorp sweeps

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a bounded absolute number of Thorp coordinate sweeps drives the entire permutation law to uniform in total variation. Because each sweep contains logarithmically many physical shuffles, the claim gives optimal logarithmic mixing order rather than a constant physical-step mixing time.

Archive: unresolved no match. Network and rotor routing are different algorithms from the Thorp shuffle.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0472 | Family 238

Row-column symmetry and contraction of coordinate sweeps

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Thorp full-permutation mixing takes order log deck size in physical shuffles, with a bounded number of coordinate sweeps sufficient. The paper uses row-column symmetry and contraction to control the whole law, preserving the distinction between sweeps and individual shuffling steps.

Archive: unresolved no match. Row-column symmetry of knot homology and discrepancy bounds are different targets.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0473 | Family 238

Random-subspace tests and trace smoothing for coordinate sweeps

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that a fixed number of Thorp coordinate sweeps gives full-deck total-variation error at most one-half times deck size to the negative fifth power. The quantitative error bound holds uniformly over initial decks and implies optimal-order logarithmic physical mixing, stronger than merely proving a vanishing error.

Archive: unresolved no match. Smoothing of free coordinates or random curves does not identify the shuffle's permutation-error target.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0474 | Family 238

Compatibility entropy and the spectrum of a Thorp sweep

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Thorp full-permutation total variation tends to zero after sufficiently many but fixed coordinate sweeps as the binary deck exponent grows. Compatibility entropy and sweep-spectrum analysis yield linear-in-binary-logarithm physical mixing; the theorem concerns the complete permutation rather than individual card positions.

Archive: unresolved no match. Sweep-map bijectivity and operator Renyi compatibility concern different processes.

Score basis: C0 estimate weighs uniform algorithmic guarantees, explicit resource accounting, and exact success criteria; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0475 | Family 238

Signed tensor densities and diagram budgets for the Thorp shuffle

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that Thorp permutation densities approach uniform in squared L2 distance after a fixed number of coordinate sweeps. This stronger norm controls total variation and yields optimal logarithmic physical mixing, while signed tensor-density and diagram accounting provide the stated analytic framework.

Archive: unresolved no match. Cluster-variable tensor diagrams and tractor densities do not establish this permutation-density contraction.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0476 | Family 238

Conditional coordinate sweeps and analytic transfer

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that conditioned coordinate-sweep laws satisfy a Schatten-moment bound sufficient to transfer optimal logarithmic mixing to binary Thorp shuffles. Line laws must be near uniform and their power-of-two sizes remain in a fixed interval; conditioning is restricted to feasible prescribed card trajectories.

Archive: unresolved no match. Anosov conditional mixing and log-concave coordinate tails are different stochastic or geometric settings.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0477 | Family 238

A strict four-row permanent inequality and permutation moments

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that near-uniform four-permutation laws with exactly uniform coordinate marginals satisfy a permanent inequality with exponent below two. Applying the inequality recursively gives full-deck Thorp mixing in logarithmically many physical steps; the finite inequality's proximity hypothesis is part of the argument.

Archive: related target. Pate's permanent challenge concerns a different inequality, so an identical permanent-based target is not established.

Score basis: C0 estimate weighs specialized technical arguments and precise hypotheses, with some checkable intermediate structure; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40353516 | [Family formalization notes](#)

OAI-M0478 | Family 239

The sharp exponential rate of singularity for symmetric Bernoulli matrices

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that symmetric matrices with independent fair upper-triangle signs have singularity probability with sharp exponential base one-half. Symmetry couples the lower triangle, and the expression allows subexponential corrections rather than specifying an exact finite-size probability or polynomial prefactor.

Archive: lexical candidate. The sharp symmetric-Bernoulli title aligns, but the archive excerpt does not expose all entry and diagonal conventions.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40324772

OAI-M0479 | Family 239

The sharp singularity rate for biased symmetric sign matrices

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that symmetric sign matrices with fixed nonfair bias have singularity probability with exponential base p -squared plus one-minus- p squared. Coincident rows attain the proposed rate; the bias stays strictly between zero and one and excludes the separately treated fair case.

Archive: related target. The fair symmetric-Bernoulli record does not automatically cover this biased distribution.

Score basis: C0 estimate weighs a sharp combinatorial threshold and interacting local-to-global constraints; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40324772

OAI-M0480 | Family 240

A CH obstruction to a prescribed categoricity threshold

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript reports that the continuum hypothesis permits an abstract elementary class with Loewenheim-Skolem number $\aleph_0$ that is eventually categorical but has two nonisomorphic models at cardinality $\beth_{(\omega_2)}$. The result obstructs a prescribed downward-transfer bound and yields a relative nonprovability conclusion, rather than refuting every eventual categoricity formulation.

Archive: lexical candidate. The threshold-transfer candidate is plausible, but the original infinitary-sentence categoricity record has a different model class and cannot supply an inherited score.

Score basis: C0 estimate weighs a longstanding structural bottleneck, universal quantifiers, and demanding independent proof checking; historical specialist effort is unobserved.

[Read manuscript](#) | OPDP ID 40353953 | [Family formalization notes](#)

OAI-M0481 | Family 240

Eventual categoricity for abstract elementary classes

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Shelah's eventual categoricity theorem for arbitrary abstract elementary classes in ZFC. For each bound on the Lowenheim-Skolem number, a single cardinal threshold works uniformly: categoricity anywhere above it forces categoricity everywhere above it. Amalgamation, tameness, and absence of maximal models are not assumed. The proposed result therefore addresses a global classification question rather than a transfer theorem within a restricted class.

Archive: lexical candidate. The named Shelah conjecture is retrieved, but its MathDB excerpt does not expose the complete quantifiers and threshold condition.

Score basis: Uniform categoricity transfer without structural hypotheses requires extensive model-theoretic context and long nonlocal arguments.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0482 | Family 241

Rigidity of the Turing degrees

D 619/1000 | p 1 | m 1 | Conditional points 383.161

Published original CFSD-1000 target difficulty

This paper claims that every order automorphism of the entire partially ordered set of Turing degrees is the identity. Its scope is global, extending beyond previous rigidity results above particular jump degrees or under jump-preservation assumptions. The question concerns whether relative computability alone distinguishes every degree without naming additional structure. If the argument holds, the automorphism group sought by the archive's problem is precisely the trivial group.

Archive: same target. The archived record asks for the automorphism group of the full Turing-degree order; the claimed identity theorem completely specifies that group.

Score basis: The archived automorphism-group question and the claimed rigidity theorem have the same full target.

Scoring target: Determine the full automorphism group of the partial order of Turing degrees.

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 7500003 | [Family formalization notes](#)

OAI-M0483 | Family 242

Single-fold Diophantine representations

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a single-fold Diophantine representation for every recursively enumerable subset of every finite power of the natural numbers. An integer polynomial is required to have exactly one complete auxiliary tuple for an input belonging to the set and none otherwise. This uniqueness is substantially stronger than ordinary existential representability. The asserted theorem would also imply finite-fold representability, linking computability with exact control of arithmetic witnesses.

Archive: unresolved no match. Retrieved Diophantine-tuple problems concern square products among integers and do not establish a match to recursively enumerable set representations.

Score basis: Uniqueness-preserving Diophantine encoding combines computability, arithmetic construction, and difficult global witness control.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0484 | Family 243

Choiceless polynomial time with counting does not capture polynomial time

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that choiceless polynomial time with counting cannot express every polynomial-time query on unordered finite structures. Its separating example is consistency of a linear system over the three-element field, encoded in a fixed vocabulary of eight binary relations. The limitation applies to the full counting formalism rather than a bounded fragment. It addresses whether computation invariant under relabelling can avoid making arbitrary choices without losing expressive power.

Archive: unresolved no match. The retrieved counting-algorithm questions concern matroid bases or linear extensions, rather than the expressive power of choiceless computation.

Score basis: Separating a rich invariant logic from polynomial-time computation requires expressive-power lower bounds rather than running a finite search.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0485 | Family 243

Witnessed symmetric choice is strictly stronger than choiceless polynomial time with counting

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that adding witnessed symmetric choice strictly enlarges choiceless polynomial time with counting. A fixed sentence using one witnessed-choice operator gives a Boolean answer on every finite input, yet no sentence of the original formalism defines the same query. The comparison permits unrestricted finite-rank hereditarily finite sets in the weaker language. The result isolates a precise source of additional expressive power within invariant finite computation.

Archive: unresolved no match. The candidate records share words about counting or strength but describe different mathematical objects and conclusions.

Score basis: Expressiveness separation needs uniform arguments against every weaker sentence, with substantial formal-language and symmetry bookkeeping.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0486 | Family 244

The Partition Principle does not imply Choice

D 621/1000 | p 1 | m 1 | Conditional points 385.641

Published original CFSD-1000 target difficulty

The manuscript claims a relative-consistency separation between the Partition Principle and the Axiom of Choice. Assuming ZF is consistent, it asserts consistency of the Partition Principle together with ordinal-indexed choice and failure of full choice. A separate construction over countable transitive ZFC models preserves ordinals and introduces no new countable sequences of ground elements. The claimed countermodel would give a negative answer to the archive's implication question.

Archive: same target. The archive asks whether PP implies AC; a model satisfying PP and not AC settles that exact implication negatively, subject to the stated consistency qualification.

Score basis: The relative-consistency countermodel directly answers the archived PP-implies-AC question.

Scoring target: Does the Partition Principle imply the Axiom of Choice over ZF?

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 1347 | [Family formalization notes](#)

OAI-M0487 | Family 245

Weak and strong normalization in pure type systems

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that weak beta normalization implies strong beta normalization for every pure type system, without a functionality assumption. Both properties quantify over all legal expressions in all valid contexts, and reduction includes type annotations. The claim is therefore a theorem about whole systems, not the false assertion that any individual term admitting a normal form terminates along every reduction path. It addresses the beta Barendregt-Geuvers-Klop conjecture.

Archive: unresolved no match. The retrieved pure local-system and graph-type questions are lexical collisions and provide no meaningful matching target.

Score basis: Whole-system normalization involves delicate substitution and reduction invariants across terms, types, and contexts.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0488 | Family 246

A Modulus Proof of Cannon's Conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Cannon's conjecture through a modulus-based argument. Every hyperbolic group with boundary homeomorphic to a two-sphere is asserted to act properly and cocompactly by isometries on hyperbolic three-space, with a finite kernel allowed. Orientation-reversing isometries are included, and torsion-freeness is not required. The proposed conclusion connects a group's topological boundary with a concrete geometric realization, extending beyond results for previously controlled boundary geometries.

Archive: related target. The retrieved Cannon record has the expected boundary hypothesis, but its excerpt truncates the action conclusion; exact equivalence is not established here.

Score basis: Recovering a geometric action from boundary topology requires advanced geometric analysis and global group-theoretic control.

[Read manuscript](#) | OPDP ID 40392685 | [Family formalization notes](#)

OAI-M0489 | Family 247

An infinite finitely presented residually finite 2-group

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims residual finiteness for a particular infinite, ordinarily finitely presented periodic Steinberg group. A finite-index kernel is then asserted to be an infinite finitely presented residually finite group in which every element has finite power-of-two order, with those orders unbounded. Finite degree truncations of the associated algebra are used to detect elements, including a central kernel. The construction strengthens existence of periodic groups by controlling finite quotients.

Archive: related target. The archived torsion-group existence question also imposes a condition on every subgroup, which this reported theorem does not supply.

Score basis: Simultaneous finite presentation, periodicity, infinitude, and residual finiteness require interacting algebraic constructions and quotient detection.

[Read manuscript](#) | OPDP ID 30001582 | [Family formalization notes](#)

OAI-M0490 | Family 247

An infinite finitely presented periodic group

D 665/1000 | p 1 | m 1 | Conditional points 442.225

Published original CFSD-1000 target difficulty

The paper claims an infinite group with an ordinary finite presentation in which every element has finite order. Its example is a Steinberg group over a constructed algebra; periodicity does not mean a single uniform bound on element orders. Companion algebraic conclusions include an infinite-dimensional finitely presented nil algebra over the two-element field. The group construction would directly refute the archive's question asking whether finite presentation forces periodic groups to be finite.

Archive: same target. Ordinary discrete-group finite presentation and periodicity match the archived formulation; the additional algebra results are separate consequences.

Score basis: The infinite periodic finitely presented group is a direct counterexample to the archived universal finiteness question.

Scoring target: Is every ordinarily finitely presented periodic group finite?

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 1195 | [Family formalization notes](#)

OAI-M0491 | Family 248

Thompson's group F is nonamenable

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that Thompson's group F is nonamenable, thereby giving a negative answer to its amenability problem. The proposed argument must go beyond the usual obstruction from a nonabelian free subgroup, because F contains none, and beyond its known failure of elementary amenability. The target concerns an invariant mean for the full group, not a property of a particular action or generating set. Acceptance would resolve a central question about this explicit group.

Archive: related target. The retrieved record asks whether F is automatic; that concerns a distinct property and is not resolved merely by nonamenability.

Score basis: Amenability here resists familiar free-subgroup tests and requires global analytic or combinatorial obstructions.

[Read manuscript](#) | OPDP ID 30003838 | [Family formalization notes](#)

OAI-M0492 | Family 249

A finitely generated counterexample to the Eilenberg-Ganea conjecture

D 622/1000 | p 1 | m 1 | Conditional points 386.884

Published original CFSD-1000 target difficulty

This paper claims a finitely generated residually finite group with integral cohomological dimension two but geometric dimension three. The example is a height kernel in a right-angled Artin group built from a finite acyclic flag complex. Its asserted obstruction rules out every two-dimensional classifying space, regardless of the number of cells. Such a group would directly contradict the Eilenberg-Ganea statement recorded in the atlas while retaining useful finiteness and quotient properties.

Archive: same target. The archived question requires a two-dimensional $K(G, 1)$ for every group of cohomological dimension two; the claimed example violates that exact requirement.

Score basis: The cohomological-dimension-two example with no two-dimensional classifying space directly refutes the archived conjecture.

Scoring target: Does every group of integral cohomological dimension two admit a two-dimensional classifying space?

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 1451 | [Family formalization notes](#)

OAI-M0493 | Family 250

Finite algebraic envelopes and the Boone-Higman conjecture

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the Boone-Higman equivalence: a finitely generated group has decidable word problem exactly when it embeds in a finitely presented simple group. In the constructive direction, the input group needs a finite generating set and a word-problem decider, but need not itself be finitely presented. The conclusion therefore strengthens classical embedding results by simultaneously requiring finite presentation and simplicity of the containing group, without bounding the algorithm's running time.

Archive: same target. The complete displayed Boone-Higman sentence has the same finitely generated input, decidable word problem, and finitely presented simple overgroup.

Score basis: Encoding arbitrary word-problem deciders while forcing finite presentation and simplicity demands substantial constructive group theory.

[Read manuscript](#) | OPDP ID 40373846 | [Family formalization notes](#)

OAI-M0494 | Family 250

Simple Finiteness overgroups of groups with decidable word problem

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims that every finitely generated group with decidable word problem embeds in a nontrivial simple group of type F-infinity. The stronger finiteness requirement means a classifying space has a finite skeleton in each dimension; it does not impose finite dimension or torsion-freeness. No finite presentation or runtime bound is required for the input. The proposed theorem strengthens the Boone-Higman existence statement by controlling all higher finiteness properties.

Archive: unresolved no match. The leading candidates concern aperiodic shifts or Engel-word maps, rather than simple overgroups with all higher finiteness properties.

Score basis: Maintaining simplicity together with finiteness in every dimension adds an unbounded topological construction to the embedding problem.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0495 | Family 250

A universal group of type Finiteness

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a single group of type F-infinity containing every finitely presented group as a subgroup. It further characterizes its finitely generated subgroups, up to isomorphism, as precisely the recursively presented finitely generated groups. The universal containing group therefore combines a common embedding target with finite classifying-space skeletons in all dimensions. This addresses a higher-dimensional version of Higman's embedding question, rather than constructing a separate envelope for each input group.

Archive: unresolved no match. The retrieved group-type and representation-theoretic candidates have different targets; no reliable archive identity was established.

Score basis: A universal embedding construction must accommodate all presentations while preserving finiteness through infinitely many dimensions.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0496 | Family 251

Unitarizability implies amenability for discrete groups

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that a discrete group is amenable exactly when every uniformly bounded complex Hilbert-space representation is similar to a unitary representation. For nonamenable groups it asserts nonsimilar examples with norm bound arbitrarily close to one, and separable Hilbert spaces when the group is countable. The proposed result settles Dixmier's unitarizability question in the discrete setting. Its quantification over all representations is stronger than testing any chosen finite collection.

Archive: same target. The archived record explicitly asks whether every unitarizable discrete group is amenable, which is the new direction claimed here.

Score basis: The converse demands constructing bounded representations that evade every similarity transformation, with substantial operator-theoretic control.

[Read manuscript](#) | [OPDP ID 40402524](#) | [Family formalization notes](#)

OAI-M0497 | Family 251

Strong Ulam Stability Characterizes Amenability

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that strong Ulam stability characterizes amenability for countable discrete groups. Here every sufficiently accurate unitary almost representation, on any complex Hilbert space, must be uniformly close in operator norm to a genuine representation on the same space. The asserted new direction constructs arbitrarily small-defect maps for nonamenable groups that remain uniformly separated from all genuine representations. Infinite-dimensional Hilbert spaces are essential, so finite-dimensional stability alone is outside this conclusion.

Archive: related target. The candidate asks whether a conjunction of two stability properties characterizes amenability; its truncated excerpt does not specify the second property.

Score basis: Uniform perturbation control over arbitrary Hilbert spaces creates a demanding combination of group theory and operator analysis.

[Read manuscript](#) | OPDP ID 40396247 | [Family formalization notes](#)

OAI-M0498 | Family 252

A torsion-free hyperbolic group that is not residually finite

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a torsion-free word-hyperbolic group that is not residually finite. Its group comes from a finite Euclidean triangle complex, and an argument places one member of a fixed finite family of nontrivial elements in every finite-quotient kernel. The proof is existential about which member works. This would refute residual finiteness even in the torsion-free subclass, without asserting a similar counterexample among hyperbolic Kleinian groups or solving their rank problem.

Archive: related target. The rank-decidability candidate is restricted to hyperbolic Kleinian groups and asks an algorithmic question, not residual finiteness of arbitrary hyperbolic groups.

Score basis: The construction must reconcile negative-curvature constraints with an obstruction surviving every finite quotient.

[Read manuscript](#) | OPDP ID 40393629 | [Family formalization notes](#)

OAI-M0499 | Family 253

An infinite finitely presented simple amenable group

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims existence of an infinite finitely presented simple amenable group. The example is an alternating subgroup of a polygon-exchange group on finitely many squares; both generators and relators are finite, although no uniform size bound is imposed. The claimed construction combines simplicity with amenability and ordinary finite presentation. It does not assert that the resulting group has two generators, a distinct strengthening that appears among the archive's nearby questions.

Archive: related target. The candidate asks for a two-generated infinite simple amenable group; the reported construction supplies finite presentation but no two-generator conclusion.

Score basis: Finite presentation must coexist with amenability and simplicity, imposing conflicting global structural constraints.

[Read manuscript](#) | OPDP ID 40385119 | [Family formalization notes](#)

OAI-M0500 | Family 254

Harmonic heights and the Artin $K(\pi,1)$ conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that the standard Salvetti complex of every finite-rank Artin group is aspherical. Equivalently, its universal cover is contractible and all higher homotopy groups vanish. The statement permits arbitrary finite and infinite Coxeter labels, as well as disconnected diagrams, and includes the full higher-dimensional complex. Its proposed harmonic-height argument addresses the general finite-rank Artin $K(\pi,1)$ conjecture rather than only familiar spherical or otherwise restricted families.

Archive: related target. The archived Artin $K(\pi,1)$ record is closely related, but its excerpt omits the full rank and conclusion conventions needed for exact identity.

Score basis: Controlling higher-dimensional cells for arbitrary Coxeter diagrams requires broad geometric group theory and long global homotopy arguments.

[Read manuscript](#) | OPDP ID 40333905 | [Family formalization notes](#)

OAI-M0501 | Family 254

An Artin group with no geometric $CAT(0)$ action

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit Artin group with 116 generators and off-diagonal Coxeter labels drawn from two, three, and infinity that admits no geometric action on any nonempty proper $CAT(0)$ space. A geometric action means a proper cocompact isometric action. The obstruction is independent of the dimension or chosen metric model. It would contradict the usual proper-space version of the Artin $CAT(0)$ conjecture, with the space convention needing careful comparison.

Archive: related target. The archive's sentence says $CAT(0)$ space without explicitly specifying properness; the reported obstruction has a proper-space convention.

Score basis: A universal obstruction to every proper nonpositively curved model is harder than ruling out one natural complex.

[Read manuscript](#) | OPDP ID 30002538 | [Family formalization notes](#)

OAI-M0502 | Family 254

Parabolic intersections in Artin groups

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims closure of parabolic subgroups under intersection in every finite-rank Artin group. Its theorem includes arbitrary conjugates, nonspherical parabolics, and diagrams with infinite Coxeter labels. In the pairwise statement, the intersection is explicitly another conjugate of a standard parabolic subgroup; established reduction results then extend this to arbitrary intersections and parabolic closure. The claim addresses a structural property across the whole finite-rank class, rather than a restricted diagram family.

Archive: related target. The named intersection conjecture is retrieved, but its excerpt does not reveal the complete ambient-rank and subgroup conventions.

Score basis: Arbitrary conjugates and unrestricted diagrams demand uniform structural arguments beyond computations in individual presentations.

[Read manuscript](#) | OPDP ID 40370268 | [Family formalization notes](#)

OAI-M0503 | Family 255

Quasi-isometric recognition of virtually polycyclic groups

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every finitely generated group quasi-isometric to a finitely generated virtually polycyclic group is itself virtually polycyclic. No algebraic condition beyond finite generation is imposed on the comparison group. The equivalent lattice-recognition formulation allows the solvable Lie group in the conclusion to differ from the original one. Thus the proposed theorem recognizes a broad algebraic class from coarse geometry, without providing a complete isomorphism classification inside that class.

Archive: related target. The archived abelian-by-cyclic classification question has additional matrix hypotheses and potentially finer conclusions than recognition of virtual polycyclicity.

Score basis: Extracting algebraic structure from coarse geometry requires global rigidity machinery and broad solvable-group context.

[Read manuscript](#) | OPDP ID 40376940 | [Family formalization notes](#)

OAI-M0504 | Family 256

Nonsingular systems of equations over arbitrary groups

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that every finite nonsingular system of equations over an arbitrary group has a simultaneous solution in an overgroup. Nonsingularity means the exponent-sum matrix has full row rank over the rationals. Equivalently, adjoining variables and the equation relators does not collapse the original group inside the resulting quotient. The statement imposes no torsion or finiteness hypothesis on the coefficient group and addresses Howie's conjecture for systems, rather than only one equation.

Archive: related target. The candidate uses the word independent; its excerpt omits the definition needed to identify that condition with rational full row rank.

Score basis: Universal injectivity for groups with arbitrary coefficients calls for a substantial argument beyond checking an exponent matrix.

[Read manuscript](#) | OPDP ID 40347844 | [Family formalization notes](#)

OAI-M0505 | Family 256

The Kervaire theorem for groups

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that the free product of any nontrivial group with an infinite cyclic group cannot be normally generated by a single element. Its stronger unimodular-word theorem asserts that the original coefficient group injects into the one-relator quotient, without restrictions on torsion, cardinality, or presentation. Other exponent sums are handled through a cyclic quotient. This would settle the group-theoretic Kervaire conjecture while separating it from similarly named manifold-dimension questions.

Archive: related target. The normal-generation record is closely related, but its cyclic factor is corrupted in the excerpt; the exact source statement remains necessary.

Score basis: The coefficient group is completely unrestricted, so a uniform quotient-injectivity argument carries the main difficulty.

[Read manuscript](#) | OPDP ID 40333437 | [Family formalization notes](#)

OAI-M0506 | Family 257

A hyperbolic group with no geometric CAT(0) action

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a word-hyperbolic group with a finite classifying space that has no geometric action on a proper complete CAT(0) space. The accompanying two-dimensional aspherical complex has a linear combinatorial disk-filling inequality, yet no finite locally CAT(0) or locally CAT(-1) homotopy model with the stated metric conventions. This distinguishes large-scale negative curvature from the existence of a finite nonpositively curved model, with properness and completeness part of the obstruction.

Archive: related target. The archive asks for geometric CAT(-1) actions; the reported CAT(0) obstruction applies under explicit proper-complete-space conventions.

Score basis: The example needs both strong combinatorial filling bounds and a geometric obstruction valid for every candidate model.

[Read manuscript](#) | OPDP ID 40399694 | [Family formalization notes](#)

OAI-M0507 | Family 258

Baumslag-Solitar-free one-relator groups are hyperbolic

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims Gersten's hyperbolicity conjecture for finitely generated one-relator groups. If such a group contains no subgroup isomorphic to any Baumslag-Solitar group with nonzero integer parameters, it is asserted to be word-hyperbolic. Both signs are allowed, and the exclusion includes the rank-two free abelian group. The conclusion converts absence of a specified family of subgroup obstructions into a global linear-filling property of the full one-relator presentation.

Archive: lexical candidate. The leading archived conjecture says every such group, but its excerpt omits the defining subgroup-exclusion hypothesis.

Score basis: A subgroup-exclusion criterion must imply global geometry across arbitrary one-relator presentations.

[Read manuscript](#)

OAI-M0508 | Family 258

Virtual compact specialness of hyperbolic one-relator groups

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that every word-hyperbolic one-relator group is virtually compact special. The conclusion provides a finite-index subgroup realized by a compact nonpositively curved cube complex locally mapping into a right-angled Artin Salvetti complex. Combining this with existing work yields a virtually free-by-cyclic description, with the free kernel allowed infinite rank. Compactness is an explicit part of the main assertion, stronger than merely finding an abstract embedding in a right-angled Artin group.

Archive: related target. The graph-of-groups candidate concerns an equivalence of separability and specialness conditions, rather than this universal one-relator theorem.

Score basis: Building a compact special cubical model requires coordinated finite-index and hyperplane control.

[Read manuscript](#) | OPDP ID 40348397

OAI-M0509 | Family 259

A group without fixed price

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a finitely generated group admitting two essentially free probability-measure-preserving actions with different costs. Its group is an amalgam involving a rank-100 free group and a direct product. A Bernoulli action is asserted to have cost strictly above one, while finite-height extensions have costs approaching one. Choosing a sufficiently large height gives a concrete cost separation for the same group, answering the general fixed-price question negatively.

Archive: same target. The archive directly asks whether cost depends on the particular free probability-measure-preserving action; two different costs answer that target.

Score basis: Comparing infima over measurable graphings needs nonlocal lower bounds and delicate measured-equivalence-relation analysis.

[Read manuscript](#) | OPDP ID 40396582

OAI-M0510 | Family 260

The spacetime Penrose inequality with charge and original-data rigidity

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a charged spacetime Penrose upper-area inequality for one-ended three-dimensional data under specified energy, decay, and trapping assumptions. With invariant mass m , total charge magnitude Q , and minimum-enclosing-area radius r , it asserts m at least Q and r at most m plus the square root of m squared minus Q squared. The familiar polynomial rearrangement needs r greater than Q . Rigidity identifies qualifying equality data with Reissner-Nordstrom slices.

Archive: partial or special case. The archive's general spacetime target does not specify these charged, one-ended, decay, and minimum-enclosing-area conventions; the scopes are not interchangeable.

Score basis: Nonmaximal charged geometric inequalities and original-data rigidity require broad PDE, geometric-analysis, and relativity machinery.

[Read manuscript](#) | OPDP ID 50000040 | [Family formalization notes](#)

OAI-M0511 | Family 260

A Charged Reduction of the Spacetime Penrose Inequality in Spatial Dimensions at Least Four

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims a purely electric charged Penrose upper-area bound in every spatial dimension at least four. It reduces to a neutral theorem within its stated strong-decay and trapping scope, allowing arbitrary interior topology and componentwise trapping signs. Enclosing area positivity and timelike ADM energy-momentum are asserted consequences. Equality, under additional connected future-horizon conditions, reconstructs a Reissner-Nordstrom-Tangherlini slice; the classification is restricted to slices with vanishing induced magnetic two-form.

Archive: related target. The retrieved Minkowski-surface inequality is a different geometric setting and does not supply an exact initial-data target.

Score basis: The reduction couples high-dimensional constraint geometry with careful decay, flux, area, and equality hypotheses.

[Read manuscript](#) | OPDP ID 40325143 | [Family formalization notes](#)

OAI-M0512 | Family 260

Spacetime Penrose inequalities: enclosing area, charge, and rigidity

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims several sharp spacetime Penrose inequalities relating invariant ADM mass to minimum enclosing area in three and four spatial dimensions. Neutral statements permit momentum, nonzero second fundamental form, disconnected trapping boundaries, and specified multiple-end conventions. Equality reconstructs Schwarzschild or Schwarzschild-Tangherlini exterior slices under additional horizon hypotheses. Charged three-dimensional results and a local anti-de Sitter perturbation result have separate assumptions. The collection's conclusions must be read theorem by theorem rather than merged into one unrestricted inequality.

Archive: related target. The surface-in-Minkowski candidate shares the Penrose name but differs from the manuscript's initial-data inequalities and rigidity statements.

Score basis: A large theorem portfolio creates unusually high context and verification burden across distinct geometric and asymptotic regimes.

[Read manuscript](#) | OPDP ID 40325143 | [Family formalization notes](#)

OAI-M0513 | Family 260

The Kerr-Newman Penrose Inequality for Axisymmetric Electrovacuum Extensors

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims the Kerr-Newman Penrose inequality for a specified axisymmetric electrovacuum exterior class, without maximality. It assumes one end, zero ADM momentum, a connected outermost area-minimizing future horizon, Coulomb electromagnetic asymptotics, and the physical-area branch. Angular momentum includes its electromagnetic contribution, and both electric and magnetic charges enter. On the strict area branch, equality is asserted to characterize admissible slices of subextremal dyonic Kerr-Newman spacetime, including smooth horizon attachment.

Archive: partial or special case. The candidate's full energy, asymptotic, momentum, and angular-momentum conventions are truncated; this is a restricted exterior theorem.

Score basis: Angular momentum, electromagnetic corrections, and equality rigidity create substantial geometric and analytic dependence on precise hypotheses.

[Read manuscript](#) | OPDP ID 40371885 | [Family formalization notes](#)

OAI-M0514 | Family 260

Electromagnetic tails and the Kerr-Newman Penrose inequality

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims counterexamples to a Kerr-Newman Penrose formulation using bare gravitational ADM angular momentum and electromagnetic fields with only inverse-square decay. Angularly varying leading tails produce violations even when total electric and magnetic charges vanish. The examples retain a connected outermost area-minimizing future horizon and satisfy the strict physical-area branch. Claimed equality examples also defeat the corresponding rigidity statement. These constructions do not address formulations with conserved total angular momentum or stronger Coulomb asymptotics.

Archive: related target. The archived excerpt does not fix the angular-momentum or electromagnetic-tail conventions, so these counterexamples cannot be declared a refutation of that record.

Score basis: Constructing constraint-satisfying counterexamples with controlled horizons requires delicate geometric perturbations and asymptotic bookkeeping.

[Read manuscript](#) | OPDP ID 40371885 | [Family formalization notes](#)

OAI-M0515 | Family 260

The nonmaximal anti-de Sitter Penrose Inequality and original-data rigidity

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a nonmaximal spacetime Penrose inequality for one-ended three-dimensional asymptotically anti-de Sitter initial-data exteriors. It requires the dominant energy condition, stated decay and integrability, a weakly future trapped boundary, and a future-timelike hyperbolic metric mass flux. Area means the infimum over full enclosing cuts. Under additional connected outermost horizon assumptions, equality identifies original data with Schwarzschild-anti-de Sitter slices; no evolution or auxiliary solvability assumption is asserted.

Archive: related target. The archived AdS instability question concerns nonlinear time evolution, whereas this paper concerns a stationary initial-data mass-area inequality.

Score basis: Nonmaximal asymptotically hyperbolic constraints and global equality reconstruction impose a long geometric-analysis horizon.

[Read manuscript](#) | OPDP ID 30003912 | [Family formalization notes](#)

OAI-M0516 | Family 260

The Penrose inequality for maximal asymptotically hyperbolic initial data

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a sharp Penrose inequality for maximal, three-dimensional asymptotically hyperbolic initial data with spherical conformal infinity. Under its energy, decay, integrability, and timelike mass assumptions, the invariant mass is bounded by the Schwarzschild-anti-de Sitter mass associated with minimum enclosing area. Disconnected future-trapped boundaries and arbitrary compact topology are allowed. The coefficient is asserted sharp through time-symmetric Schwarzschild-anti-de Sitter examples, while maximality distinguishes this theorem from a fully nonmaximal formulation.

Archive: related target. The archived asymptotically hyperbolic conjecture has a truncated scalar-curvature and boundary formulation; exact equivalence is unresolved.

Score basis: The maximal restriction reduces scope, but asymptotic mass and trapped-boundary analysis remain technically demanding.

[Read manuscript](#) | OPDP ID 40327531 | [Family formalization notes](#)

OAI-M0517 | Family 260

A local Penrose inequality for conformal perturbations of Schwarzschild-anti-de Sitter data

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims a local Penrose inequality for sufficiently small maximal vacuum conformal perturbations of positive-mass Schwarzschild-anti-de Sitter exterior data. The theorem fixes a decaying transverse-traceless seed and a qualifying solution branch. It uses the marginally trapped boundary's own area, without outermostness or area-minimization assumptions. For small positive parameters, radial seeds are asserted to give equality and nonradial seeds strict inequality, providing a perturbative result rather than a global arbitrary-data theorem.

Archive: related target. The candidate asks nondegeneracy of a linearized operator; this is related perturbative background, not the same inequality or equality conclusion.

Score basis: A local perturbative regime offers a route, but boundary and second-variation control require specialized PDE and geometric preparation.

[Read manuscript](#) | OPDP ID 40337436 | [Family formalization notes](#)

OAI-M0518 | Family 260

The spacetime Penrose inequality and enclosing area

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims the sharp spacetime Penrose inequality for smooth one-ended initial-data exteriors in every spatial dimension at least three. Its area is the infimum over full enclosing cuts in the original metric; energy, differentiated decay, trapping, and positive-area assumptions are specified. These imply timelike ADM energy-momentum. In dimensions three and four, weaker differentiability suffices and area positivity is also derived. The asymptotic, area, and end conventions delimit the claimed generality.

Archive: partial or special case. The archive's apparent-horizon formulation does not state the same enclosing-area, end, and decay conventions; full scope equivalence has not been established.

Score basis: All-dimensional nonmaximal initial-data inequalities require deep geometric analysis and careful singularity and decay control.

[Read manuscript](#) | OPDP ID 50000040 | [Family formalization notes](#)

OAI-M0519 | Family 260

Conformal flow and the Riemannian Penrose inequality with minimizing frontiers

D 825/1000 | p 0 | m 1 | Conditional points 0

Historical independent-proof burden estimate (C0)

This manuscript presents a claimed detailed conformal-flow proof of the numerical Riemannian Penrose inequality in every dimension at least three. It treats asymptotically flat exteriors with nonnegative scalar curvature and compact frontiers that are outer minimizing and locally perimeter minimizing; the metric extends smoothly through possible frontier singularities. Spin and connectedness are not required. The paper explicitly follows existing conformal-flow work and develops numerical conclusions under stronger local-minimization assumptions, so novelty requires separate assessment.

Archive: partial or special case. The Bray-Iga target concerns conformally flat data; the paper's frontier hypotheses require comparison, and its introduction acknowledges an existing all-dimensional result.

Score basis: Developing a proof within an existing flow framework is more tractable than an unrestricted new spacetime theorem, though singular-frontier analysis remains substantial.

Progress: The numerical theorem is already covered by acknowledged prior work. A new proof or exposition does not establish a new open-problem resolution; no new formalization or M2 recognition is asserted.

[Read manuscript](#) | OPDP ID 40320379 | [Family formalization notes](#)

OAI-M0520 | Family 260

Equality and rigidity in the spacetime Penrose inequality

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims original-data rigidity for equality in a spacetime Penrose inequality. Under specified outermostness and decay assumptions, equality is asserted to identify the metric and second fundamental form with a spacelike Schwarzschild-Tangherlini exterior slice attached smoothly to its horizon. The theorem covers all dimensions under strong decay and dimensions three and four under weaker decay. In dimension three, finite-component hypotheses additionally force one spherical horizon and exclude disconnected equality configurations.

Archive: related target. The Schwarzschild-surface inequality candidate is related background but does not state equality classification of arbitrary initial data.

Score basis: Equality requires recovering full initial data from sharp numerical estimates, adding substantial global geometric and PDE rigidity work.

[Read manuscript](#) | OPDP ID 40325144 | [Family formalization notes](#)

OAI-M0521 | Family 260

Boundary graph deformations for the spacetime Penrose inequality

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims graph and conformal deformations for trapped initial-data exteriors in three and four dimensions. The constructed metrics have nonnegative scalar curvature, negative inner mean curvature, lower bounds protecting every enclosing cut, and arbitrarily small ADM-energy increase. A separate maximal-vacuum construction in dimension four retains a decaying noncompact second fundamental form. These are proposed boundary routes to numerical Penrose bounds, with deformation guarantees providing the principal technical target.

Archive: related target. The archived Schwarzschild-surface inequality is related motivation rather than the same deformation-existence theorem.

Score basis: Controlling all enclosing cuts and asymptotic energy through simultaneous deformations requires substantial PDE and geometric bookkeeping.

[Read manuscript](#) | OPDP ID 40325144 | [Family formalization notes](#)

OAI-M0522 | Family 260

Area-controlled end replacement and the Bondi Penrose inequality in the CKS class

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims an area-controlled replacement of a three-dimensional Cha-Khuri-Sakovich hyperboloidal end by asymptotically flat ends. For strictly future-timelike initial-data charge, the replacements preserve the dominant energy condition and every fixed compact interior, lose asymptotically no enclosing area, and have ADM masses approaching invariant Bondi mass. A companion Penrose theorem then yields a sharp Bondi bound for possibly disconnected future-trapped boundaries. Schwarzschild exteriors are asserted to attain equality.

Archive: related target. The archived target is asymptotically flat; this paper's principal target replaces a hyperboloidal end and derives a Bondi-mass inequality.

Score basis: End surgery must preserve energy conditions, area control, and the limiting relation between two asymptotic masses.

[Read manuscript](#) | OPDP ID 50000040 | [Family formalization notes](#)

OAI-M0523 | Family 261

Absolutely Continuous Spectrum for Weak-Disorder Anderson Models in Dimensions at Least Three

D 712/1000 | p 1 | m 1 | Conditional points 506.944

Published original CFSD-1000 target difficulty

The manuscript claims purely absolutely continuous spectrum on a nonempty open energy interval for weak-disorder Anderson operators on the integer lattice in every fixed dimension at least three. Site potentials are independent and uniform, and disorder is sufficiently small but fixed during the spectral assertion. The energy interval may depend on dimension while remaining independent of disorder strength. This provides the extended-state energy range requested by the archive's Anderson-model problem, without claiming delocalization throughout the entire spectrum.

Archive: same target. The archive asks for purely absolutely continuous spectrum in some energy range at suitable disorder width; the reported theorem supplies that range in every requested dimension.

Score basis: The claimed energy interval answers the archived suitable-disorder extended-states question for the standard uniform-law Anderson model.

Scoring target: Prove a purely absolutely continuous energy range for the Anderson model in dimensions at least three at suitable disorder width.

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 7900001 | [Family formalization notes](#)

OAI-M0524 | Family 261

Pure-Point Spectrum for the Two-Dimensional Anderson Model at Every Positive Disorder

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims pure-point spectral type throughout the spectrum of the two-dimensional nearest-neighbor Anderson model for every fixed positive disorder strength. Potentials follow the independent uniform single-site law, and the probability-one event may depend on that strength. The assertion includes arbitrarily weak disorder and yields a complete orthonormal eigenbasis. It concerns spectral type; it does not, by itself, assert quantitative eigenfunction decay or dynamical localization bounds, or cover trimmed randomness models.

Archive: related target. The trimmed-Anderson candidate randomizes a different operator class; the standard fully random uniform-site theorem is not an exact match.

Score basis: Arbitrarily weak disorder across the full two-dimensional spectrum requires demanding multiscale and probabilistic spectral control.

[Read manuscript](#) | OPDP ID 40328913 | [Family formalization notes](#)

OAI-M0525 | Family 262

Equality cases in the sharp one-dimensional matrix Lieb-Thirring inequality

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a complete equality classification for sharp one-dimensional matrix Lieb-Thirring inequalities when the exponent lies strictly between one-half and three-halves. Potentials are measurable, Hermitian, nonnegative matrices with the required trace-integrability condition. Equality is asserted exactly for direct sums of scalar one-bound-state solitons and zero channels in one constant unitary basis, with independent centers and scales. This describes extremizers for every finite matrix size, extending beyond merely identifying a scalar sharp constant.

Archive: related target. The archive's scalar one-bound-state conjecture concerns optimal constants; full matrix equality cases form a distinct stronger classification target.

Score basis: Equality classification must rule out spatially varying channel mixing while handling rough matrix potentials.

[Read manuscript](#) | OPDP ID 40394063 | [Family formalization notes](#)

OAI-M0526 | Family 262

Sharp one-dimensional Lieb-Thirring inequalities for matrix potentials

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims sharp one-dimensional Lieb-Thirring bounds for every finite matrix size and every exponent strictly between one-half and three-halves. The best constant is asserted to equal the scalar one-bound-state constant independently of matrix dimension. The result includes all negative eigenvalues and measurable nonnegative Hermitian potentials with the stated trace integrability, allowing varying rank and noncommuting values at different positions. It therefore controls genuine coupling between internal channels, beyond diagonal scalar reductions.

Archive: related target. The candidate asks whether optimizers exist when constants are unknown; this manuscript establishes a specific matrix inequality and optimal constant.

Score basis: Dimension-independent sharpness for noncommuting channels requires a structural argument rather than separate scalar estimates.

[Read manuscript](#) | OPDP ID 40395571 | [Family formalization notes](#)

OAI-M0527 | Family 262

Sharp one-dimensional Lieb-Thirring constants

D 711/1000 | p 1 | m 1 | Conditional points 505.521

Published original CFSD-1000 target difficulty

This manuscript claims the optimal scalar one-dimensional Lieb-Thirring constants for every exponent strictly between one-half and three-halves. The proposed constants are those predicted by a one-bound-state potential, with explicit soliton extremizers. The estimate includes rough unbounded nonnegative potentials of the required integrability and the entire negative-eigenvalue moment sum, even with infinitely many eigenvalues. This directly answers the archive's parameter-range question rather than proving only an endpoint or a nonsharp upper bound.

Archive: same target. The archive explicitly requests optimal one-dimensional constants for the same open exponent interval; the claimed theorem determines them throughout that interval.

Score basis: The exponent interval and sharp-constant target match the published original archive problem.

Scoring target: Determine the optimal scalar one-dimensional Lieb-Thirring constants for exponents strictly between one-half and three-halves.

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 7900015 | [Family formalization notes](#)

OAI-M0528 | Family 263

Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a universal excess-charge bound for full nonrelativistic Coulomb molecules with two electron spin states. With M fixed nuclei and total charge Z , strict binding is asserted to permit at most Z plus a universal constant times M electrons, uniformly over distinct nuclear positions and real nuclear charges at least one. For neutral integer-charge atoms, it also gives universal upper and lower bounds on first ionization energy and a specified outer-electron radius.

Archive: partial or special case. The archive asks the maximal excess charge; a universal constant bound does not determine the exact optimal excess charge.

Score basis: Many-body Coulomb control must be uniform in particle number, nuclear charges, and arbitrarily arranged nuclei.

[Read manuscript](#) | OPDP ID 30004466 | [Family formalization notes](#)

OAI-M0529 | Family 263

Generalized ionization energies for full Coulomb atoms

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims the energy component of the generalized ionization conjecture for full nonrelativistic Coulomb atoms with two spin states. Removing m electrons costs asymptotically the Thomas-Fermi constant times m to the seven-thirds power when m diverges and nuclear charge divided by m also diverges. The paper further asserts matching upper and lower iterated limits. It explicitly does not require convergence in nuclear charge for each fixed m , distinguishing two limiting formulations.

Archive: related target. The retrieved ionization-conjecture excerpt does not expose its complete bundle of charge, energy, and radius statements.

Score basis: A sharp many-body asymptotic across two interacting large parameters requires long semiclassical and correlation estimates.

[Read manuscript](#) | OPDP ID 40355206 | [Family formalization notes](#)

OAI-M0530 | Family 263

Generalized outer-electron radii of neutral Coulomb atoms

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims the radius component of generalized ionization for neutral Coulomb atoms with two electron spin states. For a radius enclosing all but an expected exterior mass of m electrons, both upper and lower large-nuclear-charge limits are asserted to scale as m to the minus one-third power with the explicit Thomas-Fermi constant. The conclusion holds for every choice of ground states, without assuming radial densities or a unique fixed- m limit.

Archive: unresolved no match. Outer-billiard atoms and algebraic Coulomb branches are unrelated lexical candidates; no appropriate archive identity was established.

Score basis: Uniform outer-density control over all ground states requires substantial many-body estimates and careful iterated limits.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0531 | Family 264

Generic Future Inextendibility with Square-Integrable Connection Near a Fixed Kerr Spacetime

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims generic future inextendibility near each fixed rotating subextremal Kerr bridge at continuous-metric, locally square-integrable-connection regularity. Nearby smooth two-ended vacuum initial data admitting such an extension of their full maximal globally hyperbolic development form a meagre set in a weighted smooth topology. Only one specified finite-order seminorm needs to be small, while genericity tests all orders. The extension metric is required continuous and nondegenerate, so the precise regularity threshold matters.

Archive: related target. The candidate asks for polynomial tails of generic Kerr perturbations; inextendibility is a related consequence-type question with a different conclusion.

Score basis: Generic nonlinear Einstein evolution and weak-regularity singularity formation have exceptionally long analytic and geometric dependencies.

[Read manuscript](#) | OPDP ID 40325783

OAI-M0532 | Family 264

Generic C1 Future Inextendibility Near Rotating Subextremal Kerr Spacetimes

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims that a dense countable intersection of open sets of vacuum data near each fixed rotating subextremal Kerr bridge develops no future continuously differentiable extension. Initial data are smooth, complete, and two-ended, with nearness and genericity measured in a weighted smooth topology. Proposed extensions may be nonvacuum. The theorem addresses generic future singularity at a specific extension regularity, rather than proving exterior stability or forbidding every continuous extension.

Archive: related target. The archived Kerr stability conjecture asks control of nearby developments, rather than generic future inextendibility at differentiability class one.

Score basis: Nonlinear global developments and generic extension obstructions require substantial Einstein-equation machinery.

[Read manuscript](#) | OPDP ID 40373827

OAI-M0533 | Family 264

Quantitative Near-Kerr Evolution and Generic C2 Future Inextendibility

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims generic failure of twice continuously differentiable future extensions for vacuum developments near every fixed rotating subextremal Kerr bridge. The extendible data form a meagre set in a neighborhood requiring smallness of one finite-order weighted seminorm. All higher orders remain part of the topology, with arbitrary symbol-bounded asymptotically flat tails and no symmetry imposed. This is a precise differentiability-threshold statement about the full maximal development, not a claim of instability for all initial data.

Archive: related target. The polynomial-tail candidate concerns asymptotic decay; its excerpt does not identify this full-development extension question.

Score basis: Quantitative Kerr perturbation estimates and generic curvature obstructions impose a long PDE reasoning horizon.

[Read manuscript](#) | OPDP ID 40325783

OAI-M0534 | Family 265

A two-dimensional area law from a global spectral gap

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a two-dimensional entanglement-entropy area law from a uniform gap of the full Hamiltonian alone. It treats unique ground states of finite-range Hamiltonians on arbitrary finite induced square-lattice domains, with fixed local dimension, range, and interaction-strength bounds. Entropy of every subset is bounded by a uniform constant times crossing boundary edges, independently of domain size and shape. No separate gap assumption for subsystems is part of the stated target.

Archive: related target. The AKLT candidate asks for a hexagonal-lattice spectral gap; this paper assumes a gap on square-lattice domains and derives a different property.

Score basis: Turning one global spectral gap into uniform entanglement bounds requires nonlocal many-body control.

[Read manuscript](#) | OPDP ID 40400500

OAI-M0535 | Family 265

Polynomial PEPS approximation of gapped square-grid ground states

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims polynomial-bond-dimension projected entangled-pair approximations for unique ground states of uniformly gapped nearest-neighbor Hamiltonians on square grids. After normalization, the approximation has global vector error at most the inverse lattice side length. Constants are uniform with fixed local dimension, interaction strength, and full-Hamiltonian gap. The assertion is an existence theorem, not an efficient algorithm for constructing the tensors or a guarantee derived from gaps on smaller subsystems.

Archive: related target. Approximate Haag duality is a distinct property of gapped ground states, with no equivalence supplied by the retrieved excerpt.

Score basis: Global approximation error and polynomial bond dimension require rigorous entanglement compression across a two-dimensional lattice.

[Read manuscript](#) | OPDP ID 40405163

OAI-M0536 | Family 266

The maximum number of mutually unbiased bases in dimension six

D 598/1000 | p 1 | m 1 | Conditional points 357.604

Published original CFSD-1000 target difficulty

The manuscript claims that precisely three mutually unbiased orthonormal bases can exist in complex dimension six. Its upper bound is computer-assisted: a documented complete pipeline excludes four arbitrary complex bases under specified binary64 arithmetic and compiler conditions. Since constructions of three bases are available, that exclusion determines the maximum. The stronger no-four conclusion also implies the archive's requested no-seven alternative, but checking the computational certificate and its arithmetic assumptions remains a separate validation task.

Archive: same target. The archive permits either constructing at least four bases or proving no seven; exclusion of four supplies its negative branch and a stronger conclusion.

Score basis: The claimed no-four result implies the no-seven branch of the published original target; its original score is preserved.

Scoring target: Construct at least four mutually unbiased bases in dimension six or prove that seven cannot exist.

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 8200002 | [Family formalization notes](#)

OAI-M0537 | Family 266

Exact Fourier certificates for complex Hadamard matrices of order six

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims an exact Fourier-vanishing property for every order-six complex Hadamard matrix outside Tao's cubic equivalence class. Character sums vanish at each permutation of three positive and three negative unit exponents. A separate exact rational-and-integer certificate excludes seven mutually unbiased bases in dimension six; a completion theorem then lowers the upper bound to five. The paper includes a verifier, so its two certificates have different logical targets and neither classifies all Hadamard matrices.

Archive: related target. The candidate requests complete classification of order-six Hadamard matrices; a Fourier identity does not by itself supply that classification.

Score basis: Exact certificates offer substantial tool feedback, but continuous matrix families and verifier coverage require careful structural validation.

[Read manuscript](#) | OPDP ID 40383293 | [Family formalization notes](#)

OAI-M0538 | Family 267

Bose-Einstein condensation at positive temperature in the dilute hard-sphere gas

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims positive-temperature Bose-Einstein condensation for the exact three-dimensional dilute hard-sphere gas. At every sufficiently small fixed density and fixed exclusion distance, some strictly positive temperature independent of volume gives a positive constant-orbital condensate fraction in the canonical thermodynamic limit. The statement fixes density and temperature before taking volume to infinity. It establishes existence of a condensed positive-temperature regime, without locating a critical temperature or replacing the interacting Gibbs state by an approximation.

Archive: related target. The hard-sphere candidate asks a zero-temperature ground-energy correction, a different observable and temperature regime.

Score basis: Controlling macroscopic coherence in the exact interacting thermal state requires demanding thermodynamic-limit many-body analysis.

[Read manuscript](#) | OPDP ID 40404988 | [Family formalization notes](#)

OAI-M0539 | Family 267

Quantum Depletion and Momentum Distribution in the Dilute Hard-Sphere Bose Gas

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims the full scaled Bogoliubov momentum distribution of depleted particles in the zero-temperature hard-sphere Bose gas. The thermodynamic limit is taken first at fixed density, followed by the dilute limit, uniformly over pure and mixed ground states. Its total depleted fraction has leading coefficient eight divided by three times the square root of pi, multiplying the square root of density times scattering-length cubed. All thermodynamic accumulation values share that asymptotic.

Archive: related target. The archived target concerns a ground-energy expansion; momentum distributions and depletion are distinct conclusions for the same gas.

Score basis: Momentum-resolved many-body asymptotics require more information than a scalar energy estimate and careful control of limit order.

[Read manuscript](#) | OPDP ID 40404988 | [Family formalization notes](#)

OAI-M0540 | Family 267

Quantum Depletion for Fixed Bounded Repulsive Potentials

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims the Bogoliubov depletion asymptotic for a three-dimensional Bose gas with a fixed bounded, nonnegative radial interaction of finite range and positive scattering length. Volume tends to infinity at fixed density before the dilute limit. The depleted fraction has the usual square-root-density leading coefficient uniformly over all ground-state density matrices and thermodynamic accumulation values. No unique occupation limit at each density is required, and the potential is held fixed rather than rescaled.

Archive: unresolved no match. The retrieved repulsive-potential smoothing and quantum-transport-metric questions concern different operators and observables.

Score basis: Uniform depletion estimates for exact interacting states need substantial analysis even with bounded repulsive potentials.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0541 | Family 267

A density-uniform condensate bound for dilute Bose gases

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a positive constant-orbital condensate fraction for fixed bounded nonnegative radial finite-range interactions in three dimensions. The lower bound is uniform over sufficiently small densities and temperatures between zero and density squared. Density and interaction remain fixed while volume tends to infinity, with the volume threshold permitted to depend on them. This establishes a uniform condensation regime rather than a ground-energy expansion.

Archive: related target. The Lee-Huang-Yang candidate concerns ground-energy asymptotics, not a condensate-fraction lower bound.

Score basis: Density-uniform coherence at positive temperatures demands control of interacting Gibbs states and thermodynamic limits.

[Read manuscript](#) | OPDP ID 40403919 | [Family formalization notes](#)

OAI-M0542 | Family 267

Ground-state condensation in the dilute hard-sphere gas

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims Bose-Einstein condensation in every ground state of the three-dimensional dilute hard-sphere gas. At each sufficiently small fixed gas parameter, the constant orbital retains a positive thermodynamic fraction, with a common lower bound independent of that parameter. Arbitrary complex ground eigenvectors are included. The assertion concerns occupation uniformly over ground-state choices, beyond estimates for the energy alone.

Archive: related target. The archived hard-sphere question asks a second-order energy bound, a different observable for the same model.

Score basis: Uniform condensation over all ground states requires strong many-body density control beyond an energy upper bound.

[Read manuscript](#) | OPDP ID 40404988 | [Family formalization notes](#)

OAI-M0543 | Family 268

The periodic spin-one Haldane gap

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a uniform positive spectral gap for the pure spin-one antiferromagnetic Heisenberg chain on even periodic rings. It supplies explicit positive bounds for sufficiently large rings and asserts uniqueness of the ground state. This establishes the positive even-periodic spin-one formulation of the Haldane conjecture. It does not establish the full integer-spin versus half-integer-spin dichotomy across all spins or boundary conditions.

Archive: partial or special case. The archive's Haldane target asks the full integer-versus-half-integer dichotomy; the reported theorem addresses spin one on even periodic rings.

Score basis: A uniform thermodynamic gap for the pure interacting chain is difficult, but the spin and boundary regime are fixed.

[Read manuscript](#) | OPDP ID 50000111

OAI-M0544 | Family 268

A boundary-field gap for the spin-one Heisenberg chain

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a uniform gap for odd open spin-one antiferromagnetic Heisenberg chains with equal endpoint magnetic fields of strength three-fifths. Chains with at least the stated size have gap greater than log ten divided by 392. Applying Tasaki's index theorem then gives index minus one for subsequential local limits of the selected ground states. The boundary-field selection is an essential part of this statement.

Archive: unresolved no match. The candidates concern different spin-chain spectra or Bethe-root classifications, not this endpoint-field gap.

Score basis: Boundary selection supplies a tractable route, while uniform finite-volume gap control remains technically demanding.

[Read manuscript](#)

OAI-M0545 | Family 269

Uniform Stability of the Spherical Laughlin Gap

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims stability of the spherical fermionic Laughlin gap under weak bounded real scalar one-body potentials projected into the lowest Landau level. At filling one-third and flux three times particle number minus one, the ground state stays unique and uniformly gapped for all sufficiently large systems. Both gap and perturbation threshold are uniform over allowed potential profiles. The proof uses a companion unperturbed gap theorem.

Archive: unresolved no match. Homological stability and Plateau-cluster stability are lexical collisions, not Laughlin gap targets.

Score basis: Uniform perturbative stability over all particle numbers and profiles adds substantial control beyond the unperturbed spectral bound.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0546 | Family 269

A Fock-space inequality and the Laughlin spectral gap

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a positive uniform gap for the spherical fermionic Laughlin interaction at filling one-third, exceeding one twenty-fifth for sufficiently large systems. Its stronger operator inequality holds on the full lowest-Landau-level Fock space independently of particle number at large flux. In sectors without zero modes, that inequality bounds energy away from zero; it does not automatically separate their two lowest eigenvalues.

Archive: unresolved no match. Aldous and AKLT spectral-gap candidates concern distinct stochastic or spin Hamiltonians and do not identify the Laughlin target.

Score basis: Uniform Fock-space positivity needs a strong many-body operator inequality and careful sector interpretation.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0547 | Family 270

The unique threshold bound state of the $SU(N)$ BFSS model

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims exactly one normalizable zero-energy state for undeformed special-unitary BFSS matrix quantum mechanics at every finite matrix size at least two. The center-of-mass degrees of freedom are removed, and square integrability is imposed on the entire relative configuration space. The state is asserted spin-rotation invariant and even in fermion grading. This addresses uniqueness and existence of a threshold bound state, beyond a formal supersymmetric index calculation.

Archive: unresolved no match. The leading threshold and self-similar-state candidates concern unrelated mathematical models.

Score basis: Threshold states in noncompact supersymmetric quantum mechanics require difficult global spectral and decay analysis.

[Read manuscript](#)

OAI-M0548 | Family 270

Positive eigenvalues of the relative $SU(2)$ BFSS Hamiltonian

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims infinitely many normalizable positive-energy eigenstates of the relative special-unitary size-two BFSS Hamiltonian, with eigenvalues tending to infinity. The operator is specified through closure of its gauge-invariant supercharge form. This would contradict the original exclusion of normalizable positive-energy states at that matrix size. The claim concerns embedded point spectrum in a model with nonconfining directions, not a replacement for its zero-energy uniqueness theorem.

Archive: unresolved no match. Affine Gaudin and semiclassical eigenvalue candidates are different Hamiltonians and do not establish a match.

Score basis: Finding positive-energy eigenstates despite open nonconfining directions requires delicate operator-domain and spectral arguments.

[Read manuscript](#)

OAI-M0549 | Family 271

Bloch's Law for Finite-Range Heisenberg Ferromagnets in Three Dimensions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims Bloch's temperature-to-the-three-halves law for spontaneous magnetization in three-dimensional finite-range quantum Heisenberg ferromagnets at every fixed positive spin. Couplings are nonnegative, symmetric, and generate the full integer lattice; anisotropy is allowed. The leading coefficient depends on the quadratic magnon-dispersion determinant. Volume goes to infinity before the right zero-field derivative, with low temperature taken last. This supplies a sharp asymptotic in a particular ordered regime.

Archive: partial or special case. The archived bundle includes other dimensions and a robust nontranslation-invariant proof; this theorem treats a three-dimensional translation-invariant coupling class.

Score basis: An exact thermal asymptotic requires controlling interactions beyond an ideal spin-wave approximation.

[Read manuscript](#) | OPDP ID 7800004 | [Family formalization notes](#)

OAI-M0550 | Family 271

The first lattice correction to Bloch's law

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims the first lattice correction to Bloch's law for the nearest-neighbor three-dimensional quantum Heisenberg ferromagnet at each fixed positive spin. The magnetization deficit matches the full ideal-magnon density up to an error smaller than inverse temperature to the five-halves power. It identifies both leading and correction coefficients, with the thermodynamic limit preceding the zero-field derivative. The target is finer than existence of magnetic order.

Archive: unresolved no match. Bloch-Kato conjectures are unrelated namesakes in arithmetic geometry, not quantum magnetization targets.

Score basis: Resolving the next asymptotic coefficient requires interaction-error bounds at a finer scale than the leading law.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0551 | Family 271

The spherical magnetization law for the three-dimensional quantum Heisenberg ferromagnet

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a spherical magnetization law for the three-dimensional nearest-neighbor isotropic quantum Heisenberg ferromagnet at sufficiently low fixed positive temperature. On growing even periodic cubes, zero-field magnetization moments approach a uniform direction with deterministic positive magnitude, equal to the pressure's right zero-field derivative. Self-adjoint linear combinations formulate the law without a joint measurement of noncommuting components. It describes the symmetry-restored distribution rather than only one selected magnetized state.

Archive: related target. The candidate concerns a two-dimensional free-energy lower bound, differing in dimension and observable.

Score basis: Moment convergence and identification of macroscopic magnetization require strong thermodynamic and noncommutative probabilistic control.

[Read manuscript](#) | OPDP ID 30004243 | [Family formalization notes](#)

OAI-M0552 | Family 271

Spontaneous magnetization in the quantum Heisenberg ferromagnet

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims spontaneously magnetized equilibrium states for the nearest-neighbor isotropic quantum Heisenberg ferromagnet in every dimension at least three and every positive quantum spin. At sufficiently low positive temperature, a translation-invariant zero-field KMS state has magnetization at least one-quarter of the spin. The theorem constructs an ordered state; averaging rotations can remove its magnetization. It does not supply the archive's two-dimensional ground-state or translation-robust extensions.

Archive: partial or special case. The result addresses part A of the archived three-part long-range-order target; parts B and C have different regimes and assumptions.

Score basis: Existence of ordered exact equilibrium states demands global interacting thermal control, even with a fixed lattice model.

[Read manuscript](#) | OPDP ID 7800004 | [Family formalization notes](#)

OAI-M0553 | Family 272

Entanglement with zero distillable secret key in local dimension ten

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit entangled state in local dimension ten with zero distillable secret key for its specified local-instrument protocols. Those protocols allow joint processing of copies and unlimited two-way public communication, while an eavesdropper retains purification and transcript. Separate channel constructions address PPT composition and entanglement breaking. Identifying the protocol definition with the standard unrestricted key-distillation target is essential before calling the example a complete counterexample to the archived question.

Archive: partial or special case. The archive asks about all bipartite entangled states under standard asymptotic key distillation; the manuscript's detailed protocol class must be checked for exact correspondence.

Score basis: An obstruction to all permitted distillation protocols needs rigorous resource and adversary bookkeeping.

[Read manuscript](#) | OPDP ID 50000274 | [Family formalization notes](#)

OAI-M0554 | Family 273

The entropy photon-number inequality

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims the entropy photon-number inequality for independent bosonic inputs with finite mean energy and any finite mode count. Modes within either input may be entangled. Consequences include exact minimum output entropy at prescribed input entropy for thermal attenuator tensor powers and an exact degraded pure-loss broadcast-channel capacity region. Independence between the two inputs, rather than between their individual modes, defines the main inequality's scope.

Archive: related target. The thinned-entropy candidate concerns discrete classical ultra-log-concave variables, a related but different inequality.

Score basis: Multimode bosonic entropy control must accommodate internal entanglement and unbounded Hilbert spaces.

[Read manuscript](#) | OPDP ID 40318439 | [Family formalization notes](#)

OAI-M0555 | Family 274

Product-projection localization and the QAC0 parity lower bound

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a parity lower bound for constant-depth quantum circuits with arbitrary one-qubit unitaries and unbounded-arity Toffoli gates. Polynomially many total qubits cannot achieve any fixed positive worst-case success advantage over guessing. Ancillas start in zero, a single output qubit is measured, and other registers may be discarded without restriction. This addresses Moore's measured-output conjecture, without requiring clean final ancillas or exact computation.

Archive: related target. The archive's depth-four Boolean-circuit lower bound uses a classical model rather than this quantum gate set.

Score basis: Quantum interference and arbitrary garbage prevent direct use of classical switching-lemma bounds.

[Read manuscript](#) | [OPDP ID 40404436](#) | [Family formalization notes](#)

OAI-M0556 | Family 274

Regular trajectories, pruning and quantum parity

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This alternative paper claims the same measured-output parity obstruction through regular trajectories and pruning. For each fixed depth and polynomial qubit bound, sufficiently large input size has some input whose success probability is below one-half plus any fixed positive advantage. Arbitrary one-qubit gates and unbounded Toffoli gates are allowed, zero ancillas are required initially, and final garbage is unrestricted. The manuscript supplies an alternative argument rather than a new gate-model target.

Archive: unresolved no match. Quantum cohomology and Rabi-model parity candidates are unrelated uses of parity, not circuit lower bounds.

Score basis: A trajectory-based lower bound still requires uniform cancellation control against every allowed quantum circuit.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0557 | Family 275

Continuum Coulomb hardness with binary nuclear charges

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims QMA-hardness of approximating a continuum electronic Coulomb spectral infimum when positive nuclear charges are binary encoded. Nuclei occupy distinct rational positions, electron number is unary, and minimization includes all antisymmetric spinful continuum states. The classical deterministic reduction has polynomial bit length and energy thresholds separated by at least one, while charges may be exponentially large. Those encoding choices are central to the claimed hardness.

Archive: unresolved no match. Binary-LWE and quadratic Diophantine hardness candidates are different promise problems and do not establish a target match.

Score basis: A continuum Hamiltonian reduction requires rigorous gadget and scale control, while binary encoding supplies large physical parameters.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0558 | Family 275

QMA-hardness of continuum Coulomb energy with unit nuclear charges

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims QMA-hardness for continuum Coulomb energy with every clamped nuclear charge equal to one. Its reduction produces polynomially many nuclei and electrons, distinct rational positions, polynomial bit lengths, and threshold separation at least one. No orbital basis, magnetic field, supplementary external potential, or binding assumption is supplied. Thus the proposed result removes the exponentially large-charge resource used in a related hardness construction.

Archive: unresolved no match. The Coulomb-gas curve and continuum-mechanics candidates have different energies and computational targets.

Score basis: Unit-charge restriction requires stronger geometric simulation while maintaining control over unrestricted continuum states.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0559 | Family 276

Classical capacity and entropy inequalities for generalized amplitude damping

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims exact unassisted classical capacities for every qubit generalized amplitude-damping channel, across damping strengths and thermal occupations. Capacity equals a one-use Holevo optimum expressible as a one-variable maximum, attained by a binary pure-state ensemble and its independent products. It additionally asserts additivity of Holevo capacity, minimum output entropy, and regularized capacity with every finite-dimensional partner channel. The latter quantifiers substantially strengthen a calculation for isolated channels.

Archive: related target. The candidate concerns complete contraction coefficients, a different channel functional from capacity and minimum output entropy.

Score basis: Universal partner-channel additivity requires entropy inequalities beyond numerical optimization of a qubit channel.

[Read manuscript](#) | OPDP ID 40364676 | [Family formalization notes](#)

OAI-M0560 | Family 277

Threshold parallel repetition for finite-dimensional entangled games

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims threshold parallel repetition for every finite two-player entangled game of value below one. Winning above value plus a fixed positive margin in independent repetitions has exponentially small probability, uniformly over all finite-dimensional joint strategies and correlated question distributions. The universal exponent scales with the fifth power of the margin divided by answer-alphabet logarithms; fixed question distributions also receive a cubic-rate bound. All-win exponential decay follows as a special threshold case.

Archive: same target. Choosing a threshold below one makes winning all repetitions a subset of the bounded event, answering the archived all-win exponential-decay question.

Score basis: Entangled joint strategies require a uniform information-theoretic lower-tail argument rather than independence of outcomes.

[Read manuscript](#) | OPDP ID 40315470 | [Family formalization notes](#)

OAI-M0561 | Family 278

A Coulomb ground-state density without Kohn-Sham ensemble representation

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a three-electron Coulomb molecule whose interacting ground-state density has no noninteracting ground-state ensemble representation in any single real spin-independent local potential of the specified integrability class. Two equal positive integer nuclear charges are fixed by an exact formula. This challenges a density-representability assumption underlying Kohn-Sham constructions, without contradicting uniqueness of an interacting density's external potential.

Archive: related target. The Hohenberg-Kohn candidate concerns density-potential uniqueness, whereas this paper concerns existence of a noninteracting ensemble representation.

Score basis: The counterexample needs rigorous density obstruction together with exact Coulomb spectral and ensemble control.

[Read manuscript](#) | OPDP ID 40334356

OAI-M0562 | Family 279

Exact quantum factoring over a fixed finite gate set

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims exact quantum prime factorization with probability one, polynomial worst-case gate and qubit bounds, and a fixed finite bounded-arity gate set. The circuit family is generated uniformly in polynomial time from the integer input length. The target strengthens bounded-error quantum factoring through exact success and finite gates; it does not yield deterministic classical polynomial-time factorization, even on a positive-density subset.

Archive: related target. The archived positive-density target asks deterministic classical complexity; a quantum circuit theorem does not resolve it.

Score basis: Exact success with a fixed finite gate set imposes arithmetic and uniformity constraints beyond standard bounded-error factoring.

[Read manuscript](#) | OPDP ID 8400006 | [Family formalization notes](#)

OAI-M0563 | Family 280

Strongly rational unitary vertex operator algebras and conformal nets

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every simple unitary strongly rational vertex operator algebra generates a completely rational conformal net. Its modules are unitarizable, fusion forms positive, and the stated functor gives a braided unitary tensor equivalence with finite-index sectors. Extension theorems identify associated local extensions. Strong rationality is essential here; the claim does not cover all unitary algebras or reconstruct every conformal net.

Archive: partial or special case. The archive asks strong locality for every unitary vertex operator algebra and a reverse construction for every irreducible net; this theorem treats a restricted forward class.

Score basis: Bridging algebraic and operator-theoretic conformal structures requires extensive tensor-category, locality, and analytic preparation.

[Read manuscript](#) | OPDP ID 30002836 | [Family formalization notes](#)

OAI-M0564 | Family 281

QAOA attains the SK ground-state energy in the thermodynamic-first limit

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims eventual Parisi-optimality of QAOA for the Gaussian zero-field Sherrington-Kirkpatrick model when system size tends to infinity before circuit depth increases. For each accuracy, finite depth and deterministic disorder-independent angles achieve the limiting expected ground-energy density. No depth bound or efficient angle-selection algorithm is supplied. The conclusion therefore establishes an asymptotic existence result rather than a uniform efficient optimization procedure.

Archive: unresolved no match. Crystal bulk limits and guided-state energy estimation do not identify the same SK QAOA optimization target.

Score basis: Exchanging physical approximations for a rigorous thermodynamic-first variational limit needs substantial spin-glass and quantum-circuit analysis.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0565 | Family 281

Full support of the zero-temperature Sherrington-Kirkpatrick order parameter

D 825/1000 | p 0 | m 1 | Conditional points 0

Historical independent-proof burden estimate (C0)

The manuscript gives an alternative claimed proof that every admissible zero-temperature Parisi minimizer for the pure zero-field Sherrington-Kirkpatrick model has full Stieltjes support below overlap one. Its argument excludes gaps using a coercive identity and a strict variation. The introduction explicitly attributes the same conclusion, together with stronger smoothness, to Chen. Thus this manuscript's full-support theorem is a reconstruction burden, not a newly resolved open support question.

Archive: related target. The archived excerpt already labels the related structure as a theorem; the source explicitly acknowledges Chen's prior full-support result.

Score basis: The score estimates reproducing a demanding variational argument within an acknowledged known-result framework.

Progress: The source acknowledges the claimed full-support conclusion in prior work. The scored burden concerns an alternative proof; no new formalization or accepted M2 recognition is asserted.

[Read manuscript](#) | OPDP ID 40315484 | [Family formalization notes](#)

OAI-M0566 | Family 282

Scale and conformal symmetry in four-dimensional operational quantum field theory

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that scale symmetry implies local conformal symmetry for a specified four-dimensional unitary positive-energy quantum-field-theory class. Assumptions include discrete scaling dimensions, finite field scaling support, compatible local-net realizations, and a physical dilatation current with Ward identities. A symmetric conserved traceless stress-tensor improvement preserves spacetime charges. Global integration to a conformal action on the entire bounded net is explicitly left separate.

Archive: unresolved no match. Constructing interacting four-dimensional theories and lower-dimensional correspondence questions are different targets.

Score basis: Operational field reconstruction and stress-tensor improvement require substantial analytic context and delicate locality hypotheses.

[Read manuscript](#)

OAI-M0567 | Family 283

Polynomial-Time Unitary Synthesis from a Boolean Oracle

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a polynomially generated quantum oracle circuit that approximates every n -qubit unitary channel to diamond-norm error at most one-half after an appropriate Boolean oracle is chosen. Gates, qubits, query count, and query length have polynomial bounds. The oracle depends on the target unitary, and no efficient classical construction of it is asserted. This answers a constant-error oracle-synthesis question, not ordinary gate-only synthesis.

Archive: unresolved no match. The polynomial-evaluation and physical-oracle candidates have different computational inputs and output goals.

Score basis: Oracle encoding and channel-norm approximation require a rigorous resource distinction and nontrivial synthesis design.

[Read manuscript](#)

OAI-M0568 | Family 284

A Nearly Quartic Separation Between Randomized and Quantum Query Complexity

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims total Boolean functions giving nearly quartic separation between bounded-error randomized and quantum query complexity. Its examples rule out every universal exponent smaller than four, matching a known quartic upper bound and disproving a proposed cubic bound. Both models measure worst-case bit queries with error at most one-third on every input. This is a query separation, not a two-party communication-complexity separation.

Archive: related target. The archive's target is two-party communication complexity; bit-query complexity does not resolve that question.

Score basis: Constructing total-function lower bounds while retaining small quantum query complexity requires sophisticated composition and adversary arguments.

[Read manuscript](#) | OPDP ID 50000228

OAI-M0569 | Family 285

A torsion-free counterexample to reduced Baum-Connes injectivity

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a finitely generated torsion-free group whose coefficient-free reduced Baum-Connes assembly map has an infinite-order kernel class in degree zero. A torus-derived K-homology class has nonzero integral pairing but vanishing reduced analytic image. The example would defeat rational injectivity, hence the archived isomorphism conjecture, without relying on arbitrary coefficient algebras or torsion in the constructed group.

Archive: same target. The archived torsion-free coefficient-free reduced assembly map isomorphism statement is directly refuted by a nonzero kernel class.

Score basis: Coordinating group construction, K-homology detection, and vanishing reduced assembly requires broad nonlocal operator K-theory machinery.

[Read manuscript](#) | OPDP ID 30000299

OAI-M0570 | Family 285

A torsion-free counterexample to the Kadison-Kaplansky projection conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a finitely generated torsion-free group whose reduced group C-star algebra contains a nontrivial projection. Its canonical trace lies strictly between zero and one-half, certifying that the projection is neither zero nor the identity. The group is constructed using a graphical presentation and a free product. This would refute the projection version of Kadison-Kaplansky, rather than the distinct group-ring zero-divisor statement.

Archive: same target. The candidate's displayed conjecture explicitly prohibits nontrivial reduced-algebra projections for torsion-free groups.

Score basis: The example must realize analytic projections in a reduced algebra while retaining torsion-free group structure.

[Read manuscript](#) | OPDP ID 40333798

OAI-M0571 | Family 285

An irrational-trace counterexample to reduced Baum-Connes

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a finitely generated discrete group with a projection of irrational canonical trace in its reduced group C-star algebra. The projection's K-zero class is asserted outside the coefficient-free reduced assembly image. The obstruction concerns surjectivity for countable discrete groups, not the torsion-free injectivity example in its companion. Using trivial coefficients would make this stronger than merely constructing a failure with specially chosen coefficient algebras.

Archive: related target. The broader locally compact Baum-Connes record has a truncated assembly formulation; the exact coefficient and topology conventions remain unconfirmed.

Score basis: An irrational trace must be rigorously constructed and separated from every assembly-image class.

[Read manuscript](#) | OPDP ID 40316140

OAI-M0572 | Family 286

Arithmeticity of twisted finite correspondences for lattices over local fields

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims arithmetic exhaustion of finite correspondences between twisted group factors associated with property-T lattices over characteristic-zero local fields and arbitrary countable ICC groups. Every correspondence is a summand of finitely many models built from finite-index subgroup isomorphisms and projective representations of cocycle ratios. Products, mixed places, and specified rank-one factors are included. This classifies operator-algebraic correspondences, not arithmeticity of all complex-hyperbolic lattices.

Archive: unresolved no match. The $PU(n,1)$ arithmeticity question concerns a different lattice class and geometric conclusion; no exact archive target was identified.

Score basis: Twisted correspondence classification combines deformation rigidity, lattice structure, projective representations, and extensive categorical bookkeeping.

[Read manuscript](#)

OAI-M0573 | Family 286

The arithmetic category and stable recovery of lattice factors

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims stable canonical recovery of groups and scalar cocycles from twisted factors in the stated property-T lattice class. A stable isomorphism to an arbitrary countable ICC-group factor forces equal amplification scales, a group isomorphism, and cocycle agreement up to cochain. It also computes arithmetic correspondence operations and finite-index neighbors. The conclusion is stronger than merely detecting an abstract unamplified factor isomorphism.

Archive: unresolved no match. Stable homotopy categories and Tachikawa module categories are unrelated uses of category and stability.

Score basis: Canonical recovery and full correspondence-category operations demand broad operator-algebra and representation-theoretic context.

[Read manuscript](#)

OAI-M0574 | Family 287

An isomorphism of the free group factors

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a trace-preserving normal star isomorphism between the free group factors of ranks two and three. Consecutive higher ranks are first connected by a freely generating tuple construction. The classical all-or-none alternative then identifies every interpolated free group factor, including infinite rank, and gives the full positive-real fundamental group. This answers the archived distinct-rank isomorphism question affirmatively rather than comparing their underlying discrete groups.

Archive: same target. The candidate explicitly asks whether free group von Neumann algebras of distinct ranks are isomorphic; the reported rank-two/rank-three result answers that via the stated classical alternative.

Score basis: Constructing isomorphisms between factors with subtle free-probability invariants requires exceptionally deep structural control.

[Read manuscript](#) | OPDP ID 40402658 | [Family formalization notes](#)

OAI-M0575 | Family 288

Kadison's similarity theorem through uniform derivation estimates

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that every bounded complex-linear unital homomorphism from a unital C-star algebra into bounded Hilbert-space operators is similar to a star homomorphism. Separability, nuclearity, faithfulness, normality, and finite generation are not assumed. A companion conclusion gives a universal hyperreflexivity constant for unital von Neumann algebras. The similarity assertion addresses Kadison's problem in its standard unital formulation, rather than only well-behaved algebra subclasses.

Archive: same target. The archive's similarity question is the standard representation target; the reported unital form is the customary equivalent reduction.

Score basis: Uniform derivation control across arbitrary C-star algebras and Hilbert spaces requires substantial operator-theoretic machinery.

[Read manuscript](#) | OPDP ID 50000231 | [Family formalization notes](#)

OAI-M0576 | Family 289

Universal strong Kadison-Kastler stability

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims universal strong Kadison-Kastler stability for unital von Neumann algebras. Given any desired closeness of a conjugating unitary to identity, one tolerance for the algebras' unit-ball distance works across all algebra types, representations, and Hilbert-space cardinalities. The conclusion supplies both exact spatial conjugacy and a small implementer. It does not make the same assertion for norm-separable C-star algebras, which form a separate setting.

Archive: related target. The candidate's von Neumann distance definition is visible, but the excerpt truncates its exact conjectural conclusion and uniformity.

Score basis: A tolerance uniform over all operator-algebra types creates a stronger perturbation problem than class-specific conjugacy.

[Read manuscript](#) | OPDP ID 40324581 | [Family formalization notes](#)

OAI-M0577 | Family 289

Near Inclusions of von Neumann Algebras Without Small Spatial Embeddings

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims examples of one-sided near inclusions of von Neumann algebras with gaps tending to zero, yet no spatial embedding implemented by a unitary close to identity. The algebras act on separable Hilbert spaces and share an identity. Spatial embeddings do exist; every implementing unitary remains a fixed positive distance from identity. This distinguishes a failure of small implementation from failure of embedding itself.

Archive: unresolved no match. The retrieved Atiyah and primeness questions concern different operator-algebra invariants rather than near inclusions.

Score basis: The construction must separate normwise one-sided approximation from a uniform implementer-distance obstruction.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0578 | Family 289

Close Separable C^* -Algebras Without Spatial Conjugacy

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims arbitrarily close unital norm-separable C^* -star algebras on the same separable Hilbert space that are not conjugate by any unitary. They share both identity and von Neumann closure. The example would refute the spatial C^* -star form of Kadison-Kastler, not just the stronger demand for an implementer near identity. It therefore differs from earlier close nuclear examples that remained spatially conjugate.

Archive: same target. The candidate asks whether sufficiently close separable C^* -star algebras are spatially isomorphic; the claimed nonconjugate pairs directly negate that universal tolerance.

Score basis: The counterexample must preserve small two-sided unit-ball distance while obstructing every spatial conjugacy.

[Read manuscript](#) | OPDP ID 40398093 | [Family formalization notes](#)

OAI-M0579 | Family 290

Expected amenable subalgebras preserving core commutants

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that every expected inclusion of von Neumann algebras with separable preduals contains an expected amenable subalgebra preserving the larger inclusion's relative commutant in the continuous core. Amenability here means injectivity, and faithful normal conditional expectations are required. The result targets a relative bicentralizer formulation. It does not, from this statement alone, require the chosen amenable subalgebra to be maximal abelian.

Archive: related target. The candidate specifies a maximal-abelian relative bicentralizer target; exact comparison requires its full statement and the paper's consequences.

Score basis: Continuous-core commutants and expected amenable subalgebras require deep modular and relative operator-algebra analysis.

[Read manuscript](#) | OPDP ID 40359705 | [Family formalization notes](#)

OAI-M0580 | Family 290

Bounded recovery for modular spectral averages

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims bounded spectral recovery for a von Neumann algebra with a faithful state whose centralizer is scalar. Positive averaged norm on shrinking modular spectral bands is recovered on uniformly bounded algebra elements with shrinking support. Applications include intertwining rigidity and an alternative bicentralizer-triviality proof. The independent analytic recovery theorem is the scored target, rather than treating every subsequent alternative proof as a separate new open-problem resolution.

Archive: unresolved no match. Hilbert eigenvariety halos and modular-network spectra are unrelated lexical matches; no matching recovery statement was identified.

Score basis: Recovering Hilbert-space spectral information inside bounded algebra elements demands delicate modular analytic approximation.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0581 | Family 291

Equivariant Jiang-Su Stability for Amenable Actions in the Unital Stably Finite Case

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims automatic equivariant Jiang-Su stability for actions of countable discrete amenable groups on simple separable unital nuclear stably finite algebras already absorbing Jiang-Su. The action absorbs the trivial Jiang-Su action up to cocycle conjugacy, without restricting its behavior on the trace simplex. This settles the stated unital stably finite case, while the more general nonunital and nonfinite action classes require separate comparison.

Archive: partial or special case. The candidate's displayed algebra class does not impose unitality or stable finiteness; the reported theorem is a restricted case.

Score basis: Uniform cocycle absorption over arbitrary trace actions requires advanced classification and equivariant approximation machinery.

[Read manuscript](#) | OPDP ID 40342541 | [Family formalization notes](#)

OAI-M0582 | Family 291

Cuntz comparison and Jiang-Su absorption

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that strict comparison in the extended-functional sense forces Jiang-Su absorption for every separable simple nuclear non-elementary C-star algebra. Nonunital algebras and unbounded traces are included. A broader result uses almost-unperforation and full almost-divisibility of the Cuntz semigroup. The principal theorem addresses the strict-comparison direction of regularity, rather than every additional decomposition-rank statement sometimes grouped under Toms-Winter.

Archive: same target. The curated archive statement explicitly requests this implication in the full nonunital separable simple nuclear setting.

Score basis: The general nonunital comparison-to-absorption implication needs broad Cuntz-semigroup and central-sequence control.

[Read manuscript](#) | OPDP ID 50000379 | [Family formalization notes](#)

OAI-M0583 | Family 291

Nuclear dimension and Jiang-Su stability without elementary subquotients

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims equivalence of finite nuclear dimension and Jiang-Su stability for separable nuclear algebras without elementary ideal subquotients. Their finite nuclear dimension is then at most one. More generally, tensoring any separable nuclear algebra with Jiang-Su gives nuclear dimension at most one. This extends regularity beyond simple algebras, with ideal structure and the absence of elementary subquotients explicitly shaping the equivalence.

Archive: partial or special case. The candidate asks stabilization for every nuclear algebra without an explicit separability restriction; the stated theorem treats separable inputs.

Score basis: Nonsimple regularity requires controlling approximation maps across arbitrary ideal structures.

[Read manuscript](#) | OPDP ID 40398117 | [Family formalization notes](#)

OAI-M0584 | Family 291

Tracial projection methods and uniform property Gamma

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that real rank zero in a specified uniform tracial ultrapower implies uniform property Gamma for simple separable unital nuclear stably finite algebras with traces. A half-trace central projection is constructed. Separate theorems derive Jiang-Su absorption from strict comparison and address bounded-trace stably projectionless algebras. The scored target is the tracial real-rank criterion, keeping these distinct hypotheses and conclusions separate.

Archive: unresolved no match. The probabilistic-projection and mesa-flow candidates concern different objects; no matching real-rank criterion was identified.

Score basis: Uniform trace-splitting projections require substantial ultrapower and tracial-completion machinery.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0585 | Family 292

An explicit obstruction to nuclear norm-ultrapower embeddings

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims an explicit separable unital full group algebra that cannot embed unitaly into any norm ultrapower of a nonzero unital nuclear C-star algebra. The group is fixed independently of target algebra and free ultrafilter. In particular, the obstruction applies to a Cuntz-algebra ultrapower. The stated unital embedding convention is important when comparing this negative result with broader formulations of Kirchberg's embedding question.

Archive: related target. The archived question does not explicitly impose unital embeddings; that convention requires comparison before declaring exact identity.

Score basis: The explicit algebra must obstruct every nuclear ultrapower rather than one chosen target.

[Read manuscript](#) | OPDP ID 20001458 | [Family formalization notes](#)

OAI-M0586 | Family 293

Invariant-projection counterexamples for every irrational rotation

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a continuous nonnegative weighted irrational rotation with no nontrivial invariant projection in its containing hyperfinite finite factor, for every prescribed irrational angle. The weight has one zero and divergent negative logarithmic integral; the operator is nonzero and norm-quasinilpotent. Arbitrary Hilbert-space invariant subspaces are not excluded, because their projections may lie outside that factor. No Diophantine restriction on the angle is asserted.

Archive: unresolved no match. Fractal-measure projection and invariant-state ergodicity candidates have different projection and dynamical notions.

Score basis: Constructing a uniform-angle operator obstruction requires careful weight design and finite-factor analysis.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0587 | Family 293

Backward intertwiners and a transitive commutant

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a nonzero bounded norm-quasinilpotent operator on every infinite-dimensional separable complex Hilbert space with no nonzero proper closed hyperinvariant subspace. Its commutant is asserted to be a proper strongly closed transitive unital algebra. Hyperinvariance means invariance under the entire commutant, so this obstruction is not automatically a negative answer to the ordinary invariant-subspace problem for one operator.

Archive: unresolved no match. The Volterra-commutant and finite multiqudit candidates do not identify the general hyperinvariant-subspace target.

Score basis: A transitive commutant requires simultaneous control of all commuting operators, beyond one shift's invariant subspaces.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0588 | Family 294

A counterexample to Kaplansky's quasitrace conjecture and failure of tensor-product stable finiteness

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a separable unital C-star algebra admitting normalized two-quasitraces but no tracial state. Its quasitraces have a quantified failure of additivity on specified positive contractions, refuting the quasitrace conjecture. A consequence constructs simple stably finite tensor factors whose minimal tensor product is properly infinite, even with one reduced free-group factor fixed. This concerns analytic traces and finiteness, not group-ring zero divisors.

Archive: unresolved no match. Morita-equivalence and Markov-category tensor questions are unrelated tensor-product uses.

Score basis: Quasitrace nonadditivity and tensor-product infiniteness require delicate analytic constructions across distinct finiteness notions.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0589 | Family 295

Vanishing of higher bounded Hochschild cohomology

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that every bounded Hochschild cocycle of degree at least two on any complex von Neumann algebra, valued in itself, has a bounded primitive. Together with the established degree-one inner-derivation theorem, all positive-degree bounded cohomology vanishes in this setting. The target uses bounded analytic cochains, rather than unrestricted algebraic Hochschild cohomology of finite-dimensional algebras or singular derived-category constructions.

Archive: unresolved no match. Happel and Keller candidates concern different algebra classes and cohomology theories.

Score basis: Uniform bounded primitives across arbitrary von Neumann algebras need substantial operator-cohomological analysis.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0590 | Family 296

Relative generation and the generator problem for finite factors

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims single generation of every finite type-II-one factor with separable predual, equivalently generation by two self-adjoint operators. A preceding result makes unitary relative generators a dense countable intersection of open sets for irreducible inclusions. Established direct-integral reductions then address the separable-predual generator problem more generally. Separability is explicit in the theorem and must be retained when reading a one-line archive question.

Archive: partial or special case. The candidate says every type-II-one factor without displaying a separability convention; the paper explicitly assumes separable predual.

Score basis: Generic relative generation must imply a finite generating pair through subtle finite-factor structure.

[Read manuscript](#) | OPDP ID 40402530 | [Family formalization notes](#)

OAI-M0591 | Family 297

A counterexample to Naimark's problem in ZFC

D 925/1000 | p 0 | m 1 | Conditional points 0

Historical independent-proof burden estimate (C0)

This paper gives an alternative claimed ZFC construction of a simple unital infinite-dimensional C-star algebra with faithful trace and one irreducible-representation class. It explicitly credits Tanaka's September 22 preprint with the same existence conclusion. The new argument organizes selective bridges, reduced-HNN reordering, and countable determination. Its construction therefore targets a historically difficult theorem already publicly claimed before this release, rather than a first ZFC resolution.

Archive: same target. The archive's ZFC existence target matches, but the paper expressly attributes prior resolution to Tanaka, arXiv:2609.26930v1.

Score basis: The score estimates the substantial transfinite and operator-algebraic construction burden, not a new existence frontier.

Progress: The main existence conclusion is explicitly acknowledged as present in external prior work. The alternate construction is scored for difficulty but receives zero new open-problem-resolution credit; no new formalization is asserted.

[Read manuscript](#) | OPDP ID 40396234 | [Family formalization notes](#)

OAI-M0592 | Family 298

A finite-entropy separation of microstates and nonmicrostates free entropy

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a bounded self-adjoint tuple with finite microstates and nonmicrostates free entropies separated by at least one-half. Microstates entropy uses its original norm cutoff and matrix-size upper limit. A large but fixed number of variables is used. This disproves equality of the two entropy quantities in the finite-entropy regime, rather than comparing free entropy dimensions or defining conditional nonmicrostates entropy.

Archive: related target. The candidate compares free Stein dimension with nonmicrostates entropy dimension, not the entropy quantities in this theorem.

Score basis: Strict finite-entropy separation requires precise asymptotic matrix-volume and nonmicrostates estimates.

[Read manuscript](#) | OPDP ID 40402666 | [Family formalization notes](#)

OAI-M0593 | Family 299

The Kirchberg-Rørdam character criterion

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that a nonzero unital separable C-star algebra absorbs Jiang-Su exactly when its norm central-sequence algebra has no characters. The criterion is asserted for every free ultrafilter, without simplicity, nuclearity, trace, or comparison assumptions. It also gives Jiang-Su stability of infinite minimal tensor powers of algebras without characters. The norm-ultrapower convention distinguishes this criterion from tracial central-sequence statements.

Archive: unresolved no match. Calogero-Moser and character-variety candidates refer to unrelated notions of character.

Score basis: A character obstruction must control absorption across arbitrary separable algebra structure without nuclearity.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0594 | Family 300

Approximation Paving over Arbitrary Maximal Abelian Subalgebras

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims approximation paving over every maximal abelian subalgebra of a complex von Neumann algebra. A self-adjoint operator is approximated strongly by self-adjoint operators of at most triple norm, each norm-pavable with at most a universal constant times the inverse sixth power of error projections. No separability or expectation is assumed. The relative error is measured against each approximant's own norm.

Archive: unresolved no match. The Pedersen and bicentralizer candidates concern other maximal-abelian criteria rather than paving.

Score basis: Universal approximation paving needs quantitative control across arbitrary maximal abelian inclusions and representations.

[Read manuscript](#)

OAI-M0595 | Family 300

Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims quadratic strong-operator paving over arbitrary maximal abelian subalgebras. Every self-adjoint element has a paving with at most five hundred million times the inverse square of error projections. The number is fixed independently of representation, algebra inclusion, operator, and strong neighborhood; no separability or conditional expectation is assumed. This improves the projection-count exponent in the specified topology, rather than asserting ordinary norm paving of every original operator.

Archive: unresolved no match. The displayed candidates ask maximal-abelian classification or bicentralizer questions, not this quantitative paving property.

Score basis: Obtaining quadratic quantitative bounds uniformly over arbitrary inclusions requires a strong analytic paving mechanism.

[Read manuscript](#)

OAI-M0596 | Family 301

The trace cone classifies Razak-Jacelon stabilizations

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims classification of separable nuclear algebras after Razak-Jacelon and compact-operator stabilization by their canonical topological cones of extended lower-semicontinuous tracial weights. The resulting isomorphism realizes the prescribed cone map, allowing arbitrary ideal structure and no density assumption on finite weight domains. Topology and extended values are part of the invariant, so replacing this cone by a simpler collection of bounded traces would change the target.

Archive: related target. The named Robert conjecture is retrieved, but its excerpt truncates the cone topology and complete stabilization conclusion.

Score basis: Trace-cone realization across arbitrary ideals requires extensive classification and approximation machinery.

[Read manuscript](#) | OPDP ID 40324586

OAI-M0597 | Family 302

Filtered products and boundary-preserving compression in complex cobordism

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims coherent stable vanishing on unions of large smash-power subcomplexes when a finite-pair restriction vanishes in complex cobordism. The union requires a fixed positive proportion of restricted factors, not separate vanishing on each piece alone. Applications compress cube-valued maps while preserving every existing boundary slot. A strict vector-bundle rank inequality enables compression, with an equality example showing that strictness cannot be discarded.

Archive: unresolved no match. Filtered Seiberg-Witten maps and quantum-game compression use different theories and do not identify this complex-cobordism target.

Score basis: Coherent union-level nullhomotopies and boundary-preserving compression require substantial stable-homotopy and bundle bookkeeping.

[Read manuscript](#)

OAI-M0598 | Family 302

Radius of comparison equals half the mean dimension

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that every minimal homeomorphism of an infinite compact metrizable space has crossed-product radius of comparison equal to half its mean dimension, including infinite values. Zero mean dimension is then equivalent to small boundary property, Jiang-Su stability, and finite nuclear dimension, which is at most one. This quantitatively links dynamical complexity to an operator-algebra comparison invariant for integer actions.

Archive: related target. The power-scaling candidate may be related through mean-dimension scaling, but its truncated formula does not establish exact target identity.

Score basis: An exact dynamical-operator invariant equality needs both sharp comparison bounds and regularity classification.

[Read manuscript](#) | OPDP ID 40333206

OAI-M0599 | Family 303

Weak pure infiniteness and O-infinity absorption

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that proper infiniteness of every positive element implies strong pure infiniteness for arbitrary complex C-star algebras. For exact algebras, proper infiniteness after one fixed finite amplification is also reduced to individual proper infiniteness. Consequently, separable nuclear algebras with that amplification property absorb the infinite Cuntz algebra without simplicity or unitality. The cross-term diagonalization statement is stronger than checking separate positive elements.

Archive: related target. The groupoid candidate asks an algebraic Steinberg converse, which this analytic C-star implication does not supply.

Score basis: Uniform simultaneous diagonalization requires controlling arbitrary cross terms and comparison beyond individual infiniteness.

[Read manuscript](#) | OPDP ID 40401247 | [Family formalization notes](#)

OAI-M0600 | Family 304

The Hilbert-Smith conjecture in every finite dimension

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims Hilbert-Smith in every finite dimension for locally compact second-countable Hausdorff groups acting faithfully and jointly continuously on connected second-countable topological manifolds without boundary. Noncompactness, nonorientability, and nontriangulability are permitted. The group is asserted to be Lie, with arbitrary stabilizers and fixed points allowed. The explicit countability and manifold conventions are retained rather than silently equating this statement with an abbreviated archive formulation.

Archive: partial or special case. The original archive sentence omits second-countability, connectedness, and boundary conventions; an exact full-scope reduction was not verified here.

Score basis: A general topological-action theorem without smoothness requires deep transformation-group and manifold arguments.

[Read manuscript](#) | OPDP ID 1482

OAI-M0601 | Family 305

A boundary-only obstruction to four-dimensional disk embedding

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a four-dimensional disk-embedding obstruction determined only by boundary circles. A compact smooth oriented manifold contains immersed disks with framed algebraic dual spheres, yet their boundaries have no disjoint locally flat fillings even without prescribed output homotopy classes or framings. It further asserts that the rank-two free group, and every group containing it, is not good in Freedman-Quinn's sense.

Archive: unresolved no match. The retrieved vector-field and crystal-cluster disk questions are unrelated lexical matches.

Score basis: Four-dimensional topological embedding obstructions require subtle fundamental-group and intersection control beyond smooth local constructions.

[Read manuscript](#)

OAI-M0602 | Family 305

A marked tensor obstruction to four-dimensional disk embedding

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims immersed disks with framed algebraic dual spheres satisfying the usual equivariant intersection and reduced self-intersection conditions, but no disjoint locally flat replacements preserving boundary maps and normal framings. Replacement relative homotopy classes are not fixed. The marked tensor obstruction therefore refutes an unrestricted framed disk-embedding formulation; it is distinct from the stronger companion obstruction excluding all unframed boundary fillings.

Archive: unresolved no match. Packing disks and symplectic polydisk corners are different notions; no archive identity was established.

Score basis: Controlling the marked obstruction through arbitrary locally flat replacements requires substantial four-dimensional topology.

[Read manuscript](#)

OAI-M0603 | Family 305

A PD4 group without an aspherical manifold model

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a finitely presented integral Poincare-duality group of dimension four with a finite classifying space but no closed aspherical topological four-manifold realization. Its finite oriented aspherical Poincare complex has the required group-ring cohomology, yet its fundamental group cannot arise from any such manifold. This would negate the universal manifold-realization statement even within finite complexes, rather than merely exhibiting a nonmanifold model with another manifold model available.

Archive: same target. The archived universal aspherical-Poincare-complex manifold statement is refuted by the asserted finite counterexample.

Score basis: Distinguishing algebraic duality from every possible manifold realization requires global surgery and group-theoretic obstruction machinery.

[Read manuscript](#) | OPDP ID 40369305

OAI-M0604 | Family 306

Purely cosmetic surgery on knots in the three-sphere

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that distinct Dehn-surgery slopes on any nontrivial smooth knot in the three-sphere never produce orientation-preservingly homeomorphic filled manifolds. The homeomorphism need not preserve a surgery core or the original knot exterior. This is the purely cosmetic surgery target; orientation-reversing coincidences are not excluded. Including all rational slopes and the meridional slope makes the assertion broader than a numerical slope-bound result.

Archive: related target. The named cosmetic-surgery candidate displays its setting and orientation convention, but truncates the complete conjectural conclusion.

Score basis: All-knot surgery rigidity requires combining three-manifold topology with invariants powerful enough to detect every slope.

[Read manuscript](#) | OPDP ID 40352653

OAI-M0605 | Family 307

Failure of rational injectivity for maximal coarse assembly

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims failure of rational injectivity for maximal coarse assembly on a bounded-geometry graph union. Its uniformly discrete space is a coarse disjoint union of finite connected graphs with uniformly bounded vertex degree, and a degree-one coarse K-homology class of infinite order maps to zero. Maximal assembly is stronger here than the reduced counterpart treated separately, so the completion convention belongs to the target.

Archive: related target. The candidate's map notation does not display maximal versus reduced completion, so exact comparison remains unresolved.

Score basis: The kernel class must survive coarse homology while vanishing under maximal analytic assembly.

[Read manuscript](#) | OPDP ID 40360615 | [Family formalization notes](#)

OAI-M0606 | Family 307

A counterexample to the coarse Novikov conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims a counterexample to ordinary coarse Novikov injectivity for uniformly discrete bounded-geometry spaces. A coarse disjoint union of finite bounded-degree graphs has an infinite-order degree-one coarse K-homology class whose reduced locally compact Roe-algebra assembly image vanishes. Thus bounded geometry alone is asserted insufficient. The result does not refute injectivity theorems with additional Hilbert-embedding or other geometric hypotheses.

Archive: related target. The archived coarse-geometric map is truncated; its geometric assumptions and assembly completion are not fully visible.

Score basis: Constructing a persistent coarse homology class with vanishing reduced analytic index requires deep coarse K-theory.

[Read manuscript](#) | OPDP ID 40379176 | [Family formalization notes](#)

OAI-M0607 | Family 308

Finite Smith-Toda Complexes at Varying Primes

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a finite Smith-Toda complex at every chromatic height when the prime may vary with height. Its Brown-Peterson homology is the canonical quotient by the prime and the first height-many periodicity generators, each killed to its first power, with generator in degree zero. Finiteness holds before chromatic localization. The result does not claim one prime supports all heights or replace the generators by higher powers.

Archive: unresolved no match. Varying-prime Iwasawa invariants and Cech complexes are different targets despite shared words.

Score basis: Realizing exact homology quotients by finite spectra at arbitrarily high height requires substantial obstruction control.

[Read manuscript](#)

OAI-M0608 | Family 308

A finite Smith-Toda complex $V(4)$ at the prime 1009

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims a finite Smith-Toda complex of height four at the specified prime 1009. Its Brown-Peterson homology is the quotient by 1009 and the first four periodicity generators with canonical comodule structure and degree-zero generator. The construction takes place in the ordinary finite local stable category, before localization. It establishes a concrete prime-height realization rather than a statement covering every prime.

Archive: unresolved no match. The Toda-prime arithmetic candidates are unrelated to Smith-Toda finite spectra.

Score basis: A fixed large prime lowers uniformity demands, while realizing the exact finite-spectrum comodule still requires substantial homotopical construction.

[Read manuscript](#)

OAI-M0609 | Family 309

The Kervaire invariant problem at the prime three

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a complete survival classification of standard prime-three Kervaire classes in the mod-three Adams spectral sequence. Exactly indices zero, two, and three survive, in stems ten, 106, and 322, each with an additive-order-three representative. Other specified classes die. This is an odd-prime analogue of a Kervaire problem; it does not decide existence of the mod-two invariant-one class in stem 126.

Archive: related target. The archived dimension-126 target is mod two, whereas the manuscript's classification is at prime three.

Score basis: All-index survival and non-survival require difficult spectral-sequence and chromatic obstruction arguments.

[Read manuscript](#) | OPDP ID 11100009

OAI-M0610 | Family 310

Rational homology and Quillen's conjecture

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims nonvanishing augmented reduced rational homology of the elementary-abelian subgroup poset whenever a finite group's largest normal prime-subgroup is trivial. All primes and finite groups are included, with an explicit augmented-homology convention for the empty poset. The rational statement implies the usual noncontractibility form of Quillen's conjecture. In particular, it supplies the prime-two rational-homology case listed in the atlas.

Archive: same target. The curated prime-two rational statement is directly included in the all-prime theorem; no parent score is inherited.

Score basis: A universal finite-group homology theorem requires classification-sensitive structural control despite finite input descriptions.

[Read manuscript](#) | OPDP ID 51000162

OAI-M0611 | Family 311

Stabilizer orbits and thick tensor ideals of dualizable $K(n)$ -local spectra

D 648/1000 | p 1 | m 1 | Conditional points 419.904

Published original CFSD-1000 target difficulty

The manuscript claims classification of open-stabilizer-invariant prime and radical ideals in the Lubin-Tate deformation ring at every positive height and prime. It derives exactly height plus two thick tensor ideals among dualizable Morava- K -local spectra and a Balmer spectrum forming a chain of height plus one points. This determines the expected Hovey-Strickland filtration, distinguishing dualizable objects from the much smaller class of compact objects in the local category.

Archive: same target. The archived question explicitly concerns tensor ideals of dualizable $K(n)$ -local spectra, the category classified here.

Score basis: The complete thick-ideal filtration answers the published Hovey-Strickland target; its original score is preserved.

Scoring target: Classify thick tensor ideals of dualizable spectra in the Morava- K -local category by the expected height filtration.

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 11100015

OAI-M0612 | Family 312

The Grothendieck homotopy hypothesis via elementary expansions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims the homotopy hypothesis for every Grothendieck coherator in the specified Ara-Henry convention. Weak globular infinity-groupoids recover the homotopy theory of spaces. Its technical theorem asserts that adjoining an elementary cell expansion to a cellular groupoid preserves components and all homotopy groups, resolving a pushout conjecture. The coherator and model conventions matter when comparing this with other infinity-groupoid formulations.

Archive: related target. The archive's equivalence-of-categories statement truncates its precise groupoid and homotopy conventions.

Score basis: Uniform coherence and pushout control require lengthy higher-categorical construction rather than finite-dimensional examples.

[Read manuscript](#) | OPDP ID 40323658 | [Family formalization notes](#)

OAI-M0613 | Family 313

Finite generation for the $K(n)$ -local sphere

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims degreewise finite generation over the p -adic integers for homotopy groups of every positive-height Morava- K -local sphere at every prime. Equivalently, it asserts the same finiteness after localizing any finite prime-local spectrum. This controls each integer degree, rather than asserting finitely many nonzero degrees or ordinary finite generation over the integers. Continuous coefficient and integral recovery arguments form the analytic-algebraic background of the target.

Archive: unresolved no match. Polynomial automorphism and singularity finite-generation candidates are unrelated categories and invariants.

Score basis: All-height degreewise integral finiteness requires deep continuous-cohomology, chromatic, and inverse-limit control.

[Read manuscript](#)

OAI-M0614 | Family 314

Cyclic length and chromatic fixed-point loss

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that optimal chromatic fixed-point loss for a subgroup of a finite prime-group equals the shortest subnormal-chain length with cyclic quotients. The equality holds at every nonnegative height and prime, making loss independent of height. Lower bounds use finite genuine equivariant spectra distinguishing the two geometric fixed-point homologies. The embedding of the subgroup matters, not only its abstract isomorphism type.

Archive: unresolved no match. The classical single-fixed-point manifold candidate and cyclic-sum question do not specify this chromatic invariant.

Score basis: Sharp equivariant loss requires both universal upper bounds and constructed counterexamples at every smaller loss.

[Read manuscript](#)

OAI-M0615 | Family 315

The Singer conjecture in dimension four

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims Singer vanishing outside middle degree for universal covers of closed aspherical topological four-manifolds. It proves the stronger same-dimensional statement for finite connected aspherical integral Poincare complexes, including arbitrary orientation character. Thus only degree two may have nonzero squared-integrable Betti number. This addresses dimension four rather than the conjecture simultaneously in every dimension, and does not determine the middle-degree value.

Archive: partial or special case. The archive's Singer question ranges over arbitrary dimensions; this theorem establishes its four-dimensional case.

Score basis: Dimension four retains significant duality and analytic topology difficulty despite a fixed-dimensional scope.

[Read manuscript](#) | OPDP ID 40345707

OAI-M0616 | Family 316

The Stable Hurewicz Image of the Sphere at Two

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims Curtis's stable mod-two Hurewicz-image conjecture in every positive degree. The image is spanned by the three Hopf-invariant-one detections and whichever Kervaire-invariant-one classes exist. A consequence proves Eccles's conjecture for positive-dimensional spheres. The formulation deliberately leaves existence of potentially unresolved Kervaire classes conditional, so it does not by itself construct the dimension-126 class or decide every stable homotopy group.

Archive: same target. The candidate states the same sphere-spectrum image restriction, including the conditional Kervaire clause.

Score basis: A universal image restriction across all degrees requires strong unstable-to-stable homology and spectral control.

[Read manuscript](#) | OPDP ID 40394754

OAI-M0617 | Family 317

Thomason Model Structures in Every Strict Higher Dimension

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims proper combinatorial Thomason model structures on small strict globular categories in every positive finite dimension and in infinite dimension. Weak equivalences and fibrations are detected by a twice-extended Street nerve, with a subdivision adjunction giving a Quillen equivalence to simplicial sets. This models spaces using strict higher categories in the chosen structure; it is not a general equivalence between arbitrary weak and strict higher categories.

Archive: related target. The weakly unital category candidate concerns modeling weak higher categories, a different categorical target.

Score basis: Constructing model axioms and Quillen equivalences uniformly through higher dimensions requires extensive categorical machinery.

[Read manuscript](#) | OPDP ID 40319478 | [Family formalization notes](#)

OAI-M0618 | Family 318

Filtered chromatic splitting at generic primes

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims an ordered filtration with two-to-the-height stages for a chromatic localization overlap at primes exceeding height plus one. Its cofibers have the classical splitting pattern, and the first-stage map is the canonical unit. A filtration does not assert a direct-sum splitting: attachment maps may be nonzero. The prime restriction also excludes the height-two prime-two case mentioned in the archived counterexample record.

Archive: related target. The candidate concerns a refuted strong wedge formula at height two and prime two, outside this generic-prime filtered theorem.

Score basis: Realizing expected layers with canonical units needs careful integral map and attachment control.

[Read manuscript](#) | OPDP ID 40399498

OAI-M0619 | Family 318

The height-three chromatic overlap: an explicit filtration and its attachments

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims an explicit eight-stage filtration of the height-three chromatic overlap at every prime at least five. Two signed fracture formulas determine attachments for chosen local maps and compatibility homotopy. A separate canonical-map theorem implies the first height-one attachment is nonzero. The result therefore assembles splitting-pattern layers without claiming they split as a wedge, and fixes more map-level information than merely listing cofibers.

Archive: unresolved no match. Burnside-ring filtrations and Mahowald-height growth are related chromatic subjects but not this overlap-filtration target.

Score basis: Explicit attachment identification requires careful coherence and localization-map computations.

[Read manuscript](#)

OAI-M0620 | Family 318

A rational obstruction to strong chromatic splitting at height three

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a nonzero rational degree-minus-three canonical map between two height-three localization overlaps at every prime at least five. This obstructs the strong chromatic wedge formula even without requiring an equivalence to preserve specified summand maps. The result is stronger than finding a poorly chosen noncanonical attachment. It distinguishes the surviving filtration pattern from a false direct-sum assertion in this prime-height range.

Archive: related target. The named strong splitting record truncates its exact formula, so this obstruction is linked without asserting full statement identity.

Score basis: Detecting a canonical rational map requires precise localization and descent calculations.

[Read manuscript](#) | OPDP ID 40341679

OAI-M0621 | Family 318

Failure of finite assembly for a chromatic overlap at the prime three

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

At prime three and height three, the manuscript claims that a chromatic overlap lies outside the thick subcategory generated by rational, height-one, and height-two local spheres. Finite sums, shifts, cofibers, and retracts cannot assemble it in the specified derived-complete module category. This obstructs even finite construction without fixing proposed summands, a weaker demand than a canonical wedge splitting but still a substantial assembly restriction.

Archive: unresolved no match. Three-chromatic graphs and representational assembly over fields are unrelated targets.

Score basis: Excluding every finite thick construction requires a robust categorical invariant rather than a failed proposed filtration.

[Read manuscript](#)

OAI-M0622 | Family 318

Counterexamples to weak chromatic splitting: sphere kernels and descent exponents

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims explicit failures of canonical weak chromatic splitting for the derived prime-complete sphere. At height equal to prime for primes at least five, and height prime plus one for primes at least seven, a named classical beta-product remains nonzero below the map but is killed by it. Thus the map has no homotopy retraction. The obstruction concerns weak splitting, not merely a detailed strong-summand formula.

Archive: related target. The Hopkins record truncates the exact splitting map, so the canonical-map convention requires full-source comparison.

Score basis: Named kernel detection across two prime-height ranges demands substantial descent and stable-stem control.

[Read manuscript](#) | OPDP ID 40341854

OAI-M0623 | Family 319

Counterexamples to the Hahn-Wilson conjecture at height two

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims height-two counterexamples to the Hahn-Wilson thick-generation conjecture at sufficiently large primes. A connective prime-complete spectrum of exact fp-type two lies outside the ordinary thick subcategory generated by the specified truncated Brown-Peterson spectrum. The same spectrum satisfies both stated finite-versus-ordinary and telescopic-versus-Morava-K localization comparisons. Those comparisons do not force the proposed thick-generation classification, separating distinct finiteness assertions.

Archive: same target. The archived universal fp-type thick-generation statement is contradicted by the asserted height-two example.

Score basis: The spectrum must satisfy precise fp-type and localization properties while escaping every finite thick construction.

[Read manuscript](#) | OPDP ID 40405264

OAI-M0624 | Family 320

Nonhomeomorphic closed aspherical four-manifolds with the same homotopy type

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims homotopy-equivalent but nonhomeomorphic closed aspherical topological four-manifolds with a common word-hyperbolic fundamental group. No homeomorphism exists at all, rather than merely no homeomorphism in a chosen homotopy class. A separate self-equivalence is not homotopic to a homeomorphism. These conclusions refute a strong base-manifold Borel formulation; they do not automatically refute a weak formulation asking only about universal covers.

Archive: related target. The retrieved weak Borel record mentions universal covers, whose homeomorphism does not follow negatively from nonhomeomorphic base manifolds.

Score basis: Base-manifold nonrigidity with hyperbolic fundamental group requires delicate surgery and topological realization obstructions.

[Read manuscript](#) | OPDP ID 40361021

OAI-M0625 | Family 321

A Counterexample to Wall's D(2) Problem

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a finite connected three-dimensional complex meeting Wall's D(2) homological and all-local-coefficient conditions but having no finite homotopy model of dimension at most two. Its fundamental group is infinite. The stated counterexample targets finite dimension reduction, with the finiteness condition on the proposed replacement explicit. A truncated statement omitting replacement finiteness needs comparison before identifying the exact archived formulation.

Archive: related target. The candidate does not display the full finite-replacement convention or every D(2) condition.

Score basis: Distinguishing cohomological dimension from finite two-dimensional homotopy realization requires substantial finiteness and group-ring obstruction theory.

[Read manuscript](#) | OPDP ID 40330058

OAI-M0626 | Family 322

A positive solution to Tingley's problem

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that every surjective isometry between unit spheres of real Banach spaces extends uniquely to a surjective real-linear isometry through radial extension. No separability, reflexivity, smoothness, or strict convexity is imposed. For complex spaces viewed as real, complex linearity is not asserted. The argument aligns extremal chords through support functionals and a fixed-point theorem, addressing Tingley's sphere-extension problem rather than arbitrary metric embeddings.

Archive: unresolved no match. Discrete Painleve and positive-edge graph solutions are unrelated lexical candidates.

Score basis: Recovering linearity from sphere distances across arbitrary Banach norms requires a global geometric argument.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0627 | Family 323

Relative independence of the separable quotient problem

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims relative independence of the separable quotient assertion for infinite-dimensional real and complex Banach spaces. A real-valued measurable continuum implies the assertion, whereas the continuum hypothesis gives counterexamples. Full independence is relative to a measurable-cardinal consistency assumption; the density-aleph-one restricted statement is independent relative to ZFC consistency alone. These foundational qualifications matter: this is not an unconditional universal quotient theorem or a ZFC counterexample.

Archive: same target. The curated archive asks whether every infinite-dimensional Banach space has the specified quotient; relative independence is the claimed answer to that assertion.

Score basis: The two directions couple delicate Banach constructions with distinct set-theoretic consistency strengths.

[Read manuscript](#) | [OPDP ID 50000392](#) | [Family formalization notes](#)

OAI-M0628 | Family 324

Lipschitz Equivalent Separable Banach Spaces Need Not Be Linearly Isomorphic

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims separable real Banach spaces that are globally bi-Lipschitz equivalent but not linearly isomorphic. Explicit Lipschitz constants control a bijection, while a block-valued linear-subspace obstruction distinguishes the spaces. One contains a linear copy of the indicated Hilbert-block c-zero space and the other does not. Thus metric equivalence need not recover linear structure even in the separable setting.

Archive: unresolved no match. The open-mapping reverse-mathematics candidate asks provability in weak arithmetic systems, not Lipschitz versus linear equivalence.

Score basis: The example must maintain global metric control while preserving a robust linear-subspace obstruction.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0629 | Family 324

Bi-Lipschitz Absorption of c_0 Without a Linear Copy of c_0

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a separable real Banach space containing no linear copy of c_0 yet bi-Lipschitz equivalent to its direct sum with c_0 . The same space contains bi-Lipschitz images of every separable metric space. This gives strong nonlinear universality and absorption while prohibiting a familiar linear subspace. It is an existence statement about the whole space, not a quantitative extension theorem for maps between chosen subsets.

Archive: related target. The candidate concerns dependence of map-extension constants, a different bi-Lipschitz property.

Score basis: Nonlinear absorption and metric universality must coexist with a carefully maintained linear exclusion.

[Read manuscript](#) | OPDP ID 40384339 | [Family formalization notes](#)

OAI-M0630 | Family 325

A direct proof of the complete Crouzeix inequality

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims the sharp constant-two numerical-range inequality for matrix-valued polynomials evaluated at finite matrices, uniformly in both base and coefficient dimensions. Degenerate point and segment numerical ranges are included. The amplified assertion is stronger than the scalar inequality, which the opening acknowledges in external prior mathematics. Uniformity over noncommuting coefficient matrices is the primary scored target, rather than assigning fresh resolution credit to its scalar consequence.

Archive: related target. Original scalar parent records 1300 and 1449 are duplicates with conflicting 603 and 659 scores; neither score labels this stronger complete target.

Score basis: Complete matrix amplification requires control of noncommuting coefficients beyond the acknowledged scalar bound.

[Read manuscript](#) | OPDP ID 1300 | [Family formalization notes](#)

OAI-M0631 | Family 325

The complete Crouzeix theorem: optimal similarity and a common positive boundary representation

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims complete constant-two Crouzeix bounds for every bounded operator on arbitrary complex Hilbert space and matrix-valued polynomials, extending to nearby holomorphic functions. Finite-matrix refinements attain optimal similarity with condition number at most two and a common positive boundary representation. Its complete operator theorem is the scored target. The scalar finite-matrix consequence is acknowledged prior mathematics and has conflicting historical scores in duplicate archive records.

Archive: related target. Scalar parent records 1300 and 1449 publish conflicting 603 and 659; they are historical anchors, not inherited labels for this complete target.

Score basis: Arbitrary-operator complete bounds and common boundary representation add substantial context and analytic rigidity beyond scalar finite matrices.

[Read manuscript](#) | OPDP ID 1300 | [Family formalization notes](#)

OAI-M0632 | Family 326

The cotype-cotype conjecture under the approximation property

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that a real Banach space with the ordinary approximation property is K-convex exactly when both it and its dual have finite Rademacher cotype, possibly at different exponents. The forward direction needs no approximation assumption; the asserted new implication uses the ordinary property rather than its bounded version. Thus approximation conventions and whether both dual cotype conditions are present delimit the comparison target.

Archive: related target. The candidate specifies bounded approximation and truncates its full cotype conclusion; the theorem's ordinary-approximation scope is explicit.

Score basis: Connecting dual cotype to K-convexity requires substantial Banach-space geometry beyond finite-dimensional inequalities.

[Read manuscript](#) | OPDP ID 40358499 | [Family formalization notes](#)

OAI-M0633 | Family 327

Nontrivial Markov Type Forces Superreflexivity

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims that every real Banach space with Markov type at some exponent greater than one is superreflexive. Together with the known converse, nontrivial Markov type becomes a characterization of superreflexivity and an equivalent uniformly smooth norm. The Markov inequality tests finite reversible walks, whereas the conclusion is linear renormability. It includes Markov type two and the more general nontrivial-type question in the archive.

Archive: same target. The candidate explicitly asks both the type-two case and the general nontrivial-Markov-type implication.

Score basis: A metric random-walk property must force uniform linear geometry through a nontrivial converse argument.

[Read manuscript](#) | OPDP ID 40384667 | [Family formalization notes](#)

OAI-M0634 | Family 328

Fixed Points of Nonexpansive Maps in Reflexive Banach Spaces

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a fixed point for every nonexpansive selfmap of a nonempty closed bounded convex subset of a real reflexive Banach space, in its given norm. No replacement by an equivalent uniformly convex norm is used. A commuting-family consequence is also stated. This extends familiar fixed-point criteria to arbitrary reflexive geometry and directly addresses the archive's fixed-point-property assertion, rather than only two-dimensional smooth spaces.

Archive: same target. The candidate's first complete sentence states the same reflexive-space nonexpansive fixed-point property.

Score basis: Reflexivity alone must supply the required fixed-point geometry without standard stronger convexity hypotheses.

[Read manuscript](#) | OPDP ID 40370397 | [Family formalization notes](#)

OAI-M0635 | Family 329

Counterexamples to the duality conjecture for metric entropy

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims failure of dimension-free entropy duality for symmetric convex bodies. For every proposed pair of universal constants, a full-dimensional body and a cube violate the polar covering-entropy upper inequality. Dimension is allowed to grow with those constants. The construction refutes a universal dimension-independent assertion; it does not assert a violation of every possible bound in a fixed dimension or every individual pair of bodies.

Archive: related target. The Pietsch candidate truncates the exact rescaling and inequality; the named relation is retained without declaring formula identity.

Score basis: Dimension-growing convex constructions must separate primal and polar covering numbers uniformly over proposed constants.

[Read manuscript](#) | OPDP ID 40400041 | [Family formalization notes](#)

OAI-M0636 | Family 330

A uniformly discrete counterexample to bounded approximation in Lipschitz-free spaces

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a countable uniformly discrete metric space whose real Lipschitz-free space has approximation property but no bounded approximation property. Distinct distances have a common positive lower bound; the space is unbounded and nonproper. Finite-rank approximations work on each compact set, but their norms admit no universal bound. Uniform discreteness strengthens earlier discrete examples without asserting that the resulting metric space is proper.

Archive: related target. The candidate asks uniqueness of Lipschitz-space preduals, not the approximation properties of the canonical free space.

Score basis: The construction must separate approximation from bounded approximation while retaining a uniform discrete metric.

[Read manuscript](#) | OPDP ID 40405176 | [Family formalization notes](#)

OAI-M0637 | Family 331

Asymptotic midpoint uniform convexity and unbounded diamond distortion in a reflexive tree space

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a separable reflexive space whose given norm is asymptotically midpoint uniformly convex but which has no asymptotically uniformly convex equivalent norm. Explicit midpoint and countably branching diamond-distortion bounds quantify the distinction. This refutes a reflexive diamond converse for asymptotic uniform convexifiability, not ordinary reflexivity or every uniform-homeomorphism assertion for superreflexive unit spheres. The original norm and possible renormings are different parts of the target.

Archive: related target. The candidate concerns superreflexive unit-sphere uniform homeomorphisms; this theorem separates asymptotic convexity properties in a reflexive space.

Score basis: Quantitative graph obstructions must rule out all equivalent asymptotically uniformly convex norms.

[Read manuscript](#) | OPDP ID 40401500 | [Family formalization notes](#)

OAI-M0638 | Family 331

Midpoint lenses in segment spaces

D 800/1000 | p 1 | m 1 | Conditional points 640

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims a tail bound for arbitrary displacements in symmetric midpoint lenses of two real segment-space models. A finitely supported center and radius control every tail by twice the square root of radius squared minus center-norm squared. The common head works for all displacements. Both norms have asymptotic midpoint convexity despite lacking asymptotically uniformly convex renormings; the finite-height forest dual is reflexive.

Archive: unresolved no match. Categorical lenses and zero-spacing midpoint processes are unrelated uses of these terms.

Score basis: A simultaneous lens-tail inequality requires careful segment-space norm decomposition, but explicit models offer a focused route.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0639 | Family 331

Distortion of countably branching diamonds from midpoint and tree energies

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims explicit lower bounds on distortion of countably branching diamond graphs into specified segment-forest and coordinate-predual spaces. A depth-k diamond requires distortion squared at least one plus k divided by four. Further path-cost and recursive norms receive power-type bounds. The argument derives these estimates from midpoint tail control and direct tree energies, linking concrete Banach geometry to increasing graph depth.

Archive: related target. The candidate concerns finite-metric Euclidean distortion under tree-metric mixtures, a different target space and hypothesis.

Score basis: Sharp depth-dependent graph distortion follows a structured energy route but needs quantitative control of infinitely branching configurations.

[Read manuscript](#) | OPDP ID 40384685 | [Family formalization notes](#)

OAI-M0640 | Family 331

Exact asymptotic moduli in a Daugavet subspace of L_1

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims exact averaged-midpoint and one-sided asymptotic moduli for certain infinite-dimensional real subspaces of integrable functions. Their unit balls are totally bounded in measure and their norms have the Daugavet property. The moduli are the specified piecewise-linear maxima, and the same weak-neighborhood geometry excludes every asymptotically uniformly convex equivalent norm. A quantitative Kadets-Werner construction supplies an example meeting both assumptions.

Archive: unresolved no match. Algebraic moduli-space compactness and Kronecker Euler-characteristics are unrelated to Banach asymptotic moduli.

Score basis: Exact moduli require matching geometric bounds, while explicit known construction frameworks improve tractability.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0641 | Family 331

Midpoint convexity from bounded tree potentials and path costs

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an explicit averaged midpoint-convexity bound for four Banach tree-potential models, while excluding every asymptotically uniformly convex renorming. On finite-height trees, a global constraint norm equals the least additive cost of Hilbert vectors and root paths. Certain outer Hilbert sums are reflexive. These constructions distinguish midpoint geometry of the given norm from asymptotic convexity achievable through any equivalent norm.

Archive: unresolved no match. The candidate path-tree and graph-width conjectures concern combinatorial graphs rather than these Banach norms.

Score basis: Explicit tree decompositions improve tractability, but a renorming-independent obstruction remains nontrivial.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0642 | Family 331

Independent products in real L1: asymptotic midpoint convexity without AUC renormings

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims midpoint-convex real subspaces of integrable functions built from products of independent exponential or Gaussian-square multipliers along countably branching tree paths. Neither admits an asymptotically uniformly convex equivalent norm. The renorming obstruction extends to every positive nonconstant mean-one multiplier with finite second moment, while the positive midpoint estimates are asserted for the two specific laws. The two quantifier ranges therefore differ.

Archive: unresolved no match. Truncated random-matrix eigenvalues and cross-Sperner products are unrelated probabilistic and combinatorial targets.

Score basis: Product independence provides a concrete analytic route, while tree geometry must rule out all equivalent convex renormings.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0643 | Family 331

Midpoint convexity from two recursive potentials

D 775/1000 | p 1 | m 1 | Conditional points 600.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims quantitative midpoint convexity for tree norms defined by two least nonnegative recursive fields. Euclidean child aggregation gives cubic bounds in root-sum models, including reflexive versions; variable aggregation exponents produce further cubic and sixth-power estimates. Every specified model lacks an asymptotically uniformly convex equivalent norm. These explicit recursive constructions address Banach asymptotic geometry, not rank-one quasiconvexity of matrix-valued variational integrands.

Archive: unresolved no match. The recursive-target and Morrey candidates involve different mathematical structures.

Score basis: Recursion makes estimates constructive, but matching tail bounds and renorming obstructions requires careful infinite-dimensional analysis.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0644 | Family 332

Metric Markov Cotype Two of ℓ_1

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims metric Markov cotype two for the real absolutely summable sequence space, with an explicit universal constant. It implies extension of every Lipschitz map from a subset of a real Hilbert space into that sequence space with a universal loss in Lipschitz constant. The theorem applies to arbitrary subsets, not merely finite-dimensional domains. Markov cotype is a specific random-walk property, distinct from ordinary cotype.

Archive: related target. The metric-cotype gap question concerns a different cotype invariant, not this Markov-cotype and extension statement.

Score basis: A uniform random-walk inequality must support unrestricted Lipschitz extension into a nonreflexive target.

[Read manuscript](#) | OPDP ID 40319464

OAI-M0645 | Family 333

Smooth isometric immersions of closed surfaces into Euclidean four-space

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every closed smooth Riemannian surface admits a smooth isometric immersion into Euclidean four-space. Orientability is not assumed, and self-intersections are permitted. Thus the result controls the metric exactly with smooth regularity but does not produce an embedding or a three-dimensional realization. Its global closed-surface scope differs from asymptotically Euclidean noncompact immersion questions and holomorphic Kähler immersion targets.

Archive: related target. The candidate is asymptotically Euclidean and does not state the same closed-surface four-space target.

Score basis: Low codimension and smooth exact metric realization require substantial geometric PDE and global compatibility control.

[Read manuscript](#) | OPDP ID 40362256 | [Family formalization notes](#)

OAI-M0646 | Family 334

A Smooth Metric with No Local Isometric Immersion into Three-Space

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a smooth positive-definite surface metric agreeing with the Euclidean metric to every order at the origin, yet admitting no smooth local isometric immersion into Euclidean three-space near that point. The obstruction therefore survives formal flatness to infinite order. It refutes an unrestricted smooth local realization statement, not necessarily piecewise-smooth or lower-regularity flexible immersion assertions.

Archive: related target. A piecewise-smooth existence conjecture permits weaker regularity than the smooth immersions excluded here.

Score basis: An infinite-order-flat metric must retain a genuine nonlinear local immersion obstruction.

[Read manuscript](#) | OPDP ID 40341696 | [Family formalization notes](#)

OAI-M0647 | Family 335

An integral scalar curvature bound for real simplicial volume

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims an all-dimensional integral scalar-curvature lower bound proportional to real simplicial volume. For every dimension at least three, a dimensional constant controls the integral of the negative scalar-curvature part to power half the dimension, for every smooth metric on a closed oriented manifold. The proof depends on a companion vanishing theorem. The scale-invariant integral statement is stronger than a qualitative nonnegative-curvature vanishing conclusion.

Archive: related target. The named quantitative comparison candidate truncates its formula and begins at dimension two; exact conventions remain unconfirmed.

Score basis: A sharp type of topological-curvature control across all metrics and dimensions needs broad geometric analysis.

[Read manuscript](#) | OPDP ID 40383303

OAI-M0648 | Family 335

Positive scalar curvature forces rational inessentiality

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that strictly positive scalar curvature forces every closed oriented smooth manifold's rational fundamental class to vanish in its fundamental-group classifying space. No spin, group, or upper-dimensional assumption is imposed. Consequences exclude positive scalar curvature on closed aspherical manifolds, classify their nonnegative metrics as flat, and give simplicial-volume vanishing. Rational inessentiality is the principal statement, stronger than a theorem confined to spin manifolds.

Archive: same target. The candidate's rational-essential-manifold prohibition is the contrapositive of the asserted principal theorem.

Score basis: Removing spin and group restrictions from scalar-curvature obstructions requires exceptionally deep global geometric-topological control.

[Read manuscript](#) | OPDP ID 40403582

OAI-M0649 | Family 336

Spectral scalar curvature and uniform Urysohn width

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims uniform codimension-two Urysohn width for complete boundaryless manifolds in dimensions at least four when negative four times Laplacian plus scalar curvature has quadratic-form lower bound one. A continuous map to a complex of dimension at most dimension minus two has entire fibers of uniformly bounded diameter in the original metric. The spectral hypothesis differs from either pointwise or ball-volume-based macroscopic curvature conditions.

Archive: related target. The macroscopic-curvature candidate has a different input condition; no equivalence with the paper's spectral lower bound is established.

Score basis: A spectral condition must yield geometric fiber-diameter control without pointwise curvature bounds.

[Read manuscript](#) | OPDP ID 40372674

OAI-M0650 | Family 336

Spectral scalar curvature and Urysohn width in dimension three

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims that a complete smooth three-manifold with positive lower quadratic-form bound for negative four times Laplacian plus scalar curvature maps continuously to a graph with fiber diameters at most 500 divided by the square root of that bound. No orientability, spin, compactness, or bounded geometry is assumed. It establishes a dimension-three spectral version rather than the archive's all-dimensional width-and-macroscopic-dimension bundle.

Archive: partial or special case. The archived scalar-curvature bundle ranges over dimensions and additional conclusions; the reported theorem treats the three-dimensional spectral case.

Score basis: The fixed dimension provides a focused route, but spectral-to-geometric width control remains substantial.

[Read manuscript](#) | OPDP ID 6700033

OAI-M0651 | Family 336

Positive scalar curvature and uniform codimension-two width

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims the quantitative continuous scalar-curvature width theorem in every dimension at least four. Complete connected boundaryless manifolds with scalar curvature at least one map to dimension-minus-two complexes with uniformly bounded entire fibers. It also gives a continuous macroscopic-dimension consequence for universal covers in dimensions at least two. Continuity, rather than stronger Lipschitz control or additional bounded-geometry assumptions, is the explicitly reported map regularity.

Archive: related target. The macroscopic-ball curvature record has different hypotheses from the pointwise scalar-curvature theorem.

Score basis: Uniform all-dimensional collapsing maps require global scalar-curvature and topological decomposition arguments.

[Read manuscript](#) | OPDP ID 40352868

OAI-M0652 | Family 337

Generalized Cartan-Hadamard isoperimetry and Euclidean equality rigidity

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims generalized Cartan-Hadamard isoperimetry in every dimension: finite-volume finite-perimeter sets in complete simply connected manifolds with sectional curvature at most a nonpositive constant have at least the equal-volume model-ball perimeter. For bounded positive-volume sets, Euclidean equality identifies a region intrinsically isometric to a round Euclidean ball. The scored target includes negative-curvature model comparison and equality rigidity beyond the abbreviated archived Euclidean inequality question.

Archive: related target. Original record 1374 publishes 631 for the Cartan-Hadamard inequality; the generalized comparison-and-rigidity scope is estimated separately.

Score basis: All-dimensional sharp comparison for rough sets and equality rigidity impose broader demands than an abbreviated scalar inequality.

[Read manuscript](#) | OPDP ID 1374 | [Family formalization notes](#)

OAI-M0653 | Family 337

Sharp integral fillings in CAT(0) spaces

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims sharp Euclidean-mass fillings for every compactly supported integral cycle of dimension at least two in a proper CAT(0) space. Integer multiplicities and ambient dimension are unrestricted, and the filling remains compactly supported. It implies a Euclidean Cartan-Hadamard consequence in dimensions at least three. The exponent is the Euclidean power law, not a linear bound requiring an asymptotic-rank restriction.

Archive: related target. The candidate asks linear fillings under asymptotic-rank and cocompactness assumptions, a different exponent and class.

Score basis: Sharp current fillings in arbitrary metric ambient dimension require substantial geometric-measure theory.

[Read manuscript](#) | OPDP ID 40404845 | [Family formalization notes](#)

OAI-M0654 | Family 338

Uniformization of complete Kähler manifolds with positive bisectional curvature

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that every complete connected noncompact Kähler manifold with strictly positive holomorphic bisectional curvature is biholomorphic to complex Euclidean space. All complex dimensions are included. The conclusion is holomorphic equivalence, not isometry to the Euclidean metric, and concerns noncompact manifolds rather than projective-space classification in the compact case. No additional growth hypothesis appears in the reported main theorem.

Archive: related target. The named Yau record truncates its properties; its precise curvature and growth assumptions require full-source comparison.

Score basis: Global holomorphic uniformization from curvature alone needs deep complex geometric analysis.

[Read manuscript](#) | OPDP ID 40372567

OAI-M0655 | Family 339

Entropy equality and local symmetry in negative curvature

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that normalized Liouville measure has maximal entropy for geodesic flow on a closed negatively curved manifold of dimension at least three exactly when the metric is locally symmetric. All rank-one symmetric types and scales are included. This is an equality-rigidity theorem for dynamical entropy, not a general volume-entropy upper bound under scalar-curvature assumptions or a dimension-two statement.

Archive: related target. The Agol-Storm-Thurston candidate concerns scalar curvature and volume entropy, not this Liouville maximal-entropy criterion.

Score basis: Entropy equality must force local geometric symmetry through subtle geodesic dynamics and curvature control.

[Read manuscript](#) | OPDP ID 40360968

OAI-M0656 | Family 340

A counterexample to the nearby Lagrangian conjecture

D 709/1000 | p 1 | m 1 | Conditional points 502.681

Published original CFSD-1000 target difficulty

This paper claims a closed exact embedded Lagrangian in the cotangent bundle of a product of spheres that is diffeomorphic to the base yet not Hamiltonian isotopic to its zero section. One sphere dimension is nine and the other depends on a sufficiently large even parameter. A single such example would refute the unrestricted nearby Lagrangian statement, without asserting failure for every base or dimension.

Archive: same target. The archive quantifies over all closed bases and exact Lagrangians, so the asserted product-of-spheres example supplies a complete negative answer.

Score basis: The counterexample directly refutes the published unrestricted nearby Lagrangian target; its original score is preserved.

Scoring target: Not every closed exact Lagrangian in a cotangent bundle is Hamiltonian isotopic to the zero section.

Progress: Conditional on mathematical acceptance, the manuscript claims a full answer to the explicitly scored target. This does not establish competition eligibility, novelty, or accepted credit for a broader related archive problem.

[Read manuscript](#) | OPDP ID 11300005

OAI-M0657 | Family 341

Deforming hypersymplectic four-manifolds to hyperkähler triples

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a cohomology-preserving deformation of every smooth normalized positive triple of closed two-forms on a closed oriented four-manifold to a hyperkähler triple. The path remains hypersymplectic, and each individual cohomology class is fixed. Normalization of the intersection matrix is explicit and cannot be dropped casually when fixing all classes. The assertion is a deformation theorem, stronger than merely proving existence of some unrelated hyperkähler structure.

Archive: related target. The named Donaldson candidate has a truncated setup; the normalization convention needs confirmation before exact identity.

Score basis: Preserving positivity and three cohomology classes throughout a global deformation requires substantial nonlinear geometry.

[Read manuscript](#) | OPDP ID 40335870

OAI-M0658 | Family 342

Taming implies compatibility on four-manifolds

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that an almost complex structure on a closed four-manifold tamed by some symplectic form is compatible with some symplectic form. The almost complex structure remains fixed; the compatible form may differ from the taming one. This resolves an existence question, not every prescribed-volume Calabi-Yau equation or all a priori estimates appearing in related Donaldson conjectures.

Archive: related target. The candidate concerns Calabi-Yau estimates with extra prescribed data rather than existence of a compatible form.

Score basis: Constructing compatibility for a fixed almost complex structure requires global symplectic and elliptic control.

[Read manuscript](#) | OPDP ID 40396388 | [Family formalization notes](#)

OAI-M0659 | Family 343

Symplectic Ball Packings in Higher Dimensions

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a complete criterion for disjoint symplectic packing of finitely many closed balls into a larger open ball in real dimensions at least six. Total capacity-to-the-half-dimension volume sum and each pairwise capacity sum must be strictly smaller than their target bounds, and those conditions are asserted sufficient. Embeddings extend to neighborhoods of closed source balls, making strict inequalities and source conventions essential.

Archive: related target. The curated packing target is closely related; its full open-versus-closed and strictness conventions need comparison.

Score basis: Proving sufficiency beyond volume and two-ball obstructions needs constructive higher-dimensional symplectic flexibility.

[Read manuscript](#) | OPDP ID 51000185 | [Family formalization notes](#)

OAI-M0660 | Family 344

The metric Blaschke theorem

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims the metric Blaschke theorem: a connected closed smooth Riemannian manifold with global injectivity radius equal to diameter is, up to scale, a standard compact rank-one symmetric space. Every positive dimension is included. The conclusion is metric isometry, not only homeomorphism or cohomology classification. It is distinct from projective-connection variants whose hypotheses concern embedded geodesics rather than the injectivity-radius equality.

Archive: same target. The candidate's complete sentence states isometric compact-rank-one-symmetric classification of Blaschke manifolds.

Score basis: Metric rigidity from a global radius equality requires controlling geodesic and curvature structure in arbitrary dimensions.

[Read manuscript](#) | OPDP ID 40399337

OAI-M0661 | Family 345

Infinitely many closed geodesic images on every Riemannian sphere

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims infinitely many prime closed geodesics with distinct images on every smoothly metrized standard sphere of dimension at least two, including degenerate higher-dimensional metrics. The two-sphere case is acknowledged classical. Existing finite-cover and three-manifold results extend the conclusion to every nonempty closed smooth three-manifold. The new higher-sphere target concerns closed image-distinct trajectories, not nonclosed geodesics in a complete plane.

Archive: related target. The archived plane question concerns properly embedded nonclosed geodesics on a noncompact surface.

Score basis: Infinitude despite degeneracy requires strong variational and iteration control beyond producing one closed geodesic.

[Read manuscript](#) | OPDP ID 40401157

OAI-M0662 | Family 346

A Codimension-One Bound for the Singular Set of a Stationary Integral Varifold

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that the singular set of every stationary integral varifold in a Euclidean open set has Hausdorff dimension at most one less than its own positive dimension. Ambient codimension is arbitrary. The asserted bound is sharp and imposes neither minimizing nor stability hypotheses. It strengthens almost-everywhere regularity by quantitatively bounding exceptional-set dimension, with singular-set conventions needing comparison to the recorded integer-rectifiable formulation.

Archive: related target. The named codimension-one record truncates its singular-set definition and ambient conventions.

Score basis: Stationarity alone must yield sharp exceptional-set control without minimizing or stability machinery.

[Read manuscript](#) | OPDP ID 40367349

OAI-M0663 | Family 346

Almost-everywhere regularity of stationary integral varifolds

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims zero same-dimensional measure for the singular set of every stationary integral varifold in Euclidean open sets, in arbitrary positive dimension and codimension. Almost every support point has a neighborhood equal to a constant integer multiple of a smooth embedded minimal submanifold. The result also covers round spheres. The theorem concerns full integral stationarity rather than assuming minimizing behavior or codimension one.

Archive: related target. The original question says parametrized integer stationary varifolds; correspondence of its parametrized convention to the reported integral class requires verification.

Score basis: General stationary-varifold regularity requires replacing the stronger hypotheses that usually enable local smoothness.

[Read manuscript](#) | OPDP ID 30003908

OAI-M0664 | Family 347

Hamiltonian Fixed Points Below the Stable Morse Number in Dimension Twenty-Two

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims simply connected closed Kähler manifolds in fixed real dimension twenty-two with nondegenerate Hamiltonian fixed-point counts below stable Morse number. For parameter m , the asserted counts are $80 + 1952m$ versus stable Morse number $80 + 1968m$. All orbit loops are contractible, giving an unbounded deficit in one dimension. The target is the stable-Morse variant, not every weaker Betti-number bound.

Archive: related target. The curated Arnold record does not fix which critical-point lower-bound version it intends; the manuscript explicitly targets stable Morse number.

Score basis: The construction needs exact fixed-point counts together with integral Morse-theoretic lower bounds.

[Read manuscript](#) | OPDP ID 50000166 | [Family formalization notes](#)

OAI-M0665 | Family 347

Sharpness of the Cyclic Integral Floer Bound Below the Stable Morse Number

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a real-dimension-3332 simply connected closed Kähler example attaining the cyclic integral Floer bound while falling exactly 32 below stable Morse number. All 1,872,232 fixed points are nondegenerate with contractible loops, and the minimal Chern number is one. This separates two integer lower bounds even when the weaker one is sharp. The large dimension belongs to the construction, not a universal restriction on possible examples.

Archive: unresolved no match. The braid-homology and number-field bound candidates have different invariants and objects.

Score basis: Exact integral-homology torsion and fixed-point enumeration require substantial but explicit construction bookkeeping.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0666 | Family 347

Hamiltonian Fixed Points Below the Stable Morse Number

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims a simply connected closed Kähler manifold with a nondegenerate Hamiltonian diffeomorphism having sixteen fewer fixed points than stable Morse number. Every periodic orbit is contractible. The construction combines a Hamiltonian involution, integral homology torsion at two primes, and complex blowups. It targets a strengthened integral Morse bound, without claiming that every ordinary homological version of Arnold's fixed-point inequality is false.

Archive: related target. The archive's general critical-point language does not specify stable versus ordinary Morse or homological bounds.

Score basis: Coordinating blowups, torsion-sensitive Morse estimates, and nondegenerate fixed-point counts creates the main difficulty.

[Read manuscript](#) | OPDP ID 50000166 | [Family formalization notes](#)

OAI-M0667 | Family 347

A nondegenerate counterexample to the Morse-number Arnold bound

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims closed symplectic twelve-manifolds whose nondegenerate Hamiltonian diffeomorphisms have fewer fixed points than ordinary Morse number, with deficits arbitrarily large within that dimension. All Hamiltonian orbit loops are contractible. This directly targets the Morse-number form of Arnold's assertion. It does not settle the separate Arnold-Givental intersection question or automatically contradict lower bounds stated only through Betti numbers or cup length.

Archive: related target. The original 665-score record bundles Arnold and Arnold-Givental targets without displaying the exact Morse-number variant; its score is not inherited.

Score basis: A finite-dimensional geometric construction must distinguish ordinary Morse complexity from all its weaker homological bounds.

[Read manuscript](#) | OPDP ID 4900001 | [Family formalization notes](#)

OAI-M0668 | Family 347

Three fixed points on the symplectic quadric threefold

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a smooth Hamiltonian diffeomorphism of the complex quadric threefold with exactly three fixed points, at least one degenerate. Its critical number and unit-inclusive rational cup length are asserted to equal four. Thus the example refutes those unrestricted count conventions. The distinction between ordinary cup length and a unit-inclusive count is essential: three fixed points alone do not violate a convention assigning cup length three.

Archive: related target. The original record writes $cl(M)$ without exposing whether the count includes the unit; that convention determines whether this is a counterexample.

Score basis: The example requires exact degenerate fixed-point counts and careful topological normalization of the lower bound.

[Read manuscript](#) | OPDP ID 30004183 | [Family formalization notes](#)

OAI-M0669 | Family 348

Zero-Plane Rigidity for Einstein Four-Manifolds

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims zero-plane rigidity for closed positive-Einstein four-manifolds with nonnegative sectional curvature. If one sectional plane has zero curvature, the universal Riemannian cover is a product of two round two-spheres; under Ricci normalization three, both radii are inverse square root three. This identifies the boundary case of the curvature cone, leaving strictly positive sectional curvature to a separate classification theorem.

Archive: partial or special case. The candidate asks full nonnegative-curvature Einstein classification, while this theorem addresses its zero-plane case.

Score basis: Equality analysis of coupled Weyl blocks requires delicate four-dimensional curvature rigidity.

[Read manuscript](#) | OPDP ID 40340403 | [Family formalization notes](#)

OAI-M0670 | Family 348

An L^2 Einstein Gap for Nonnegatively Curved Four-Manifolds

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a universal scale-invariant squared-integrable gap for trace-free Ricci curvature on simply connected closed nonnegatively curved four-manifolds, assuming a stated Einstein-classification premise. Sufficiently small energy forces the smooth manifold to be a four-sphere, complex projective plane, or two-sphere product. No extra geometry bounds are imposed. The conclusion is diffeomorphism classification, not isometry of the original near-Einstein metric.

Archive: related target. The candidate adds a circle action; this paper instead imposes small Ricci energy and an Einstein-classification premise.

Score basis: Gap compactness without auxiliary geometry bounds is demanding, and the classification premise is an explicit dependency.

[Read manuscript](#) | OPDP ID 40317829 | [Family formalization notes](#)

OAI-M0671 | Family 348

Positively curved Einstein four-manifolds

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that every closed connected Einstein four-manifold with strictly positive sectional curvature is, up to scale and isometry, round four-sphere, Fubini-Study complex projective plane, or round real projective four-space. Orientability is not assumed. This handles the interior positive-curvature class, distinct from the zero-plane nonnegative boundary case and from classification of arbitrary nonnegative metrics without the Einstein equation.

Archive: partial or special case. The candidate includes nonnegative curvature and simple connectivity; the reported strict-positive classification has a different class.

Score basis: Positive Einstein curvature must force full metric rigidity, requiring detailed four-dimensional curvature estimates.

[Read manuscript](#) | OPDP ID 40340403 | [Family formalization notes](#)

OAI-M0672 | Family 349

The Solomon-Yau least-volume theorem

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims the Solomon-Yau least-volume bound for every closed connected minimal hypersurface immersion into a unit round sphere in dimensions at least two. A non-totally-geodesic image has volume at least the smallest minimal Clifford product. Volume is computed on the immersion domain, retaining covering multiplicities. This is a spherical hypersurface comparison, not directly an unrestricted classification of all least-density singular cones.

Archive: related target. The Simons-cone candidate concerns hypercone density and possible area minimization, rather than the same spherical immersion class.

Score basis: An all-dimensional sharp minimal-volume gap requires global geometric and spectral control.

[Read manuscript](#) | OPDP ID 40396471

OAI-M0673 | Family 350

Sharp nodal length on smooth surfaces

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a constant times square-root-eigenvalue upper bound for nodal length of every nonzero real Laplace eigenfunction on a fixed smooth closed connected surface. Combining it with an acknowledged known lower bound gives two-sided comparability in this setting. The constant depends on the metric, not on the eigenfunction or eigenvalue. Smooth surfaces are the target, rather than analytic metrics alone or higher-dimensional nodal volume.

Archive: same target. The candidate's full sentence requests square-root-eigenvalue comparability for surface nodal lengths, the assertion supplied here.

Score basis: Sharp smooth-metric upper bounds require fine eigenfunction oscillation and geometric nodal control.

[Read manuscript](#) | OPDP ID 40403290 | [Family formalization notes](#)

OAI-M0674 | Family 350

Smooth counterexamples to Yau's nodal upper bound in dimensions three and four

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims smooth fixed-metric counterexamples to the square-root-eigenvalue nodal upper bound in dimensions three and four. One metric lies arbitrarily smoothly close to the round three-sphere metric; another is on a two-sphere times two-torus. Exact real eigenfunction sequences have unbounded normalized nodal measures. Each sequence uses one metric, so the effect is not merely a family of metric-dependent constants diverging across different metrics.

Archive: related target. The general nodal-volume record is truncated; the dimension-two surface statement is not contradicted by these higher-dimensional examples.

Score basis: Fixed-metric spectral constructions must combine exact eigenfunctions with rapidly growing nodal geometry.

[Read manuscript](#) | OPDP ID 40375943 | [Family formalization notes](#)

OAI-M0675 | Family 350

Power-law violations of Yau's nodal upper bound

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a smooth metric on four-sphere times circle with eigenfunctions whose nodal four-volume exceeds eigenvalue to one-half plus a fixed positive exponent by an unbounded ratio. The metric is held fixed along the sequence. This gives a power-law violation in dimension five, stronger than merely unbounded square-root normalization, and refutes allowing an arbitrarily small positive power loss for that unrestricted smooth upper-bound formulation.

Archive: related target. The nodal-volume candidate omits its complete smoothness and exponent formulation; it is retained as a related parent.

Score basis: Power-law fixed-metric violations require stronger spectral-geometric growth than a logarithmic excess.

[Read manuscript](#) | OPDP ID 40375943 | [Family formalization notes](#)

OAI-M0676 | Family 351

A closed Ricci flow with bounded scalar curvature and finite-time curvature blowup

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a closed-manifold Ricci flow in sufficiently high dimension with bounded scalar curvature but full curvature diverging at a finite maximal time. In one fixed high dimension, examples have two-sided power-law blowup with arbitrarily large exponents. Thus the unrestricted scalar-curvature extension criterion is asserted false, while lower-dimensional versions require separate treatment. Bounded scalar trace alone need not control the full curvature tensor.

Archive: related target. The Hamilton scalar-blowup record truncates its dimensional and quantitative formulation; the reported examples are high-dimensional.

Score basis: Singularity construction must decouple bounded scalar trace from unbounded tensor curvature through a nonlinear flow.

[Read manuscript](#) | OPDP ID 40375188

OAI-M0677 | Family 351

Bounded scalar curvature and smooth extension of four-dimensional Ricci flow

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims smooth extension of Ricci flow on any closed real four-manifold through a finite time whenever scalar curvature remains uniformly bounded. The extension stays on the same manifold, not merely through a generalized singular-space continuation. This establishes the four-dimensional criterion while a companion presents high-dimensional counterexamples. The dimensional restriction therefore matters to both correctness and comparison with an all-dimensional archived conjecture.

Archive: partial or special case. The archived criterion is stated for general dimension; the reported positive theorem is four-dimensional.

Score basis: Converting scalar control into full smooth extension in dimension four requires demanding blowup and compactness analysis.

[Read manuscript](#) | OPDP ID 40375188

OAI-M0678 | Family 351

Path selection and an elliptic inequality on degenerating Ricci-flat trees

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims a sequential elliptic estimate for the renormalized Einstein-Hilbert functional on a fixed finite tree of four-dimensional Ricci-flat spaces. As neck lengths diverge and a weighted Ricci error tends to zero, the functional is little-o of that error. Root-end decay exceeds one and joined quotient groups are nontrivial. A multivariable asymptotic path-selection theorem supplies the route, rather than a general arbitrary-tree estimate.

Archive: unresolved no match. Parse-word trees and Kähler expansion coefficients are different targets.

Score basis: Degenerating neck geometry and path-dependent asymptotic expansions require substantial analytic coherence and weight control.

[Read manuscript](#)

OAI-M0679 | Family 352

A finite-time singularity of Calabi flow on projective space

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a smooth Kähler metric in the Fubini-Study class of complex projective ten-space whose Calabi flow develops unbounded scalar curvature in finite time. That class contains a constant-scalar-curvature metric. A single such singular solution would defeat global smooth existence for every initial metric in a qualifying class, independently of whether convergence could have been proved had the flow remained smooth.

Archive: same target. The curated target asserts global existence and convergence in every class containing a constant-scalar-curvature metric; finite-time failure in such a class contradicts it.

Score basis: A fourth-order geometric-flow singularity within a canonical stable class needs rigorous nonlinear construction.

[Read manuscript](#) | OPDP ID 50000407

OAI-M0680 | Family 353

A Smooth Nonquadratic Entire Affine Maximal Graph in Dimension Ten

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a smooth nonquadratic entire solution of the classical affine maximal graph equation in dimension ten with Hessian positive definite at every point. It refutes a smooth entire-graph quadraticity assertion without additional completeness assumptions. No global uniform Hessian bound or completeness of the affine metric is asserted. Consequently, a Bernstein formulation requiring that completeness cannot be declared refuted simply from the constructed graph.

Archive: related target. The candidate explicitly begins a completeness condition absent from the reported example.

Score basis: Constructing an entire smooth convex nonlinear solution while excluding quadraticity requires delicate fourth-order analytic design.

[Read manuscript](#) | OPDP ID 40392282 | [Family formalization notes](#)

OAI-M0681 | Family 353

The affine Bernstein theorem in dimensions three through nine

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims affine Bernstein rigidity in dimensions three through nine. A smooth locally uniformly convex affine-maximal graph with complete induced Euclidean metric must be an elliptic paraboloid. A second theorem drops the graph assumption for affine-complete immersed hypersurfaces in the same dimensions. Euclidean completeness and affine completeness are distinct hypotheses attached to different statements; neither extends this asserted dimension range to every dimension.

Archive: partial or special case. The archived entire-graph question has a truncated completeness convention and begins at dimension two; exact full-range identity is not established.

Score basis: Fourth-order rigidity requires global convexity and completeness analysis across a bounded dimension range.

[Read manuscript](#) | OPDP ID 40392282 | [Family formalization notes](#)

OAI-M0682 | Family 354

The Isoperimetric Conjecture for the Cubic Flat Three-Torus

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims the isoperimetric profile and complete minimizer classification of the cubic flat three-torus. Minimizers are balls, circular tubes around shortest closed geodesics, coordinate slabs, and their complements, with transition volumes explicitly given by four pi over 81 and one over pi. The profile minimizes the three corresponding perimeter expressions. Equality at transitions is included, rather than just proving candidates optimal within separate symmetry classes.

Archive: same target. The candidate explicitly lists spheres, cylinders, and slabs as cubic-torus optima, the classification supplied by the reported theorem.

Score basis: Complete finite-perimeter classification requires ruling out every other topology and shape across all volumes.

[Read manuscript](#) | OPDP ID 40377552 | [Family formalization notes](#)

OAI-M0683 | Family 355

Uniqueness of tangent flows at the first singular time of embedded surface mean-curvature flow

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims unique tangent flows at every point of the first singular time of compact embedded surface mean-curvature flow in three-space. Every fixed-center rescaling converges smoothly on negative-time compact intervals to one multiplicity-one self-shrinker, with position and axes fixed in ambient coordinates. No mean-convexity or prescribed model is assumed. Later generalized singularities and higher-dimensional flows remain outside the reported scope.

Archive: partial or special case. The archive asks tangent uniqueness for mean-curvature singularities generally; the theorem concerns embedded surfaces at their first singular time.

Score basis: Unique multiplicity-one blowup with no mean-convexity requires strong compactness and nonlinear singularity control.

[Read manuscript](#) | OPDP ID 40397731

OAI-M0684 | Family 356

Gigli's distributional curvature characterization of Alexandrov spaces

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims Gigli's equivalence between lower Alexandrov curvature and a specified distributional sectional-curvature bound in dimensions at least two. The converse uses a full-support synthetic Ricci-lower-bound space with Hausdorff reference measure and Gigli's original global test classes. Those measure and test conventions are substantive, not interchangeable with arbitrary weighted spaces or a local smooth interpretation. The result characterizes nonsmooth sectional curvature through analytic data.

Archive: related target. The named RCD-Alexandrov record truncates its complete measure and distributional-test conventions.

Score basis: Nonsmooth curvature equivalence requires matching global test calculus with metric triangle comparison.

[Read manuscript](#) | OPDP ID 40341787 | [Family formalization notes](#)

OAI-M0685 | Family 356

Weak Hessian bounds along every geodesic in RCD spaces

D 875/1000 | p 1 | m 1 | Conditional points 765.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that a bounded globally Lipschitz function with distributional Hessian bounded above by a bounded continuous function satisfies the corresponding second-derivative inequality along every minimizing geodesic in a full-support finite-dimensional RCD space. The word every is essential: weak measure-based calculus must control possibly exceptional geodesics. The theorem is a Hessian-to-geodesic implication, not an assertion about constancy of essential dimension.

Archive: related target. The candidate's RCD dimension-constancy problem concerns a distinct geometric invariant.

Score basis: Extending weak Hessian information to all geodesics needs careful representative and approximation control.

[Read manuscript](#) | OPDP ID 40402514 | [Family formalization notes](#)

OAI-M0686 | Family 357

Bi-Lipschitz Coordinates at Regular Points of Noncollapsed RCD Spaces

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a bi-Lipschitz coordinate neighborhood around every regular point of a noncollapsed RCD space in every integer dimension at least two. The neighborhood uses restricted ambient distance, and the distortion bound depends only on dimension, while neighborhood size may depend on the point and geometry. The result is an open-neighborhood theorem, stronger than rectifiability on almost-everywhere subsets, and does not address gluing along singular boundaries.

Archive: related target. The doubling candidate concerns boundary spaces, not regular-point coordinate neighborhoods.

Score basis: Upgrading tangent regularity to uniform-distortion open coordinates requires substantial nonsmooth geometric control.

[Read manuscript](#) | OPDP ID 40364475

OAI-M0687 | Family 358

A Three-Manifold Without Conjugate Points and Without a Nonpositively Curved Metric

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a closed orientable smooth three-manifold admitting a smooth metric without conjugate points but no smooth metric with nonpositive sectional curvature. This separates a geodesic-Jacobi-field condition from existence of a globally curvature-controlled metric on the same underlying manifold. The obstruction excludes all alternative nonpositive metrics, not merely showing that the constructed no-conjugate-point metric has some positive sectional curvature.

Archive: unresolved no match. Harmonic-map-flow existence and even-dimensional Hopf Euler characteristics do not identify this three-manifold existence target.

Score basis: Metric construction must coexist with a topological obstruction to every alternative nonpositive-curvature metric.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0688 | Family 359

A negatively pinched Kähler threefold without bounded holomorphic coordinates

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a contractible domain in complex dimension three with a complete Kähler metric whose real sectional curvature is bounded between two negative constants. No bounded holomorphic map into complex three-space has everywhere nonzero Jacobian, so the domain is not biholomorphic to a bounded domain. Both upper and lower negative pinching are retained. The conclusion concerns holomorphic coordinate uniformization, not bounded cohomological fundamental classes.

Archive: related target. The candidate asks a bounded-fundamental-class property on pinched negatively curved manifolds, a different invariant.

Score basis: Negative pinching must be reconciled with a global obstruction to bounded holomorphic coordinates.

[Read manuscript](#) | OPDP ID 40369506 | [Family formalization notes](#)

OAI-M0689 | Family 359

One-sided negative sectional curvature and the holomorphic Liouville property

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims a complete Kähler domain in some fixed finite complex dimension, diffeomorphic to Euclidean space, with real sectional curvature at most minus one and only constant bounded holomorphic functions. Curvature is unbounded below, so no two-sided pinching is asserted. This gives a one-sided negative-curvature Liouville counterexample and differs from compact canonical-bundle ampleness or two-sided negatively pinched bounded-domain questions.

Archive: unresolved no match. The compact holomorphic-curvature candidates impose different compactness and bundle conclusions.

Score basis: One-sided curvature control must support a global holomorphic Liouville obstruction without accidental lower pinching.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0690 | Family 360

Global Support and Convex Injectivity Domains under Weak MTW

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims convexity of every tangent injectivity domain on a smooth compact boundaryless Riemannian manifold satisfying weak Ma-Trudinger-Wang curvature. Conjugate cut points are allowed. More generally, squared-distance-potential subgradients are minimizing velocities with global supporting mountains, without density assumptions. The theorem addresses global geometric support from a weak cost-curvature condition rather than regularity obtained only for particular transport densities.

Archive: related target. The Villani record gives the central implication but truncates ambient compactness and exact weak-curvature conventions.

Score basis: Cut-locus geometry and global support must follow from weak fourth-order cost curvature.

[Read manuscript](#) | OPDP ID 40355190 | [Family formalization notes](#)

OAI-M0691 | Family 360

Uniform Bi-Holder Transport from Weak MTW

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims uniform Hölder regularity of optimal transport maps and their inverses under weak Ma-Trudinger-Wang curvature on a fixed compact boundaryless manifold. Positive upper and lower density bounds suffice, with no further density smoothness. Maps have homeomorphic representatives, and conjugate cut points are permitted. The common exponent and constant depend on geometry and density bounds, not each individual pair of densities.

Archive: related target. The original question asks regularity when MTW fails; this theorem assumes weak MTW and does not answer that failure regime.

Score basis: Uniform forward-and-inverse regularity requires substantial cut-locus and transport-potential estimates.

[Read manuscript](#) | OPDP ID 30001863 | [Family formalization notes](#)

OAI-M0692 | Family 361

A counterexample to integer-degree harmonic dimension comparison

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a complete smooth nonnegatively Ricci-curved metric on some even-dimensional Euclidean space with more integer-growth-degree harmonic functions than the Euclidean polynomial count. Dimension is at least eight, and growth degree at least two. The metric is Euclidean near the origin, has positive volume ratio below one, and nonunique tangent cones at infinity. Thus the claimed violation concerns fixed integer-degree dimension comparison, not all asymptotic growth formulas.

Archive: unresolved no match. Algebraic dynamical degrees and mosaic harmonic degrees are unrelated invariants.

Score basis: Nonunique asymptotic cones must yield a rigorous excess of global harmonic functions while preserving Ricci nonnegativity.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0693 | Family 361

A Three-Dimensional Counterexample to Integer-Degree Harmonic Dimension Comparison

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims three-dimensional nonnegative-Ricci metrics with a fixed-factor excess over the Euclidean degree-k harmonic count. For each allowed volume ratio and excess factor, sufficiently large k has a metric with at least that factor times the square of k plus one independent harmonic functions. Metrics may depend on k and can approach Euclidean geometry bi-Lipschitzly. It does not assert one metric works at every large degree.

Archive: unresolved no match. Functorial seminorm and projective dynamical-degree questions do not match harmonic dimension comparison.

Score basis: Preserving Ricci bounds and prescribed volume ratio while constructing excess harmonic functions requires careful geometric spectral design.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0694 | Family 362

Global classical solutions of the three-dimensional relativistic Vlasov-Maxwell system

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims global smooth existence and uniqueness for arbitrary admissible large initial data of the three-dimensional one-species relativistic Vlasov-Maxwell system. Particle density is initially compactly supported; electromagnetic fields have finite energy and bounded derivatives at every order. No smallness or symmetry is imposed, and smoothness persists on every finite interval. The species count and initial support and field assumptions delimit the reported global regularity theorem.

Archive: partial or special case. The curated regular-finite-energy target does not display the same species, support, and derivative conventions; the abbreviated original Vlasov-Maxwell score is not inherited.

Score basis: Large-data kinetic-field control requires long nonlinear estimates preventing momentum-support and field blowup.

[Read manuscript](#) | OPDP ID 50000191 | [Family formalization notes](#)

OAI-M0695 | Family 363

Nonuniqueness with local conservation for the hard-sphere Boltzmann equation

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims two distinct global entropy solutions of the periodic three-dimensional hard-sphere Boltzmann equation from one nonnegative finite-entropy density with bounded velocity support. Both satisfy exact local mass, momentum, and kinetic-energy conservation, strong integrable-norm continuity, and polynomially weighted collision integrability on bounded intervals. The conservation strengthening distinguishes this example from solutions carrying only global energy inequalities; it does not assert nonuniqueness of smooth classical solutions.

Archive: related target. The candidate concerns an annihilated Boltzmann hierarchy, not this nonlinear periodic equation and entropy-solution class.

Score basis: Maintaining exact local conservation through a weak-solution nonuniqueness construction is particularly demanding.

[Read manuscript](#) | OPDP ID 40381459 | [Family formalization notes](#)

OAI-M0696 | Family 363

Nonuniqueness for the periodic hard-sphere Boltzmann equation

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims two distinct global renormalized hard-sphere Boltzmann solutions on the three-torus with the same nonnegative finite-mass, finite-energy, finite-entropy initial density. They conserve local mass and total momentum and satisfy energy and entropy-dissipation inequalities. Strong integrable-norm continuity and collision integrability are guaranteed on a common initial interval. These admissibility requirements differ from the stronger all-time local conservation asserted by its companion.

Archive: related target. The annihilated-hierarchy uniqueness candidate concerns a different equation hierarchy and function class.

Score basis: Renormalized kinetic nonuniqueness requires constructing distinct admissible limits without violating entropy control.

[Read manuscript](#) | OPDP ID 40381459 | [Family formalization notes](#)

OAI-M0697 | Family 364

The Boltzmann-Grad limit for stable radial potentials on regular kinetic intervals

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims a Boltzmann-Grad derivation for a grand-canonical interacting Newtonian gas with initial pair exclusion and stable finite-range radial potentials. Attractive wells and formed clusters are allowed. Fixed-order rescaled marginals converge in integrable norm throughout each finite interval on which the classical kinetic solution retains Gaussian bounds. This is a regular-lifespan conditional limit, not an unconditional all-time derivation or a fixed-scatterer Lorentz-gas theorem.

Archive: related target. The original Lorentz-gas question has fixed-scatterer dynamics, unlike the interacting Newtonian gas here.

Score basis: Clusters, recollisions, and attractive interactions require extensive uniform microscopic-to-kinetic estimates.

[Read manuscript](#) | OPDP ID 30001644

OAI-M0698 | Family 364

Hard-sphere fluctuations on the regular Boltzmann lifespan

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This manuscript claims finite-dimensional Gaussian fluctuations away from equilibrium for a deterministic grand-canonical hard-sphere gas during its regular Boltzmann lifespan. Smooth initial densities have the stated summable Gaussian bounds. The empirical measure is centered by exact microscopic expectation, and the limiting field solves a linear fluctuating Boltzmann equation. Finite-dimensional convergence does not itself establish convergence in every path-space topology or arbitrary long-time behavior.

Archive: related target. The candidate asks mean Boltzmann-limit convergence; fluctuations and microscopic centering form a different target.

Score basis: Fluctuation cumulants need finer recollision control than a law-of-large-numbers kinetic limit.

[Read manuscript](#) | OPDP ID 40398655

OAI-M0699 | Family 365

Determination of a metric and a unitary connection from one boundary patch

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims simultaneous recovery of a smooth Riemannian metric and unitary connection on a trivial rank-two Hermitian bundle from zero-frequency measurements on one arbitrary nonempty boundary patch. Both inputs and observations use that patch. Recovery is modulo a diffeomorphism and unitary gauge equal to identity there. Dimension is at least three. This combines anisotropic geometric and gauge identification, rather than only reconstructing boundary topological products.

Archive: related target. The mixed cup-product candidate asks a different recovered invariant and truncates its boundary-data convention.

Score basis: Simultaneous metric and connection recovery from local data requires deep elliptic uniqueness and gauge control.

[Read manuscript](#) | OPDP ID 40318968 | [Family formalization notes](#)

OAI-M0700 | Family 365

Smooth anisotropic uniqueness in the Calderón problem from one boundary patch

D 950/1000 | p 1 | m 1 | Conditional points 902.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims smooth anisotropic Calderon uniqueness from arbitrary same-patch boundary measurements in dimensions at least three. The zero-frequency Dirichlet-to-Neumann energy form, with both input and observation restricted to one nonempty open patch, determines the metric up to a diffeomorphism fixing that patch. No real-analytic metric assumption is stated. Gauge and measurement conventions distinguish this strong local-data theorem from an abbreviated full-boundary conjecture.

Archive: related target. The candidate truncates its Cauchy-data and boundary-gauge conclusion; exact equivalence with the stronger same-patch formulation is unconfirmed.

Score basis: Arbitrary smooth anisotropic recovery from a single patch requires exceptionally strong elliptic inverse-problem control.

[Read manuscript](#) | OPDP ID 40381165 | [Family formalization notes](#)

OAI-M0701 | Family 365

Nonuniqueness for bounded measurable scalar conductivities in three dimensions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims two distinct uniformly positive bounded measurable scalar conductivities on a three-dimensional ball with identical full Dirichlet-to-Neumann operators. Both equal one near the boundary. Thus the proposed failure is genuinely interior nonuniqueness at bounded measurable regularity, not an anisotropic diffeomorphism gauge or boundary-value discrepancy. One such example gives a negative answer to unrestricted real isotropic uniqueness in dimensions at least three.

Archive: same target. The curated target explicitly asks uniqueness or a counterexample for real bounded measurable uniformly elliptic isotropic conductivities.

Score basis: Uniform ellipticity and identical full boundary operators must coexist with a genuine scalar interior difference.

[Read manuscript](#) | OPDP ID 50000326 | [Family formalization notes](#)

OAI-M0702 | Family 366

Interior regularity of planar Mumford-Shah minimizers

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims interior regularity of reduced absolute planar Mumford-Shah minimizers with bounded fidelity data. Every interior discontinuity point locally lies on a regular arc, a crack-tip arc, or three regular arcs meeting at 120 degrees. Only finitely many global components meet relatively compact open sets. The statement concerns interior reduced minimizers, not arbitrary stationary configurations or an unqualified boundary-regularity theorem.

Archive: related target. The candidate refers to a structure described above without exposing its complete minimizer and boundary conventions.

Score basis: Crack-tip and triple-junction exclusion of other singularities requires substantial variational blowup and topology control.

[Read manuscript](#) | OPDP ID 40359694

OAI-M0703 | Family 367

The critical dimension for one-phase Bernoulli minimizers

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that seven is the critical dimension for the one-phase Bernoulli variational problem. Nonzero one-homogeneous global minimizers are flat through dimension six, while a nonflat minimizer exists in seven. Dimension reduction gives smooth local free boundaries below seven and sharp singular dimension at most ambient dimension minus seven thereafter; in seven the singular set is locally finite. The claim specifies minimizing, not merely stationary, solutions.

Archive: partial or special case. The original record collects open Bernoulli questions; this theorem addresses its critical-dimension and associated regularity component.

Score basis: The exact threshold requires both universal cone classification and a matching nonflat construction.

[Read manuscript](#) | OPDP ID 30005205 | [Family formalization notes](#)

OAI-M0704 | Family 368

Strong diffeomorphic approximation in three dimensions for $1 \leq p \leq 2$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims strong first-order Sobolev approximation of every three-dimensional homeomorphism between arbitrary bounded domains by smooth diffeomorphisms onto the same target, for exponents from one through two. Both maps and derivatives converge in the corresponding norm. The fixed-target requirement preserves surjectivity rather than merely smoothing locally. It addresses the low-exponent Ball-Evans regime, not analytic complex-domain approximation or Kakeya maximal estimates.

Archive: unresolved no match. Fatou-Bieberbach and Kakeya candidates concern different approximation and maximal-operator targets.

Score basis: Smoothing rough homeomorphisms while preserving global injectivity and the exact target requires careful geometric construction.

[Read manuscript](#)

OAI-M0705 | Family 368

Strong diffeomorphic approximation in three dimensions for $p > 2$

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion manuscript claims the same fixed-target strong diffeomorphic approximation in three dimensions for every finite Sobolev exponent greater than two. Each approximating map is a smooth diffeomorphism between the original bounded domains and belongs to the required Sobolev class. Together with the low-exponent companion, it covers all finite exponents, while no infinite-exponent assertion is made. The higher-integrability regime has its own proof requirements.

Archive: unresolved no match. The retrieved holomorphic approximation and Kakeya questions do not identify this Ball-Evans target.

Score basis: Higher Sobolev control must survive injectivity-preserving surgery and fixed-target approximation.

[Read manuscript](#)

OAI-M0706 | Family 369

Strict hot spots and absence of interior critical points on smooth simply connected planar domains

D 900/1000 | p 1 | m 1 | Conditional points 810

Provisional OPDP / CFSD-1000 reconstruction (C0)

The paper claims that every nonzero first-positive Neumann eigenfunction on a smooth bounded simply connected planar domain has no interior critical point. Consequently, global maxima and minima are strictly on the boundary. Convexity and symmetry are not required, and eigenvalue multiplicity is allowed. This is stronger than merely locating one extremum, but the planar smooth simply connected class does not include every higher-dimensional convex or rough-boundary version.

Archive: related target. The candidate combines simply connected and convex hot-spots regimes with truncated setup; the reported theorem fixes the planar smooth class.

Score basis: Eliminating every interior gradient zero across arbitrary planar geometry requires strong eigenfunction topology and nodal analysis.

[Read manuscript](#) | OPDP ID 40360825 | [Family formalization notes](#)

OAI-M0707 | Family 370

The Subcritical Hénon-Lane-Emden Conjecture

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims nonexistence of strictly positive entire solutions to the weighted Henon-Lane-Emden system whenever the sum of dimension-plus-weight divided by power-plus-one exceeds dimension minus two. Dimensions at least two, positive powers, and real weights are allowed, with continuity at the origin and classical regularity elsewhere but no infinity condition. In the specified weight range, known radial existence completes the complementary classification.

Archive: related target. The weighted-system candidate truncates its equation and regularity conventions, so the exact statement remains source-dependent.

Score basis: All-dimensional weighted coupled Liouville nonexistence requires strong elliptic estimates without asymptotic assumptions.

[Read manuscript](#) | OPDP ID 40372125 | [Family formalization notes](#)

OAI-M0708 | Family 371

Stable self-similar blowup for a supercritical defocusing Schrödinger equation on the torus

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims stable self-similar finite-time blowup for a sufficiently high odd-power defocusing nonlinear Schrödinger equation on the twelve-torus. Blowup data contain an open set in a specified high Sobolev space. Certain Gaussian Fourier data of arbitrarily high regularity therefore have positive blowup probability. The theorem concerns one supercritical dimension-power regime, not every defocusing model or a mere norm-inflation sequence across varying initial scales.

Archive: related target. The candidate addresses supercritical regularity loss with a truncated model and initial-data family, not the exact torus stable-blowup statement.

Score basis: Stability of a defocusing supercritical singularity requires detailed profile and perturbation control.

[Read manuscript](#) | OPDP ID 40394740 | [Family formalization notes](#)

OAI-M0709 | Family 372

Global Uniqueness for the Smooth Isotropic Elasticity Inverse Problem

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims that full static displacement-to-traction boundary data uniquely determine both smooth real Lamé parameters on every bounded smooth connected three-dimensional domain. Positivity requires shear modulus positive and three times the first parameter plus twice the shear modulus positive on the closure. This is an inverse identification theorem for linear isotropic elasticity, not uniqueness of nonlinear equilibrium under prescribed displacement in the original elasticity collection.

Archive: related target. The original 710-score record asks uniqueness of equilibrium solutions, not determination of material moduli from a boundary operator; its score is not inherited.

Score basis: Recovering two coupled smooth coefficients needs substantial elliptic-system inverse-problem control.

[Read manuscript](#) | OPDP ID 2000008 | [Family formalization notes](#)

OAI-M0710 | Family 373

A counterexample to the Monge ansatz for the three-marginal Coulomb cost

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a smooth compactly supported identical marginal for three-body Coulomb transport whose optimum cannot be realized by two measure-preserving maps. Its square-root density is also smooth. Nevertheless, preserving maps approach the Kantorovich optimum, so equality of infima survives despite nonattainment. The construction extends to inverse-power interactions in every dimension at least two. Thus the failure is of optimal Monge realization, not relaxation value.

Archive: same target. The candidate asks whether Monge-type Coulomb minimizers always exist; the claimed three-marginal smooth example negates that universal existence.

Score basis: An explicit smooth marginal must forbid all Monge attainers while admitting near-optimal preserving maps.

[Read manuscript](#) | OPDP ID 40403037 | [Family formalization notes](#)

OAI-M0711 | Family 374

Sharp One-Third Stability of Brenier Maps

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims sharp one-third Hölder stability of quadratic Brenier maps in integrable-square norm when their source is uniform on a compact convex body and targets lie in a fixed compact set. Distance between targets is measured in quadratic Wasserstein distance. Arbitrary atomic or singular targets are allowed. A three-atom family disproves a proposed uniform square-root exponent, so improved target smoothness is not silently assumed.

Archive: related target. The candidate's compact-target setup is visible, but the proposed exponent and full inequality are truncated.

Score basis: Sharp map-stability exponents need matching uniform estimates and a precise counterexample family.

[Read manuscript](#) | OPDP ID 40403036 | [Family formalization notes](#)

OAI-M0712 | Family 375

A positive resolution of De Giorgi's conjecture in dimension eight

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims that every entire eight-dimensional Allen-Cahn solution valued strictly between minus and plus one with a positive directional derivative is a planar heteroclinic. It also classifies stable seven-dimensional solutions without an energy-growth assumption as constant wells or planar heteroclinics. The two dimensional and stability conditions are different. The reported local equation is not a nonlocal fractional De Giorgi variant.

Archive: related target. The up-to-eight archive record truncates equation and boundedness details; the nonlocal stable candidate is a different operator.

Score basis: Borderline-dimension rigidity requires universal stable-profile classification without standard growth control.

[Read manuscript](#) | OPDP ID 40395135

OAI-M0713 | Family 376

Finite Instructions and Solenoidal Shear Flows

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims realization of finite reciprocal affine instruction maps by smooth incompressible shear flows on a three-torus. A reversible recorder then makes a fixed particle enter a fixed strip exactly when a chosen machine halts, from zero velocity and pressure, with mean-zero solenoidal forcing periodic after initialization. Whole closed rectangles and intermediate trajectories are controlled, preventing an endpoint-only simulation from bypassing unintended detector entries.

Archive: related target. The candidate asks physical vortex-coherence mechanisms, not effective machine simulation by tailored forcing.

Score basis: Effective coding must preserve incompressibility and observation equivalence during every intermediate trajectory.

[Read manuscript](#) | OPDP ID 40394139 | [Family formalization notes](#)

OAI-M0714 | Family 376

A Fixed Particle Test for Computation in a Forced Viscous Flow

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims a fixed-particle halting test in forced viscous flow on a flat three-torus at every fixed positive computable viscosity. Initial velocity is zero, and mixed derivatives of forcing and velocity are bounded and square-integrable in time. Ordinary machine instructions are recorded in a third coordinate; increasingly fine spatial gates use longer time steps. The construction addresses reachability under designed forcing, not spontaneous turbulence or unforced regularity.

Archive: unresolved no match. Particle-method benchmarks and forced mean-curvature transport do not identify this Navier-Stokes computation target.

Score basis: All-time effective simulation must balance fine spatial coding with derivative integrability and viscous compensation.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0715 | Family 376

Universal Computation with Eventually Stationary Navier-Stokes Forcing

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims universal halting detection by a fixed particle in a torus Navier-Stokes flow whose smooth mean-zero forcing becomes stationary after time one. The fluid starts at rest at any fixed positive computable viscosity. A reversible recording table and spatial clock give a finite effective description. Separate variants use forcing periodic immediately or integrable time derivatives. Eventually stationary forcing does not mean the entire resulting velocity field is stationary.

Archive: related target. Leray's stationary boundary problem asks existence for different data and domains, not computation under an eventually stationary force.

Score basis: A finite spatial clock must sustain unbounded computation while producing an eventually stationary admissible force.

[Read manuscript](#) | OPDP ID 40397953 | [Family formalization notes](#)

OAI-M0716 | Family 376

Geometric Programs for Solenoidal Forcing

D 825/1000 | p 1 | m 1 | Conditional points 680.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This companion paper claims mean-zero solenoidal forcing periodic from time zero that realizes fixed-particle halting detection on a three-torus with zero initial velocity and pressure. Geometric subroutines implement diagonal maps on coding sheets with normal area compensation, and reciprocal maps on thick boxes. Initialization, inverses, and intermediate-path bounds are explicitly part of the simulation, distinguishing immediate periodicity from periodicity achieved only after a loading stage.

Archive: unresolved no match. Solenoidal Hodge theory and graph zero forcing are unrelated uses of solenoidal and forcing.

Score basis: Periodic initialization and full-domain incompressible instruction maps require precise geometric bookkeeping.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0717 | Family 376

Prefix Instructions and Incompressible Flows

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims smooth incompressible implementations of prefix instructions that can change area or erase information by storing their history. The main torus forcing is periodic after loading, starts the fluid at rest, and gives a fixed open-strip detector for machine halting. Alternative constructions use boxes, normal compensation, invariant planes, and spatial clocks. Effective cutoffs and derivative bounds specify the fields rather than encoding an unevaluated halting oracle.

Archive: related target. Bressan's mixing-cost target asks quantitative mixing bounds, not these effective instruction realizations.

Score basis: Information-preserving histories and full-domain incompressible transport must maintain effective control at arbitrarily fine scales.

[Read manuscript](#) | OPDP ID 40326502 | [Family formalization notes](#)

OAI-M0718 | Family 376

Computation under Rapidly Vanishing Navier-Stokes Forcing

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims halting detection in viscous flows from rest while every mixed derivative of force and velocity decays faster than any inverse time power. Three memory schemes preserve prior records on separated spatial scales, with one physical time unit per machine step. An exact open detector accommodates a shrinking signal. Viscosity is fixed and positive, so rapid temporal decay must not be confused with a vanishing-viscosity limit.

Archive: related target. The archived target sends viscosity to zero and studies boundary-flow limits, a different asymptotic regime.

Score basis: Rapid decay must coexist with indefinitely growing effective memory and an exact reachability detector.

[Read manuscript](#) | OPDP ID 40402295 | [Family formalization notes](#)

OAI-M0719 | Family 376

Velocity-Field Detection of Computation in Forced Navier-Stokes Flows

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims halting detection through the velocity field rather than a tracked particle. A torus construction tests one vertical component using shrinking stirring intervals; an open-cylinder construction uses an integral detector and expanding translation regions with globally bounded force derivatives. Vertical velocity obeys an advection-diffusion equation, so short-burst and escaped-mass estimates control detector errors. These are observer variants within explicitly designed smooth forcing classes.

Archive: related target. Batchelor-scale dissipation asks a statistical scale law, not effective velocity-field halting detection.

Score basis: Diffusion errors must be bounded strongly enough to preserve exact machine-observation equivalence.

[Read manuscript](#) | OPDP ID 40371823 | [Family formalization notes](#)

OAI-M0720 | Family 376

Scalar Potentials and Slow Clocks for Forced Fluid Computation

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims compact-spatial-support forcing of three-dimensional viscous flow from rest whose particle at the origin detects halting by entering a fixed half-space. Mixed derivatives decay at inverse time to one plus time-derivative order, using an unbounded logarithmic clock. Three scalar potentials lift, execute, and lower coded rectangles. Separate processors give other time regimes and observer variants, but the principal target combines fixed support with quantified decay.

Archive: unresolved no match. Complex-potential curves and packing universal-optimality questions are unrelated scalar-potential targets.

Score basis: Slow-clock rescaling must preserve both effective instruction transport and spatial-support and derivative estimates.

[Read manuscript](#) | [Family formalization notes](#)

OAI-M0721 | Family 376

Incompressible Box Transport and Finite Computation

D 850/1000 | p 1 | m 1 | Conditional points 722.5

Provisional OPDP / CFSD-1000 reconstruction (C0)

The manuscript claims effective incompressible transport of entire solid rational boxes under positive diagonal determinant-one maps. Source and target families are internally disjoint but may overlap each other; evacuation and detours handle that overlap. A balanced recorder then gives compact-support Navier-Stokes forcing periodic after loading and a fixed open-cube halting detector. Further clock and observer variants have separate bounds and comparison classes.

Archive: related target. The mailing-transport candidate concerns optimal-plan graph structure, not incompressible box transport or machine coding.

Score basis: Whole-box inverses, overlapping transport, and all-time detector control impose substantial geometric and effective-construction dependencies.

[Read manuscript](#) | OPDP ID 40402747 | [Family formalization notes](#)

OAI-M0722 | Family 377

Uniform Interior $C^{1,\alpha}$ Estimates for Infinity-Harmonic Functions

D 925/1000 | p 1 | m 1 | Conditional points 855.625

Provisional OPDP / CFSD-1000 reconstruction (C0)

This paper claims uniform interior differentiability with a positive Hölder gradient exponent for bounded infinity-harmonic functions in every dimension at least three. The exponent depends only on dimension, lies at most one-third, and is not explicit. Gradient size and seminorm are controlled by oscillation on a larger ball. Neither endpoint exponent nor boundary regularity is claimed; the assertion concerns Euclidean interior estimates rather than constructing a general Riemannian theory.

Archive: related target. The candidate asks existence of a theory for general Riemannian metrics, not this Euclidean interior regularity theorem.

Score basis: Uniform higher-dimensional gradient regularity requires delicate degenerate nonlinear PDE estimates.

[Read manuscript](#) | OPDP ID 40348274